Youth Competition Times

# Chemical Engineering

## Previous Years
## Chapterwise Objective
## Solved Paper

**Useful for :** **UPPSC Poly.Lect, UPPSC AE, UPSC Poly.Lect, ASSAM PSC, APPSC, BPSC, CGPSC, GATE, GUJRAT PSC, HP PSC, ISRO SCIENTIST/ENGINEER, JPSC, J&K PSC, KERLA PSC, KARNATAKA PSC, MAHARASHTRA PSC, MPPSC, OPSC, PUNJAB PSC, RPSC, TNPSC, TRB Poly. Lect., TELANGANA PSC, BHEL ET, CIL MT, RRB SSE, RRB JE, SJVNL ET, VIZAG STEEL MT.**

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Youth Competition Times
12, Church Lane Prayagraj-211002
**Mob. : 9415650134**
**Email : yctap12@gmail.com**
**website : www.yctbooks.com**

**Publisher Declaration**

Edited and Published by A.K. Mahajan for YCT Publications Pvt. Ltd.
and printed by Roop Printing Press.
In order to publish the book, full care has been taken by the editor and the publisher, still your suggestions and queries are welcomed.
**In the event of any dispute, the Judicial area will be Prayagraj.**

**Rs. : 595/-**

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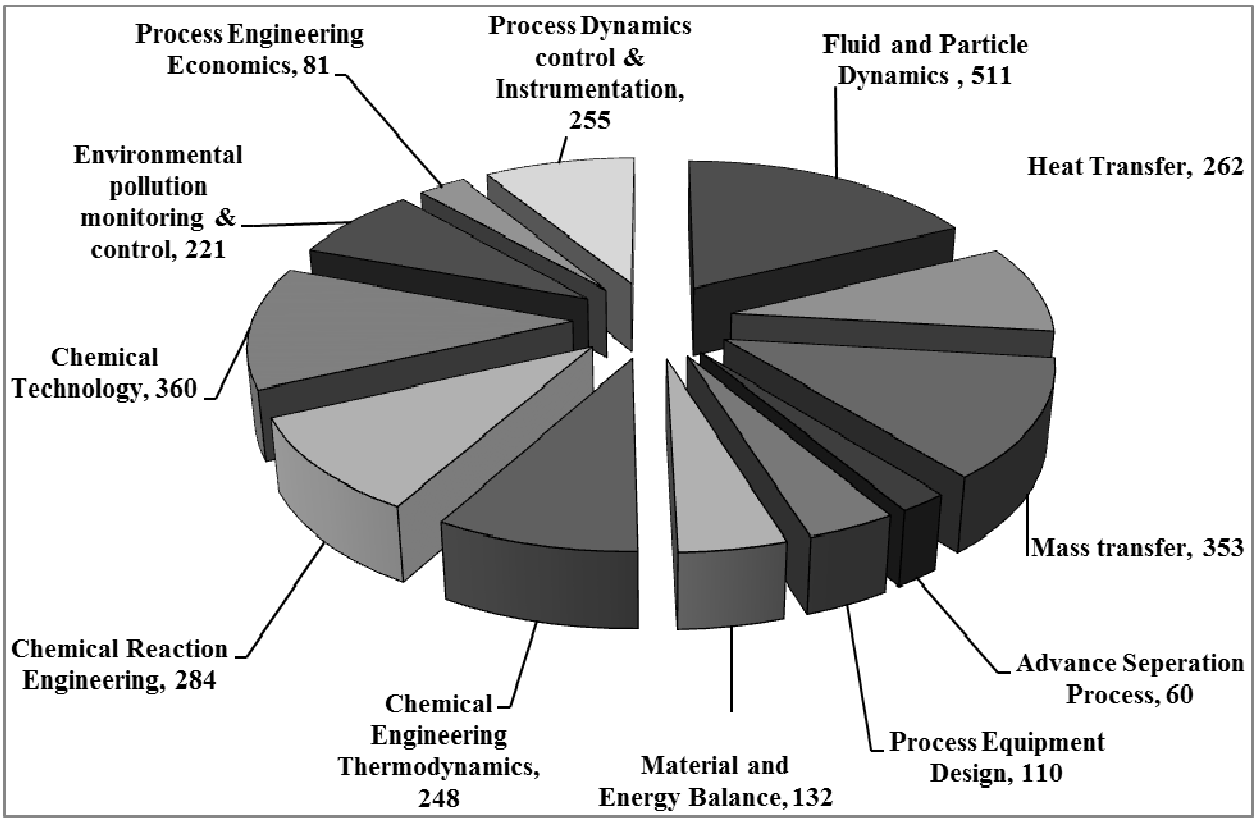
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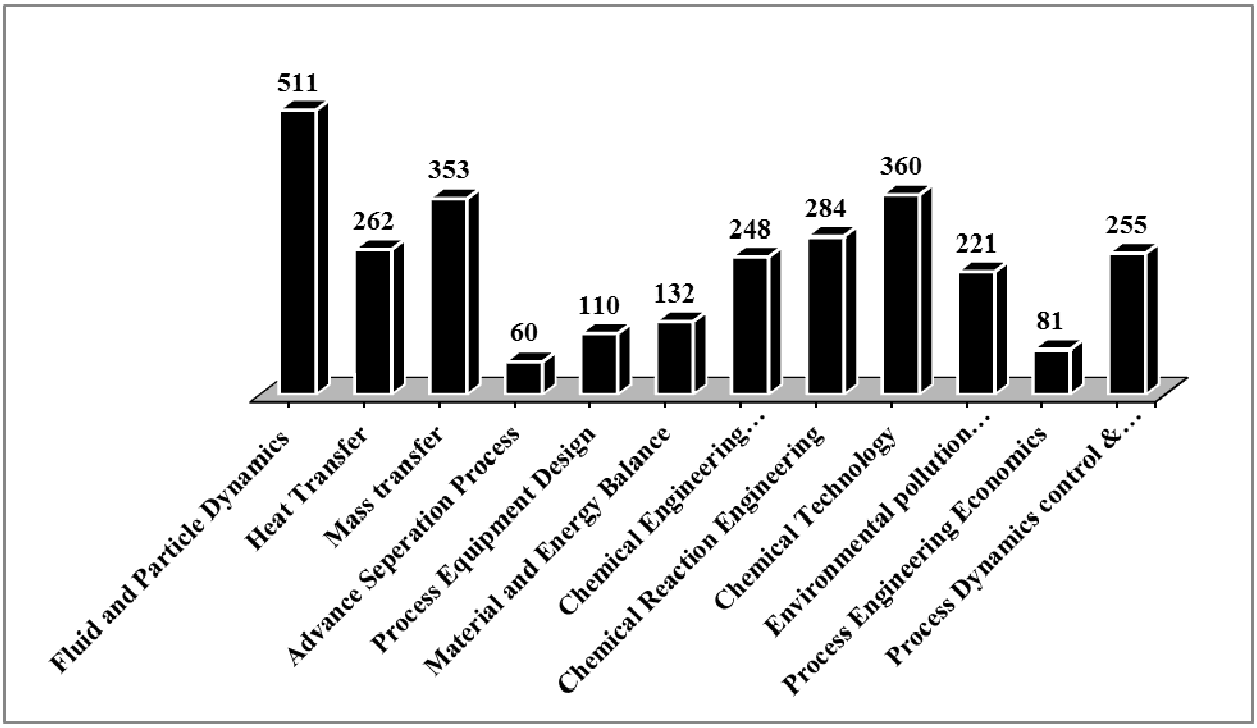
# Chemical Engineering
# PREVIOUS YEAR PAPERS ANALYSIS CHART

| S. No | Exam | Proposed Year | Question Paper | Total Question |
|---|---|---|---|---|
| **Assam PSC** | | | | |
| 1. | Assam PSC CCE Pre | 2015 | | 100 |
| **Andhra Pradesh PSC** | | | | |
| 2. | APPSC Poly. Lect. | 14.03.2022 | | 150 |
| 3. | APPSC IOF | 2009 | | 150 |
| **Fact Management Trainee** | | | | |
| 4. | Fact Management Trainee | 2022 | | 100 |
| **Graduate Aptitude Test in Engineering (GATE)** | | | | |
| 5. | GATE | 2011 | | 50 |
| 6. | GATE | 2012 | | 50 |
| 7. | GATE | 2013 | | 50 |
| 8. | GATE | 2014 | | 50 |
| 9. | GATE | 2015 | | 50 |
| 10. | GATE | 2016 | | 50 |
| 11. | GATE | 2017 | | 50 |
| 12. | GATE | 2018 | | 50 |
| 13. | GATE | 2019 | | 50 |
| 14. | GATE | 2020 | | 50 |
| 15. | GATE | 2021 | | 50 |
| 16. | GATE | 2022 | | 50 |
| **Gujarat Public Service Commission** | | | | |
| 17. | GPSC Technical Advisor (Chemical) Class-1, | 28.04.2022 | | 200 |
| **Haryana PSC** | | | | |
| 18. | Haryana Poly. Lect. | 2022 | | 100 |
| **Hindustan Petroleum Corporation Limited** | | | | |
| 19. | HPCL Engineering | 01.11.2022 | | 80 |
| **ISRO Scientist/Engineer** | | | | |
| 20. | ISRO Scientist/Engineer | 08.02.2015 | | 80 |
| 21. | ISRO Scientist/Engineer | 12.02.2016 | | 80 |
| 22. | ISRO Scientist/Engineer | 13.03.2016 | | 80 |
| 23. | ISRO Technical/Assistant | 06.08.2017 | | 80 |

| | | | | |
|---|---|---|---|---|
| 24. | ISRO Scientist/Engineer | 08.04.2018 | | 80 |
| 25. | ISRO Scientist/Engineer | 25.02.2018 | | 80 |
| 26. | ISRO Technical/Assistant | 25.02.2018 | | 80 |
| 27. | ISRO Scientist/Engineer | 2021 | | 80 |
| 28. | ISRO Scientist/Engineer | 04.06.2022 | | 80 |
| 29. | ISRO Technical/Assistant | 04.06.2022 | | 80 |
| **Karnataka Public Service Commission** | | | | |
| 30. | Karnataka PSC ADF | 2015 | | 100 |
| **Kerala PSC** | | | | |
| 31. | Kerala PSC Asstt. Prof. | 2017 | | 100 |
| 32. | Kerala PSC Poly. Lect. | 2015 | | 100 |
| **MP PSC** | | | | |
| 33. | MP PSC Forest Service | 2014 | | 150 |
| **Odisha Public Service Commission** | | | | |
| 34. | OPSC Poly. Lect. | 2018 | Paper-I | 100 |
| 35. | OPSC Poly. Lect. | 2018 | Paper-II | 100 |
| **Rajasthan Public Service Commission** | | | | |
| 36. | RPSC ACF | 2011 | | 200 |
| 37. | RPSC Lect. | 15.01.2016 | | 100 |
| 38. | RPSC ACF & FRO | 24.02.2021 | | 120 |
| **Tamilnadu PSC** | | | | |
| 39. | TNPSC AE | 2022 | | 200 |
| 40. | TNPSC AE | 2019 | | 200 |
| 41. | TNPSC AE | 2018 | | 200 |
| 42. | TNPSC AE | 2007 | | 200 |
| **Union Public Service Commission** | | | | |
| 43. | UPSC Poly. Lect. | 2019 | | 100 |
| 44. | UPSC JWM | 2017 | | 100 |
| **Uttar Pradesh Service Commission** | | | | |
| 45. | UPPSC AE | 13.12.2020 | Paper-I | 100 |
| 46. | UPPSC AE | 13.12.2020 | Paper-II | 100 |
| 47. | UPPSC AE | 29.05.2022 | Paper-I | 100 |
| 48. | UPPSC AE | 29.05.2022 | Paper-II | 100 |
| **WB PSC** | | | | |
| 49. | WB PSC Scientific Assistant | 2019 | | 100 |
| | | | **Total** | **4850** |

# Trend Analysis of Previous Year Chemical Engineering Papers Through Bar Graph and Pie Chart



# 01.

# Fluid and Particle Dynamics

## A. Fluids and their properties

**1. When the fluid is at rest, the shear stress is:**

(a) Maximum (b) Minimum
(c) Zero (d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**2. Density of water is maximum at:**

(a) $0^oC$ (b) $20^oC$
(c) $4^oC$ (d) $100^oC$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**3. Dynamic viscosity has the dimensions as:**

(a) $MLT^{-2}$ (b) $ML^{-1}T^{-1}$
(c) $ML^{-1}T^{-2}$ (d) $MLT^{-1}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**4. Dilatant and Pseudoplastic fluids follow a power law**

$$\tau_v\, g_c = K' \left(\frac{du}{dy}\right)^{n'}$$

**Where constant K' is**

(a) flow consistency index
(b) non newtoniun flow correction factor
(c) flow behaviour index
(d) shear stress correction factor

**TNPSC AE-2018**

**Ans. (a)**

**5. For pseudoplastic liquids with intermediate range (between 10 to 100), the power consumed by six blade turbine compared to newtoniun fluid will be**

(a) less (b) more
(c) equal (d) not predicted

**TNPSC AE-2018**

**Ans. (a)**

**6. _____fluids break down under continued shear and on mixing give lower shear stress for a given shear rate.**

(a) Pseudoplastic (b) Rheopectic
(c) Thixotropic (d) Newtonian

**TNPSC AE-2019**

**Ans. (c)**

**7. For laminar flow of a shear-thinning fluid in a pipe if the volumetric flow rate is doubled, the pressure gradient will increase by a factor of**

(a) $\frac{1}{2}$ (b) $< 2$
(c) $> 2$ (d) 2

**TNPSC AE-2019**

**Ans. (d)**

**8. Dilatant fluids are said to be shear rate**

(a) thinning (b) thickening
(c) both (A) and (B) (d) independent

**TNPSC AE-2019**

**Ans. (b)**

**9. Match the correct sequences:**

| | | | |
|---|---|---|---|
| **I** | Sewage sludge | **P** | Pseudoplastic |
| **II** | Rubber latex | **Q** | Dilatant |
| **III** | Quick sand | **R** | Newtonian |
| **IV** | Water | **S** | Binghamplastic |

**Code:**

(a) I – Q, II – S, II – R, VI – P
(b) I – Q, II – P, III – R, IV – S
(c) I – R, II – P, III – Q, IV – S
(d) I – S, II – P, II – Q , IV – R

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**10. A lubricant 100 times more viscous than water would have a viscosity in Pa – s**

(a) 10 (b) 0.01
(c) 1 (d) 0.1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**11. Falling drops of rain acquire spherical shape on account of**

(a) Surface tension (b) Temperature
(c) Compressibility (d) Viscosity

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**12. Viscosity of gases increases with**

(a) increase in temperature
(b) decrease in temperature
(c) independent of temperature
(d) first decreases upto 150 $^oC$ and then increases with temperature

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**13. An example of Bingham plastic is**

(a) Gas
(b) Non-colloidal solution
(c) Sewage sludge
(d) Rubber latex

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**14. The apparent viscosity of a fluid is given by $0.007\left(\frac{dV}{dY}\right)^{0.3}$, where $\left(\frac{dV}{dY}\right)$ is the velocity gradient. The fluid is**

(a) Bingham plastic
(b) dilatant
(c) pseudoplastic
(d) thixotropic

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**15. The dynamic viscosity of water at room temperature will be close to unity in which of the following units?**

(a) Poise (b) Centistoke
(c) Milli N-s/$m^2$ (d) Pa-s

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**16. At a shear rate of $10s^{-1}$, the apparent viscosity of a non-Newtonian liquid was found to be 1 Pa s. At a shear rate of $100s^{-1}$, the apparent viscosity of the same liquid was found to be 0.5 Pa s. If the liquid follows power law behavior, the apparent viscosity (in Pa s) at a shear stress of 10 N $m^{-2}$ is ____(rounded off to two decimal place).**

**GATE-2018**

**Ans. (0.98 to 1.02)**

**17. Mach, number is the ratio of the speed of the**

(a) Fluid of that of the light
(b) Light of that of the fluid
(c) Fluid of that of the sound
(d) Sound of that of the fluid

**RPSC ACF-2011**

**Ans. (c)**

**18. Viscosity of a fluid with specific gravity 1.3 is measured to be 0.0034 Ns/$m^2$. Its kinematic viscosity, in $m^2/s$ is**

(a) $2.6 \times 10^{-6}$ (b) $4.4 \times 10^{-6}$
(c) $5.8 \times 10^{-6}$ (d) $7.2 \times 10^{-6}$

**RPSC ACF-2011**

**Ans. (a)**

**19. The shear stress at a point in a glycerine mass in motion if velocity gradient is 0.25 m/s/m, will be**

(a) 0.0236 kg/$m^2$ (b) 0.02036 kg/$m^2$
(c) 0.0024 kg/$m^2$ (d) None of these

**RPSC ACF-2011**

**Ans. (*)**

**20. A piece of metal of specific gravity 7 floats in mercury of specific gravity 13.6 what fraction of its volume is under mercury?**

(a) 0.5 (b) 0.4
(c) 0.515 (d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**21. The unit of viscosity in CGS system is commonly known as Poise. The dimension of Poise is**

(a) g/cm.$s^2$ (b) g.cm/s
(c) g.$cm^2$/s (d) g/cm.s

**RPSC ACF-2011**

**Ans. (d)**

**22. Newton's law of viscosity relates to.**

(A) Shear stress and Viscosity
(B) Velocity gradient and pressure intensity
(C) Shear stress and rate of angular deformation in a fluid
(D) Pressure gradient and rate of angular deformation

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**23. If two capillary tubes of diameter 0.5 mm and 1 mm are dipped in a pot containing mercury, then rise of mercury is.**

(A) Same in both the tubes
(B) Greater in 1 mm diameter tube
(C) Greater in 0.5 mm diameter tube
(D) Zero in both the tubes

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**24. The value of flow behaviour index 'n' in Power law equation, $\tau = k\left(\frac{du}{dy}\right)^n$, for Pseudoplastic fluid is**

(a) n = 1 (b) n < 1
(c) n > 1 (d) n = 0

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**25. Tooth paste is a**

(a) Bingham plastic (b) Pseudo plastic
(c) Newtonian liquid (d) Dilatant liquid

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**26. Eddy viscosity is a turbulent transfer of ——**

(a) Fluid (b) Heat
(c) Momentum (d) Pressure

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**27. Which of the following statements are correct?**

**P : For a rheopectic fluid, the apparent viscosity increases with time under a constant applied shear stress**

**Q : For a pseudoplastic fluid, the apparent viscosity decreases with time under a constant applied shear stress.**

**R : For a Bingham plastic, the apparent viscosity increases exponentially with the deformation rate.**

**S : For a dilatants fluid, the apparent viscosity increases with increasing deformation rate.**

(a) P and Q only (b) Q and R only
(c) R and S only (d) P and S only

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**28. Newton''s law of viscosity relates the**

**A. Shear stress and strain**
**B. Velocity gradient and pressure intensity**
**C. Shear stress and rate of angular deformation in fluid**
**D. Pressure gradient and rate of angular deformation**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**29. Unit of viscosity is**

**A. Poise** **B. Centipoise**
**C. $Ns/m^2$** **D. All of the above**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**30. Viscosity has the dimensions**

**A. $ML^{-1}T^{-1}$** **B. $M^{-1}LT$**
**C. $ML^{-1}T^{-3}$** **D. $MLT^{-3}$**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**31. Principle of fluid mechanics works on the utilization of……**

**A. Velocity**
**B. Accelerating mass**
**C. Volume**
**D. Work**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**32. A fluid is said to be ideal if it is**

(a) Viscous and compressible
(b) Viscous and incompressible
(c) Inviscous and compressible
(d) Inviscous and incompressible

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**33. Density of water is maximum at**

(a) 0ºC (b) 4ºC
(c) 20ºC (d) 100ºC

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**34. An example of non-Newtonian fluid may be**

(a) ideal gas
(b) thin liquid
(c) thick, long-chain hydrocarbon
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**35. The viscosity of atmospheric air may be about ____ centipoise.**

(a) 1.5 (b) 15
(c) 150 (d) 0.015

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**36. With increase in the temperature, viscosity of a liquid**

(a) increases (b) decreases
(c) remains constant (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**37. The dimension of dynamic viscosity is**

(a) $ML^{-1}T^{-1}$ (b) $L^2T^{-1}$
(c) $LT^{-2}$ (d) $ML^{-1}T^{-2}$

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

**38. Viscosity of the gases**

(a) Increases with temperature and pressure
(b) Decreases with temperature
(c) Decreases with pressure
(d) None of the above

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**39. The ratio of dynamic viscosity to density is called**

(a) Absolute viscosity (b) Intrinsic viscosity
(c) Kinematic viscosity (d) Viscosity index

**ISRO-12.02.2016**

**Ans. (c)**

**40. Which of the following is correct?**

(a) Rate = Resistance / Driving force
(b) Driving force = Rate / Resistance
(c) Resistance = Driving force × Rate
(d) Rate = Driving Force / Resistance

**ISRO-12.02.2016**

**Ans. (b)**

**41. Time-dependent fluids are**

(a) Thixotropic and Rheopectic liquids
(b) Newtonian fluids
(c) Both (a) and (b)
(d) None of the above

**ISRO-12.02.2016**

**Ans. (a)**

**42. The general equation relating shear stress and shear rate for steady-state laminar flow of a fluid is given by: Shear stress = –μ (Shear rate)$^n$. Depending upon the nature of fluid n may vary from less than 1 to more than 1. What will be its value for 1% solution of carboxy methyl cellulose in water?**

(a) > 1 (b) 1
(c) < 1 (d) 0

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**43. In an incompressible fluid, density**

(a) Is greatly affected by moderate changes in pressure
(b) Is greatly affected by moderate changes in temperature
(c) Remains unaffected with moderate changes in temperature and pressure
(d) Is dependent on changes in both the temperature and pressure

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**44. An ideal fluid is**

(a) Frictionless and incompressible
(b) One which obeys Newton's law of viscosity
(c) Highly viscous
(d) None of the above

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**45. Newton's law of viscosity, Fourier's law of heat conduction and Fick's law of diffusion have a common term that is similar and has the same unit in all three cases. What is this term?**

(a) Driving force
(b) Resistance
(c) Flux
(d) Proportionality constant

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**46. The catalyst used in the ammonia synthesis is :**

(a) Aluminium oxide (b) Iron oxide
(c) Zeolite (d) Copper oxide

**AP PSC IOF-2009**

**Ans. (a)**

**47. Tooth paste is an example of a fluid that exhibits a behaviour of :**

(a) Pseudoplastics (b) Bingham plastics
(c) Newtonian (d) Dilatant

**AP PSC IOF-2009**

**Ans. (b)**

**48. The dimensions of kinematic viscosity :**

(a) $L^2T^{-1}$ (b) MLT
(c) $ML^{-1}T$ (d) $L^2T^{-2}$

**AP PSC IOF-2009**

**Ans. (a)**

**49. Which of the following statements are CORRECT?**

P. For a rheopectic fluid, the apparent viscosity increases with time under a constant applied shear stress
Q. For a pseudoplastic fluid, the apparent viscosity decreases with time under a constant applied shear stress
R. For a Bingham plastic, the apparent viscosity increases exponentially with the deformation rate
S. For a dilatant fluid, the apparent viscosity increases with increasing deformation rate

(a) P and Q only
(b) Q and R only
(c) R and S only
(d) P and S only

**GATE-2014**

**Ans. (d)**

**50. The distance between two parallel plates is 0.00914 m, the lower plate is being pulled at a relative velocity of 0.366 m/s greater than the top plate. The fluid used is soyabean oil with viscosity of $0.4 \times 10^{-2}$ P*a*.s at 303 K. The shear stress will be nearly:**

(a) 0.12 $N/m^2$ (b) 0.14 $N/m^2$
(c) 0.16 $N/m^2$ (d) 0.18 $N/m^2$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**51. When the momentum flux is less than the value of $\tau_0$, the velocity gradient for a Bingham fluid, will be:**

(a) Two (b) One
(c) Half (d) Zero

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**52. The viscosity of a gas:**
(a) Increases with increase in temperature
(b) Decreases with increase in temperature
(c) Remains unaffected with change in temperature
(d) Is greater than the viscosity of a liquid

**MPPSC Forest Service Exam-2014**

**Ans. (a) : Increases with increase in temperature**1.b Fluid statics

**53. A vacuum of 100mm Hg is equivalent to an absolute pressure of:**
(a) 100 mm Hg
(b) 660 mm Hg
(c) 860 mm Hg
(d) 100/760 mm Hg

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 660 mm Hg

## B. Fluid statics

**54. The hydrostatic law states that rate of increase of pressure in a vertical equation is equal to:**
(a) Density of the fluid
(b) Specific weight of the fluid
(c) Weight of the fluid
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**55. Atmospheric pressure held in terms of water column is:**
(a) 7.6 m (b) 9.81 m
(c) 8.23 m (d) 10.30 m

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**56. Which of the following is used for low range pressure Measurement.**
(a) Venturimeter (b) Orifice meter
(c) U-tube manometer (d) Pitot tube

**TNPSC AE-2018**

**Ans. (c)**

**57. The hydrostatic law states that the rate of increase of pressure in a vertical direction is equal to**
(a) density of the fluid
(b) specific weight of the fluid
(c) viscosity
(d) surface tension
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**58. Depth of oil having specific gravity 0.6 to produce a pressure of 3.6 kg/cm$^2$ will be**
(a) 40 cm (b) 36 cm
(c) 50 cm (d) 60 cm

**RPSC ACF-2011**

**Ans. (d)**

**59. Pressure at an elevation of 2000 m in an isothermal atmosphere assuming T = 20 ºC & $P_{atm}$ = 100 kPa is**
(a) 87 kPa (b) 82 kPa
(c) 79 kPa (d) 21 kPa

**RPSC ACF-2011**

**Ans. (c)**

**60. A dam of width 50 m is used to hold water in a reservoir. If the water height is 10 m from the bottom of the dam, what is the total force (F) acting on the dam due to the water? (Assume g = 10 m/sec$^2$ and the fluid density is 1000 kg/m$^3$**
(A) F = 12.5 * $10^6$ N
(B) F = 25 * $10^6$ N
(C) F = 50 * $10^6$ N
(D) F = 5 * $10^6$ N
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**61. A U tube manometer filled with mercury is connected between points in a pipeline. If the manometer reading is 20 cm of mercury (difference in mercury column height). Calculate the pressure difference between the points when water is flowing in the pipe.**
**Density of mercury = 13600 kg/m$^3$**
**Density of water = 1000 kg/m$^3$**
**g can be taken as 10m/s$^2$**
(a) 27.2 KN/m$^2$ (b) 25.2 KN/m$^2$
(c) 2 KN/m$^2$ (d) 20.2 KN/m$^2$

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**62. Differential manometer measures the**
**A. Absolute pressure**
**B. Gauge pressure**
**C. Pressure difference**
**D. Pressure gradient**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**63. Which of the following is the statement of Pascal's law?**
(a) Under static conditions pressure above datum on left limb is equal to pressure above datum on right limb.
(b) The force acting at a point in a static fluid is same in all directions.
(c) The pressure acting at a point in a static fluid is given by P = hρg.
(d) Absolute pressure is equal to gauge pressure plus atmospheric pressure.

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**64. Absolute pressures are _____.**

(a) Always positive
(b) Always negative
(c) May be positive or negative
(d) Measured with respect to local atmospheric pressure

**Karnataka PSC ADF - 2015**

**Ans. (a)**

**65. The centre of pressure is**

(a) Always above the centroid of the area.
(b) Always below the centroid of the area.
(c) Always at the centroid of the submerged area.
(d) Always a point on the line of action of the resultant force.

**Karnataka PSC ADF - 2015**

**Ans. (d)**

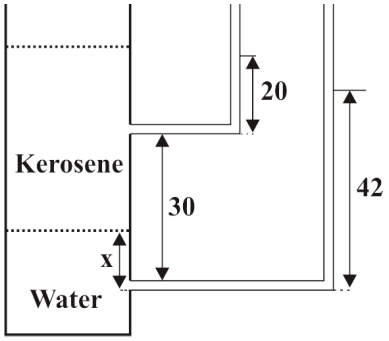
**66. The pressure intensity is the same in all directions at a point in fluid**

(a) Only when the fluid is frictionless.
(b) When there is no motion of one fluid layer relative to an adjacent layer.
(c) Only when the fluid is at rest having zero velocity.
(d) Regardless of the motion of one fluid layer relative to an adjacent layer.

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**67. A vertical cylindrical vessel has a layer of kerosene (of density 800 kg/m$^3$) over a layer of water (of density 1000 kg/m$^3$). L-shaped glass tubes are connected to the column 30 cm apart. The interface between the two layers lies between the two points at which the L-tubes are connected. The levels (in cm) to which the liquids rise in the respective tubes are shown in the figure below.**




**The distance (x in cm, rounded off to the first decimal place) of the interface from the point at which the lower L-tube is connected is ——**

**GATE-2016**

**Ans. (9.9 to 10.1)**

## C. Fluid Kinematics

**68. Study of fluid motion with the forces causing the flow is known as:**

(a) Kinematics of fluid flow
(b) Dynamics of fluid flow
(c) Statics of fluid flow
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**69. Continuity equation deals with the law of conservation of:**

(a) Mass (b) Momentum
(c) Energy (d) All of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**70. The equation of continuity applies to**

(a) Incompressible fluids
(b) Compressible fluids
(c) Highly viscous fluids
(d) Both incompressible and compressible fluids

**RPSC ACF-2011**

**Ans. (d)**

**71. The point of view of fluid flow that is concerned with the field of flow is called:**

(a) Lagrangian description
(b) Reynolds flow
(c) Newtonian description
(d) Eulerian method of description

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**72. Equation of continuity is a mathematical expression for**

(a) Law of conservation of mass
(b) Law of conservation of momentum
(c) Law of conservation of energy
(d) None of these

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**73. Continuity equation is based on :**

(a) Newton's law of viscosity
(b) Newton's law of cooling
(c) The law of conservation of mass
(d) Newton's law of motion

**AP PSC IOF-2009**

**Ans. (c)**

**74. In a steady incompressible flow, the velocity distribution is given by $\vec{V} = 3x\hat{i} - Py\hat{j} + 5z\hat{k}$, where, V is in m/s and x, y and z are in m. In order to satisfy the mass conservation, the value of the constant P (in s$^{-1}$) is ———**

**GATE-2014**

**Ans.** (7.99 to 8.01)

**75. The continuity equation in fluid mechanics is a mathematical statement using:**

(a) The principle of conservation of momentum
(b) The principle of conservation of mass
(c) The principle of conservation of energy
(d) The principle of conservation of mass and energy

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** The principle of conservation of mass

## D. Fluid Dynamics

**76. Ratio of maximum to average velocity in case of streamline flow between parallel plates is:**

(a) 1 (b) 1.5
(c) 2 (d) 2.5

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**77. Bernoulli's equation for steady frictionless flow states that, along a streamline:**

(a) Total pressure is constant
(b) Total velocity is constant
(c) Total mechanical energy is constant
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**78. For an ideal fluid the Reynold's number is :**

(a) Infinity (b) Zero
(c) one (d) < 2100

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**79. Navier-Stokes equation is of the form**

(a) $\rho\frac{DV}{Dt} = -\nabla_P + \rho g$

(b) $\rho\frac{DV}{Dt} = -\nabla_P + \mu\nabla^2 V + \rho g$

(c) $\frac{D\rho}{Dt} = -\rho(\nabla.V)$

(d) $\rho\frac{\partial\rho}{\partial t} = -(\nabla.\rho \vee V) - \nabla_P - \nabla.\tau + \rho g$

**TNPSC AE-2018**

**Ans. (b)**

**80. Flow profile is laminar is**

(a) Batch reactor (b) Tubular reactor
(c) Plug-flow reactor (d) Stirred tank reactor

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**81. Bernoulli's equation is a mathematical expression of**

(a) the ratio of kinetic to viscous forces in a flow stream
(b) friction loss as fluid moves through a rough pipe
(c) potential and kinetic energies in a flow stream
(d) fluid density and compressibility in a restriction

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**82. Path followed by water jet issuing from the bottom of a water tank will be a**

(a) Parabola (vertex being at the opening)
(b) Hyeperbolic
(c) Horizontal straight line
(d) zig-zig path (which is geometrically undefined)

**RPSC ACF-2011**

**Ans. (a)**

**83. Bernoulli's equation cannot be applied when the flow is**

(a) Rotational (b) Turbulent
(c) Unsteady (d) All of the above

**RPSC ACF-2011**

**Ans. (d)**

**84. For an ideal fluid flow, the Reynolds number is**

(a) 2100 (b) 100
(c) 0 (d) Infinity

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**85. Bernoulli's equation for steady frictionless flow states that along a streamline.**

(a) Total pressure is constant
(b) Total mechanical energy is constant
(c) Velocity head is constant
(d) None of the above

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**86. Water is flowing at 1.5 lit/s in a 0.05 m diameter tube. What is the average velocity of the water in the pipe?**

(a) 0.764 m/s (b) 3.056 m/s
(c) 764 m/s (d) 3056 m/s

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**87. Which formula correctly represents the maximum discharge over a broad crested wier?**

(a) $Q_{max} = 1.84\ C_d\ b\ H^{5/2}$
(b) $Q_{max} = 1.84\ C_d\ b\ H^{3/2}$
(c) $Q_{max} = 1.705\ C_d\ b\ H^{3/2}$
(d) $Q_{max} = C_d\ b\ H^{5/2}$

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**88. By german tower process, styrene is polymerized through:**

(a) Solution polymerization
(b) Suspension polymerization
(c) Bulk polymerization
(d) Emulsion polymerization

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**89. During visual interpretation of Bernoulli's equation the hydraulic grad line (HGL) shows:**

(a) The height of the total Bernoulli constant
(b) The sum of the energy grade line (EGL) and velocity head
(c) The half of the height to which liquid would rise in a piezometer tube attached to the flow
(d) The height corresponding to elevation and pressure head

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**90. A 10-cm fire hose with a 3-cm nozzle discharges 1.5 $m^3$/min to the atmosphere. Assuming frictionless flow, find the change in velocity of fluid between the two points, point 1 at the hose & point 2 at nozzle?**

(a) 32.2m/s (b) 322 m/s
(c) 3.22m/s (d) 22.3m/s

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**91. The water flows through a pipe at a velocity of 2 m/s. The pressure gauge reading is 2 bar. The datum head is 2 m. Find the piezometric head. (Assume all Bernoulli's assumptions, Density of water = 1000 kg/$m^3$, g = 9.8 m/$s^2$).**

(a) 18.4 m (b) 28.6 m
(c) 22.4 m (d) 20.6 m

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**92. Converging-diverging nozzles are also known as……,**

**A. Pascal nozzle**
**B. Bernoulli's nozzle**
**C. Torricelli''s nozzle**
**D. De laval' nozzle**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**93. Laminar flow of a Newtonian fluid ceases to exist, when Reynolds number exceeds**

(a) 4000 (b) 2100
(c) 1500 (d) 3000

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**94. Turbulent flow generally occurs in cases involving**

(a) highly viscous fluid
(b) very narrow passages
(c) very slow motion
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**95. Hydraulic radius is the ratio of**

(a) wetted perimeter to flow area
(b) flow area to wetted perimeter
(c) flow area to square of wetted perimeter
(d) square root of flow area to wetted perimeter

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**96. Bernoulli's equation cannot be applied when the flow is**

(a) rotational
(b) turbulent
(c) unsteady
(d) all of the above

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**97. At high Reynolds number**

(a) Inertial forces control and viscous forces are unimportant
(b) Viscous forces predominate
(c) Inertial forces are unimportant and viscous forces control
(d) None of these

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

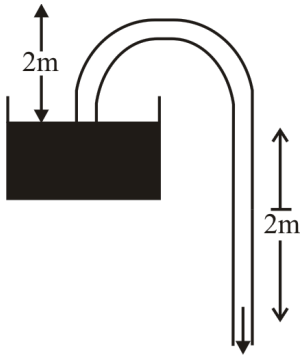
**98. Bernoulli's equation cannot be applied when the flow is**

(a) Turbulent (b) Rotational
(c) Unsteady (d) All of the above

**ISRO Scientist Engineer-06.08.2017**

**Ans. (d)**

**99. A siphon tube having a diameter of 2 cm draws water from a large open reservoir and discharges into the open atmosphere as shown in the figure. Assume incompressible fluid and neglect frictional losses. (g = 9.8 m/$s^2$)**




**The volumetric flow rate (in 1/s) of water at the discharge is**

(a) 3.11 (b) 6.67
(c) 30.77 (d) 42.99

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**100. Reynolds number for flow of water at room temperature through 4 cm diameter pipe at an average velocity of 5 cm/sec. is**

(a) 20 (b) 200
(c) 2,000 (d) 20,000

**ISRO-12.02.2016**

**Ans. (c)**

**101. Volumetric flow rate is equal to**

(a) The product of mass flow rate and density
(b) The product of velocity and area of cross section
(c) The product of diameter and length
(d) The velocity divided by cross sectional area

**ISRO-12.02.2016**

**Ans. (b)**

**102. The state of steady flow occurs when**

(a) Conditions change steadily with time
(b) Conditions are the same at the adjacent points at any instant
(c) Conditions do not change with time at any point
(d) Rate of change of velocity is constant

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**103. Match List – I with List – II and select the correct answer using the code given below the lists.**

| List-I | List-II |
|---|---|
| A. Dulong equation | 1. Mechanical energy balance |
| B. Kirchhoff equation | 2. Gross calorific value of coal |
| C. Bernoulli equation | 3. Variation of heat reaction with temperature of constant pressure |
| D. Antonie equation | 4. Vapour pressure of liquids at a given temperature |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 2 | 1 | 4 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 4 | 1 | 2 | 3 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**104. In a steady and incompressible flow of a fluid (density = 1.25 kg $m^{-3}$), the difference between stagnation and static pressures at the same location in the flow is 30 mm of mercury (density = 13600 kg $m^{-3}$). Considering gravitational acceleration as 10 m $s^{-2}$, the fluid speed (in m $s^{-1}$) is ———**

**GATE-2014**

**Ans. (79 to 82)**

## E. Pipe flow and flowmetery

**105. Orifice-meter is used to measure:**

(a) Pressure at a point (b) Velocity at a point
(c) Discharge (d) Average velocity

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**106. The ratio of the area of jet of water at vena-contracta to the area of the orifice is known as:**

(a) Coefficient of discharge
(b) Coefficient of velocity
(c) Coefficient of contraction
(d) Coefficient of viscosity

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**107. Notch is a device used for measuring**

(a) Rate of flow through pipes
(b) Rate of flow through a small channel
(c) Velocity through a pipe
(d) Velocity through a small channel

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**108. When the pipes are connected in parallel, the total loss of head:**

(a) Is equal to the sum of the loss of head in each pipe
(b) Is same as in each pipe
(c) Is equal to the reciprocal of the sum of loss of head in each pipe
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**109. Pitot tube is used for the measurement of :**

(a) Pressure (b) Flow
(c) Velocity at a point (d) Discharge

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**110. Losses for flow through valves and fittings are expressed in terms of :**

(a) Shape factor
(b) Roughness factor
(c) Drag coefficient
(d) Equivalent length of a straight pipe

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**111. For the laminar flow of a fluid in a circular pipe of radius R, the Hagen-Poiseuille equation predicts that volumetric flow rate to be proportional to:**

(a) R (b) $R^2$
(c) $R^4$ (d) $R^{0.5}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**112. Vena contracta is defined as the cross section of minimum area, at which the streamlines of fluid changes from**

(a) An expansion to a contraction
(b) Entrance to exit of pipe
(c) A contraction to an expansion
(d) Laminar region to turbulent region

**TNPSC AE-2018**

**Ans. (c)**

**113. Orifice meter is used for measurement of**

(a) temperature (b) pressure
(c) rate of flow (d) level

**TNPSC AE-2019**

**Ans. (c)**

**114. The local velocity of a fluid along a streamline flow can be measured using**

(a) Pitot tube (b) Venturimeter
(c) Rotameter (d) Orificemeter

**TNPSC AE-2019**

**Ans. (a)**

**115. For a Newtonian fluid flowing in a circular pipe under steady state conditions in fully developed laminar flow, the fanning friction factor is**

(a) $0.046\ Re^{-0.2}$ (b) $0.0014 + \frac{0.125}{Re^{0.32}}$
(c) $\frac{16}{R_e}$ (d) $\frac{24}{R_e}$

**TNPSC AE-2019**

**Ans. (c)**

**116. The velocity distribution with respect to radius is_____with the open at the centerline of pipe for the flow of Newtonian fluid under laminar condition.**

(a) Log (b) Parabola
(c) Linear (d) Hyperbola

**TNPSC AE-2019**

**Ans. (b)**

**117. Fluid is flowing in a pipe with velocity 0.1 m/s, viscosity of fluid 0.04 kg/m/s, diameter of pipe 0.2 m, kinematic viscosity of fluid $4 \times 10^{-5}$ $m^2/s$. What is the friction factor of pipe ?**

(a) 0.064 (b) 0.032
(c) 0.008 (d) 0.016

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**118. For a well designed venturimeter, the venturi coefficient is about**

(a) 0.98 (b) 0.4
(c) 0.9 (d) 0.8

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**119. Hagen -Poiseuille equation applied for flow conditions**

(a) Compressible (b) Laminar
(c) Viscous (d) Turbulent

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**120. For laminar flow of a shear thinning liquid in pipe if the volumetric flow rate is doubled, the pressure gradient will increase by a factor of**

(a) 1/2 (b) 2
(c) >2 (d) <2

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**121. The turbulent flow through conduits, the velocity gradient is**

(a) small near the wall
(b) large near the wall
(c) large in the turbulent core
(d) small in the buffer zone
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**122. The relationship between flow (q) and head (h) is applicable for flow through a rectangular-shaped weir is**

(a) $q \propto h$ (b) $q \propto h^{1/2}$
(c) $q \propto h^{2/3}$ (d) $q \propto h^{3/2}$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**123. Water is flowing under laminar conditions in a pipe of length L. If the diameter of the pipe is doubled for a constant volumetric flow rate, the pressure drop across the pipe**

(a) decreases 2 times (b) decreases 16 times
(c) increases 2 times (d) increases 16 times
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**124. A Newtonian fluid (with $\rho$ -density, $\mu$ -viscosity) is flowing with velocity 'v' in a circular tube of diameter D and length L. For Laminar flow $\Delta P$ is proportional to**

(a) $\frac{L\rho v^2}{D}$ (b) $\frac{L\mu v}{D^2}$
(c) $\frac{D\rho v^2}{L}$ (d) $\frac{\mu v}{L}$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**125. If the flow through a circular clarifier is 500 cubic meter per hour and that the surface overflow rate is 50 $m^3/m^2$ day. What should be the clarifier diameter?**

(a) 17.48 m (b) 16.77 m
(c) 19.24 m (d) 20.25 m

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**126. For the annular flow, surface of zero momentum is closer to**

(a) Outer wall (b) Inner wall
(c) Exactly in between (d) Close to outer wall

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**127. For laminar flow in a pipe, the value of momentum correction factor (β)**

(a) 3/4 (b) 4/3
(c) 0 (d) 1

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**128. Reynolds analogy is**

(a) [Nu/(RePr)] + f/2 (b) [Nu/(RePr)] + 2/f
(c) [Re/(NuPr)] + f/2 (d) [Pr/(ReNu)] + f/2

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**129. For the fully developed laminar flow through the parallel plates, the ratio of maximum velocity to that of average velocity is**

(a) 2 (b) $\sqrt{2}$
(c) 4/3 (d) 3/2

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**130. For the flow measurement of a canal, which device is used?**

(a) Rectangular weir (b) Venturimeter
(c) Pitot tube (d) Rotameter

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**131. Water flows through a circular pipe having diameter 5 cm with a pressure of 2 bars and velocity of 2m/s. The kinetic head of water is equal to**

(a) 20.4 m (b) 0.101 m
(c) 0.204 m (d) 10.1 m

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**132. The pressure drop across an orifice meter is overestimated by 5%. What will be the percentage error while measuring the flow rate?**

(a) +2.47 (b) 5
(c) –2.47 (d) –5

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**133. In which of the following cases, it is possible for flow to occur from low pressure to high pressure?**

(a) Flow of liquid in a diverging pipe
(b) Flow of liquid in a converging pipe
(c) Flow of air downward in a pipe
(d) In a constant cross-section conduit

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**134. Pitot tube is used to measure**

(a) liquid level in a tank
(b) flow velocity at a point
(c) angular deformation
(d) vorticity

**GATE-2018**

**Ans. (b)**

**135. A venturi meter is installed to measure the flow rate of water in a 178 mm diameter (ID) pipe. The throat diameter is 102 mm, The differential pressure measured using a discharge coefficient = 0.98; water density = 1000 $kg/m^3$.**

**The volumetric flow rate to water (in $m^3/s$) is-—.**

**GATE-2018**

**Ans. (0.145 to 0.155)**

**136. Oil is being delivered at a steady flowrate through a circular pipe of radius $1.25 \times 10^{-2}$ m and length 10 m. The pressure drop across the pipe is 500 Pa.**

**The shear stress at the pipe wall, rounded to 2 decimal places, is _____ Pa.**

**GATE-2017**

**Ans. (0.30 to 0.33)**

**137. In a venturi meter, $\Delta P_1$ and $\Delta P_2$ are the pressure drops corresponding to volumetric flowrates $Q_1$ and $Q_2$. If $Q_2/Q_1 = 2$, then $\Delta P_2/\Delta P_1$ equals**

(a) 2 (b) 4
(c) 0.5 (d) 0.25

**GATE-2017**

**Ans. (b)**

**138. A simple pitot tube measures**

(a) Average velocity (b) Maximum velocity
(c) Point velocity (d) Static pressure

**RPSC ACF-2011**

**Ans. (c)**

**139. Shear stress in a fluid flowing in a round pipe**

(a) varies parabolically across the cross-section
(b) remains constant over the cross-section
(c) is zero at the center and varies linearly with the radius
(d) is zero at the wall and increase linearly to the center

**RPSC ACF-2011**

**Ans. (c)**

**140. Power loss in an orifice meter is**

(a) less than that in a venturi meter
(b) same as that in a venturi meter
(c) more than that in a venturi meter
(d) data insufficient, cannot be predicted

**RPSC ACF-2011**

**Ans. (c)**

**141. Which of the following is called 'domestic water meter' for measurement of flow?**

(a) Reciprocating piston type meter
(b) Dry gas meter
(c) Nutating disc meter
(d) Sealed drum meter

**RPSC ACF-2011**

**Ans. (c)**

**142. A rotameter, through which air at room temperature and atmospheric pressure is flowing, gives a certain reading for a flow rate of 100 cc/s. If helium (Molecular weight 4) is used and the rotameter shows the same reading, the flow rate is**

(a) 26 cc/s (b) 42 cc/s
(c) 269 cc/s (d) 325 cc/s

**RPSC ACF-2011**

**Ans. (c)**

**143. For turbulent fluid flow through pipes, the kinetic energy and momentum correction factors are practically equal to**

(a) 0.5 (b) 1
(c) 2 (d) 4

**RPSC ACF-2011**

**Ans. (b)**

**144. The flow meter which can be used to measure the flow rate of hostile acids or alkalis is**

(a) Rotameter
(b) Magnetic flow meter
(c) Orifice meter
(d) Venturi meter

**RPSC ACF-2011**

**Ans. (b)**

**145. For laminar flow in a pipe, V is equal to**

(a) $U_{max}$ (b) 0.5 $U_{max}$
(c) 0.25 $U_{max}$ (d) 2 $U_{max}$

**RPSC ACF-2011**

**Ans. (b)**

**146. If the pressure at the inlet of a pipe is 90 Kg/cm$^2$ & pressure drop over the pipe line is 10 Kg/cm$^2$, then efficiency of transmission is**

(a) 66.6% (b) 77.7%
(c) 55.5% (d) 88.8%

**RPSC ACF-2011**

**Ans. (d)**

**147. For laminar flow in a pipe, the fanning friction factor depends on Reynolds number according to the relation**

(a) f = 24/ Re (b) f = 18.5/ Re
(c) f = 16/ Re (d) None of the above

**RPSC ACF-2011**

**Ans. (c)**

**148. For laminar flow of Newtonian fluids in pipes, the ratio of average velocity to maximum velocity is equal to**

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

**RPSC ACF-2011**

**Ans. (d)**

**149. The head loss due to sudden expansion is.**

(A) $\left(V_1^2 - V_2^2\right)/2g_c$
(B) $\left(V_1 - V_2\right)^2/2g_c$
(C) $\left(V_1 - V_2\right)/2g_c$
(D) $\left(V_1^2 - V_2^2\right)/g_c$

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**150. Two pipes of same length and diameters d & 2d respectively are connected in series. The diameter of an equivalent pipe of same length is.**

(A) Less than d
(B) Between d & 1.5d
(C) Between 1.5d & 2d
(D) Greater than 2d

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**151. Oil is pumped through a 10 cm ID pipe at the rate of 3.14 × 10$^{-3}$ m$^3$/s. The pipe is 50 m long. Estimate the pressure drop assuming laminar flow.**

**Density of oil = 900 kg/m$^3$**
**Viscosity of oil = 1.2 Poise**

(a) 307 N/m$^2$ (b) 76800 N/m$^2$
(c) 3070 N/m$^2$ (d) 7680 N/m$^2$

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**152. The equivalent diameter of an annulus of inner and outer radii $R_1$ and $R_0$ respectively is.**

(a) $4(R_0 - R_1)$ (b) $\sqrt{(R_0.R_1)}$
(c) $2(R_0 - R_1)$ (d) $(R_0 + R_1)$

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**153. Assume that H is the total head at the inlet of pipe, $h_f$ is head lost due to friction in the pipe and Q is discharged fluid per second, The power transmitted through pipe is given by:**

(a) WQH/75 (b) WQ (H – $h_f$)/75

(c) WQ(H–$h_f$) (d) WQH

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**154. Identify the flow measurement device in which the fluid is passed through two U-shaped curved tubes that are vibrated at their natural frequency. This creates an alternating force that produces small elastic deformation in the tubes, From the magnitude of deformation, the mass flow rate may be calculated.**

(a) Vortex shedding meter

(b) U tube manometer

(c) Insertion meter

(d) Coriolis meter

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**155. The ratio of average fluid velocity to the maximum velocity in case of laminar flow of a Newtonian fluid in a circular pipe is**

(a) 0.66 (b) 1

(c) 2 (d) 0.5

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**156. The hydraulic radius for flow in a rectangular duct of cross-sectional dimensions H and W is**

(a) $\sqrt{\frac{HW}{\pi}}$ (b) $\frac{(HW)^2}{4(H+W)}$

(c) HW/[2(H + W)$^2$] (d) 2HW/(H + W)

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**157. A venturi meter (discharge coefficient = 0.98) is installed to measure the flow rate of water (density = 1000 kg/m$^3$) in a 200 mm diameter (ID) pipe. The throat diameter is 100 mm. The differential pressure measured by using a manometer is 150 kN/m$^2$. The volumetric flow rate of water (in m$^3$/s) is**

(a) 0.14 (b) 0.20

(c) 0.41 (d) 0.24

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**158. A circular pipe of I.D. 4 m is bifurcated into two pipes of I.D. 2 m each. If the average velocity of water flowing through the main pipe is 5 m/s, then the average velocity through the bifurcated pipe is**

(a) 20 m/s (b) 10 m/s

(c) 7.07 m/s (d) 5 m/s

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**159. The equivalent diameter for pressure drop is___ that for heat transfer.**

**A. Smaller than**

**B. Greater than**

**C. Equal to**

**D. Non related with**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**160. The ratio of average fluid velocity to the maximum velocity in case of laminar flow of a Newtonion fluid in a circular pipe is**

(a) 1 (b) 0.5

(c) 0.66 (d) 2

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**161. Shear stress in a fluid flowing in a round pipe**

(a) Varies parabolically across the cross-section

(b) Remains constant over the cross-section

(c) Is zero at the centre and varies linearly with the radius

(d) Is zero at the wall and increases linearly to the centre

**ISRO Scientist/Engineer-25.02.2018**

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**162. The ratio of maximum to average velocity in case of streamline flow between parallel plates is**

(a) 1 (b) 1.5

(c) 2 (d) 2.5

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**163. Drag force on the float of a rotameter**

(a) $\propto Q$ (b) $\propto \sqrt{Q}$

(c) $\propto Q^2$ (d) is constant

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**164. Flow measurement in open channels is done by**

(a) venturi meter (b) orifice meter

(c) weir (d) rotameter

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**165. Vena contracta formed during flow of a liquid through an orifice meter has**

(a) minimum liquid cross-section

(b) more diameter compared to orifice diameter

(c) minimum velocity of fluid stream

(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**166. If more than two branches of pipes are to be connected at the same point, then which of the following is used?**

(a) Elbow (b) Union
(c) Tee (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**167. The local velocity of a fluid along a streamline can be measured by**

(a) Pitot tube
(b) Venturi meter
(c) Rota meter
(d) Orifice meter

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**168. In case of a pressure driven laminar flow of a Newtonian fluid of viscosity (mu) through a horizontal circular pipe, the velocity of the fluid is proportional to**

(a) mu
(b) mu^ 0.5
(c) mu^ (−1)
(d) mu^ (−0.5)

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**169. Water is flowing under laminar conditions in a pipe of length L. If the diameter of the pipe is doubled, for a constant volumetric flow rate, the pressure drop across the pipe**

(a) Decreases 2 times (b) Decreases 16 times
(c) Increases 2 times (d) Increases 16 times

**ISRO Scientist/Engineer-13.03.2016**
**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**170. If more than two branches of pipes are to be connected at the same point, then use a/an**

(a) elbow (b) tee
(c) union (d) reducer

**ISRO-12.02.2016**

**Ans. (b)**

**171. Which of the following flow metering instruments is an area meter**

(a) Venturimeter
(b) Rotameter
(c) Pitot tube
(d) Hot wire anemometer

**ISRO-12.02.2016**

**Ans. (b)**

**172. For fully developed laminar flow of a fluid through a tube, the Hagen- Poiseuille equation gives the relation for average flow velocity in terms of tube radius, pressure drop across a given length and viscosity of fluid. If the maximum velocity is equal to $\Delta P R^2/4\mu L$, then the average velocity is given by**

(a) $V_{av} = V_{max}/4$ (b) $V_{av} = V_{max}/6$
(c) $V_{av} = V_{max}/2$ (d) $V_{av} = (V_{max})^{1/2}$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**173. With reference to fluid flow, which of the statements is/are correct?**

**1. For laminar flow of Newtonian fluid through a pipe, Fanning friction factor becomes half on doubling the volumetric flow rate.**

**2. Fanning friction factor accounts for both skin friction and form friction.**

**Select the correct answer using the code given below.**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**174. Equivalent length of a pipe fitting is**

(a) Length of a straight pipe having same heat transfer rate as the fitting.
(b) Length of pipe having same frictional loss as the pipe fitting
(c) Length exhibiting same change in velocity as the pipe fitting
(d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**175. What will be the equivalent hydraulic diameter of a rectangular duct of sides a and b?**

(a) $(a \times b)^2/(a + b)^2$ (b) $2(a \times b)/(a + b)$
(c) $(a \times b)/(a - b)$ (d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**176. Variable area meter used for a flow measurement :**

(a) Orifice meter (b) Rotameter
(c) Venturimeter (d) Manometer

**AP PSC IOF-2009**

**Ans. (b)**

**177. The hydraulic diameter of an annulus of inner and outer radii, $R_i$ and $R_o$ respectively is :**

(a) $4(R_o - R_i)$ (b) $\sqrt{R_o - R_i}$
(c) $2(R_o - R_i)$ (d) $R_o + R_i$

**AP PSC IOF-2009**

**Ans. (c)**

**178. Water is flowing at 50° F through a long horizontal plastic pipe of 3 inch inside diameter at a velocity of 8 ft/s. The density of water is 62.42 lb/ft$^3$ and the viscosity is $8.8 \times 10^{-4}$ lb/ft.s. The Reynolds number is :**

(a) $2 \times 10^6$ (b) $1.4 \times 10^5$
(c) $5.32 \times 10^3$ (d) $1.53 \times 10^4$

**AP PSC IOF-2009**

**Ans. (b)**

**179. The ratio of $V_{av}$ to $V_{max}$ for a laminar flow of Newtonian fluid in a circular tube is :**

(a) 0.75 (b) 2
(c) 0.5 (d) 0.66

**AP PSC IOF-2009**

**Ans. (c)**

**180. An incompressible fluid is flowing through a contraction section of length L and has a 1–D (x-direction) steady state velocity distribution, $u = u_0\left(1+\frac{2x}{L}\right)$. If $u_0$ = 2 m/s and L = 3m, the convective acceleration (in $m/s^2$) of the fluid at L is ——**

**GATE-2014**

**Ans. (7.99 to 8.01)**

**181. In case of a pressure driven laminar flow of a Newtonian fluid of viscosity (μ) through a horizontal circular pipe, the velocity of the fluid is proportional to**

(a) μ
(b) $\mu^{0.5}$
(c) $\mu^{-1}$
(d) $\mu^{-0.5}$

**GATE-2014**

**Ans. (c)**

**182. Two different liquids are flowing through different pipes of the same diameter. In the first pipe, the flow is laminar with centerline velocity, $V_{max.1}$, whereas in the second pipe, the flow is turbulent. For turbulent flow, the average velocity is 0.82 times the centerline velocity, $V_{max.2}$. For equal volumetric flow rates in both the pipes, the ratio $V_{max.1}$ /$V_{max.2}$ (up to two decimal place) is_____.**

**GATE-2015**

**Ans. (1.6 to 1.7)**

**183. The ratio of average velocity in a pipeline to the velocity at the centre of the pipe for laminar flow is**

(a) 2 (b) 0.5
(c) 0.87 (d) 0.7

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**184. Discharge of water ($m^3/s$) from 60 cm inner pipe diameter at 3m/s will be**

(a) 0.818 (b) 0.828
(c) 0.838 (d) 0.848

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**185. Vena-contracta formed during flow of a liquid through an orificemeter has**

(a) Minimum liquid cross-section
(b) More diameter compared to orifice diameter
(c) Minimum velocity of fluid stream
(d) Minimum viscosity of fluid stream

**Karnataka PSC ADF - 2015**

**Ans. (a)**

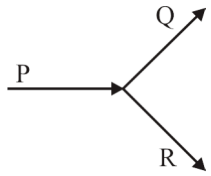
**186. Lower BWG means**

(a) Lower thickness tube
(b) Lower cross-section of tube
(c) Inner diameter of tube
(d) Outer diameter of tube

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**187. A liquid mixture of ethanol and water is flowing as inlet stream P into a stream splitter. It is split into two streams, Q and R, as shown in the figure below.**




**The flowrate of P, containing 30 mass% of ethanol, is 100 kg/h. What is the least number of additional specifications required to determine the mass flowrates and compositions (mass%) of the two exit streams?**

(a) 0 (b) 1
(c) 2 (d) 3

**GATE-2016**

**Ans. (b)**

**188. To avoid separation of flow in the divergence section of a venturimeter the limiting cone angle for the section is:**

(a) $8^0$ (b) $12^0$
(c) $15^0$ (d) $18^0$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $8^0$

**189. Which of the following is not a variable head meter:**

(a) V-notch (b) Venturimeter
(c) Rota meter (d) Pitot tube

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Rota meter

**190. For laminar flow of Newtonian fluid in a pipe of circular cross-section, the ratio of the maximum velocity to the average velocity is:**

(a) 2 (b) 0.8
(c) 0.5 (d) 1.2

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 2

## F. Boundary layer theory

**191. A fluid flowing over a flat plate has the following properties. Dynamic viscosity: 25 × $10^{-6}$ kg/ms, Specific heat: 2.0 kJ/kgK, Thermal conductivity : 0.05 W/MK. The hydrodynamic boundary layer thickness is measured to be 0.5 mm. The thickness of thermal boundary layer would be:**

(a) 0.1 mm (b) 0.5 mm
(c) 1.0 mm (d) 0.023 mm

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**192. The boundary layer separation takes place if:**

(a) Pressure gradient is zero
(b) Pressure gradient is positive
(c) Pressure gradient is negative
(d) Both (A) and (C)

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**193. When the hydrodynamic and thermal boundary layers of a fluid flowing over a heated plate is identical, then the value of prandtl number of a gas is**

(a) <1 (b) 1.0
(c) >1 (d) 0.9

**TNPSC AE-2019**

**Ans. (b)**

**194. The boundary layer thickness in turbulent flow over a flat plate**

(a) increases as $x^{5/4}$ (b) Increases as $x^{1/5}$
(c) increases $x^{4/5}$ (d) increases as $x^{1/2}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**195. Thickness of thermal boundary layer is more compared to that of hydrodynamic boundary layer, when the value of Prandtl number is**

(a) 1 (b) > 1
(c) < 1 (d) > 5

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**196. For flow of fluid over a heated plate, the following fluid properties are known:**
**Viscosity = 0.001 Pa-s; Specific heat at constant pressure = 1 kJ/kg-K; Thermal conductivity = 1 W/m-K.**
**The hydrodynamic boundary layer thickness at a specified location on the plate is 1 mm. The thermal boundary layer thickness at the same location is**

(a) 0.001 mm (b) 0.01 mm
(c) 1 mm (d) 1000 mm

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**197. In a roll crusher, rolls of diameter 1 m each are a set in such a manner that minimum clearance between the crushing surfaces is 15 mm. If the angle of nip 31°, the maximum diameter of the particle (in mm) which can be crushed is ___(rounded off to third decimal place).**

**GATE-2018**

**Ans. ( 52 to 54)**

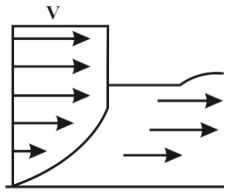
**198. The thickness of laminar boundary layer over a flat plate varies along the distance from the leading edge of the plate. As the distance increases, the boundary layer thickness**

(a) increases
(b) decreases
(c) initially increases and then decreases
(d) initially decreases and then increases

**GATE-2017**

**Ans. (a)**

**199. Considering the parallel flow of a fluid over a flat plate, boundary layer thickness, δ, is typically defined as:**




(a) Distance from the surface at which fluid velocity = v
(b) Distance from the surface at which fluid velocity >v
(c) Distance from the surface at which fluid velocity = 0.01v
(d) Distance from the surface at which fluid velocity = 0.99 v

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**200. If the thermal boundary layer is thinner than hydrodynamic boundary layer for a flat plate in a fluid, Prandtl number is:**

(a) Equal to 1 (b) Between 0.5 & 1
(c) More than 1 (d) Less than 0.5

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**201. The concept of boundary layer that provides important links between ideal-fluid flow and real-fluid flow was given by**

(a) Reynolds (b) Prandtl
(c) Mach (d) Hagen

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**202. If the thermal boundary layer has the same thickness as that of the hydrodynamic boundary layer for a flat plate in a fluid, Prandtl number is:**

(a) 1 (b) 1.5
(c) 0.5 (d) 0.2

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

**203. For uniform laminar flow over a flat plate, the thickness of the boundary layer, δ, at a distance x from the leading edge of the plate follows the relation:**

(1) $\delta(x) \propto x^{-1}$
(2) $\delta(x) \propto x$
(3) $\delta(x) \propto x^{1/2}$
(4) $\delta(x) \propto x^{-1/2}$

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**204. A Newtonian fluid which is in fully developed flow has the actual velocity distribution at the entrance to the heated section and the theoretical distribution throughout the tube. They both are:**

(a) Parabolic
(b) Circular
(c) Linear
(d) Elliptic

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

## G. Pumps blowers & fans

**205. Boiler feed water pump is usually a:**

(a) Reciprocating pump
(b) Gear pump
(c) Multistage centrifugal pump
(d) Diaphragm pump

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**206. During the suction stroke of a reciprocating pump, the separation may take place;**

(a) At the end of the suction stroke
(b) In the middle of the suction stroke
(c) In the beginning of the suction stroke
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**207. The flow of water, leaving the impeller in a centrifugal casing is:**

(a) Forced vortex flow (b) Free vortex flow
(c) Centrifugal flow (d) Centripetal flow

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**208. A centrifugal pump with air in its casing is air bound and displacing this air by drawing liquid into the suction line by an independent source to operate the pump is called as**

(a) Pump hammering (b) Pump priming
(c) NPSH of pump (d) Pump cavitation

**TNPSC AE-2018**

**Ans. (b)**

**209. The discharge line of a_____ pump can be completely closed without damage.**

(a) Reciprocating pump (b) Centrifugal pump
(c) Screw pump (d) Piston pump

**TNPSC AE-2019**

**Ans. (b)**

**210. Cavitation can be prevented by**

(a) Maintain the discharge head greater than the vapour pressure
(b) Maintaining the suction head sufficiently greater than the vapour pressure
(c) Maintaining suction head = developed head
(d) Maintaining suction head lower that the vapour pressure

**TNPSC AE-2019**

**Ans. (b)**

**211. Match the items in the following columns :**

| | |
|---|---|
| **(i)** Gear pump | **A.** Suspension |
| **(ii)** Air lift pump | **B**. Concentrated $H_2So_4$ |
| | **C.** Viscous oil |
| | **D.** Toluene |

(a) i – D, ii – B (b) i – C, ii – A
(c) i – B, ii – A (d) I – A, ii – C

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**212. With increasing flow rate, the hydraulic efficiency of centrifugal pump**

(a) Increases and then decreases
(b) Monotonically decreases
(c) Remains unchanged
(d) Decreases and then increases

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**213. To pump a corrosive fluid, which of the following pump should be chosen?**

(a) Diaphram pump (b) Centrifugal pump
(c) Piston pump (d) Plunger pump

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**214. Identify the pump used for leak proof towards handling dangerous liquids**

(a) Piston pump
(b) Plunger pump
(c) Canned-rotor centrifugal pump
(d) Gear pump
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**215. The Mechanical efficiency of a Reciprocating Pump varies from _____ to _____ for small pumps.**

(a) 70 to 90 % (b) 50 to 70 %
(c) 40 to 50 % (d) 30 to 40 %
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**216. To discharge large quantities of liquid in multistage centrifugal pump, the impellers are connected**

(a) in parallel
(b) in series
(c) in both parallel and series
(d) tangentially
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**217. A centrifugal pump having an impeller of diameter 120 mm, delivers a power of 12 HP. If the diameter of impeller is halved, what is the power, if other parameters are kept constant?**

(a) 6 HP (b) 2 HP
(c) 4 HP (d) 3 HP

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**218. Boiler feed pump is usually a**

(a) Reciprocating pump
(b) Gear pump
(c) Multi stage centrifugal pump
(d) Diaphragm pump

**RPSC ACF-2011**

**Ans. (c)**

**219. With increase in pump speed, its NPSH requirement**

(a) Decreases
(b) Increases
(c) Remains unaltered
(d) Can either increase or decrease, depends on other factors

**RPSC ACF-2011**

**Ans. (b)**

**220. Cavitation will not occur if the sum of the velocity and pressure heads at the suction is**

(a) Zero
(b) Much larger than the vapour pressure of the liquid
(c) Much smaller than the vapour pressure of the liquid
(d) Equal to the vapour pressure of the liquid.

**RPSC ACF-2011**

**Ans. (c)**

**221. Cavitation in pumps occur when NPSH is**

(a) $\leq 0$
(b) $\geq 0$
(c) Infinity
(d) Doesn't related to NPSH

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**222. In which equipment the mechanical, kinetic and potential energies do not change appreciably?**

(a) Centrifugal pump
(b) IC engine
(c) Blowers and compressors
(d) Boiler

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**223. While considering the performance of a centrifugal pump, the ratio of discharge (Q) with sum of discharge and the loss of fluid due to leakage ($Q + Q_L$) is termed as:**

(a) Mechanical efficiency
(b) Volumetric efficiency
(c) Hydraulic efficiency
(d) Overall efficiency

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**224. Circulation pump is located below the evaporator to**

**A. Avoid cavitation**
**B. Avoid frequent priming**
**C. Create more suction head**
**D. None of the above**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**225. Cavitation in a centrifugal pump results from**

**A. High discharge pressure**
**B. Low barometric pressure**
**C. High discharge velocity**
**D. High discharge rate**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**226. While designing the centrifugal blower, handling pressure is limited to**

(a) 0.5 psi (b) 5 psi
(c) 50 psi (d) 500 psi

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**227. In a centrifugal pump, the liquid enters the pump**

(a) at the centre (b) at the bottom
(c) at the top (d) from the sides

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**228. A turbine is used to convert**

(a) kinetic energy to mechanical energy
(b) mechanical energy to hydraulic energy
(c) hydraulic energy to mechanical energy
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**229. Priming is needed in a**

(a) reciprocating pump (b) gear pump
(c) centrifugal pump (d) diaphragm pump

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**230. Head developed by a centrifugal pump depends on its**

(a) speed (b) impeller diameter
(c) Both (a) and (b) (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**231. Rubber-lined pumps can be used to pump**

(a) caustic soda (b) chlorinated brine
(c) hypochloric acid (d) All of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**232. Which of the following produces maximum pressure difference for transportation of gases?**

(a) Vacuum pump (b) Blower
(c) Fan (d) Compressor

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**233. Plunger pumps are used for.**

(a) higher pressure (b) slurries
(c) viscous mass (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**234. A centrifugal pump does not pump liquid when the casing is filled with air because**

(a) Developed head is less.
(b) The increase in pressure of fluid is less.
(c) Increase in pressure is more.
(d) There is no increase in the pressure of fluid

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**235. The characteristics curve (Head-Capacity relationship) of a centrifugal pump is represented by the equation $\Delta H_{pump} = 43.8 - 0.19Q$, where $\Delta H_{pump}$ is the head developed by the pump (in m) and Q is the flowrate (in $m^3/h$) through the pump. This pump is to be used for pumping water through a horizontal pipeline. The frictional head loss $\Delta H_{piping}$ (in m) is related to the water flowrate $Q_L$ (in $m^3/h$) by the equation $\Delta H_{piping} = 0.0135 Q_L^2 + 0.045 Q_L$. The flowrate (in $m^3/h$, rounded off to the first decimal place) of water pumped through the above pipeline, is——**

**GATE-2016**

**Ans. (48.5 to 49.5)**

**236. Water (density = 1000 kg $m^{-3}$) is pumped at a rate of 36 $m^3/h$, from a tank 2 m below the pump, to an overhead pressurized vessel 10 m above the pump. The pressure values at the point of suction from the bottom tank and at the discharge point to the overhead vessel are 120 kPa and 240 kPa, respectively. All pipes in the system have the same diameter. Take acceleration due to gravity, g = 10 $ms^{-2}$. Neglecting frictional losses, what is the power (in kW) required to deliver the fluid?**

(a) 1.2 (b) 2.4
(c) 3.6 (d) 4.8

**GATE-2016**

**Ans. (b)**

**237. Two geometrically similar pump are run at the same speed. One pump lifts water at the rate of 20 litres per second against a head of 15 meters. The head of the other pump in metres to deliver half the discharge is**

(a) 4.73 (b) 0.5
(c) 9.45 (d) 2.38

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**238. The maximum depth from which a centrifugal pump can draw water is**

(a) 34 feet (b) 760 mm
(c) dependent on the speed of the pump
(d) dependent on the power of the pump

**ISRO-12.02.2016**

**Ans. (a)**

**239. For a centrifugal pump, capacity, head and power consumed are proportional to**

(a) $N^3$, N and $N^2$
(b) N, $N^2$ and $N^3$
(c) (1/N), ($1/N^2$) and ($1/N^3$)
(d) ($1/N^3$), (1/N) and ($1/N^2$)

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**240. Match the following**

| Group 1 | | Group 2 | |
|---|---|---|---|
| **P.** | Turbulence | I. | Reciprocating pump |
| **Q.** | NPSH | II. | Packed bed |
| **R.** | Ergun equation | III. | Fluctuating velocity |
| **S.** | Rotameter | IV. | Impeller |
| **T.** | Power number | V. | Vena contracta |

(a) P-III, R-II, T-IV
(b) Q-V, R-II, S-III
(c) P-III, R-IV, T-II
(d) Q-III, S-V, T-IV

**GATE-2014**

**Ans. (a)**

**241. Slurries are most conveniently pumped by a**
(a) syringe pump
(b) diaphragm pump
(c) vacuum pump
(d) gear pump

**GATE-2014**

**Ans. (b)**

**242. A centrifugal pump delivers water at the rate of 0.22 $m^3/s$ from a reservoir at ground level to another reservoir at a height H, through a vertical pipe of 0.2 m diameter. Both the reservoirs are open to atmosphere. The power input to the pump is 90 kW and it operates with an efficiency of 75%**
**Data:**
**Fanning fraction factor for pipe flow is $f$ = 0.004. Neglect other head losses.**
**Take gravitational acceleration, g = 9.8 $m/s^2$ and density of water is 1000 $kg/m^3$.**
**The height H, in meters, to which the water can be delivered (up to one decimal place) is _____**

**GATE-2015**

**Ans. (25 to 27)**

**243. Which one of the following pumps can be used for higher pressure delivery and metering?**
(a) Centrifugal pump
(b) Rotary pump
(c) Reciprocating pump
(d) Steam pump

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**244. Cavitation in pump reduces capacity and causes severe erosion. This occurs when:**
(a) Suction pressure is greater than the vapour pressure of the liquid
(b) Suction pressure is equal to vapour pressure
(c) Suction pressure is only greater than the vapour pressure
(d) Suction pressure is less than vapour pressure

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Suction pressure is only greater than the vapour pressure

**245. The purpose providing check valve in the discharge line of a centrifugal pump is:**
(a) To control the flow of fluid when pump is running
(b) To prevent back flow when pump is stopped
(c) To control the discharge pressure
(d) To control the discharge temperature

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** To prevent back flow when pump is stopped

**246. The compression ratio is maximum for:**
(a) Fans
(b) Blowers
(c) Compressors
(d) Vacuum pumps

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Vacuum pumps

## H. Flow past immersed bodies

**247. The drag force exerted by a fluid on a body immersed in the fluid is due to:**
(a) Pressure and viscous force
(b) Pressure and gravity force
(c) Pressure and turbulence force
(d) Gravity force and surface tension

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**248. When a falling body has attained terminal velocity, the weight of the body is equal to :**
(a) Drag force minus buyoant force
(b) Buyoant force minus drag force
(c) Drag force plus buyoant force
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**249. The theoretical time taken by a particle to pass between entry and exit of a sedimentation basin is called.**
(a) Setting period (b) Screening period
(c) Detention period (d) Cleaning period

**TNPSC AE-2018**

**Ans. (c)**

**250. In settling in the stokes law range, the terminal velocity $u_+$ is proportional to**
(a) $D_p$ (b) $D_p^{0.5}$
(c) $D_p^{1.5}$ (d) $D_p^2$

**TNPSC AE-2019**

**Ans. (d)**

**251. For free settling of a spherical particle, Stoke's law is valid for Reynold's number**
(a) < 2100 (b) < 1
(c) < 112 (d) < 10

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**252. Creeping flow is obtained only at Re of the order of**
(a) One thousand or more
(b) One thousand or less
(c) One or more
(d) One or less

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**253. The terminal velocity of a spherical particle in gravitational settling under Stokes' regime varies**
(a) linearly with the particle diameter
(b) linearly with the viscosity of the liquid
(c) directly with the square of particle diameter
(d) inversely with the density of particle

**GATE-2018**

**Ans. (c)**

**254. At low Reynolds number**
(a) Viscous forces are unimportant
(b) Viscous forces control
(c) Viscous forces control and inertial forces are unimportant
(d) Gravity forces control

**RPSC ACF-2011**

**Ans. (c)**

**255. A suspension of uniform particles in water at a concentration of 500 kg of solids per cubic meter of slurry is settling in a tank. Density of the particles is 2500 kg/m$^3$ and terminal velocity of a single particle is 20 cm/s. What will be the settling velocity of suspension? Richardson- Zaki index is 4.6**
(a) 20 cm/s (b) 14.3 cm/s
(c) 7.16 cm/s (d) 3.58 cm/s

**RPSC ACF-2011**

**Ans. (c)**

**256. As predicted by stokes' law, the drag coefficient, $C_D$, is given by**
(a) $16 / N_{Re}, p$ (b) $8 / N_{Re}, p$
(c) $24 / N_{Re}, p$ (d) $32 / N_{Re}, p$

**RPSC ACF-2011**

**Ans. (c)**

**257. Terminal velocity is**
(a) Constant velocity with no acceleration
(b) A fluctuating velocity
(c) Attained after moving one-half of total distance
(d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**258. In hindered settling, particles are**
(a) Placed farther from the wall
(b) Near each other
(c) Not affected by other particles and the wall
(d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**259. Forces acting on a particle settling in fluid are**
(a) Gravitational and buoyant forces
(b) Centrifugal and drag forces
(c) Gravitational or centrifugal, buoyant and drag forces
(d) External, drag and viscous forces

**Assam PSC-CCE-Pre-2015**
**RPSC ACF-2011**

**Ans. (c)**

**260. Stokes' law is valid (strictly theoretically) when particle Reynolds number is $N_{Re.p}$ is**
(a) Much less than one
(b) Equal to one
(c) Much greater than one
(d) None of the foregoing

**RPSC ACF-2011**

**Ans. (a)**

**261. Two spherical particles, one of dia $d_1$ and the other of dia $d_2$, settle freely through a pool of liquid and the settling is in accordance with Stoke's law. $d_1:d_2 = 1 : 2$. Therefore, $u_1 : u_2$ is equal to**
(a) 1 : 2 (b) 2 : 1
(c) 4 : 1 (d) 1 : 4

**RPSC ACF-2011**

**Ans. (d)**

**262. A particle 'A' of diameter 10 μ settles in an oil of specific gravity 0.9 and viscosity of 10 Poise under Stokes Law. A particle 'B' of diameter 20 μ settling in the same oil will have settling velocity,**
(A) Same as that of A
(B) $1/4^{th}$ that of A
(C) Twice as that of A
(D) 4 times as that of A
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**
**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**263. Terminal velocity of a particle settling in a fluid happens when**
(a) Gravity force = Buoyancy force – Drag force
(b) Buoyancy force = Gravity force
(c) Gravity force = Buoyancy force + Drag force
(d) Gravity force = Drag force

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**264. Consider a rigid solid sphere falling with a constant velocity in a fluid. The following data are known at the conditions of interest: viscosity of the fluid = 0.1 pa s, acceleration due to gravity =10 $ms^{-2}$, density of the particle = 1180 kg $m^{-3}$ and density of the fluid = 1000 kg $m^{-3}$. The diameter (in mm, rounded off to the second decimal place) of the largest sphere that settles in the Stokes' law regime ( Reynolds number ≤ 0.1), ——**

**GATE-2016**

**Ans. (2.05 to 2.25)**

**265. Drag coefficient for flow past immersed body is the ratio of _____ to the product of velocity head and density.**

(a) Shear stress
(b) shear force
(c) friction factor
(d) average drag per unit projected area

**ISRO-12.02.2016**

**Ans. (d)**

**266. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).**

**Assertion (A) :** The rate of settling increases with increased initial concentration of the solids in the slurry.

**Reason (R) :** The effective density and viscosity of the medium through which the particles settle increase with increased initial concentration of the solids in the slurry.

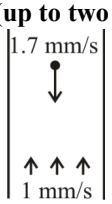
**Choose the correct answer from the code given below:**

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(c) (A) is true, but (R) is false
(d) (A) is false, but (R) is true

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**267. A spherical solid particle of 1 mm diameter is falling with a downward velocity of 1.7 mm/s through of liquid (viscosity 0.04 Pa. s) at a low Reynolds number (Stokes regime). The liquid is flowing upward at a velocity of 1 mm/s. All velocities are with respect to a stationary reference frame. Neglecting the wall effects, the drag force per unit projected area of the particle, in Pa, (up to two decimal places) is ___**




**GATE-2015**

**Ans. (1.25 to 1.35)**

**268. Consider a spherical droplet of water with diameter 2 μm, the viscosity of air η is 0.0172 g / m.s.; and the density of water is $10^6$ g / $m^3$. Its settling velocity will be nearly:**

(a) $3.3\times10^{-4}$ m/s
(b) $2.5\times10^{-4}$ m/s
(c) $1.3\times10^{-4}$ m/s
(d) $0.5\times10^{-4}$ m/s

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**269. For motion of spherical particles in a stationary fluid, the drag coefficient in hindered settling is:**

(a) Always equal to that in free settling
(b) Always less than in free settling
(c) Always greater than in free settling
(d) Equal to or less than in free settling

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Always greater than in free settling

## I. Friction in Flow

**270. Fanning equation is applicable for_____ region flow.**

(a) transition
(b) laminar
(c) turbulent
(d) laminar and turbulent

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**271. The following table provides four sets of fanning Friction factor data, for different values of Reynolds number (Re) and roughness factor $\left(\frac{K}{D}\right)$.**

| | Re | $10^2$ | $10^3$ | $10^5$ | $10^5$ |
|---|---|---|---|---|---|
| | $\left(\frac{K}{D}\right)$ | 0 | 0.001 | 0 | 0.001 |
| Set I | $f$ | 0.16 | 0.016 | $16\times10^{-5}$ | $16\times10^{-5}$ |
| Set II | $f$ | 0.016 | 0.16 | 0.0055 | 0.0045 |
| Set III | $f$ | 0.16 | 0.016 | 0.0045 | 0.0055 |
| Set IV | $f$ | 0.0045 | 0.0055 | 0.016 | 0.16 |

**Which of the above sets of friction factor data is correct?**

(a) Set I (b) Set II
(c) Set III (d) Set IV

**GATE-2017**

**Ans. (c)**

**272. Consider two pipes of same length and diameter through which water is passed at the same velocity. The friction factor for rough pipe is $f_1$ and that for smooth pipe is $f_2$. Pick out the correct statement.**

(a) $f_1 = f_2$
(b) $f_1 < f_2$
(c) $f_1 > f_2$
(d) Data not sufficient to relate $f_1$ and $f_2$

**RPSC ACF-2011**

**Ans. (c)**

**273. Which of the following analogies gives St = f / 2?**
(a) Reynolds analogy
(b) Prandtl analogy
(c) Chilton-Colburn analogy
(d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**274. Fanning friction factor is——**
(a) 0.25 times Darcy friction factor
(b) Same as Darcy friction factor
(c) 2 times Darcy friction factor
(d) Independent

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**275. For fanning friction factor *f* (for flow in pipes) and drag co-efficient $C_D$ (for flow over immersed bodies) which of the following statements are true?**
**P : *f* accounts only for the skin friction**
**Q : $C_D$ accounts only for the skin friction**
**R : $C_D$ accounts for both skin friction and form friction**
**S : Both *f* and $C_D$ depend on the Reynolds number**
**T : For laminar flow through a pipe, *f* doubles on doubling the volumetric flow rate**
(a) R, S, T (b) P, Q, S
(c) P, R, S (d) P, Q, S, T

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**276. $f = \frac{16}{N_{Re}}$ is valid for**
(a) turbulent flow
(b) laminar flow through an open channel
(c) steady flow
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**277. The slope of the line in friction factor chart for a NRe value of < 2100 in a pipe flow is**
(a) 16 (b) –1
(c) –16 (d) 0

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**278. Water flows through a smooth circular pipe under turbulent conditions. In the viscous sub-layer, the velocity varies linearly with the distance from the wall. The Fanning friction factor is defined as, $f = \frac{\tau_w}{\rho \bar{u}^2/2}$ where $\tau_w$ is the shear stress at the wall of the pipe, ρ is the density of the fluid and $\bar{u}$ is the average velocity in the pipe. Water (density = 1000 kg $m^{-3}$, viscosity = $1\times10^{-3}$ kg $m^{-1}$ $s^{-1}$) flows at an average velocity of 1 m $s^{-1}$ through the pipe. For this flow condition, the friction factor *f* is 0.005. At a distance of 0.05 mm from the wall of the pipe (in the viscous sub-layer), the velocity (in m $s^{-1}$, rounded off to the third decimal place), is**

**GATE-2016**

**Ans. (0.12 to 0.13)**

**279. For a flow through a smooth pipe, the Fanning friction factor (*f*) is given by $f = mRe^{-0.2}$ in the turbulent flow regime, where Re is the Reynolds number and m is a constant. Water flowing through a section of this pipe with a velocity 1 m/s results in a frictional pressure drop of 10 kPa. What will be the pressure drop across this section (in kPa), when the velocity of water is 2 m/s?**
(a) 11.5 (b) 20
(c) 34.8 (d) 40

**GATE-2016**

**Ans. (c)**

**280. Losses for flow through valves and fittings are expressed in terms of**
(a) drag coefficient
(b) equivalent length of a straight pipe
(c) shape factor
(d) roughness factor

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**281. Losses for flow through valves and fittings are expressed in terms of**
(a) Drag coefficient
(b) Equivalent length of a straight pipe
(c) Shape factor
(d) Roughness factor

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**282. For Fanning fraction factory *f* (for flow in pipes) and drag coefficient $C_D$ (for flow over immersed bodies), which of the following statements are true?**

**P. $f$ accounts only for the skin fraction**
**Q. $C_D$ accounts only for the form fraction**
**R. $C_D$ accounts for both skin fraction and form fraction**
**S. Both $f$ and $C_D$ depend on the Reynolds number**
**T. For laminar flow through a pipe, $f$ doubles on doubling the volumetric flow rate.**

**(1) R, S, T** **(2) P, Q, S**
**(3) P, R, S** **(4) P, Q, S, T**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**283. The tube is said to be hydraulically smooth:**
(a) If the friction factor is zero for all values of Reynolds number
(b) If the friction for a given Reynolds number does not change by further smoothing of pipe
(c) If the friction factor for a given Reynolds number changes by further smoothing of pipe
(d) If the friction factor is independent of Reynolds number

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** If the friction for a given Reynolds number does not change by further smoothing of pipe

**284. The friction factor:**
(a) Is dimension less
(b) Has the units of energy per unit weight of fluid
(c) Is inversely proportional to Reynolds number
(d) Appears in pressure drop relation for laminar flow through pipes

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Is dimension less

## J. Dimensional Analysis & Dimensionless Numbers

**285. Chilton-Colburn analogy of momentum, heat and mass transfer is applicable**
(a) When there is only skin friction
(b) When there are both skin friction and form drag
(c) When there is only form drag
(d) When heat transfer happens by radiation

**TNPSC AE-2018**

**Ans. (a)**

**286. Compressible flow at ordinary densities and high velocities, the more basic parameter required is**
(a) Reynolds number
(b) Mach number
(c) Froude number
(d) Pressure co-efficient

**TNPSC AE-2018**

**Ans. (b)**

**287. Which of the following equation refers Graetz number?**

(a) $\frac{mCp}{KL}$ (b) $\frac{KL}{mCp}$
(c) $\frac{mCp}{K\mu}$ (d) $\frac{KL}{mCp\Delta T}$

Where m is mass flow rate: Cp is heat capacity : K is thermal conductivity and L is length

**TNPSC AE-2019**

**Ans. (a)**

**288. Peclet number is the product of_____number and _____number**
(a) Reynolds, Schmidt (b) Prandtl, weber
(c) Prandtl, Schmidt (d) Reynolds, Prandtl

**TNPSC AE-2019**

**Ans. (d)**

**289. _____number can be used for convective heat transfer.**
(a) Mach (b) Froude
(c) Nusselt (d) Power

**TNPSC AE-2019**

**Ans. (c)**

**290. Choose the correct relationship from the following power equation used in power calculation in agitated vassels.**

(a) $N_p = \psi(R_e, F_o)$ (b) $N_p = \psi(R_e, F_r)$
(c) $N_p = \psi(R_e, S_e)$ (d) $N_p = \psi(R_e, S_h)$

**TNPSC AE-2019**

**Ans. (b)**

**291. Match the items in the following columns**

**(i). Stanton number** **A.** $\frac{hd}{k}$

**(ii) Prandtl number** **B.** $\frac{h}{c_p^{\rho v}}$

**C.** $\frac{c_p k}{\mu}$

**D.** $\frac{c_p \mu}{k}$

(a) i – B, ii – D (b) i – A, ii – B
(c) i – B, ii – C (d) i – A, ii – D

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**292. The dimentionless group in mass transfer that is equivalent to Prandtl number in heat transfer is**

(a) Stanton number (b) Nusselt number
(c) Schmidt number (d) Sherwood number

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**293. The Stanton number (St) for mass transfer is defined as**
**Where**
**Sh – Sherwood number**
**Re – Reynold's number**
**Sc – Schmidt number**

(a) $\frac{Sh}{Re \cdot Sc}$ (b) $\frac{Sh}{Re Sc^{1/3}}$

(c) $\frac{Re}{Sc}$ (d) Re . Sc

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**294. Dimensionless number used to estimate the dispersion in mechanically agitated lagoons is**

(a) Peclet number (b) Reynolds number
(c) Froude number (d) Grashof number
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**295. Bond number is the ratio of**

(a) Gravity force to surface tension force
(b) Pressure force to surface tension force
(c) Gravity force to pressure force
(d) Viscous force to surface tension force

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**296. Stanton number for mass transfer is defined as**

(a) Schmidt number/(Sherwood Number × Re)
(b) Re/(Schmidt Number × Sherwood Number)
(c) (Re × Sherwood Number)/Schmidt Number
(d) Sherwood Number/(Re × Schmidt Number)

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**297. The dimensionless group in mass transfer that is equivalent to Prandtl number in heat transfer is**

(a) Nusselt number (b) Sherwood number
(c) Schmidt number (d) Stanton number

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**298. Prandtl number for air is**

(a) 0.65 (b) 0.75
(c) 6.5 (d) 65

**RPSC ACF-2011**

**Ans. (b)**

**299. Corresponding to Prandtl number in heat transfer, the dimensionless group in mass transfer is**

(a) Reynolds number (b) Sherwood number
(c) Peclet number (d) Schmidt number

**RPSC ACF-2011**

**Ans. (d)**

**300. Schmidt number is a ratio of**

(a) Momentum diffusivity to mass diffusivity
(b) Thermal diffusivity to mass diffusivity
(c) Thermal diffusivity to momentum diffusivity
(d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**301. The ratio of surface convection resistance to internal conduction resistance is called.**

(A) Grashof number
(B) Biot number
(C) Stanton number
(D) Prandtl number

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

302. **Buckingham Π- theorm states that in any physical problem including 'n' quantities having 'm' dimensions, the quantities can be arranged into____independent dimensionless parameter.**

(A) m (B) n
(C) n-m (D) n/m

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**303. The dimensionless parameter describing flow-induced boiling represented by $[p_a - p_v)/(.5\rho V^2)]$ where**
**$p_a$ = ambient pressure**
**$p_v$ = vapor pressure**
**V = characteristic flow velocity**
**$\rho$ = fluid density**

(a) Cavitation number (b) Knudsen number
(c) Prandtl number (d) Capillary height

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**304. The prandtl Number is the ratio of**

(a) Momentum diffusivity of thermal diffusivity
(b) Thermal diffusivity of momentum diffusivity
(c) Shear stress to thermal diffusivity
(d) Thermal diffusivity to kinematic viscosity

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**305. The weber number may be defined as the ratio of**

**A. Inertial forces to surface tension forces**
**B. Surface tension forces to pressure forces**
**C. Pressure forces to viscous forces**
**D. Viscous forces to gravity forces**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**306. Which is not a dimensionless parameter?**

(a) Reynolds number (b) Froude number
(c) Atomic number (d) Mach number

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**307. Match the dimensionless numbers in Group-1 with the ratios in Group-2**

| Group-1 | Group-2 |
|---|---|
| P Biot number | I. $\frac{\text{buoyancy force}}{\text{viscous force}}$ |
| Q Schmidt number | II. $\frac{\text{internal thermal resistance of a solid}}{\text{boundary layer thermal resistance}}$ |
| R Grash of number | III. $\frac{\text{momentum diffusivity}}{\text{mass diffusivity}}$ |

(a) P-II, Q-I, R-III
(b) P-I, Q-III, R-II
(c) P-III, Q-I, R-II
(d) P-II, Q-III, R-I

**GATE-2016**

**Ans. (d)**

**308. Prandtl number is the ratio of**

(a) mass diffusivity to thermal diffusivity
(b) Momentum diffusivity to thermal diffusivity
(c) Thermal diffusivity to mass diffusivity
(d) thermal diffusivity to momentum diffusivity

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**309. Froude number is defined as the ratio of,**

(a) Pressure force to surface tension
(b) Pressure force to inertial force
(c) Inertial force to gravity force
(d) Gravity force to inertial force

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (c)**

**310. Corresponding to Nusselt number in heat transfer, the dimensionless group in mass transfer is the _____ number.**

(a) Peclet (b) Stanton
(c) Schmidt (d) Sherwood

**ISRO-12.02.2016**

**Ans. (d)**

**311. Match the following dimensionless numbers with the appropriate ratio of forces:**

| Dimensionless numbers | Ratio of forces |
|---|---|
| (A) Power Number | P. Conductive heat transfer/ Conductive heat of transfer |
| (B) Weber Number | Q. Inertial force /Surface tension force |
| (C) Nusselt Number | R. Inertial force /gravitational force |
| (D) Froude Number | S. Drag stress/Inertial stress |

(a) A-P, B-R, C-Q, D-S (b) A-S, B-Q, C-R, D-P
(c) A-S, B-Q, C-P, D-R (d) A-R, B-S, C-Q, D-P

**ISRO-12.02.2016**

**Ans. (c)**

**312. Prandtl number is**

(a) $C_p\mu / k$
(b) The ratio of momentum diffusivity to thermal diffusivity
(c) $N_{pr}$ / Dimensionless number
(d) All (a), (b) and (c)

**ISRO-12.02.2016**

**Ans. (a)**

**313. Match List – I with List – II and select the correct answer using the code given below the lists.**

| List-I | List-II |
|---|---|
| A. Grashoff number | 1. Compressible flow |
| B. Froude number | 2. Free convection |
| C. Euler number | 3. Free surface flow |
| D. Mach number | 4. Pressure variations in flow |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 1 | 4 | 3 |
| (b) | 4 | 3 | 2 | 1 |
| (c) | 2 | 3 | 4 | 1 |
| (d) | 4 | 1 | 2 | 3 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**314. Which of the following is NOT correctly matched?**

(a) Stanton number for mass transfer: $\frac{Sh}{Re\,Sc}$
(b) Colburn j factor for mass transfer : $\frac{Sh}{Re\,Sc^{2/3}}$
(c) Lewis number : $\frac{Sc}{Pr}$
(d) Peclet number for mass transfer : Re Sc

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**315. Which of the following is NOT correctly matched?**

(a) Friction factor: Shear force/inertial force

(b) Ratio of Nusselt number to Biot number: $\frac{\text{Conductive resistance of fluid}}{\text{Conductive resistance of solid}}$

(c) Colburn j factor for heat transfer: $\frac{Nu}{Re\,Pr^{1/3}}$

(d) Lewis number: $\frac{\text{Momentum diffusivity}}{\text{Mass diffusivity}}$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**316. Lewis number is of relevance to**

(a) absorption (b) adsorption

(c) humidification (d) distillation

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**317. The ratio of the thermal boundary layer thickness to the concentration boundary layer thickness is proportional to**

(a) Nusselt number (b) Lewis number

(c) Sherwood number (d) Prandtl number

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**318. The Schmidt number is the ratio of :**

(a) The mass diffusivity to the momentum diffusivity

(b) The momentum diffusivity to the mass diffusivity

(c) The thermal diffusivity to the mass diffusivity

(d) The momentum to thermal diffusivity

**AP PSC IOF-2009**

**Ans. (b)**

**319. For turbulent mass transfer in pipes the Sherwood number depends upon the Reynolds number (Re) as :**

(a) $Re^{0.33}$ (b) $Re^{0.53}$

(c) $Re^{0.83}$ (d) Re

**AP PSC IOF-2009**

**Ans. (c)**

**320. The dimensionless number associated with natural convection is :**

(a) Nusselt number (b) Prandtl number

(c) Grashoff number (d) Reynolds number

**AP PSC IOF-2009**

**Ans. (c)**

**321. The ratio of inertial force to gravity force is called :**

(a) Reynolds number (b) Schmidt number

(c) Froude number (d) Weber number

**AP PSC IOF-2009**

**Ans. (c)**

**322. The flow condition is identified by :**

(a) Froude number (b) Prandtl number

(c) Reynolds number (d) Weber number

**AP PSC IOF-2009**

**Ans. (c)**

**323. Froude number $N_{Fr}$ is :**

(a) $\frac{V^2}{gL}$ (b) $\frac{P}{\rho V^2}$

(c) $\frac{\mu}{\rho D_{AB}}$ (d) $\frac{kL}{D_{AB}}$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**324. The dimensionless group in mass transfer that is equivalent to Prandtl Number in heat transfer is:**

(a) Nusselt number

(b) Sherwood number

(c) Schmidt number

(d) Stanton number

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Schmidt number

## K. Solid-solid separation

**325. Generally, particle size are expressed in different units, choose the correct one.**

(a) Ultra fine particles in nanometers

(b) Very fine particles in square meters per gram

(c) Fine particles in millimeters

(d) Coarse particles in inches or millimeters

**TNPSC AE-2019**

**Ans. (d)**

**326. The specific surface of particles means**

(a) total surface area of a unit size of particles

(b) total surface area of a unit volume of particles

(c) total surface area of a unit mass of particles

(d) total surface area of a unit density of particles

**TNPSC AE-2019**

**Ans. (c)**

**327. In a equation $v_p = \alpha D_p^3$ where α is the volume shape factor, for a cube, the value of α is**

(a) 0.5 (b) 1.0

(c) 1.5 (d) 2.0

**TNPSC AE-2019**

**Ans. (b)**

**328. If $d_p$ is the equivalent diameter of a non spherical particle, $v_p$ is its volume and $s_p$ is its surface area, then sphericity, $\phi_s$ is**

(a) $d_p s_p/v_p$ (b) $6v_p/d_p s_p$

(c) $6\, d_p s_p/v_p$ (d) $v_p/d_p s_p$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**329. In a cumulative screen analysis $x_F = 0.5$ , $x_D = 0.8$ and $x_B = 0.2$. What is the ratio of overflow to feed? Where $x_F$, $x_D$ and $x_B$ are cumulative mass fraction of feed, overflow and underflow ?**

(a) 0.1 (b) 0.3
(c) 0.5 (d) 0.7
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**330. The expression $\frac{1}{\sum_{i=1}^{n}(x_i/\bar{D}p_i)}$ represents**

**Where $x_i$ = mass fraction in given increment**
**$D\bar{p}_i$ = average particle diameter**

(a) mass mean diameter
(b) volume mean diameter
(c) arithmetic mean diameter
(d) volume-surface mean diameter
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**331. Sorting classifiers used to separate the particles based on differing**

(a) densities (b) sizes
(c) shapers (d) viscosities
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**332. Colligative means depending**

(a) on the number of particles
(b) on the nature of particles
(c) on both number and nature of particles
(d) on identity of individual particles

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**333. Wet sieving is employed, when the product contains**

(a) abrasive materials in large quantity
(b) coarse materials in large quantity
(c) non-sticky materials in large quantity
(d) very fine materials in large quantity

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**334. Size analysis was carried out on a sample of gravel. The data for mass fraction ($x_i$) and average particle diameter ($D_{pi}$) of the fraction is given in the table below.**

| $xi$ | $D_{pi}$ (mm) |
|---|---|
| 0.2 | 5 |
| 0.4 | 10 |
| 0.4 | 20 |

**The mass mean diameter of the sample, to the nearest integer, is _____ mm,**

**GATE-2017**

**Ans. (12.99 to 13.01)**

**335. For non spherical particle, the sphericity**

(a) is defined as the ratio of surface area of a sphere having the same volume as the particle to the actual surface area of the particle
(b) has a dimension of length
(c) is always less than 1
(d) is the ratio of volume of a sphere having the same surface area as the particle to the actual volume of the particle

**RPSC ACF-2011**

**Ans. (a)**

**336. With increase in the capacity of screens, the screen effectiveness**

(a) Remains unchanged
(b) Increases
(c) Decreases
(d) Decreases exponentially

**RPSC ACF-2011**

**Ans. (c)**

**337. The ratio of the actual mesh dimension of Taylor series to that of the next smaller screen is**

(a) 2 (b) $\sqrt{2}$
(c) 1.5 (d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**338. For spheres and cubes, sphericity is equal to one. For a cylinder whose length is equal to diameter, sphericity is equal to-**

(a) 0.28 (b) 0.65
(c) 0.73 (d) 1.0

**RPSC ACF-2011**

**Ans. (c)**

**339. The ratio of the actual mesh dimension of Taylor series to that of the next smaller screen.**

(A) 2 (B) $\sqrt{2}$
(C) 1.5 (D) $\sqrt{3}$
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**340. Industrial screen are not made from**

(a) Plastic cloth (b) Metal bar
(c) Silk (d) Wood

**ISRO Technical Asstt.-25.02.2018**

**Ans. (*)**

**341. Mass of the particles fed per unit time per unit area of the screen is termed as**

(a) Efficiency (b) Capacity
(c) Material flow (d) None of the above

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**342. Which one of the following analysis belongs to the particle size distribution curve?**
(a) Differential analysis (b) Numerical analysis
(c) Bernoulli's analysis (d) None of the above
**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**343. Value of Sphericity ($\phi_s$) for Sphere**
(a) 0 (b) 1
(c) Infinity (d) –1
**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**344. Formula for calculating mass mean diameter**
(a) $\sum(x_i/D_{pi})$ (b) $\sum(x_i D_{pi})^2$
(c) $\sum(x_i D_{pi})$ (d) $1/\sum(x_i/D_{pi})$
**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**345. Shape factor value for cylinder (height = diameter)**
(a) 1 (b) 0.785
(c) 0.524 (d) 0
**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**346. Ratio of actual mesh dimension of any standard screen to that of the next-smaller screen**
(a) 1.414 (b) 1.732
(c) 1.189 (d) 1.234
**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**347. _____ analysis can be used for diameters greater than about 44 μm**
(a) Wet-screen
(b) Dry-screen
(c) Logarithmic mean screen
(d) Tyler-screen
**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**348. Which size measurement technique/analysis is useful for sizing particles with diameters greater than about 100 μm?**
(a) Dry screening (b) Coulter counter
(c) Gravity sedimentation (d) Wet screening
**HPCL Engineer-01.11.2022**

**Ans. (a)**

**349. With increase in the capacity of screens, the screen effectiveness**
**A. Remains unchanged**
**B. Increases**
**C. Decreases**
**D. Decreases exponentially**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**350. Sphericity of a cubical particle, when the equivalent diameter is taken as the height of the cube is**
**A. 0.5** **B. 1**
**C. $\sqrt{2}$** **D. $\sqrt{3}$**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**351. Mesh indicates the number of holes per**
**A. Square inch**
**B. Linear inch**
**C. Square foot**
**D. Linear foot**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**352. Formula for calculating Sphericity ($\Phi_s$)**
(a) $(6/V_P)/(D_P/S_P)$ (b) $(6/D_P)/(V_P/S_P)$
(c) $(6/D_P)/(S_P/V_P)$ (d) $(6/S_P)/(V_P/D_P)$
**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**353. Formula for calculating volume-surface mean diameter**
(a) $\Sigma(x_i/D_{pi})$ (b) $\Sigma(x_i D_{pi})^2$
(c) $\Sigma(x_i D_{pi})$ (d) $1/\Sigma(x_i/D_{pi})$
**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**354. Shape factor value for sphere**
(a) 1 (b) 0.785
(c) 0.524 (d) 0
**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**355. Ratio of actual mesh dimension of any standard screen to next smaller intermediate screen**
(a) 1.410 (b) 1.732
(c) 1.189 (d) 1.234
**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**356. Size distribution is carried out first to find the size of mixture of particles. Among the following operations, which is not a method of size distribution?**
(a) Sieving (b) ICI Sedimentation
(c) Beaker decantation (d) Filtration
**Karnataka PSC ADF - 2015**

**Ans. (d)**

**357. Separation of particles of various sizes, shape and densities by allowing them to settle in a fluid is called**
(a) Froth floatation (b) Thickening
(c) Screening (d) Classification
**Karnataka PSC ADF - 2015**

**Ans. (d)**

**358. Sphericity of a cubical particle, when its equivalent diameter is taken as the height of the cube is**

(a) 0.5 (b) 1.0

(c) $\sqrt{2}$ (d) $\sqrt{3}$

**ISRO-12.02.2016**

**Ans. (b)**

**359. Arrange the sphericity of the following in increasing order.**

**1. Solid cylinder of diameter D and length D**

**2. Solid cylinder of diameter D and length D/2**

**3. Solid cylinder of diameter D and length 2D**

**4. Solid cube of side D**

**Select the correct answer from the code given below :**

(a) 1, 2, 3, and 4 (b) 4, 3, 2 and 1

(c) 1, 3, 2 and 4 (d) 4, 2, 3 and 1

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**360. The diameters of sand particles in a sample range from 50 to 150 microns. The number of particles of diameter x in the sample is proportional to $\frac{1}{50+x}$. The average diameter, in microns, (up to one decimal place) is_____**

**GATE-2015**

**Ans. (93.3 to 95.3)**

**361. The sphericity of a solid particle of cubical shape is:**

(a) $\left(\frac{\pi}{6}\right)^{1/3}$ (b) $\left(\frac{\pi}{6}\right)$

(c) $\left(\frac{\pi}{6}\right)^{1/2}$ (d) $\left(\frac{\pi}{3}\right)$

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**362. Calculate the effectiveness of a screen having $x_F$ = 0.49, $x_D$ = 0.94 and $x_B$ = 0.023. The answer is:**

(a) 81.2% (b) 91.3%

(c) 76% (d) 87.2%

**Kerala PSC Lect. 05.02.2015**

**Ans. (b)**

## L. Size Reduction

**363. If the radii of the ball mill and the ball are R and r, respectively, the critical speed ($n_c$) of the mill is given by**

(a) $n_c = \frac{1}{2}\sqrt{\frac{g}{R-r}}$ (b) $n_c = \frac{1}{2\pi}\sqrt{\frac{g}{R-r}}$

(c) $n_c = \frac{1}{\pi}\sqrt{\frac{g}{R-r}}$ (d) $n_c = \frac{1}{2\pi}\sqrt{\frac{R-r}{g}}$

**TNPSC AE-2018**

**Ans. (b)**

**364. Match the size reduction equipments with the principles:**

| | | | |
|---|---|---|---|
| **(a)** | **Ball mill** | **(i)** | **compression** |
| **(b)** | **jaw crushers** | **(ii)** | **attrition** |
| **(c)** | **Ultra fine grinders** | **(iii)** | **cutting** |
| **(d)** | **knife cutter** | **(iv)** | **impact** |

| | **(a)** | **(b)** | **(c)** | **(d)** |
|---|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 4 | 2 | 1 | 3 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 1 | 3 | 4 | 2 |

**TNPSC AE-2018**

**Ans. (c)**

**365. Sphericity for a non-spherical particle is given by _____ Where $D_P$ – Equivalent diameter of particle, $S_P$ – Surface area of one particle, $V_P$ – Volume of one particle**

(a) $\frac{6V_P}{D_P S_P}$ (b) $\frac{V}{6D_P S_P}$

(c) $\frac{D_P S_P}{V_P}$ (d) $\frac{V_P}{D_P S_P}$

**TNPSC AE-2018**

**Ans. (a)**

**366. The constants $K_R$, $K_B$ and $K_k$ in the laws of crushing depends on**

(a) feed material

(b) compression

(c) finer products

(d) capacity of machine

**TNPSC AE-2018**

**Ans. (a)**

**367. For size reduction of a large plastic sheet into small units of fixed dimensions, which equipment is the most useful.**

(a) Jaw crusher (b) Roll crusher

(c) Rod mill (d) Rotary knife cutter

**TNPSC AE-2018**

**Ans. (d)**

**368. Fluid Energy Mill is an example of**

(a) Crusher (b) Grinder

(c) Ultra fine grinder (d) Cutting machine

**TNPSC AE-2018**

**Ans. (c)**

**369. Choose the correct relationship used in work index. If $D_p$ is in mm, P in kW, and m in tons per hour.**

(a) $K_b = 0.2162\ W_i$ (b) $K_b = 0.3162\ Wi$

(c) $K_b = 0.4162\ Wi$ (d) $W_b = 0.5162\ Wi$

**TNPSC AE-2019**

**Ans. (b)**

**370. According to bond law, which of the following is correct ?**

(a) $\frac{P}{m} \propto D_p$ (b) $\frac{P}{m} \propto \frac{1}{D_p}$

(c) $\frac{P}{m} \propto \sqrt{D_p}$ (d) $\frac{P}{m} \propto \frac{1}{\sqrt{D_p}}$

**TNPSC AE-2019**

**Ans. (d)**

**371. Choose the correct relationship**

(a) $\frac{s_p}{v_p} = \frac{3}{\phi_s D_p}$ (b) $\frac{s_p}{v_p} = \frac{6}{\phi_s D_p}$

(c) $\frac{s_p}{v_p} = \frac{9}{\phi_s D_p}$ (d) $\frac{s_p}{v_p} = \frac{3}{\phi_s D_p^2}$

**TNPSC AE-2019**

**Ans. (b)**

**372. Cutting size reduction equipment gives**

(a) coarse reduction and hard solids
(b) coarse, medium or fine products
(c) fine products
(d) definite size

**TNPSC AE-2019**

**Ans. (d)**

**373. Value of $\phi_s$ for a spherical particle is**

(a) 0 (b) <1
(c) >1 (d) 1

**TNPSC AE-2019**

**Ans. (d)**

**374. Jaw Crusher is operated by**

(a) Cutting (b) Compression
(c) Impact (d) Attrition

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**375. The critical speed of the ball mill depends on**

(a) The ratio of the mill and the ball radii
(b) The radii of the mill and the ball
(c) The radius and number of balls
(d) The number of balls used

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**376. The power required to crush 1000 tons/h of limestone if 80% of the feed passes a 50.88 mm screen and 80% of the product passes a 3.175 mm screen, is: (Work index for limestone = 12.74)**

(a) 11169 kW (b) 169.6 kW
(c) 1189 kW (d) 1696 kW

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**377. Tumbling mills operates at the critical speed in percentage of**

(a) 0 – 20 % (b) 20 – 40 %
(c) 65 – 80 % (d) 80 – 95 %
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**378. The sphericity is defined as**
**Where**
**Vp = Volume of particle**
**Dp = Diameter of particle**
**Sp = Surface area of particle**

(a) $\frac{6Vp}{DpSp}$ (b) $\frac{6Dp}{VpSp}$

(c) $\frac{6Sp}{Dp \cdot Vp}$ (d) $\frac{6Vp}{Sp}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**379. Energy consumed for crushing one ton of material ranges from**

(a) 0.01 to 0.1 kWh (b) 0.5 to 1.5 kWh
(c) 2 to 3.5 kWh (d) 4 to 5 kWh

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**380. Colloid mills achieve size reduction mainly by**

(a) impact (b) attrition
(c) cutting (d) compression

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**381. Size reduction of coarse hard solids using a crusher is accomplished by**

(a) attrition (b) compression
(c) cutting (d) impact

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**382. Trommels operate at**

(a) high speed (b) low speed
(c) intermediate speed (d) very high speed

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**383. Match the equipment in Column A with the corresponding process in Column B**

| Column A | | Column B | |
|---|---|---|---|
| (P) | Centrifugal sifter | (I) | Mixing |
| (Q) | Bowl mill | (II) | Sedimentation |
| (R) | Gravity thickener | (III) | Screening |
| (S) | Two-arm kneader | (IV) | Grinding |

(a) P-I, Q-IV, R-II, S-III
(b) P-III, Q-IV, R-II, S-I
(c) P-IV, Q-I, R-II, S-III
(d) P-IV, Q-III, R-I, S-II

**GATE-2018**

**Ans. (b)**

**384. Critical speed of a ball mill depends on**

(a) the radius of the mill (shell) and the radius of the particles

(b) the radius of the mill (shell) and the density of the particles

(c) the radius of the balls and the radius of the particles

(d) the radius of the balls and the radius of the mill (shell)

**GATE-2018**

**Ans. (d)**

**385. Which of the following is the correct sequence of equipment for size reduction of solids?**

(a) Fluid energy mill → Jaw Crusher → Hammer mill

(b) Hammer mill → Jaw Crusher → Fluid energy mill

(c) Fluid energy mill → Hammer mil → Jaw crusher

(d) Jaw Crusher → Hammer mill → Fluid energy mill

**GATE-2017**

**Ans. (d)**

**386. Kick's law relates to**

(a) Energy consumption

(b) Final particle size

(c) Feed size

(d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**387. The reduction ratio for fine grinders is**

(a) $5 \pm 10$ (b) $20 \pm 40$

(c) $10 \pm 20$ (d) as high as 100

**RPSC ACF-2011**

**Ans. (d)**

**388. The main size reduction operation in ultra fine grinders is**

(a) Cutting (b) Compression

(c) Attrition (d) Impact

**RPSC ACF-2011**

**Ans. (c)**

**389. The reduction ratio for grinders is defined as ($D_f$ = Feed particle Diameter; $D_p$ = Product particle diameter).**

(A) $D_f/D_p$ (B) $D_p/D_f$

(C) $D_f - D_p$ (D) $D_f * D_p$

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**390. Ultrafine particles are generally measured in,**

(a) Inches (b) Millimetre

(c) Centimeter (d) Area per unit mass

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**391. Which of the following law is used for crushing ?**

(a) Kirchoff's law (b) Dalton's law

(c) Raoult's law (d) Bond's law

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**392. For the ball milling operation, speed of the mill should be _____ than critical speed.**

(a) Equal (b) Higher

(c) Lesser (d) Zero

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**393. Example for ultrafine grinders**

(a) Fluid energy mill (b) Tumbling mill

(c) Roller mill (d) Pebble mill

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**394. What rotational speed in evolutions per second would you recommend for a ball mill 1080 mm in diameter charged with 100 mm balls?**

(a) Between 0.75 to 1 (b) Less than 0.5

(c) Greater than 1 (d) Between 0.5 to 0.75

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**395. Identify the correct name of the operation of paper production in which Debarking, Chipping, Cutting, Screening are done.**

(a) Washing and screening

(b) Pulping

(c) Raw Material Preparation

(d) Acquisition of Raw Material

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**396. Select the correct name of the operation of paper production in which Debarking, Chipping, Cutting, Screening are done.**

(a) Washing and screening

(b) Acquisition of Raw Material

(c) Pulping

(d) Raw Material preparation

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**397. For which equipment the energy requirement (per unit mass of material crushed/ground) is highest?**

(a) Fluid energy mill (b) Ball mill
(c) Rod mill (d) Jaw crusher

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**398. The work required for crushing of solid is proportional to the new surface created. This is known as**

(a) Bond's law (b) Rittinger's law
(c) Kick's law (d) None of these

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**399. In ball mill operation, the feed size ($D_f$ in meters) and the ball diameter ($D_b$ in meters) are related as (where, k= Grindability constant)**

**A. $D_b^2 = k D_f$**
**B. $D_b = k D_f$**
**C. $D_b^3 = k D_f$**
**D. $D_b^2 = k D_f^2$**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**400. Centrifuging in revolving mills refer to**

(a) Falling of grinding medium over particle
(b) Collision of grinding medium during grinding.
(c) Slipping and rolling of grinding medium to reach bottom of the heap.
(d) The carryover of grinding medium right round in contact with wall

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**401. The tenacity or hardness with which the bond holds the cutting point or abrasives in place of a grinding wheel is defined by the term?**

(a) Structure (b) Grit size
(c) Grain size (d) Grade

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**402. Size reduction of coarse hard solids using a crusher is accomplished by**

(a) Attrition
(b) Compression
(c) Cutting
(d) Impact

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**403. A _____ mixer resembles a ball mill without balls.**

(a) Banbury (b) Pug mill
(c) Tumbling (d) Pan

**ISRO-12.02.2016**

**Ans. (c)**

**404. Pick out the wrong statement**

(a) Magnetic separation method can be employed to treat both dry and wet ores
(b) Gyratory crusher is used for coarse crushing
(c) Reduction ratio in crushing operation is defined as the ratio of minimum feed size to the maximum product size
(d) Screens are of stationary, moving and vibratory types

**ISRO-12.02.2016**

**Ans. (c)**

**405. Rittinger's law states that the work required in crushing is proportional to**

(a) The number of particles obtained
(b) The new surface created
(c) The surface area of the feed particles
(d) The number of particles crushed

**ISRO-12.02.2016**

**Ans. (b)**

**406. Match List-I with List – II and select the correct answer using the code given below the lists.**

| List-1 | List-II |
|---|---|
| A. Ball mill | 1.Employs compression |
| B. Fluid energy mill | 2.Employs impact |
| C. Gyratory crusher | 3.Employs interparticle attrition |
| D. Granulator | 4.Does not employ compression, impact or attrition |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 2 | 4 | 1 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 1 | 3 | 2 | 4 |
| (d) | 2 | 4 | 1 | 3 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**407. Arrange the following equipments in the increasing order of energy requirement per unit mass of material crushed/ground**

**1. Jaw crusher 2. Rod mill**
**3. Ball mill 4. Fluid energy mill**

**Select the correct answer from the code given below**

(a) 1, 2, 3 and 4
(b) 1, 3, 2 and 4
(c) 4, 3, 2 and 1
(d) 4, 2, 3 and 1

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**408. For crushing of solids the Rittinger's law states that the work required for crushing is proportional to:**
(a) The size reduction ratio
(b) The new surface created
(c) The change in volume due to crushing
(d) The number of particles crushed

**AP PSC IOF-2009**

**Ans. (b)**

**409. In order to produce fine solid particles between 5 and 10 μm, the appropriate size reducing equipment is**
(a) fluid energy mill
(b) hammer mill
(c) jaw crusher
(d) smooth roll crusher

**GATE-2014**

**Ans. (a)**

**410. Which one of the following unit operations can be used to make tablets from powders of medicinals?**
(a) Crushing
(b) Grinding
(c) Solid blending
(d) Pelletizing

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**411. The critical speed of a ball mill of radius, R which contains ball of radius r is proportional to:**
(a) $(R - r)^{1/2}$
(b) $R - r$
(c) $(R - r)^{-1/2}$
(d) $(R - r)^{-1}$

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**412. The crushing energy required to create new surface is given by:**
(a) Fick's law
(b) Rittinger's law
(c) Fourier's law
(d) Kopp's law

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Rittinger's law

## M. Mixing and Agitation (Liquid/solid/paste)

**413. The masticators that disintegrate scrap rubbers are called as**
(a) Intensive mixers (b) Extensive mixers
(c) Agitator (d) kneader

**TNPSC AE-2018**

**Ans. (a)**

**414. The power number, $N_P$ is defined by**
(a) $n^3 Da^5 \rho / Pg_c$ (b) $q / n Da^3$
(c) $Pg_c / n^3 Da^5 \rho$ (d) $n Da^3 / q$

**TNPSC AE-2018**

**Ans. (c)**

**415. Choose the correct relationship used in flow calculations in agitated vessels**
(a) $R_e = D_a^2 \rho / \mu$ (b) $R_e = D_t^2 \rho / \mu$
(c) $R_e = D_a V \frac{\rho}{\mu}$ (d) $R_e = nD_a^2 \rho / \mu$

**TNPSC AE-2019**

**Ans. (d)**

**416. Choose a correct proportion followed in the design of agitated vessel $D_a$ – Dimeter of agitator, $D_t$ = diameter of tank**
(a) $\frac{D_a}{D_t} = \frac{1}{2}$ (b) $\frac{D_a}{D_t} = \frac{1}{3}$
(c) $\frac{D_a}{D_t} = \frac{1}{4}$ (d) $\frac{D_a}{D_t} = \frac{1}{5}$

**TNPSC AE-2019**

**Ans. (b)**

**417. In which of the following condition, the circulatory flow and swirling is larger in agitated vessels.**
(a) impeller can be mounted off center
(b) shaft is moved away from the centerline of the tank
(c) impeller and shaft on the centerline of the tank
(d) agitator mounted in the side of the tank

**TNPSC AE-2019**

**Ans. (c)**

**418. The purpose of a adding filter aid in feed slurry is**
(a) to increase the porosity of cake
(b) to decrease the porosity of cake
(c) to increase the density of cake
(d) to decrease the density of cake

**TNPSC AE-2019**

**Ans. (a)**

**419. The overall flow pattern in an agitated vessel is not depend on_____velocity component.**
(a) radial (b) centrifugal
(c) longitudinal (d) tangential
**TNPSC AE-2019**

**Ans. (b)**

**420. Choose a relationship among the following for calculating power requirement in Agitated Vessels**
**Where**
**Re – Reynold's Number**
**Sn – Sherwood Number**
**Fr – Froude Number**
**Sc – Schmidt Number**
(a) $N_P = \psi$ (Re, Sh) (b) $N_P = \psi$ (Fr, Sh)
(c) $N_P = \psi$ (Re, Sc) (d) $N_P = \psi$ (Re, Fr)
(e) Answer not known
**TNPSC AE-2022**

**Ans. (d)**

**421. Power number is used in the design of**
(a) Pumps
(b) Agitated vessel
(c) Crusher
(d) Grinder
(e) Answer not known
**TNPSC AE-2022**

**Ans. (b)**

**422. In agitator, at Reynolds number less than 300, the power number for baffled tank is _____ unbaffled tank.**
(a) greater than (b) lesser than
(c) independent of (d) equal to
(e) Answer not known
**TNPSC AE-2022**

**Ans. (d)**

**423. A propeller (diameter D = 15 m) rotates at N = 1 revolution per second (rps). To understand the flow around the propeller, a lab-scale model is made. Important parameters to study the flow are velocity of the propeller tip ($V = \pi ND$), diameter D and acceleration due to gravity (g). The lab-scale model is $1/100^{th}$ of the size of the actual propeller.**
**The rotation speed of the lab-scale model, to the nearest integer, should be _____ rps.**
**GATE-2017**

**Ans. (9.9 to 10.1)**

**424. Propellers are**
(a) axial flow mixers
(b) low speed impeller
(c) used for mixing liquids of high viscosity
(d) radial flow mixers
**RPSC ACF-2011**

**Ans. (a)**

**425. Tur bine impeller**
(a) Produces only radial current
(b) Produces only tangential current
(c) Is effective over wide range of viscosities
(d) Does not produce tangential current
**RPSC ACF-2011**

**Ans. (c)**

**426. In which type of impeller used in liquid agitation, flow is coaxial?**
(a) Turbine (b) Propeller
(c) Paddle (d) None of these
**RPSC ACF-2011**

**Ans. (a)**

**427. Identify the process which generally accomplishes physical change, chemical change and increased rate of transportation:**
(a) agitation (b) blending
(c) filtering (d) screening
**HPCL Engineer-01.11.2022**

**Ans. (a)**

**428. Which of the following is most suitable for handling fibrous and dense slurries?**
**A. Propeller agitator**
**B. Cone type agitator**
**C. Turbine agitator**
**D. Radial propeller agitator**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**429. During agitation of liquids, the**
**A. Froude number is independent for the curves between power number and Reynolds number in baffled system**
**B. Power number becomes independent of impellers Reynolds number at high Reynolds number, but is independent on the geometry of the impeller**
**C. Froude number is used to account for the effect of surface (e.g. The centre vortex) on the power number**
**D. All A, B and C**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**430. An agitated cylindrical vessel is fitted with baffles and flat blade impellers. The power number for this system is given by $N_p = \frac{P}{\rho n^3 D^5}$ where P is the power consumed for the mixing, $\rho$ is the density of the fluid, n is the speed of the impeller and D is the diameter of the impeller. The diameter of the impeller is $1/3^{rd}$ the diameter of the tank and the height of liquid level is equal to the tank diameter. The**

**impeller speed to achieve the desired degree of mixing is 4 rpm. In a scaled up design, the linear dimensions of the equipment are to be doubled, holding the power input per unit volume constant. Assuming the liquid to be Newtonian and $N_p$ to be independent of Reynolds number, what is the impeller speed ( in rpm) to achieve the same degree of mixing in the scaled up vessel?**

(a) 0.13 (b) 1.26
(c) 2.52 (d) 3.82

**GATE-2016**

**Ans. (c)**

**431. Use of baffles in agitators help in minimizing the _____ tendency.**

(a) swirling (b) vortexing
(c) both (a) and (b) (d) neither (a) nor (b)

**ISRO-12.02.2016**

**Ans. (c)**

**432. The impeller Reynold's number for an agitated vessel is given by**

(a) $D_a^2 N\rho/\mu$ (b) $D_a N\rho/\mu$
(c) $D_a^2 N^2\rho/\mu$ (d) $D_a^2 N^3\rho/\mu$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**433. Mixing of thick dough and paste is carried out in a**

(a) Double cone mixer
(b) Ribbon mixer
(c) Propeller mixer
(d) Double arm kneader

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**434. Swirling in an agitated vessel can be prevented by :**

(a) increasing the vessel diameter
(b) reducing the impeller diameter
(c) mounting baffles in the vessel
(d) by using turbine impllers

**AP PSC IOF-2009**

**Ans. (c)**

**435. The power number is proportional to:**

(a) The ratio of the drag force acting on a unit area of the impeller to the inertial stress
(b) The ratio of internal stress to the gravitational force per unit area acting on the fluid
(c) The ratio of the pressure force acting on a unit area of the impeller to the gravitational force per unit area acting on the fluid
(d) The ratio of the drag force acting on a unit area of the impeller to the surface tension force per unit area acting on the fluid

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** The ratio of the drag force acting on a unit area of the impeller to the inertial stress

## N. Solid-fluid separation

**436. A filter aid is added to the slurry before filtration to**

(a) Decrease the porosity of the cake
(b) Increase the porosity of the cake
(c) Increase the compressibility coefficient of the cake
(d) Decrease the compressibility coefficient of the cake

**TNPSC AE-2018**

**Ans. (b)**

**437. The compressibility coefficient of incompressible cake is**

(a) 1 (b) 0
(c) –1 (d) ∞

**TNPSC AE-2018**

**Ans. (b)**

**438. For a constant pressure filtration, neglecting the filter cloth resistance, the filtration equation is**

(a) $dt/dV = K_c V$ (b) $DV/dt = K_c V$
(c) $dV/t = K_c V$ (d) $dt/V = K_c V$

**TNPSC AE-2018**

**Ans. (a)**

**439. In general cake porosity near the filter medium is**

(a) zero (b) Low
(c) High (d) Very high

**TNPSC AE-2019**

**Ans. (b)**

**440. Bag filters are used for**

(a) Liquid-Liquid separations
(b) Gas-Liquid separations
(c) Liquid-solid separations
(d) Gas-solid separations

**TNPSC AE-2019**

**Ans. (d)**

**441. Select a condition which prevails in cross of low filtration**

(a) Filter medium resistance increases
(b) Filter cake resistance increases
(c) No resistances
(d) Both resistances increase

**TNPSC AE-2019**

**Ans. (c)**

**442. Area required for thickener can be minimized by**

(a) increasing the downward velocity
(b) decreasing the downward velocity
(c) maintaining the downward velocity
(d) decreasing the solid flux

**TNPSC AE-2019**

**Ans. (a)**

**443. Filter medium resistance is important during the following stages operations**

(a) Entire period (b) Early period
(c) Ending period (d) Intermediate period

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**444. For incompressible cakes and negligible resistance of the filter medium and piping, the filtrate volume collected in time, t and at a total pressure drop of (–Δ) is $V_f$ The filtrate volume collected in time, t and at a total pressure drop of 4(–Δ) would be**

(a) $16\ V_f$ (b) $V_f$
(c) $4\ V_f$ (d) $2\ V_f$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**445. Separation factor in cyclone separator is ratio of**

**Where**

**Fc = Centrifugal force**

**$F_g$ = Gravity force**

(a) $Fc - F_g$ (b) $Fc + F_g$
(c) $Fc / F_g$ (d) $F_g / Fc$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**446. Filter aid is used to**

(a) increase the rate of pressure drop
(b) decrease the porosity of the cake
(c) increase the porosity of the cake
(d) act as a support base for septum
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**447. Particle aggregation is brought about by inducing velocity gradients is known as**

(a) Ortho kinetic flocculation
(b) Peri kinetic flocculation
(c) Micro flocculation
(d) Kinetic flocculation
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**448. Among the following _____ is called Heavy Fluid Separation.**

(a) Filtration
(b) Thickness
(c) Sink and float method classifiers
(d) Clarifiers
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**449. In the Empirical Equation for Cuter resistance $\alpha = \alpha_0 (\Delta P)^s$**

**Where α = Specific cake resistance**

**$\alpha_0$ = Constant**

**ΔP = Pressure drop**

**S = Compressibility coefficient**

**Compressibility coefficient value falls between**

(a) 0.2 and 0.8 (b) 2 and 8
(c) 0.02 and 0.08 (d) 1 and 10
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**450. Among the following, choose an equation relating flow in an agitated vessel (q) to agitator diameter ($D_a$)**

(a) $q \propto nD_a$ (b) $q \propto nD_a^{1.5}$
(c) $q \propto nD_a^2$ (d) $q \propto nD_a^3$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**451. To remove very small amount of tiny solid impurities from liquid, we use**

(a) Pressure filter (b) Vacuum filter
(c) Centrifugal filter (d) Flocculation
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**452. Solid particles separation based on difference in their flow velocities through fluids is termed as**

(a) elutriation (b) sedimentation
(c) classification (d) clarification

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**453. The most commonly used dispersing agents to prevent flocculation are**

(a) carbonates
(b) bi-carbonates
(c) sulphates
(d) silicates & phosphates

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**454. Where the density difference of the two liquid phases to be separated is very small (as in milk cream separator), the most suitable separator is a**

(a) tubular-bowl centrifuge
(b) disc-bowl centrifuge
(c) suspended batch basket centrifuge
(d) perforated horizontal basket continuous centrifuge

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**455. Dilute slurry is filtered at constant pressure to recover the solids. To reduce the time of filtration, the dilute slurry is initially evaporated to remove one-fourth of the solvent. If the resistance of the filter medium is neglected, filtration time will be reduced by a factor of**

(a) 32 (b) 16
(c) 8 (d) 4

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**456. The specific cake resistance for compressible sludges is a function of the pressure drop**

(a) over cake
(b) over medium
(c) overall
(d) over filter medium

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**457. A gas bubble (gas density $\rho_g$ = 2kg/m$^3$: bubble diameter D = $10^{-4}$ m) is rising vertically through water (density $\rho$ = 1000 kg/m$^3$, viscosity $\mu$ = 0.001 pa.s). Force balance on the bubble leads to the following equation,**

$$\frac{dv}{dt} = -g\frac{\rho g - \rho}{\rho_g} - \frac{18\mu}{\rho_g D^2}v$$

**where $v$ is the velocity of the bubble at any given time t. Assume that the volume of the rising bubble does not change. The value of g = 9.81 m/s$^2$**

**The terminal rising velocity of the bubble (in cm/s), rounded to 2 decimal places, is ____ cm/s.**

**GATE-2017**

**Ans. (0.52 to 0.56)**

**458. The speed of a rotary drum vacuum filter (in rpm) may be**

(a) 1 (b) 50
(c) 100 (d) 500

**RPSC ACF-2011**

**Ans. (a)**

**459. Diatomaceous earth is**

(a) Explosive (b) Filter aid
(c) Filter medium (d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**460. Flow of filtrate through cake in a plate and frame press is best described by**

(a) Kozency-Carman equation
(b) Hagen-Poiseuille equation
(c) Fanning's equation
(d) Kremser equation

**RPSC ACF-2011**

**Ans. (a)**

**461. The thickness of cakes formed on industrial rotary vacuum filters is usually between**

(a) 0.5 and 2mm (b) 3.2 and 38mm
(c) 80 and 100mm (d) 200 and 400mm

**RPSC ACF-2011**

**Ans. (c)**

**462. Filter aids are added to the slurry prior to filtration in order to form**

(a) Compact cakes of low porosity
(b) Cakes of increased porosity
(c) Crystalline cakes
(d) None of the above

**RPSC ACF-2011**

**Ans. (b)**

**463. Sedimentation on commercial scale occurs in**

(a) Classifiers (b) Rotary drum filters
(c) Thickeners (d) Cyclones

**RPSC ACF-2011**

**Ans. (b)**

**464. A rotary drum filter is–**

(a) A continuous vacuum filter
(b) A discontinuous pressure filter
(c) A continuous pressure filter
(d) None of the above

**RPSC ACF-2011**

**Ans. (a)**

**465. In constant – pressure filtration,**

(a) The filtrate flow rate is maximum at the start and decreases continuously to the end
(b) The filtrate flow rate is manimum at the start and increases continuously to the end
(c) The filtrate flow rate is constant throughout
(d) Increases slowly to a constant value

**RPSC ACF-2011**

**Ans. (a)**

**466. In constant pressure filtration, the rate of filtration follows the relation (v : filtrate volume; t : time; k and C : constants)**

(a) $\frac{dv}{dt} - kv + C$ (b) $\frac{dv}{dt} = kv^2$
(c) $\frac{dv}{dt} = kv$ (d) $\frac{dv}{dt} = \frac{1}{kv + C}$

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**467. A straight line is obtained on plotting reciprocal of filtration rate Vs the volume of filtrate for___flow of filtrate.**

**A. Compressible cake and laminar**
**B. Incompressible cake and laminar**
**C. Compressible cake and turbulent**
**D. Incompressible cake and turbulent**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**468. Industrially, the process of sedimentation is conducted on a large scale in equipment called**

**A. Sorting classifiers**
**B. Cyclones**
**C. Thickners**
**D. Filters**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**469. Filtrate flow-rate in case of a rotary dryer vacuum filter (in which $R_m << R_c$) is proportional to____of the cycle time**

**A. $\sqrt{\mu}$** **B. $1/\sqrt{\mu}$**
**C. $1/\mu$** **D. $1/\mu^2$**

**Where $\mu$ = Filtrate viscosity, $R_m$ = Filtrate medium, $R_c$ = Cake resistance**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**470. Which type of filter is used to remove small amount of solids to produce a clean gas or sparkling clear liquids such as beverages?**

(a) Clarifying filter (b) Cake filter
(c) Cross flow filter (d) None of the above

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**471. Primary factor to be considered in the design of filters**

(a) Filtration area (b) Cake resistance
(c) Filtrate flow (d) None of the above

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**472. In a cyclone separator used for separation of solid particles from a dust laden gas, the separation factor is defined as the ratio of the centrifugal force to the gravitational force acting on the particle $S_r$ denotes the separation factor at a location (near the wall) that is at a radial distance r from the centre of the cyclone. Which one of the following statements is INCORRECT?**

(a) $S_r$ depends on mass of the particle
(b) $S_r$ Depends on the acceleration due to gravity
(c) $S_r$ depends on tangential velocity of the particle
(d) $S_r$ depends on the radial location (r) of the particle

**GATE-2016**

**Ans. (a)**

**473. A filtration is conducted at constant pressure to recover solids from dilute slurry. To reduce the time of filtration, the solids concentration in the feed slurry is increased by evaporating half the solvent. If the resistance of the filter medium is negligible, the filtration time will be reduced by a factor of**

(a) 1 (b) 2
(c) 4 (d) 8

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**474. A centrifugal filtration unit operating at a rotational speed of ω has inner surface of the liquid located at a radial distance of R from the axis of rotation. The thickness of liquid film is δ and no cake is formed. The initial pressure drop during filtration is,**

(a) $\frac{1}{2}\omega^2 R^2 \rho_L$ (b) $\frac{1}{2}\omega^2 \delta^2 \rho_L$
(c) $\frac{1}{2}\omega^2 \delta \rho_L (2R+\delta)$ (d) $\frac{1}{2}\omega^2 R \rho_L (R+2\delta)$

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (c)**

**475. The unit of filter medium resistance is,**

(a) $cm^{-1}$ (b) $g.\ cm^{-1}$
(c) $cm.\ g^{-1}$ (d) $g^{-1}$

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**476. During filtration operation the filtrate encounters the resistance of the**

(a) filter medium
(b) cake
(c) Channel carrying the slurry to the upstream side of the cake and filtrate away from the filter medium
(d) All (a), (b) and (c)

**ISRO-12.02.2016**

**Ans. (d)**

**477. In the cyclone separator used for separation of dust from dust laden gas, the gas**

(a) Enters the cyclone form the top
(b) is admitted tangentially at high velocity
(c) develops a helical motion inside the chamber
(d) both (b) and (c)

**ISRO-12.02.2016**

**Ans. (d)**

**478. With reference to the classifiers, which of the following statements is/are correct?**

**1. Sorting classifier is a device which separate particles of differing densities.**
**2. Differential settling methods utilize the difference in terminal velocities that can exist between substances of differing density. The density of the medium is less than that of either substance.**

**Select the correct answer using the code given below:**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**479. Which of the following is NOT correctly matched?**

(a) **Clarifier:** A settler which removes virtually all the particles from a liquid
(b) **Classifier:** A device which separates the solids into two fractions
(c) **Thickener:** A device used for continuously decreasing the ratio of solid to liquid in a dilute suspension of finally divided sized particles by settling and decanting
(d) **Cyclone separator:** A settling device in which a strong centrifugal force, acting radially is used

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**480. The rate of filtration depends upon**

(a) Viscosity of filtrate
(b) Area of filtering surface
(c) Resistance of the filter cake
(d) On all of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**481. In spray column pulsed column and rotating disc contactor, the counter-current flow is produced by**

(a) Gravity (b) Pulsation
(c) Centrifugal force (d) Electrical charge

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**482. A typical batch filtration cycle consists of filtration followed by washing. One such filtration unit operating at constant pressure difference first filters a slurry during which 5 liters of filtrate is collected in 100 s. This is followed by washing, which is done for $t_w$ seconds and uses 1 liter of wash water. Assume the following relation to be applicable between the applied pressure drop ΔP, cake thickness L at time t, and volume of liquid V collected in time t.**

$$\frac{\Delta P}{L} = k_1 \frac{dV}{dt};\ L = k_2 V, \text{ if L is changing.}$$

**$k_2$ and $k_2$ can be taken to be constant during filtration and washing. The wash time $t_w$, in seconds (up to one decimal place) is _____**

**GATE-2015**

**Ans. (39.5 to 40.5)**

**483. Which of the following refers to discrete types of settling:**

(a) Type I sedimentation
(b) Type II sedimentation
(c) Type III sedimentation
(d) Type IV sedimentation

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Type I sedimentation

**484. A filter aid is added to the slurry before filtration to:**

(a) Decreases the porosity of the cake
(b) Increase the porosity of the cake
(c) Increase the compressibility coefficient of the cake
(d) Decrease the compressibility coefficient of the cake

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Increase the porosity of the cake

**485. Fluid-energy mill is an example of:**

(a) Crusher
(b) Grinder
(c) Ultrafine grinder
(d) Cutting machine

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Ultrafine grinder

## O. Transportation of solids

**486. The most suitable equipment for the transportation of 200 mesh size particles is**

(a) belt conveyor
(b) pneumatic conveyor
(c) screw conveyor
(d) bucket conveyor

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**487. Dry powdery solid material are transported by a**

(a) Belt conveyor (b) Screw conveyor
(c) Bucket elevator (d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**488. A mixer which is used for mixing non-cohesive solids and consist of a vertical tank containing a helical conveyor that elevates and circulates the material, is known is:**

(a) Tumbling mixer (b) Beater mixer
(c) Ribbon blender (d) Internal screw mixture

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**489. The conveyor suitable to transport large quantity of coarse solids to places of less elevation to a larger distance is**

(a) Screw conveyor
(b) Pneumatic conveyor
(c) Chain conveyor
(d) Belt conveyor

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**490. Which of the following can not be recommended for transportation of abrasive material?**

(a) Apron conveyor (b) Belt conveyor
(c) Flight conveyor (d) Chain conveyor

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**491. Sticky materials are transported by**

(a) Apron conveyor
(b) Screw conveyor
(c) Belt conveyor
(d) Hydraulic conveyor

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**492. Vertical transportation of materials can be done by a/an**

(a) Apron conveyor
(b) Pneumatic conveyor
(c) Bucket elevator
(d) both (b) and (c)

**ISRO-12.02.2016**

**Ans. (d)**

**493. Pneumatic transport of material is observed in :**

(a) Screw conveyor (b) Bucket elevator
(c) Vacuum cleaner (d) Belt conveyor

**AP PSC IOF-2009**

**Ans. (c)**

**494. Which one of the following unit operation is versatile; can be used to mix heat or cool; operates under pressure; and also useful for powders of sticky materials?**

(a) Pneumatic conveying
(b) Bucket elevators
(c) Screw conveyor
(d) Belt conveyor

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

## P. Fluidisation

**495. Laminar part of Ergun equation is called as.**

(a) Hagen-Poiseuille equation
(b) Stoke's equation
(c) kozeny-Cosman equation
(d) Burke-Plummer equation

**TNPSC AE-2018**

**Ans. (c)**

**496. Which of the following equations is valid for laminar flow of a fluid through a packed bed?**

(a) Fanning equation
(b) Kozney- Karman equation
(c) Hagen- Poiseuille eqation
(d) Blake-Plummer equation

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**497. For liquid flow through a packed bed, the superficial velocity as compared to average velocity through the channel in the bed is**

(a) More
(b) less
(c) equal
(d) independent of porosity

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**498. For liquid flow through a packed bed, the superficial velocity as compared to averaged velocity through the channel in the bed is**

(a) more
(b) less
(c) equal
(d) Independent of porosity

**RPSC ACF-2011**

**Ans. (b)**

**499. As the expansion of fluidized bed is concerned, the fluidized bed expands when**

(a) The superficial velocity remains constant
(b) The superficial velocity increases
(c) The superficial velocity diminishes to zero
(d) The superficial velocity decreases

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**500. Pressure drop in a packed bed for laminar flow is given by**

(a) Kozeny-Carman equation
(b) Blake-Plummer equation
(c) Leva's equation
(d) None of them

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**501. Operating velocity ($v_o$) in a packed tower is usually**
(a) twice the flooding velocity ($v_f$)
(b) half the flooding velocity
(c) equal to the flooding velocity
(d) more than the flooding velocity

**ISRO-12.02.2016**

**Ans. (b)**

**502. Fluidized beds are formed, when :**
(a) Fluid friction is zero
(b) Gravity force is less than fluid friction
(c) Pressure force is equal to gravity force
(d) Sum of fluid friction and pressure forces is equal and opposite to gravity forces

**AP PSC IOF-2009**

**Ans. (b)**

**503. If the average bed density at a given velocity is the same in all sections of the bed, then it is called :**
(a) Aggregative fluidization
(b) Particulate fluidization
(c) Minimum fluidization
(d) Bubbling fluidization

**AP PSC IOF-2009**

**Ans. (c)**

**504. The equation valid for a fluid flow through packed bed for very large Reynolds numbers :**
(a) Fanning equation
(b) Blake-plummer equation
(c) Kozeney-Carmen equation
(d) Hagen-Poiseuille equation

**AP PSC IOF-2009**

**Ans. (b)**

**505. A cylindrical packed bed of height 1 m is filled with equal sized spherical particles. The particles are nonporous and have a density of 1500 kg/m$^3$. The void fraction of the bed is 0.45. The bed is fluidized using air (density 1 kg/m$^3$). If the acceleration due to gravity is 9.8 m/s$^2$, the pressure drop (in Pa) across the bed at incipient fluidization (up to one decimal place) is_____.**

**GATE-2015**

**Ans. (8080 to 8100)**

**506. If,**

**$\in$ is void fraction**
**$\rho$ is density**
**$V$ is superficial velocity**
**$\Delta L$ is length**
**$D_P$ is diameter of packed bed**

**Burke-Plummer equation will be :**

(a) $\Delta P = \dfrac{1.75\rho(V')^2 \Delta L}{D_P}\left(\dfrac{1-\in}{\in^3}\right)$

(b) $\Delta P = \dfrac{1.25\rho(V')^2 \Delta L}{D_P}\left(\dfrac{1-\in}{\in^3}\right)$

(c) $\Delta P = \dfrac{1.75\rho(V')^3 \Delta L}{D_P}\left(\dfrac{1-\in}{\in^2}\right)$

(d) $\Delta P = \dfrac{1.25\rho(V')^3 \Delta L}{D_P}\left(\dfrac{1-\in}{\in^2}\right)$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**507. A packed bed is composed of cylinders having a diameter Dp = 0.02 m and length 1 m. For Bulk density of 962 kg/m$^3$, solid density of 100 kg/m$^3$ and with a = 288 m$^2$ the value of $D_e$ will be nearly :**
(a) 0.29 mm (b) 0.37 mm
(c) 0.45 mm (d) 0.53 mm

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**508. In 'Fluidized bed Combustion' (FBC) boiler, sulfur oxides formed during combustion react with the lime-stone (CaCO3) to:**
(a) Form solid $CaSO_4$ which falls to the bottom of the furnace
(b) Form solid CaS which falls to the bottom of the furnace
(c) Separate ash from coal
(d) Increase the combustion rate

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**509. Consider the following statements regarding catalyst and catalytic reactions:**
**1. Hot spots are developed in fluidized bed reactors**
**2. Bubbling fluidized bed is most commonly used in industries**
**3. Very little reaction occurs in the gas bubbles**
**4. Catalytic reaction and catalyst regeneration in a single unit is more costly**
**Which of the above statements are correct?**
(a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 4 only (d) 2 and 4 only

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**510. Fluidized beds are formed when**
(a) Fluid friction is zero
(b) Gravity force is less than fluid friction
(c) Pressure forces equal gravity forces
(d) Sum of fluid friction and pressure forces is equal and opposite to gravity forces

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Pressure forces equal gravity forces

**511. For a fluidized bed, with the increase in expansion of the bed up to solids carry over from the bed, the pressure drop across the bed:**
(a) Increase rapidly
(b) Decrease rapidly
(c) First increases then decrease
(d) Remains essentially constant

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Remains essentially constant

# 02.

# Heat Transfer

## A. Conduction

**1. For a given ambient air temperature with increase in the thickness of insulation of a hot cylindrical pipe, the rate of heat loss from the surface would :**

(a) Decrease
(b) First decrease and then increase
(c) Increase
(d) First increase and then decrease

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**2. A circular tube of outer diameter is 5 cm and inner diameter is 4 cm is used to convey hot fluid. The inner surface of the wall of the tube is at a temperature of 80°C while the outer surface of the wall of the tube is at 25°C. What is the rate of heat transport across the tube wall per meter length of the tube at steady state, if the thermal conductivity of the tube wall is 10 W/mK?**

(a) 13823 W/m (b) 15487 W/m
(c) 17279 W/m (d) 27646 W/m

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**3. The temperature distribution, at a certain instant of time in a concrete slab during curing is given by, $T = 3x^2 + 3x + 16$, where x is in cm and T in K. The rate of change of temperature with time is given by (assume diffusivity to be 0.0003 $cm^2/s$)**

(a) +0.0009 K/s (b) +0.0048 K/s
(c) –0.0012 K/s (d) +0.0018 K/s

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**4. The highest value of thermal conductivity is expected for:**

(a) Solid ice (b) Water
(c) Steam (d) Superheated steam

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**5. If thermal conductivity of a material of a wall varies as $k_0$ (1 + αt), then the temperature at the center of the wall as compared to that in case of constant thermal conductivity, will be:**

(a) More
(b) Less
(c) same
(d) Depend on other factors

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**6. Heat transfer by conduction through the walls of a hollow cylinder Q of the inner radius $r_1$ and outer radius $r_2$ with $t_1$ and $t_2$ temperature at radii $r_1$ and $r_2$ is equal to:**

(a) $\dfrac{2\pi lk(t_1+t_2)}{2.3\log\left[\frac{r_2}{r_1}\right]}$ (b) $\dfrac{2\pi lk(t_1-t_2)}{2.3\log\left[\frac{r_2}{r_1}\right]}$

(c) $\dfrac{2\pi k(t_1+t_2)}{2.3\log\left[\frac{r_2}{r_1}\right]}$ (d) $\dfrac{2\pi k(t_1-t_2)}{2.3\log\left[\frac{r_2}{r_1}\right]}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**7. A gas filled tube has 2 mm inside diameter and 25 cm length. The gas is heated by an electrical wire of diameter 50 microns located along the axis of the tube. Current and voltage drop across the heating element are 0.5 A and 4V respectively. If measured wire and inside tube wall temperature are 175°C and 150°C respectively, find the thermal conductivity of the gas filling the tube:**

(a) 0.156 (b) 0.184
(c) 0.123 (d) 0.188

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**8. Match the substances with their thermal conductivity**

| Substance | Thermal conductivity, W/m °C |
|---|---|
| (a) Air | 1. 17 |
| (b) Water | 2. 415 |
| (c) Stainless steel | 3. 0.5 |
| (d) Silver | 4. 0.014 |

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| (a) | 4 | 3 | 1 | 2 |
| (b) | 1 | 2 | 3 | 4 |
| (c) | 3 | 1 | 4 | 2 |
| (d) | 2 | 4 | 3 | 1 |

**TNPSC AE-2018**

**Ans. (a)**

**9. Unsteady state model is called**

(a) Stationary model
(b) transient model
(c) Distributed parameter model
(d) Lumped parameter model

**TNPSC AE-2018**

**Ans. (b)**

**10. In conduction heat transfer, the rate of heat transfer is given by $Q = K.A\frac{(T_1 - T_2)}{x}$, in which the ratio $x/K$ is called as.**

(a) Thermal conductivity
(b) Thermal diffusivity
(c) Thermal resistance
(d) Thermal gradient

**TNPSC AE-2018**

**Ans. (c)**

**11. What will be the cause for temperature drop at the composite wall interface?**

(a) Temperature difference
(b) Temperature gradient
(c) Thickness
(d) Voids

**TNPSC AE-2018**

**Ans. (d)**

**12. A glass window is 5 mm thick and the thermal conductivity of glass is 1.5 W/mK. The inner surface of the window is at 20°C and the outer surface is at 10°C. The dimension of the window is 1 m × 2 m. Calculate the rate of heat loss through the window**

(a) 60 KW (b) 600 W
(c) 60 W (d) 6000 W

**TNPSC AE-2018**

**Ans. (d)**

**13. Due to which of the following reasons most metals are good conduction of heat?**

(a) Capacity to absorb free energy of electrons
(b) Energy transport due to molecular vibration
(c) Lattice defect
(d) Lattice vibration and transfer by free electrons

**TNPSC AE-2019**

**Ans. (d)**

**14. Thermal conductivity of gases_____ with increasing temperatures and for liquids_____with increasing temperature.**

(a) increase, increase (b) increase, decrease
(c) decrease, increase (d) decrease, decrease

**TNPSC AE-2019**

**Ans. (b)**

**15. At critical thickness of insulation around a circular pipe, heat transfer is**

(a) maximum (b) minimum
(c) constant (d) none of the above

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**16. The Prandtl number for liquid metals is of the order of**

(a) $10^3$ (b) 1
(c) $10^{-2}$ (d) $10^2$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**17. The unit of resistance to heat transfer is**

(a) $(m^2 - K)/W$ (b) $J/(m^2 - K)$
(c) $W/(m^2 - K)$ (d) $J/(m - K)$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**18. According to the kinetic theory, the thermal conductivity of a monoatomic gas is proportional to (where T is absolute temperature)**

(a) $T^2$ (b) T
(c) $T^{1.5}$ (d) $T^{0.5}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**19. In transient heat conduction, the two significant dimensionless parameters are _____ number and _____ number.**

(a) Reynolds, Fourier
(b) Reynolds, Sherwood
(c) Sherwood, Fourier
(d) Biot, Fourier
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**20. Thermal diffusivity is defined as the ratio of**

(a) Thermal conductivity to thermal capacity
(b) Thermal capacity to thermal conductivity
(c) Thermal capacity to diffusivity
(d) Mass diffusivity to thermal capacity
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**21. Which of the statement is true**

(a) Thermal conductivity of substance depends on temperature
(b) Thermal conductivity of metal increases with temperature
(c) Thermal conductivity of gases are generally higher
(d) Thermal conductivity of gases are generally lower

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**22. The critical radius for the curved insulation**

(a) remains unaffected with change in thermal conductivity of insulation
(b) increases with decrease in the thermal conductivity of insulation
(c) decreases with decrease in the thermal conductivity of insulation
(d) increases linearly with decrease in the thermal conductivity of insulation

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**23. A hollow sphere of internal diameter ID = 20 cm and outer diameter OD = 30 cm contains hot fluid. What should be the critical radius of insulation for maximum rate of heat transfer? (Thermal conductivity k = 0.86 W/m-K and convection heat transfer coefficient of the outer fluid $h_o$ = 20 W/$m^2$-K)**

(a) 0.086 cm (b) 0.86 m
(c) 8.6 mm (d) 8.6 cm

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**24. A composite flat wall of a furnace is made up of two materials A and B. The thermal conductivity of A is twice that of B, while the thickness of layer of A is half that of B. If the temperatures at the two sides of the wall are 400K and 1200K, then the temperature drop (in K) across the layer of material A is**

(a) 125 (b) 133
(c) 150 (d) 160

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**25. The formula for thermal resistance ($R_k$) of a spherical shell of inner and outer radii as $r_i$ and $r_0$ respectively (where k is the thermal conductivity), is**

(a) $R_k = (r_i - r_0)/4\pi\, r_i r_0 k$
(b) $R_k = 4\pi\, r_i r_0 k/(r_i - r_0)$
(c) $R_k = (r_i - r_0)/4\pi\, k$
(d) $R_k = r_i\, r_0/4\pi\, k\,(r_i - r_0)$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**26. The wall of a pipe of radius 1 m is at a uniform temperature of 200 $^oC$, and is covered by insulation of thickness 0.1 m. The ambient air outside the insulated pipe is at 20 $^oC$ and has heat transfer coefficient of 10 W $m^{-2}$ $K^{-1}$. The thermal conductivity of the insulation material is 0.05 W $m^{-1}$ $K^{-1}$. If the heat transfer occurs at steady state, the temperature (in $^oC$) of the outer surface of insulation is _____(rounded off to second decimal plACFace).**

**GATE-2018**

**Ans. (27.2 to 28.6)**

**27. The one-dimensional unsteady heat conduction equation is**

$$\rho C_p \frac{\partial T}{\partial t} = \frac{1}{r^n \partial r}\left(r^n k \frac{\partial T}{\partial r}\right)$$

**where T – temperature, t – time, r – radial position, k – thermal conductivity, ρ – density, and $C_p$ – specific heat.**

**For the cylindrical coordinate system, the value of n in the above equation is**

(a) 0 (b) 1
(c) 2 (d) 3

**GATE-2017**

**Ans. (b)**

**28. Consider a composite wall consisting of three layers of insulation of length L1, L2, L3, and thermal conductivities $k_1$, $k_2$ and $k_3$ respectively. The insulating layers are placed in sequence 1, 2 and 3 and a certain rate of heat transfer results. If the order in now reversed to 3, 2, 1, rate of heat transfer through the wall under otherwise uniform conditions.**

(a) Will decrease
(b) Will increase
(c) Will remain unchanged
(d) Cannot be predicted, more information required

**RPSC ACF-2011**

**Ans. (c)**

**29. A furnace wall of thickness I m and of surface area 2$m^2$, is made of a material whose thermal conductivity is 1 KJ/hr m $^oC$. The temperature of inner surface wall is 1000$^oC$ and of outer surface is 200$^oC$ heat flow through the wall in KJ/hr will be**

(a) 2000 (b) 1600
(c) 1200 (d) 800

**RPSC ACF-2011**

**Ans. (b)**

**30. Ratio of the thermal conductivity of water to that of ice is approximately**

(a) 0.25 (b) 0.5
(c) 1.0 (d) 2.0

**RPSC ACF-2011**

**Ans. (a)**

**31. Transfer of heat by molecular collision is called**

(a) Conduction (b) Convection
(c) Radiation (d) All of these

**RPSC ACF-2011**

**Ans. (a)**

**32. Unit of thermal conductivity is:**

(a) Cal/cm.sec. deg C
(b) Cal/$cm^2$.sec. deg C
(c) Cal/cm.$sec^2$. deg C
(d) Cal/gm.sec. deg C

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**33. The ratio of thermal conductivity to the product of density & specific heat is known as:**

(a) Prandtl number
(b) Nusselt number
(c) Heat transfer coefficient
(d) Thermal diffusivity

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**34. If the rate of heat transfer varies with time, it is called:**

(a) Convective heat transfer
(b) Steady state conduction
(c) Unsteady state conduction
(d) None of these

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**35. As per Fourier's law of heat conduction, rate of heat flow across an isothermal surface is _____ the temperature gradient at the surface**

(a) Equal to
(b) Directly proportional to
(c) Inversely proportional to
(d) Proportional to the fourth power of

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**36. Which among the Copper, Nickel, Aluminum & Commercial bronze has the highest thermal conductivity?**

(a) Nickel (b) Aluminium
(c) Copper (d) Commercial bronze

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**37. Determine the rate of heat loss through the roof during night when the roof of an electrically heated home is 4 m long, 12 m wide, and 0.25 m thick, and is made of a flat layer of concrete whose thermal conductivity is k 50.8 W/m.K. The temperatures of the inner and the outer surfaces of the roof during night are measured to be 21 °C and 11 °C, respectively, for a period of 10 hours.**

(a) 61.9 kW (b) 1.69 kW
(c) 16.9 kW (d) 6.69 kW

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**38. Water enters a thin walled tube (L = 1 m, D = 3 mm) at an inlet temperature of 97°C and mass flow rate of 0.015 kg/s. The tube wall is maintained at a constant temperature of 27°C. Given: density of water = 1000 kg/$m^3$; viscosity of water = 489 × $10^{-6}$ pa.s, specific heat of water = 4184 J/kg/k; inside heat transfer co-efficient = 12978 W/($m^2$K). The outlet temperature of water in °C will be**

(a) 28 (b) 96
(c) 62 (d) 37

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**39. A solid at a temperature $T_1$ is kept in an evacuated chamber at a temperature of $T_2$ (> $T_1$). The rate of increase in the temperature of the body is proportional to**

(a) $T_2 - T_1$ (b) $T_2^2 - T_1^2$
(c) $T_2^3 - T_1^3$ (d) $T_2^4 - T_1^4$

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**40. The unit of resistance to heat transfer is**

(a) W/K (b) $W^{-1}$ $K^{-1}$
(c) WK (d) K/W

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**41. Let us assume two walls of same thickness and same cross-sectional area having thermal conductivities in the ratio 1 : 2. Let us say there is same temperature difference across the wall faces, the ratio of heat flow will be**

(a) 1 : 2 (b) 1 : 1
(c) 2 : 1 (d) 4 : 1

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**42. The heat flux (from outside to inside) across an insulating wall with thermal conductivity k = 0.04 W/m K and thickness 0.16 m is 10 W/$m^2$ . The temperature of the inside wall is –5°C. The outside wall temperature is**

(a) 25°C (b) 30°C
(c) 35°C (d) 40°C

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**43. Critical thickness of insulation for a sphere is steady state heat conduction given by ($h_o$=heat transfer coefficient at outer surface)**

**A. $r_c = k/h_o$** **B. $r_c = 2k/h_o$**
**C. $r_c = h_o/2k$** **D. $r_c = h_o/k$**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**44. Calculate the rate of heat flow across 3 $m^2$ surface of an insulation material of thickness 20 cm having thermal conductivity 0.04 W/m°C, if the temperature of the cold side is 10°C and that of hot side is 90°C.**

(a) 40 W (b) 48 W
(c) 53 W (d) 60 W

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

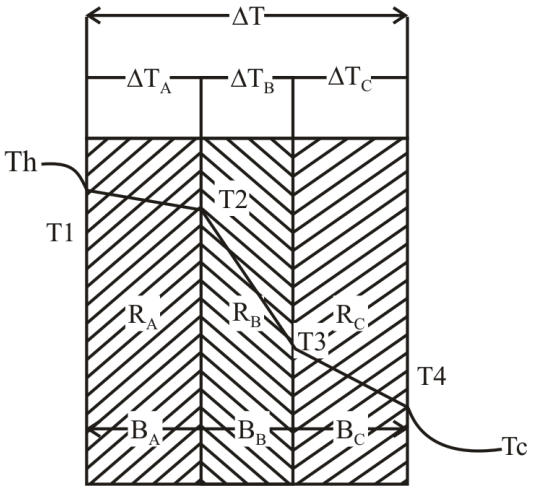
**45. A composite wall consists of two layers of different materials having thermal conductivities $k_1$ and $k_2$. For equal thickness of two layers the equivalent thermal conductivity of the slab will be**

(a) $k_1 + k_2$ (b) $k_1 k_2$
(c) $\frac{k_1 + k_2}{k_1 k_2}$ (d) $\frac{2k_1 k_2}{k_1 + k_2}$

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**46. Consider a flat wall in series, having thickness, $B_A$, $B_B$, $B_C$ and temperature difference of $\Delta T_A$, $\Delta T_B$, $\Delta T_C$, as shown below.**



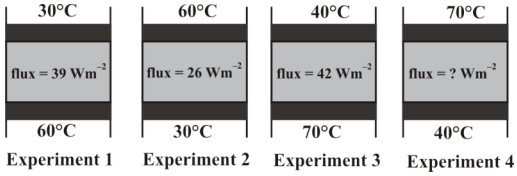

**What will be the rate of heat flow in the above case?**

(a) $q = \frac{\kappa \Delta T}{B}$ (b) $q = \frac{\kappa \Delta T}{R}$
(c) $q = \kappa A \Delta T$ (d) $q = \frac{\Delta T}{\Sigma R}$

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**47. In an experimental setup, mineral oil is filled in between the narrow gap of two horizontal smooth plates. The setup has arrangements to maintain the plates at desired uniform temperatures. At these temperatures, ONLY the radiative heat flux is negligible. The thermal conductivity of the oil does not vary perceptibly in this temperature range. Consider four experiments at steady state under different experimental conditions, as shown in the figure below. The figure shows plate temperatures and the heat fluxes in the vertical direction.**



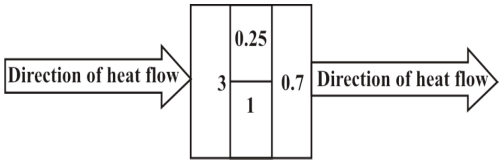

**What is the steady state heat flux (in W $m^{-2}$) with the top late at 70°C and the bottom plate at 40°C?**

(a) 26 (b) 39
(c) 42 (d) 63

**GATE-2016**

**Ans. (a)**

**48. A composite wall is made of four different materials of construction in the fashion shown below. The resistance (in K/W) of each of the sections of the wall is indicated in the diagram.**




**The overall resistance (in K/W, rounded off to the first decimal place) of the composite wall, in the direction of heat flow is——**

**GATE-2016**

**Ans. (3.88 to 3.92)**

**49. If $K_{Al}$, $K_w$ and $K_a$ are the thermal conductivities of aluminum, water and air at the same temperature, then**

(a) $K_w > K_{AI} > K_a$ (b) $K_{AI} > K_w > K_a$
(c) $K_a > K_w > K_{AI}$ (d) $K_a > K_{AI} > K_w$

**ISRO-12.02.2016**

**Ans. (a)**

**50. Which of the following has the highest thermal conductivity?**

(a) brick (b) air
(c) water (d) silver

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**51. What is thermal diffusivity and what is its basic dimension?**

(a) $(K/C_p\ \rho\ ), L^2/T$ (b) $C_p\ \rho/K, T/L^2$

(c) $C_p\ \mu/K, L/T$ (d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**52. Heat transfer rate per unit area is called**

(a) thermal conductivity

(b) heat flux

(c) heat transfer coefficient

(d) thermal diffusivity

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**53. What will be the typical range of Prandtl number of water?**

(a) 0.1 to 1 (b) 600 to 1000

(c) 0.01 to 0.02 (d) 5 to 10

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**54. A hot sphere of radius r at temperature $T_S$ is being cooled in still air at temperature $T_a$ far away from the sphere (i.e. r = ∞) so that $(Qr)/(4r^2)\ (T_s - T_a)\ k = 1$, hence Nusselt number will be**

(a) 1 (b) 2

(c) 0 (d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**55. Best tube materials from thermal conductivity point of view alone is :**

(a) Aluminium (b) Stainless steel

(c) Carbon steel (d) Copper

**AP PSC IOF-2009**

**Ans. (d)**

**56. The mechanism of heat transfer in solids is mainly by :**

(a) Convection (b) Conduction

(c) Radiation (d) Natural convection

**AP PSC IOF-2009**

**Ans. (b)**

**57. The term $\frac{Q}{A}$ is known as (in heat transfer) :**

(a) Thermal resistance

(b) Thermal conductivity

(c) Thermal loading

(d) Thermal gradient

**AP PSC IOF-2009**

**Ans. (c)**

**58. A composite wall consists of two layers of different material having conductivity $k_1$ and $k_2$. For equal thickness of the two layers, the equivalent thermal conductivity of the slab will be :**

(a) $k_1 + k_2$ (b) $\frac{2k_1k_2}{k_1 + k_2}$

(c) $\frac{k_1 + k_2}{k_1k_2}$ (d) $k^2{}_1k_2$

**AP PSC IOF-2009**

**Ans. (b)**

**59. The bottom face of a horizontal slab of thickness 6 mm is maintained at 300 °C. The top face is exposed to a flowing gas at 30 °C. The thermal conductivity of the slab is 1.5 W $m^{-1}$ $K^{-1}$ and the convective heat transfer coefficient is 30 W $m^{-2}$ $K^{-1}$. At steady state, the temperature (in °C) of the top face is ———**

**GATE-2014**

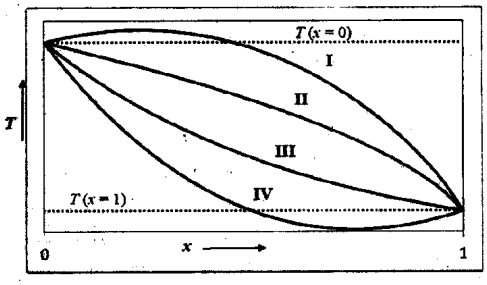
**Ans. (268 to 274)**

**60. A brick wall of 20 cm thickness has thermal conductivity of 0.7 W $m^{-1}$ $K^{-1}$. An insulation of thermal conductivity 0.2 $Wm^{-1}K^{-1}$ is to be applied on one side of the wall, so that the heat transfer through the wall is reduced by 75%. The same temperature difference is maintained across the wall before and after applying the insulation. The required thickness (in cm) of the insulation is ———**

**GATE-2014**

**Ans. (17.0 to 17.3)**

**61. Consider a solid block of unit thickness for which the thermal conductivity decreases with an increase in temperature. The opposite faces of the block are maintained at constant but different temperature: T(x = 0) > T (x = 1). Heat transfer is by steady state conduction in x-direction only. There is no source or sink of heat inside the block. In the figure below, identify the correct temperature profile in the block.**




**(1) I (2) II**

**(3) III (4) IV**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**62. Consider the following data for an insulating wall :**

**Thick fibre insulating board Δx = 25.4 mm**
**Inside temperature $T_1$ = 352.7 K**
**Outside temperature $T_2$ = 297.1 K**
**Thermal conductivity of fibre k = 0.048 W/m.K**

**The rate of heat transfer per unit area will be nearly :**

(a) 70 $W/m^2$ (b) 105 $W/m^2$
(c) 145 $W/m^2$ (d) 170 $W/m^2$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**63. On a hot summer's day a concrete highway may reach a temperature of 55° C. A stream of water is directed on the highway so that the surface temperature is suddenly lowered to 35° C. The values of k = 1.279 W/mK** $\alpha = 1.77 \times 10^{-3}\ m^2/h$ **and** $\frac{x}{2\sqrt{\alpha t}}$ **= 0.482 are for concrete. The time required to cool the concrete to 45°C at depth of 5 cm from the surface will be:**

(a) 1.26h (b) 1.52h
(c) 2.52h (d) 3.26h

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**64. The critical radius(r) of insulation on a pipe is given by:**

(a) r = 2k/h (b) r = k/h
(c) r = k/2h (d) r = h/k

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** r = k/h

**65. Heat transfer by conduction is directly proportional to:**

(a) Thermal potential difference/Thermal resistance
(b) Thermal resistance/Thermal potential difference
(c) Thermal potential difference x Thermal resistance
(d) Thermal resistance

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Thermal potential difference/Thermal resistance

**66. Fouling factor ($R_f$) is defined as:**

(a) $R_f = U_{clean} - U_{dirty}$
(b) $R_f = (1/U_{dirty}) - (1/U_{clean})$
(c) $R_f = (1/U_{clean}) - (1/U_{dirty})$
(d) $R_f = \frac{1}{\sqrt{U_{clean} \times U_{dirty}}}$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $R_f = (1/U_{dirty}) - (1/U_{clean})$

**67. The thermal conductivity of a gas:**

(a) Decreases with increasing temperature
(b) Increases with increasing temperature
(c) Is independent of temperature
(d) Is greater than the thermal conductivity of water at the same temperature

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Increases with increasing temperature

## B. Convection

**68. A fluid is flowing inside the inner tube of a double pipe heat exchange with diameter D. For a fixed mass flow rate, the tube side heat transfer coefficient for turbulent flow conditions is proportional to :**

(a) $D^{0.8}$ (b) $D^{-0.2}$
(c) $D^{-1}$ (d) $D^{-1.8}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**69. For turbulent flow in a tube, the heat transfer coefficient is obtained from the Dittus-Boelter equation, if the tube diameter is halved and the flow rate is doubled, then the heat transfer coefficient will change by a factor of :**

(a) 1 (b) 1.74
(c) 6.1 (d) 37

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**70. An electronic semiconductor device generates 38 × $10^{-3}$ kcal/h of heat. To keep surface temperature at upper same limit of 75°C, it is desirable that the heat generated should be dissipated to the surrounding environment which is at 30°C. The task is accomplished by attacking aluminum fins, 0.5 $mm^2$ square and 10 mm to the surface. Relevant data are given as k = 165 kcal/m-h-deg, h = 10.75 kcal/$m^2$-h-deg. The value of m will be:**

(a) 18.23$m^{-1}$ (b) 19.23$m^{-1}$
(c) 22.23 $m^{-1}$ (d) 3.62 $m^{-1}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**71. It is required to increase the heat dissipation rate over the surface of an electronic device of spherical shape of 5 mm radius exposed to convection with h = 10 W/m$^2$-K by encasing it in a spherical sheath of conductivity 0.04 W/m-K. For maximum heat flow, diameter of the sheath should be:**

(a) 18mm (b) 16mm
(c) 12mm (d) 8mm

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**72. Consider the development of laminar boundary layer for a moving non-reacting fluid on contact with a flat plate of length 1 along the flow direction. The average value of heat transfer coefficient can be obtained by multiplying the local heat transfer coefficient at the trailing edge by the factor:**

(a) 0.75 (b) 1.0
(c) 1.5 (d) 2.0

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**73. Air at 293 K is flowing along a heated plate at 407 K and at a velocity of 3m/s. The plate is 2m long. The heat transferred from the first 40 cm from the leading edge of the plate is 1450 W. Determine the width of the plate. Given properties of air at 350 K: $\rho = 0.998$ K/m$^3$, v = $20.76 \times 10^{-6}$ m$^2$/s, k = 0.03 W/m-K, $N_{pr} = 0.697$.**

(a) 3 m (b) 0.3 m
(c) 30m (d) 13 m

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**74. Water at 303 K enters a 25 mm I.d tube at the rate of 1200 liters per hour. Steam condenses on the surface of the tube (28 mm o.d) at a temperature of 393 K and its film heat transfer coefficient may be taken as 5800 W/m$^2$-K. Find the length of the tube required to heat the water to 343 K. Assume the thermal conductivity of metal wall as 950 W/m-K and of water as 0.63 W/m-K. The average density and viscosity of the water may be taken as 0.98 g/cm$^3$ and 0.0006 $P_a$.s respectively:**

(a) 15 m (b) 5 m
(c) 0.5 m (d) 0.05 m

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**75. Wilson plot is used to determine**

(a) film heat transfer coefficients
(b) overall heat transfer coefficients
(c) rate of heat flow
(d) thermal diffusivity

**TNPSC AE-2018**

**Ans. (a)**

**76. The overall coefficient of heat transfer is used in the problem of**

(a) Radiation
(b) Diffusivity
(c) Viscosity
(d) Conduction and conversion

**TNPSC AE-2019**

**Ans. (d)**

**77. Temperature profile in a steady state convective heat transport outside a tube is**

(a) hyperbolic (b) linear
(c) parabolic (d) non-linear

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**78. The Grashof number signifies the ratio of**

(a) Buoyancy to surface tension
(b) Buoyancy to initial force
(c) inertial to viscous forces
(d) Buoyancy to viscous forces

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**79. The Biot number is important in problems involving**

(a) transient heat conduction
(b) heat transfer by radiation
(c) heat transfer by forced convection
(d) heat transfer by natural convection

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**80. Find out the correct statement of the Colburn analogy between heat transfer and fluid friction**

(a) $jH = (St)^{2/3} (Pr)^{2/3} \phi_v^{-1} = \frac{f}{2}$

(b) $jH = (St) (Pr)^{2/3} \phi_v^{-1} = \frac{f}{2}$

(c) $jH = (St) (Pr)^{1/3} \phi_v^{-1} = \frac{f}{2}$

(d) $jH = (St)^{1/3} (Pr)^{2/3} \phi_v^{-1} = \frac{f}{2}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**81. Controlling heat transfer film coefficient is the one which offers _____ resistance to heat transfer.**

(a) no (b) the least
(c) the largest (d) lower

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**82. A cross-flow type air heater has an area of 50 $cm^2$. The overall heat transfer coefficient is 100W/$m^2$-K and heat capacity of both hot and cold steam is 1000 Wm-K. The value of NTU is**

(a) 0.2 (b) 6.0

(c) 1000 (d) 5.0

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**83. During the transient convective cooling of a solid object, the Biot number approaches 0. This indicates**

(a) uniform temperature throughout of the object

(b) negligible convection at the surface of the object

(c) significant thermal resistance within the object

(d) significant temperature gradient within the object.

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**84. Heat transfer rate is maximum in**

(a) laminar flow

(b) turbulent flow

(c) cocurrent flow

(d) countercurrent flow

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**85. An insulated storage tank contains 1000 kg liquid of specific heat 10 kJ $kg^{-1}$. The liquid is heated by saturated steam, condensing in a helical coil at a temperature of 180 $^oC$. The heat transfer area of the coil is 0.1 $m^2$. If the overall heat transfer coefficient is constant at 1000 W $m^{-2}$ $K^{-1}$, then the time (in hours) required to raise the temperature of the liquid in the tank from 20 $^oC$ to 80 $^oC$ is ____ (rounded off to second decimal place).**

**GATE-2018**

**Ans. (12.75 to 13.25)**

**86. A fluid flows over a heated horizontal plate maintained at temperature $T_w$. The bulk temperature of the fluid is $T_\infty$. The temperature profile in the thermal boundary layer is given by:**

$$T = T_w + (T_w - T_\infty)\left[\frac{1}{2}\left(\frac{y}{\delta_t}\right)^3 - \frac{3}{2}\left(\frac{y}{\delta_t}\right)\right], 0 \le y \le \delta_t$$

**Here, y is the vertical distance from the plate, $\delta_t$ is the thickness of the thermal boundary layer and k is the thermal conductivity of the fluid.**

**The local heat transfer coefficient is given by**

(a) $\frac{k}{2\delta_t}$ (b) $\frac{k}{\delta_t}$

(c) $\frac{3}{2}\frac{k}{\delta_t}$ (d) $2\frac{k}{\delta_t}$

**GATE-2017**

**Ans. (c)**

**87. Forced convection is unimportant when**

(a) $(Re.\ Pr)^{1/2} >> 1$ (b) $(Re.\ Pr)^{1/2} << 1$

(c) $(Re.\ Pr)^{1/2} \to 1$ (d) None of the above

**RPSC ACF-2011**

**Ans. (b)**

**88. Heat transfer coefficients for fluids flowing through pipes in laminar region are calculated by using the correlation of Sieder and Tate. Which one of the following is the correlation proposed by Sieder and Tate?**

(a) $Nu = 0.023\ (Re)^{0.8}\ (Pr)^{0.33}\ (\mu/\mu_w)^{0.14}$

(b) $Nu = 1.86\ [(Re)\ (Pr)\ (D/L)]^{1/3}\ (\mu/\mu_w)^{0.14}$

(c) $Nu = 2.0 + 0.60\ (Re)^{0.50}\ (Pr)^{1/3}$

(d) None of the above

**RPSC ACF-2011**

**Ans. (a)**

**89. In turbulent flow through a tube, flow velocity is increased two times. The change in heat transfer co-efficient will be**

(a) 64.1% decrease (b) 74.1% increase

(c) 74.1% decrease (d) 64.1% increase

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**90. A pipe of 25 mm diameter carries steam. The heat transfer co-efficient between the cylinder and surrounding is 5 W/$m^2$K. It is proposed to reduce the heat loss from the pipe by adding insulation having a thermal conductivity of 0.05 W/mK. Which one of the following statement is true?**

(a) The outer radius of the pipe is equal to the critical radius

(b) The outer radius of the pipe is less than the critical radius

(c) Adding the insulation will reduce the heat loss

(d) Adding the insulation will increase the heat loss

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**91. An oil cooler in a high performance engine has an outside surface area of 0.12 $m^2$ and a surface temperature of 65°C. At any time, air moves over the surface of the cooler at a temperature of 30°C and the heat transfer co-efficient is equal to 45.4 $W/m^2k$. find out the heat transfer rate.**

(a) 238.43 W (b) 190.68 W
(c) 543.67 W (d) 675.98 W

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**92. A composite wall consists of two plates A and B placed in series normal to the flow of heat. The thermal conductivities are $k_A$ and $k_B$ and the specific heat capacities are $C_{pA}$ and $C_{pB}$ , for plates A and B respectively. Plate B has twice the thickness of plate A. At steady state, the temperature difference across the plate A is greater than that across the Plate B when**

(a) $C_{pA} > C_{pB}$ (b) $C_{pA} < C_{pB}$
(c) $k_A < 0.5\ k_B$ (d) $k_A > 2k_B$

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**93. Laminar flow in a pipe, heat is transferred from hot wall to the liquid by**

**A. Conduction only**
**B. Convection only**
**C. Forced convection and conduction**
**D. Free convection and conduction**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**94. For condensation of pure vapors, if the heat transfer coefficients in film wise and drop wise condensation are respectively $h_f$ and $h_d$, then**

**A. $h_f > h_d$**
**B. $h_f < h_d$**
**C. $h_f = h_d$**
**D. $h_f$ could be greater or smaller than $h_d$**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**95. For a current carrying wire of 20 mm diameter exposed to air (h = 25 $W/m^2$), maximum heat distribution occurs when the thickness of insulation (k = 0.5 W/mK)**

(a) 20 mm (b) 1.5 mm
(c) 10 mm (d) 0 mm

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**96. Which mechanism of heat transfer is involved in heating a pot with water on a stove?**

(a) Convection
(b) Conduction
(c) Radiation
(d) Induction

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**97. The heat transfer by radiation from a mild steel surface is to be reduced by reducing the emissivity of the surface. This can be best achieved by**

(a) painting the surface black
(b) roughening the surface
(c) surface a mirror finish
(d) none of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**98. Water enters a thin walled tube (L = 1m, D = 3mm) at an inlet temperature of 97º C and mass flow rate 0.015 kg/s. The tube wall in maintained at a constant temperature of 27º C. Given the following data for water Density 'rho' = 1000 kg/' m^(3)'. Viscosity 'mu' = 489 × '10 ^ (−6)' Ns'/'m^ (2)', Specific heat 'C_ (p)' = 4184 J/kg/k**

**Inside heat transfer coefficient h = 12978 W/('m^(2)'K)**

**The outlet temperature of water in ºC is**

(a) 28 (b) 37
(c) 62 (d) 96

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**99. In pipe flow, heat is transferred from hot wall to the liquid by**

(a) conduction only
(b) forced convection only
(c) forced convection and conduction
(d) free and forced convection

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**100. The flow of air across heated radiator is an example for …….**

(a) Forced convection
(b) Natural convection
(c) Radiation
(d) Conduction

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

**101. The units of overall heat transfer coefficient are :**

(a) kcal/m/hr/°C (b) kcal/$m^2$/hr/°C

(c) kcal/hr/°C (d) kcal/$m^2$/$hr^2$/°C

**AP PSC IOF-2009**

**Ans. (b)**

**102. A heated solid copper sphere (of surface area A and volume V) is immersed in a large body of cold fluid. Assume the resistance to heat transfer inside the sphere to be negligible and heat transfer coefficient (h), density (ρ), heat capacity (C), and thermal conductivity (k) to be constant. Then, at time t, the temperature difference between the sphere and the fluid is proportional to**

(1) $\exp\left[-\frac{hA}{\rho VC}t\right]$ (2) $\exp\left[-\frac{\rho VC}{hA}t\right]$

(3) $\exp\left[-\frac{4\pi k}{\rho CA}t\right]$ (4) $\exp\left[-\frac{\rho CA}{4\pi k}t\right]$

(a) 1 (b) 2

(c) 3 (d) 4

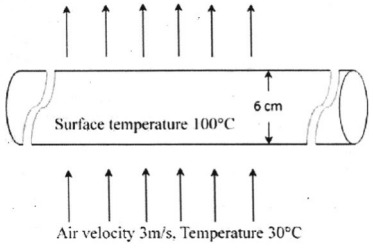
**GATE-2015**

**Ans. (a)**

**103. Air is flowing at a velocity of 3 m/s perpendicular to a long pipe as shown in the figure below. The outer diameter of the pipe is d = 6 cm and temperature at the outside surface of the pipe is maintained at 100°C. The temperature of the air far from the tube is 30°C.**

**Date for air: Kinematic viscosity, v = $18\times10^{-6}$ $m^2/s$, Thermal conductivity, k = 0.03 W/(m.K)**

**Using the Nusselt number correlation $Nu = \frac{hd}{k} = 0.024 \times Re^{0.8}$ the rate of heat loss per unit length (W/m) from the pipe to air (up to one decimal place) as_____**




**GATE-2015**

**Ans. (245 to 225)**

**104. Height Equivalent to a Theoretical Plate (HETP) is the height of packing that will give the same separation as one of the theoretical plate in gas-liquid mass transfer operations. HETP, which is experimentally determined, depends upon the**

(a) Flow rate of each fluid

(b) Type and size of packing

(c) Concentration of each fluid

(d) All of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**105. In natural convection heat transfer the correlating parameter is:**

(a) Graetz number

(b) Eckert number

(c) Grashof number

(d) Bond number

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Grashof number

**106. In forced convention, fluid moves under the influence of:**

(a) Changes in fluid pressure produced by external work

(b) Buoyant forces arising from changes in density

(c) Elastic forces

(d) Surface tension forces

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Changes in fluid pressure produced by external work

**107. Experiment conducted with a sparingly dissolving cylinder wall in a flowing liquid yielded the following correlation for the Sherwood number(Sh)**

**$Sh = 0.023\ (Re)^{0.83}(Sc)^{1/3}$**

**Assuming the applicability of the Chilton – colburn analogy, the corresponding correlation for heat transfer is:**

(a) $St = 0.023(Gr)^{0.83}\ (Pr)^{1/3}$

(b) $Nu = 0.023(Re)^{0.83}\ (Pr)^{1/3}$

(c) $H = 0.023(Re)^{0.83}\ (Pr)^{2/3}$

(d) $Nu = 0.069(Re)^{0.5}\ (Pr)^{4/3}$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $Nu = 0.023(Re)^{0.83}\ (Pr)^{1/3}$

## C. Radiation

**108. A black body at a higher temperature $T_H$ transfers energy by radiation to a black body at a lower temperature $T_L$. Initially $T_H = 1850^oC$, $T_L = 500^oC$ and the net rate of energy transfer is 25 W. After sometime, when $T_H = 1500^oC$ and $T_L = 750^oC$, what is the net rate of energy transfer ?**

(a) 8.73 W (b) 9.60 W
(c) 16.89 W (d) 11.01 W

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**109. The heat transfer by radiation from mild steel surface is to be reduced by emissivity of the surface. This can be best achieved by:**

(a) Painting the surface black
(b) Painting the surface white
(c) Giving the surface a mirror finish
(d) Roughening the surface

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**110. Two parallel surfaces each of emissivity 0.7 are maintained at different temperatures and accordingly between them. It is desired to reduce 75% of this radian heat transfer by inseating thin parallel shields of emissivity on both sides of shields should be:**

(a) One (b) Two
(c) Three (d) Four

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**111. If a body reflects all radiations on it, then it is called:**

(a) Black body (b) White body
(c) Grey body (d) Opaque body

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**112. Two large parallel plate one at $727^oC$ with emissivity $e_1 = 0.8$ and another at $227^oC$ with emissivity $e_2 = 0.4$. An aluminium radiation shield with an emissivity $e_3 = 0.05$ on both sides is placed between the plates. Calculate the percentage reduction in heat transfer rate between the two plates as a result of the shield. Use $s = 5.67 \times 10^{-8}$ W/m$^2$-K**

(a) 91.56 (b) 92.49
(c) 92.55 (d) 93.41

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**113. Ratio of emissive power of a body to the emissive power of a perfectly black body is called**

(a) emissivity (b) absorptivity
(c) transmissivity (d) reflectivity

**TNPSC AE-2018**

**Ans. (a)**

**114. A perfect black body**

(a) Is black in colour
(b) Reflects all incident radiation
(c) Absorbs all incident radiation
(d) Transmit all incident radiation

**TNPSC AE-2018**

**Ans. (c)**

**115. Planck's law holds good for bodies**

(a) Polished (b) black
(c) all coloured (d) white

**TNPSC AE-2019**

**Ans. (d)**

**116. _____is the ratio of the energy absorbed by the body to total energy falling on it.**

(a) Emissivity (b) Emissive power
(c) Absorptive power (d) Absorptivity

**TNPSC AE-2019**

**Ans. (c)**

**117. A black body has the_____ attainable emissive power at any given temperature.**

(a) minimum (b) maximum
(c) average (d) total

**TNPSC AE-2019**

**Ans. (b)**

**118. The equilibrium temperature reached by the absorber plate when no heat is being extracted from the collector is called**

(a) Saturation temperature
(b) Stagnation temperature
(c) Eutetic temperature
(d) Collecting temperature
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**119. Let $I_{b\lambda}$ be the spectral blackbody radiation intensity per unit wavelength about the wavelength $\lambda$.**
**The blackbody radiation intensity emitted by a blackbody over all wavelengths is**

(a) $\frac{dI_{b\lambda}}{d\lambda}$ (b) $\frac{d^2I_{b\lambda}}{d\lambda^2}$
(c) $\int_0^\infty I_{b\lambda}d\lambda$ (d) $\int_0^\infty \lambda I_{b\lambda}d\lambda$

**GATE-2017**

**Ans. (c)**

**120. Highest value of emissivity is of**

(a) Ice (b) Carbon

(c) Rubber (d) Paper

**RPSC ACF-2011**

**Ans. (a)**

**121. Absorptivity for a blackbody is**

(a) Zero (b) 0.5

(c) 1.0 (d) None of the above

**RPSC ACF-2011**

**Ans. (c)**

**122. Stefan Boltzmann law states that total emissive power of a black body is _____ the absolute temperature**

(a) Inversely proportional to

(b) Directly proportional to

(c) Proportional to the square of

(d) Proportional to the fourth power of

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**123. Sum of the absorptivity and reflectivity of an opaque body is:**

(a) 0.5 (b) 1

(c) 0 (d) None of these

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**124. According to Planks distribution law of heat radiation for a blackbody, the value of $(2C_1\pi^5)/(15C_2^4)$, where $C_1$ and $C_2$ are constant, is known as:**

(a) the Stefan-Boltzmann constant

(b) Fick's constant

(c) Plank's constant

(d) Fourier's constant

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**125. Select the correct option which correctly represents the below given expression**

$$C_{f.x}\frac{Re_L}{2}Nu_x Pr^{-1/3}$$

(a) Grashoff number

(b) Sieder-Tate correlation

(c) Reynolds analogy

(d) Chilton- Colburn analogy

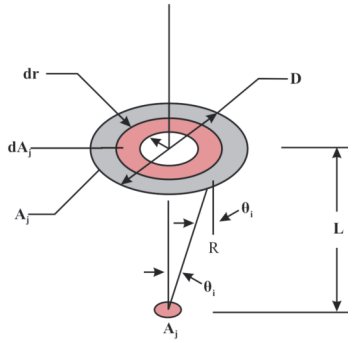
**HPCL Engineer-01.11.2022**

**Ans. (d)**

**126. What will be the view factor while considering a diffuse circular disk of diameter D and area $A_j$ and a plane diffuse surface of area $A_i << A_j$?**

**If the surfaces are parallel, and $A_i$ is located at a distance L from the center of $A_j$.**

**Also assume surfaces to be diffuse and uniform radiosity on surface $A_j$**




(a) $F_{ij} = D^2/(D^2 + 4L^2)$ (b) $F_{ij} = D^2/(\theta^2 + 4L^2)$

(c) $F_{ij} = D^2/(D^2 + 4\theta^2)$ (d) $F_{ij} = \theta^2/(D^2 + 4L^2)$

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**127. Select the correct expression among the following which can be used to represent the surface radiactive resistance.**

(a) $(1 - \varepsilon_i)/(\varepsilon_1 A_i)$ (b) $1/(\varepsilon_i A_i)$

(c) $\varepsilon_i/(\varepsilon_i A_i)$ (d) $(\varepsilon_i A_i)(1 - \varepsilon_i)$

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**128. Consider a 20-cm-diameter spherical ball at 400 K suspended in air. Assuming the ball closely approximates a blackbody, determine the total blackbody emissive power:**

(a) 1.45 kW/m$^2$ (b) 2.32 kW/m$^2$

(c) 23.2 kW/m$^2$ (d) 2.2 kW/m$^2$

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**129. According to Wein's displacement law which is the correct value of the ratio of $E_{b\lambda \cdot max}c/T^5$ is equal to:**

(a) $1.307 \times 10^{-7}$ W/(m$^3$.°K$^5$)

(b) $1.307 \times 10^{-3}$ W/(m$^3$.°K$^5$)

(c) 1.307W/(m$^3$.°K$^5$)

(d) $1.307 \times 10^{-5}$ W/(m$^3$.°K$^5$)

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**130. What is the unit of the solid angle which is denoted by ω?**

(a) Armstrong (b) Degree

(c) Steradian (d) Radian

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**131. The radiative heat loss between the top corner of flat plate solar collector at a temperature of $T_p$(K) and ambient air at a temperature $T_a$(K) is proportional to**

(a) $T_p^4 + T_a^4$ (b) $T_p^4 - T_a^4$

(c) $T_p^2 - T_a^2$ (d) $T_p + T_a$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**132. In gray body, emissivity is**

**A. Less than 1**

**B. Equal to 1**

**C. More than 1**

**D. None of the above**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**133. The space between two hollow concentric spheres of radii 0.1 m and 0.2 m is under vacuum. Exchange of radiation (uniform in all directions) occurs only between the outer surface ($S_1$) of the smaller sphere and the inner surface($S_2$) of the larger sphere. The fraction (rounded of to the second decimal place) of the radiation energy leaving $S_2$ which reaches $S_1$ is—**

**GATE-2016**

**Ans. (0.245 to 0.255)**

**134. Stefan-Boltzmann law which describes the radiation heat transfer states that, it is proportional to _____ (T- temperature in ºC, t-Absolute temperature in ºK)**

(a) $t^4$ (b) $T^4$

(c) $1/t^4$ (d) $1/T^4$

**ISRO-12.02.2016**

**Ans. (b)**

**135. If a body reflects all radiations incident on it, then it is known as a :**

(a) Black body (b) White body

(c) Grey body (d) Opaque body

**AP PSC IOF-2009**

**Ans. (b)**

**136. In a completely opaque medium, if 50% of the incident monochromatic radiation is absorbed, then which of the following statements are CORRECT?**

**P. 50% of the incident radiation is reflected**

**Q. 25% of the incident radiation is reflected**

**R. 25% of the incident radiation is transmitted**

**S. No incident radiation is transmitted**

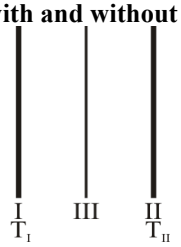
(a) P and S only (b) Q and R only

(c) P and Q only (d) R and S only

**GATE-2014**

**Ans. (a)**

**137. Two infinitely large parallel plates (I and II) are held at temperatures $T_I$ and $T_{II}$ ($T_I > T_{II}$) respectively, and placed at a distance 2d apart in vacuum. An infinitely large flat radiation shield (III) is placed in parallel in between I and II. The emissivities of all the plates are equal. The ratio of the steady state radiative heat fluxes with and without the shield is.**




**(1) 0.5** **(2) 0.75**

**(3) 0.25** **(4) 0**

(a) 1 (b) 2

(c) 3 (d) 4

**GATE-2015**

**Ans. (a)**

**138. The total hemispherical emissive power will be denoted by the symbol:**

(a) $\in$ (b) $\in_\lambda$

(c) e (d) $e_\lambda$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**139. A body at a given temperature will emit radiation of a whole range of wavelengths and not a single wavelength. The statement is related to:**

(a) Wavelength and Frequency

(b) The Origins of Radiant Energy

(c) The Distribution of Radiant Energy

(d) The Emissive Power

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**140. Which of the following laws are the laws of black-body radiation?**

(a) Planck's law, Wien's law and Fourier's law

(b) Planck's law, Fourier's law and Stefan-Boltzmann law

(c) Planck's law, Wien's law and Stefan-Boltzmann law

(d) Fourier's law, Wien's law and Stefan-Boltzmann law

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**141. A black body does not:**

(a) Emit radiation

(b) Absorb radiation

(c) Reflect radiation

(d) Follow Stefan – Boltzmann's law

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Reflect radiation

## D. Boiling and condensation

**142. The normal boiling points of water and toluene are 100 ºC and 111 ºC respectively. Toluene and water are completely immiscible in each other. The normal boiling point of equimolar mixture of toluene and water is**

(a) Less than 100ºC
(b) 100ºC
(c) between 100 and 111ºC
(d) 111ºC

**TNPSC AE-2018**

**Ans. (a)**

**143. The presence of even small amounts of noncondensing gas in a condensing vapor will**

(a) increase the rate of condensation
(b) not affect the rate of condensation
(c) reduce the rate of condensation
(d) indirectly affect the rate of condensation
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**144. Which one states that the boiling point of a given solution is a linear function of the boiling point of pure water at the same pressure**

(a) Boiling point elevation
(b) Charle's law
(c) Boyle's law
(d) Duhring's rule
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**145. Eutonic point is used in**

(a) Distillation (b) Evaporation
(c) Sublimation (d) Crystallisation

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**146. In pool-boiling, the highest heat transfer coefficient occurs in**

(a) subcooled boiling zone
(b) nucleate boiling zone
(c) partial film boiling zone
(d) film boiling zone

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**147. Vapour bubbles are formed in the nucleate boiling regime at a frequency of 10 bubbles per second per nucleation site. There are 100 nucleation sites per $m^2$ of heating area. The latent heat of vapourization and the density of vapour under the operating condition are 1000 kJ/kg and 1 kg/$m^3$ respectively. The diameter of each bubble is $10^{-3}$m. Assume that the entire heat supplied is used for vapour generation. The heat flux (in Watt per $m^2$ of heating area) is _____( rounded off to third decimal place).**

**GATE-2018**

**Ans. (0.51 to 0.54)**

**148. In nucleate boiling, the pressure inside a bubble in higher than the pressure of the surrounding liquid, Assuming that both the liquid and vapour are saturated . the temperature of the liquid will ALWAYS be**

(a) at $100^0$C
(b) lower that the temperature of the vapour
(c) equal to the temperature of the vapour
(d) higher than the temperature of the vapour

**GATE-2017**

**Ans. (b)**

**149. Which of the following accounts for maximum energy loss in a boiler**

(a) Flue gases
(b) Ash content in the fuel
(c) Incomplete combustion
(d) Unburnt carbon in flue gases

**RPSC ACF-2011**

**Ans. (a)**

**150. A 4 KW, 20 litre water heater is switched ON for 10 min. The heat capacity for water is 4 KJ/Kg-K. Assuming total electrical energy has gone into heating the water, the increase of water temperature is.**

(A) 15 ºC (B) 20 ºC
(C) 26 ºC (D) 30 ºC
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

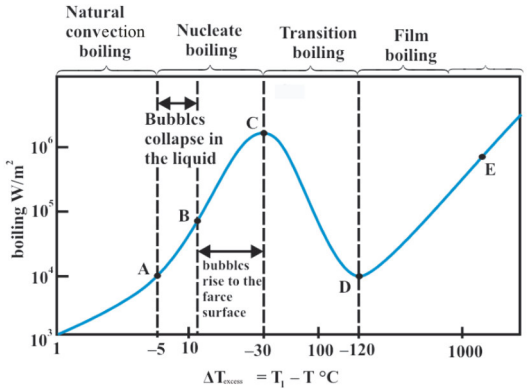
**151. Boiling point elevation of liquid in an evaporator results in:**

(a) Increased capacity
(b) Reduced capacity
(c) Increased economy
(d) No change in capacity or economy

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**152. Which point in the below given plot represents the critical heat flux?**




(a) Point B (b) Point A
(c) Point C (d) Point E

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**153. To determine Boiling Point Elevation (BPE) for strong solutions which empirical method is used?**

(a) Roult's law (b) Hagen equation
(c) Graetz number (d) Duhring's rule

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**154. Soluble silica present in boiler feed water can be removed by**

(a) Coagulation (b) Pre-heating
(c) Filtration (d) Ion exchange

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**155. Which of the following is the most detrimental for water used in high pressure boiler?**

(a) silica (b) dissolved oxygen
(c) phenol (d) turbidity

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**156. Pressure of steam from the fire tube boilers is**

(a) High (b) Moderate
(c) Low (d) All

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**157. The widely used method for conditioning of boiler feed-water is**

(a) cold lime process (b) coagulation
(c) hot lime process (d) sequestration

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**158. Boiler feed-water is treated to prevent**

(a) scaling and corrosion
(b) foaming and priming
(c) Both (a) and (b)
(d) Neither (a) nor (b)

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**159. A jacketed stirred tank with a provision for heat removal is used to mix sulphuric acid and water in a steady state flow process. $H_2SO_4(l)$ enters at a rate of 4 kg/h at 25°C and $H_2O$ ($l$) enters at a rate of 6 kg/h at 10°C. The following data are available:**

**Specific heat capacity of water = 4.2kJ $kg^{-1}K^{-1}$**

**Specific heat capacity of aqueous solution of 40 mass% $H_2SO_4$ = 2.8 kJ (kg solution)$^{-1}K^{-1}$.**

**Assume the specific heat capacities to be independent of temperature.**

**Based on reference states of $H_2SO_4(l)$ and $H_2O$ ($l$) at 25°C, the heat of mixing for aqueous solution of 40 mass% $H_2SO_4$ = –650kJ ($kgH_2SO_4$)$^{-1}$**

**If the mixed stream leaves at 40°C, what is the rate of heat removal (in kJ/h)?**

(a) 1802 (b) 2558
(c) 5702 (d) 6458

**GATE-2016**

**Ans. (a)**

**160. Steam at100°C is condensing on a vertical steel plate. The condensate flow is laminar. The average Nusselt numbers are $Nu_1$ and $Nu_2$, when the plate temperatures are 10°C and 55°C, respectively. Assume the physical properties of the fluid and steel to remain constant within the temperature range of interest. Using Nusselt equations for film-type condensation, what is the value of the ratio $\frac{Nu_2}{Nu_1}$ ?**

(a) 0.5 (b) 0.84
(c) 1.19 (d) 1.41

**GATE-2016**

**Ans. (c)**

**161. Water always boils when its**

(a) Temperature reaches 100ºC
(b) Vapour pressure equals 76 cm of Hg
(c) Saturated vapour pressure equals the external pressure on its surface
(d) Saturated vapour pressure is less than the atmospheric pressure

**ISRO-12.02.2016**

**Ans. (c)**

**162. In a pool boiling experiment, the following phenomena were observed.**

**1. Natural convection**
**2. Transition boiling**
**3. Film boiling**
**4. Nucleate boiling**

**Arrange these in the correct sequence of their occurrence. Select the correct answer from the code given below.**

(a) 1, 2, 3 and 4 (b) 1, 4, 2 and 3
(c) 4, 3, 2 and 1 (d) 1, 2, 4 and 3

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b) :**

**163. Match List- I with List – II and select the correct answer using the code given below the lists.**

| List-I | List-II |
|---|---|
| A. Multiple effect evaporator | 1. Chilling of milk |
| B. Barometric leg | 2. Heating of gases |
| C. Extended surface | 3. Steam economy |
| D. Plate heat exchanger | 4. Condensation at reduced pressure |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 4 | 3 | 2 | 1 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 4 | 3 | 1 | 2 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**164. Dropwise condensation usually occurs on :**
(a) Glazed surface (b) Coated surface
(c) Oily surface (d) Rough surface

**AP PSC IOF-2009**

**Ans. (c)**

**165. If the liquid wets the surface, the condensate flows on the surface in the form of a film, and the process is called:**
(a) Drop condensation
(b) Film condensation
(c) Film flow condensation
(d) Laminar film condensation

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**166. The peak heat flux in a boiling curve is called:**
(a) Burnout point (b) Leidenfrost point
(c) Boiling point (d) Nusselt point

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Burnout point

**167. According to Duhring's rule, the boiling point of a given solution is:**
(a) A linear function of the concentration of the solute
(b) A linear function of the boiling point of water at the same pressure
(c) A linear function of the pressure
(d) An exponential function of the boiling point of pure water at the same pressure

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** A linear function of the boiling point of water at the same pressure

## E. Heat exchangers

**168. In a double pipe heat exchanger, the inner diameter and outer diameter of the inner pipe are 4 cm and 5 cm respectively. The inner diameter of the outer pipe is 10 cm with a wall thickness of 1 cm. Then the equivalent diameters (in cm) of the annuls for heat transfer and pressure drop respectively are:**
(a) 15, 5 (b) 21, 6
(c) 6, 19 (d) 15, 21

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a) :**

**169. A chemical having specific heat of 3.3 kJ/Kg-K flowing at the rate of 20000 kg/h enter a parallel flow heat exchanger at 120°C. The flow rate of cooling water is 50000 kg/h with an inlet temperature of 20°C. The coefficient is 1050 W/m$^2$-K. Water specific heat = 4.186 kJ/Kg-K. Calculate the effectiveness of the heat exchanger:**
(a) 0.2 (b) 0.3
(c) 0.4 (d) 0.6

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**170. Automobile radiator is a heat exchanger of :**
(a) Counter flow type (b) Parallel flow type
(c) Cross flow type (d) Regenerator type

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**171. In an oil cooler, 60 g/s of hot oil enters a thin metal pipe of diameter 25 mm. An equal mass of cooling water flows through the annular space between the pipe and a large concentric pipe, the oil and water moving in opposite directions. The oil enters at 420 K and is to be cooled to 320 K. If water enters at 290 K, what length of pipe is required ? Take heat transfer coefficient of 1.6 k W/m$^2$-K on the oil side and 3.6 kW/m$^2$-K on water side. Specific heat of oil is 2.0 kJ/kg-K and that of water is 4.18 kJ/kg-K:**
(a) 0.267m (b) 2.67 m
(c) 26.7 m (d) 0.0267m

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**172. It is required to cool 250 kg/h of hot liquid with inlet temperature of 393 K using a parallel flow arrangement 1000 kg/h of cooling water is available for cooling purpose at a temperature of 283 K. Calculate the outlet temperatures of liquid and water and the effectiveness of heat exchanger if the overall heat transfer coefficient is 1160 W/m$^2$-K and the heat transfer surface of the exchanger is 0.25 m$^2$. ($C_p$ of water = 4187 J/kg-K, $C_p$ of liquid = 335 J/kg-K)**

(a) 24.21°C (b) 242.2°C
(c) 48.94°C (d) 33°C

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**173. Two iron pipes of the same nominal diameter but different schedule number will have :**
(a) The same inside diameter
(b) The same outside diameter
(c) The same wall thickness
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**174. The LMTD correction factor for multipass exchangers is always**
(a) < 1 (b) >1
(c) = 1 (d) = 0

**TNPSC AE-2018**

**Ans. (a)**

**175. With same logarithmic mean temperature difference, the heat transfer area requirement will be minimum for**
(a) Parallel flow
(b) Co-current flow
(c) Cross flow
(d) Counter-current flow

**TNPSC AE-2018**

**Ans. (d)**

**176. Why are floating heads provided in heat exchangers ?**
(a) To regulate the flow
(b) To increase the pressure drop
(c) To decrease the pressure drop
(d) To avoid deformation of tuber due to thermal expansion.

**TNPSC AE-2019**

**Ans. (d)**

**177. If the baffle spacing in a shell and tube heat exchanger increases, then the Reynolds number of the shell side fluid**
(a) decreases
(b) remain unchanged
(c) increases drastically
(d) increases

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**178. A cold fluid is heated from 40°C to 130°C by steam at 150°C the LMTD in parallel flow is**
(a) Zero
(b) Lower than LMTD in counter flow
(c) Equal to LMTD in counter flow
(d) Greater than LMTD in counter flow

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**179. The ratio of heat transfer area to the volume of heat exchanger for a compact heat exchanger should be**
(a) > 200 $ft^{-1}$
(b) = 200 $ft^{-1}$
(c) Between 101 $ft^{-1}$ and 199 $ft^{-1}$
(d) < 100 $ft^{-1}$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**180. TEMA standards specify a minimum pitch of _____ times the outside diameter of tubes for triangular pitch.**
(a) 1.25 (b) 1.50
(c) 1.75 (d) 2.0
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**181. Which of the following is used to find energy target for heating and cooling using heat exchanger ?**
**($\Delta T$ = Temperature difference)**
(a) $(\Delta T)_{max}$ (b) $(\Delta T_1 + \Delta T_2)/2$
(c) $\Delta T_1 - \Delta T_2$ (d) $(\Delta T)_{min}$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**182. In shell and tube heat exchangers, the fluid having corrosive and fouling tendency is routed through**
(a) the shell
(b) either shell or tube but at very slow velocity
(c) the shell, when the flow is counter-current; and the tube, when the flow is co-current
(d) the tube due to easier internal cleaning of the tubes.

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**183. Which tube arrangement in heat exchangers facilitates highest heat transfer rate?**
(a) Triangular pitch
(b) Square pitch
(c) Diagonal square pitch
(d) Heat transfer rate is independent of tube arrangement

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**184. A heat exchanger shell of outside radius 15 cm is to be insulated with glass wool of thermal conductivity 0.0825 W/m degree Celsius. The temperature at the surface is 280 degree Celsius and it can be assumed to remain constant after the layer of insulation has been applied to the shell. The convective film coefficient between the outside surface of glass wool and the surrounding air is estimated to be 8 $W/m^2$ degree Celsius. What is the value of a critical radius?**

(a) 9.31 mm (b) 10.31 mm
(c) 11.31 mm (d) 12.31 mm

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**185. A hot liquid is to be cooled in a 1–1 shell and tube heat exchanger from 80 $^oC$ to 50 $^oC$. Cooling water enters the tube side at 30 $^oC$ and exits at 45 $^oC$. The properties of the liquids are constant. Also, the overall heat transfer coefficient is same for counter-current and co-current modes. The percentage saving in heat transfer area for counter-current option with respect to the area of co-current option is**

(a) 22.12 (b) 36.32
(c) 27.10 (d) 44.50

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**186. Lower BWG means ____ of the tube.**

(a) lower thickness
(b) lower cross-section area
(c) outer diameter
(d) inner diameter

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**187. Segmental baffles in a 2-4 shell and tube heat exchanger**

(a) change the flow pattern of the tube side fluid and increase the overall heat transfer co-efficient
(b) increase the heat transfer coefficient in the shell side and support the tubes
(c) help to reduce the thermal expansion of the tubes and increase the heat transfer coefficient in the tube side
(d) increase the number of passes in the shell side and increase the heat transfer coefficient in the tube side

**GATE-2018**

**Ans. (b)**

**188. In a heat exchanger, the inner diameter of a tube is 25 mm and its outer diameter is 30 mm, The overall heat transfer coefficient based on the inner area is 360 $W/m^2.^\circ C$. Then the overall heart transfer coefficient based on the outer area, rounded to the nearest integer, is _____ $W/m^2.^\circ C$.**

**GATE-2017**

**Ans. (298 to 302)**

**189. Pick out the wrong statement.**

(a) Orifice baffles are never used in a shell and tube heat exchanger.
(b) Pressure drop on the shell side of a heat exchanger depends upon tube pitch also
(c) In a horizontal tube evaporator, surface blanketing by air is avoided.
(d) Split ring type and pull through type floating heads are two commonly used floating heads is heat exchanges.

**RPSC ACF-2011**

**Ans. (a)**

**190. In a fin tube heat exchanger,**

(a) Only heat transfer area is augmented
(b) Only heat transfer coefficient is augmented
(c) Both heat transfer area and heat transfer coefficient are augmented
(d) None of the above

**RPSC ACF-2011**

**Ans. (a)**

**191. The overall heat transfer co-efficient (U) for a composite wall of thickness $x_1, x_2, x_3$ and corresponding conductivities $k_1, k_2, k_3$ is given by.**

(A) $1/U = k_1/x_1 + k_2/x_2 + k_3/x_3$
(B) $U = k_1/x_1 + k_2/x_2 + k_3/x_3$
(C) $1/U = x_1/k_1 + x_2/k_2 + x_3/k_3$
(D) $U = x_1/k_1 + x_2/k_2 + x_3/k_3$

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**192. A counter flow heat exchanger is used to heat water from 20 ºC to 80 ºC by using exhaust gas entering at 140 ºC and leaving at 80 ºC. What is the LMTD for the best heat exchanger?**

(A) 80 ºC
(B) 60 ºC
(C) 110 ºC
(D) Not determinable

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**193. Multi pass heat exchangers are used**
(a) To minimize pressure drop
(b) To increase the heat transfer rate
(c) To facilitate easy fabrication
(d) None of these

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**194. Deformation of tubes due to thermal expansion in a shell and tube heat exchanger is prevented by providing:**
(a) Floating head
(b) Multiple pass construction
(c) Baffles
(d) Counter current flow

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**195. In heat exchangers, the term approach refers to:**
(a) Efficiency
(b) Economy
(c) Logarithmic mean temperature difference
(d) Terminal point temperature difference

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**196. Double pipe heat exchanges are used when,**
(a) Corrosive liquids are passed
(b) Overall heat transfer coefficient is low
(c) Heat transfer area required is law
(d) Viscous liquids are passed

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**197. In a double pipe heat exchanger, hot fluid is cooled from 300°C to 200°C in counter flow by a cold fluid entering at 150°C and heated to 200°C. The value of LMTD is?**
(a) 0°C (b) 108°C
(c) 72 °C (d) 36°C

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**198. Which kind of evaporators are will suited for concentrating heat sensitive liquids?**
(a) Circulation evaporator
(b) Multi-pass tubular evaporator
(c) Once through evaporator
(d) Agitated film evaporator

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**199. Determine the Overall Heat Transfer Coefficient for a double-tube counter-flow heat exchanger where $h_i$ (= 7663 w/m$^2$.k) and $h_0$ (=75.2w/m$^2$.k) are the convection heat transfer coefficients inside and outside the tube, respectively.**
(a) 74.5 W/m$^2$.K (b) 116.W/m$^2$.K
(c) 12.6 W/m$^2$.K (d) 712.2 W/m$^2$. K

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**200. Schedule Numbers (SN) and Birmingham Wire Gauge (BWG) are used for standard pipes and tubes, respectively. For a pipe/tube of given nominal diameter, the pipe/tube thickness (T) is related to SN and BWG as**
(a) $T \propto (SN)$ and $T \propto \frac{1}{(BWG)}$
(b) $T \propto \frac{1}{(SN)}$ and $T \propto \frac{1}{(BWG)}$
(c) $T \propto \frac{1}{(SN)}$ and $T \propto (BWG)$
(d) $T \propto (SN)$ and $T \propto (BWG)$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**201. For shell-and-tube heat exchangers, with increasing the heat transfer area, the purchased cost per unit heat transfer**
(a) Increases
(b) Passes through a maxima
(c) Remains constant
(d) Dccreases

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**202. In a counter-current double pipe heat exchanger, the hot fluid inlet and outlet temperatures are 80 °C and 60 °C. The cold fluid inlet and outlet temperatures are 20 °C and 40 °C. What is LMTD?**
(a) 20°C (b) 30°C
(c) 36.4°C (d) 40°C

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**203. In a counter-flow double pipe heat exchanger, oil (m = 2 kg/s, $C_p$ = 2.1 kJ/kg.°C) is cooled from 90 °C to 40°C by water (m = 1 kg/s, $C_p$ = 4.2 kJ/kg °C) which enters the inner tube at 10 °C. The radius of the inner tube is 3 cm and its length is 5 m. Neglecting the wall resistance, the overall heat transfer co-efficient based on the inner radius, in kW/m$^2$ K, is**
(a) 0.743 (b) 7.43
(c) 74.3 (d) 2475

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**204. A designer chooses the value of fluid flow rates and specific heats in such a manner that the heat capacities of the two fluids are equal. A hot fluid enters the counter flow heat exchanger at 100°C and leaves at 60°C. The cold fluid enters the heat exchangers at 40°C. The mean temperature difference between the two fluids is (in °C)**

**A. 60** **B. 40**

**C. 20** **D. 10**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**205. Correction is applied to LMTD for**

**A. Parallel flow**

**B. Counter flow**

**C. Cross flow**

**D. None of the above**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**206. Water is extensively used as the heat exchange medium since**

**A. It has a low dirt factor**

**B. It has a high heat capacity**

**C. It has low viscosity**

**D. It is relatively less corrosive**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**207. Standards on general design methods and fabrication materials for tubular heat exchangers is described by**

(a) TEAM (b) ASME

(c) TEMA (d) API

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**208. Correction of logarithmic mean temperature difference is applied for heat transfer rate calculation in heat exchangers operating in**

(a) Cross flow

(b) Counter current flow

(c) Parallel flow

(d) None of these

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**209. Minimum baffle spacing in a shell & tube heat exchanger is:**

(a) 1/5 of shell diameter

(b) 1/10 of shell diameter

(c) 1/3 of shell diameter

(d) 1/2 of shell diameter

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**210. In a 1–1 pass shell and tube exchanger, steam is condensing in the shell side at a temperature ($T_s$) of 135°C and the cold fluid is heated from a temperature ($T_1$) of 20°C to a temperature ($T_2$) of 90°C. The energy balance equation for this heat exchanger is**

$$\ln\frac{T_s - T_1}{T_s - T_2} = \frac{UA}{mc_p}$$

**Where U is the overall heat transfer coefficient, A is the heat transfer area, m is the mass flow rate of the cold fluid and $c_P$ is its specific heat. Tube side fluid is in a turbulent flow and the heat transfer coefficient can be estimated from the following equation:**

**$Nu = 0.023\ (Re)^{0.8}\ (Pr)^{1/3}$**

**Where Nu is the Nusselt number, Re is the Reynolds number and Pr is the Prandtl number. The condensing heat transfer coefficient in the shell side is significantly higher than the tube side heat transfer coefficient. The resistance of the wall to heat transfer is negligible. If only the mass flow rate of the cold fluid is doubled, what is the outlet temperature (in °C) of the cold fluid at steady state?**

(a) 80.2 (b) 84.2

(c) 87.4 (d) 88.6

**GATE-2016**

**Ans. (b)**

**211. Standard pipes of different schedule numbers and standard tubes of different BWG numbers are available in the market. For a pipe/ tube of a given nominal diameter, which one of the following statements is TRUE?**

(a) Wall thickness increases with increase in both the schedule number and the BWG number

(b) Wall thickness increases with increase in the schedule number and decreases with increase in the BWG number

(c) Wall thickness decreases with increase in both the schedule number and the BWG number

(d) Neither the schedule number, nor the BWG number has any relation to wall thickness

**GATE-2016**

**Ans. (b)**

**212. In a shell and tube heat exchanger, the 'tube pitch' is defined as the**

(a) O.D. of the tube for square pitch
(b) shortest distance between two adjacent tube holes
(c) shortest centre to centre distance between adjacent tubes
(d) none of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**213. LMTD in case of counter flow heat exchanger as compared to parallel flow heat exchanger is**

(a) higher
(b) lower
(c) same
(d) depends on the area of heat exchanger

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**214. The resistance to heat transfer contributed by scale formation in heat exchanger tubes is known as:**

(a) Boundary layer (b) Fouling factor
(c) Friction factor (d) Drag coefficient

**ISRO Scientist Engineer-06.08.2017**

**Ans. (b)**

**215. Correction of logarithmic mean temperature difference is applied for heat transfer rate calculation in heat exchangers operating in**

(a) Counter current flow
(b) Cross flow
(c) Parallel flow
(d) None of these

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

**216. Air is to be heated by condensing steam. Two heat exchangers are available:**
**(i) a shell and tube heat exchanger, and (ii) a finned tube heat exchanger, Tube side (inner)heat transfer area is equal in both cases. The recommended arrangement is,**

(a) Finned tube heat exchanger with air inside and steam outside
(b) Finned tube heat exchanger with air, outside and steam inside
(c) Shell and tube heat exchanger with air inside tubes and steam shell side
(d) Shell and tube heat exchanger with air on shell side and steam inside tubes

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (b)**

**217. Baffles used on shell side of a heat exchanger will**

(a) not affect heat transfer rate
(b) decrease heat transfer rate
(c) increase heat transfer rate
(d) decrease pressure drop on shell side

**ISRO-12.02.2016**

**Ans. (c)**

**218. 1- 4 shell and tube heat exchanger means**

(a) 1 shell side pass and 4 tube side passes
(b) 4 shell side passes and 1 tube side pass
(c) 1 longitudinal baffle and 4 transverse baffles
(d) 1 transverse baffles and 4 longitudinal baffle

**ISRO-12.02.2016**

**Ans. (a)**

**219. Fouling factor**

(a) is a dimensionless quantity
(b) does not provide a safety factor for design
(c) accounts for additional resistance to heat flow
(d) none of the above

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**220. In terms of inlet and outlet temperature differences $\theta_1$ and $\theta_2$, respectively, the logarithmic mean temperature difference is defined as**

(a) $ln(\theta_1/\theta_2)$ (b) $(ln\,\theta_1 + ln\,\theta_2)/2$
(c) $(\theta_1 - \theta_2)/ln(\theta_1/\theta_2)$ (d) $ln(\theta_1 + \theta_2)/2$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**221. With reference to double-pipe heat exchanger, which of the statements is/are correct?**

**1. For a fluid flowing in annulus of a double pipe heat exchanger, the equivalent diameter for heat transfer and pressure drop are the same.**
**2. A fluid is flowing inside the inner tube of a double-pipe heat exchanger with diameter d. For a fixed mass flow rate, the tube side heat transfer coefficient for turbulent flow conditions is proportional to $\alpha^{-1.8}$.**

**Select the correct answer using the code given below:**

(a) 1 Only
(b) 2 Only
(c) Both 1 and 2
(d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**222. LMTD correction factor is used for :**

(a) Double pipe exchanger
(b) Plate and frame heat exchanger
(c) Multipass shell and tube heat exchanger
(d) Counterflow of hot and cold fluids

**AP PSC IOF-2009**

**Ans. (c)**

**223. Floating head heat exchangers are used for :**

(a) Heat transfer between corrosive fluids
(b) Co-current heat transfer systems
(c) Where $\Delta T$ between the shell and the tubes is more
(d) Counter-current heat transfer

**AP PSC IOF-2009**

**Ans. (c)**

**224. In most of the shell and tube heat exchangers the tube pitch as compared to tube dia is :**

(a) Less (b) 1.25 to 1.5 times
(c) 2.5 times (d) One-fourth

**AP PSC IOF-2009**

**Ans. (b)**

**225. Triangular pitch tube layout is preferred over square pitch because :**

(a) It permits use of less tubes in a given shell dia
(b) Permits easy external cleaning
(c) More tubes can be accommodated in a given shell dia
(d) Easy to layout the tubes

**AP PSC IOF-2009**

**Ans. (c)**

**226. Log mean temperature difference in the case of counter current flow as compared to concurrent flow heat exchanger is :**

(a) more (b) less
(c) same (d) none of the above

**AP PSC IOF-2009**

**Ans. (a)**

**227. The log mean temperature difference can be replaced by the arithmetic mean when the ratio of the two terminal values of temperature differences, $\Delta T_1$ and $\Delta T_2$ does not exceed :**

(a) 10 (b) 2
(c) 4 (d) 3.5

**AP PSC IOF-2009**

**Ans. (*)**

**228. In shell and tube heat exchanger the corrosive liquid is generally passed through :**

(a) Shell side (b) Tube side
(c) Both (a) and (b) (d) None of the above

**AP PSC IOF-2009**

**Ans. (b)**

**229. A device which converts the latent or sensible heat of one fluid into the latent heat of vaporization of another is called a :**

(a) Heat exchanger (b) Recuperator
(c) Boiler (d) Regenerator

**AP PSC IOF-2009**

**Ans. (b)**

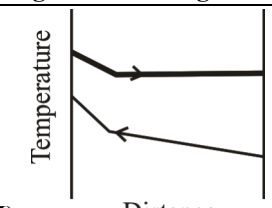
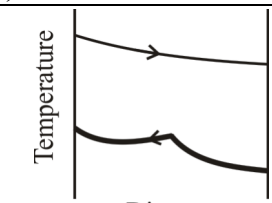
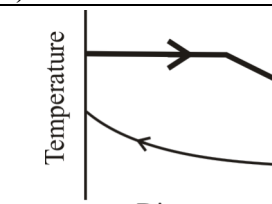
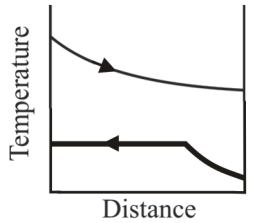
**230. Baffles are provided in heat exchanger to enhance :**

(a) Heat transfer area
(b) Heat transfer coefficient
(c) Temperature gradient
(d) Thermal resistance

**AP PSC IOF-2009**

**Ans. (b)**

**231. In the figure below, the temperature profiles of cold and hot fluids in counter current double pipe heat exchanges (in different modes of operation) are shown on the left. For each case, match the heat exchanges process for the fluid represented by the bold curve with the options given on the right.**

| | |
|---|---|
|  **(I)** | **(P) Heating of sub-cooled feed to super heated vapour.** |
|  **(II)** | **(Q) Condensation of super heated vapour** |
|  **(III)** | **(R) Boiling of sub-cooled liquid** |
|  **(IV)** | **(S) Condensation of saturated vapour followed by sub-cooling** |

**(1) I-P, II-Q, III-R, IV-S**
**(2) I-P, II-Q, III-S, IV-R**
**(3) I-Q, II-P, III-S, IV-R**
**(4) I-Q, II-S, III-P, IV-R**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**232. Choose the correct option.**
**Segmental baffles in a 2-4 shell and tube heat exchanger**
(a) Change the flow pattern of tube side fluid and increase the overall heat transfer coefficient
(b) Increase the heat transfer coefficient in the shell side and support the tube
(c) Help to reduce the thermal expansion of the tubes and increase the heat transfer coefficient in the tube side
(d) Increase the number of passes in the shell side and increase the heat transfer coefficient in the tube side

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**233. In a shell and tube heat exchanger, having square pitch, the shell side equivalent diameter is given by**
**(where, p = pitch, d = outside diameter of tube, n = number of tubes)**
(a) $\frac{4\left(p^2 - nd^2/4\right)}{nd}$ (b) $\frac{\left(p^2 - nd^2/4\right)}{nd}$
(c) $4p^2/nd$ (d) $nd/4p^2$

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**234. Pick out the wrong statement.**
(a) Safety valves are provided in heat exchangers for removal of non-condensable gases.
(b) Liquid metals and dowtherm are used for very high temperature steam.
(c) In a shell and tube heat exchanger, high pressure fluid is generally routed through the tube.
(d) A loop seal or U-seal are provided in a vertical condenser sub-cooler to prevent quick draining of condensate.

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**235. Which of the following is usually not used as the heat exchange fluid in a flat plate solar collector:**
(a) Air
(b) Fuel oil
(c) Water
(d) Ethylene glycol and water

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Fuel oil

**236. In a finned tube heat exchanger:**
(a) Only heat transfer area is augmented
(b) Only film coefficient is augmented
(c) Both heat transfer area and film coefficient are augmented
(d) None of these

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Only heat transfer area is augmented

**237. For the same process temperatures, the ratio of the LMTD in parallel flow to the LMTD in counter flow in Liquid–Liquid heat exchanger is always:**
(a) 1 (b) $< 1$
(c) $> 1$ (d) $\infty$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $< 1$

**238. What does 1-4 shell and tube heat exchanger imply:**
(a) 4 shell side passes and 4 tube side passes
(b) 4 shell side passes and 1 tube side passes
(c) 1 shell side pass and 4 tube side passes
(d) 4 tubes per pass

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** 1 shell side pass and 4 tube side passes

**239. Baffles used on shell side of a heat exchanger will:**
(a) Decrease heat transfer rate
(b) Increase heat transfer rate
(c) Not affect heat transfer rate
(d) Decrease pressure drop on shell side

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Increase heat transfer rate

## F. Evapouration

**240. Capacity of an evaporator is defined as**
(a) No. of kilograms water vaporized per kilograms of steam fed
(b) No. of kilograms of water vaporized per hour
(c) No. of kilograms of steam consumed per kilogram of water vaporized
(d) No. of kilograms of steam consumed per hour.

**TNPSC AE-2018**

**Ans. (b)**

**241. The type of evaporator used for concentrating of highly heat- sensitive materials is**

(a) Climbing film evaporators
(b) Falling film evaporators
(c) Forced circulation evaporators
(d) Agitated film evaporators

**TNPSC AE-2018**

**Ans. (b)**

**242. The economy in a single effect evaporator is**

(a) > 1 (b) 0
(c) < 1 (d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**243. Rate of evaporation decreases with**

(a) Increase in atmosphere pressure
(b) Increase in temperature
(c) Increase in movement of air above
(d) Increase in humidity of surrounding air

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**244. Which type of evaporator is preferred for the materials with heat sensitive products and liquids with suspended solids?**

(a) Long tube vertical (b) Horizontal tube
(c) Reboiler (d) Agitated thin film
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**245. The advantage of backward-feed multiple effect evaporators over forward-feed evaporators is that**

(a) heat sensitive materials can be handled
(b) there is no additional cost of pumping
(c) most concentrated liquor is at highest temperature
(d) equal heat transfer coefficients exist in various effects

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**246. In case of a multiple effect evaporator, the optimum number of effects is decided using**

(a) cost- benefit analysis
(b) floor area availability
(c) terminal parameters
(d) evaporation capacity required

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**247. Economy of evaporators used for concentrating sugarcane juice is**

(a) $\frac{\text{kg of concentrated juice produced}}{\text{kg of steam supplied}}$

(b) $\frac{\text{kg of steam supplied}}{\text{kg of sugarcane juice fed}}$

(c) $\frac{\text{kg of water vaporized}}{\text{kg of steam supplied}}$

(d) $\frac{\text{kg of sugarcane juice fed}}{\text{kg of water vaporized}}$

**GATE-2018**

**Ans. (c)**

**248. The specific heat of concentrated orange juice is 3.8 kilo-joules per kg ºC. How much heat must be removed to cool 3.0 kg of juice from 30 ºC to 0 ºC ?**

(a) 24 KJ (b) 38 KJ
(c) 340 KJ (d) 600 KJ

**RPSC ACF-2011**

**Ans. (c)**

**249. During evaporative cooling process with recirculated water, which one of the following will remain constant?**

(a) Partial pressure of vapour
(b) Relative humidity
(c) Dry bulb temperature
(d) Wet bulb temperature

**RPSC ACF-2011**

**Ans. (d)**

**250. Which type of feeding in a multiple effect evaporator does not require pumps to transfer liquid between the effects?**

(a) Forward feed (b) Backward feed
(c) Mixed feed (d) Parallel feed

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**251. In crystallizing evaporators, where a slurry of crystals and mother liquor is withdrawn, feed may be admitted directly to each effect, which is known as:**

(a) Backward feed (b) Mixed feed
(c) Parallel feed (d) Forward feed

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**252. Amount of water evaporated per unit weight of steam fed to an evaporator is known as:**

(a) Capacity (b) Efficiency

(c) Feed rate (d) Economy

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**253. For evaporation of viscous liquids, high rate of heat transfer can be obtained using:**

(a) Forced circulation evaporator

(b) Agitated film evaporator

(c) Long tube climbing film evaporator

(d) None of these

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**254. The quantity of heat required to evaporate 1 kg of a saturated liquid is called**

(a) specific heat (b) 1 kcal

(c) 1 calorie (d) latent heat

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**255. Amount of water evaporated per unit weight of steam fed to an evaporator is known as:**

(a) Capacity (b) Efficiency

(c) Economy (d) Feed rate

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**256. The number of kgs vapourised per kg of steam fed to the evaporator is defined as**

(a) Capacity

(b) Rate of evaporation

(c) Economy

(d) Rate of vapourisation

**ISRO-12.02.2016**

**Ans. (c)**

**257. Multiple effect evaporators are used to**

(a) increase the steam economy and decrease the capacity

(b) increase the steam economy and capacity

(c) decrease the steam economy and capacity

(d) decrease the steam economy and increase the capacity

**ISRO-12.02.2016**

**Ans. (b)**

**258. Fruit juice can be concentrated in**

(a) a long tube evaporator

(b) falling film evaporator

(c) high pressure evaporator

(d) none of the above

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**259. A multiple effect evaporator has a capacity to process 40,000 kg of a 10% NaOH solution per day to 25% solids. The water evaporated in kg per day :**

(a) 700 (b) 50,000

(c) 24,000 (d) 40,000

**AP PSC IOF-2009**

**Ans. (c)**

**260. Steam economy of a multiple effect evaporator system is defined as**

(a) kilogram of steam used per hour

(b) kilogram of steam consumed in all the effects for each kilogram of steam fed

(c) kilogram of steam used in all the effects for each kilogram of water vaporized per hour

(d) kilogram of water vaporized from all the effects for each kilogram of steam fed to the first effect

**GATE-2014**

**Ans. (d)**

**261. If saturated steam is used as the heating medium,**

**$\bar{V}$ = Vapour rate from reboiler**

**$\lambda_s$ = Latent heat of steam**

**$\lambda_m$ =Molal latent heat of mixture**

**The steam required at reboiler $m_s$ is:**

(a) $\frac{\bar{V}\lambda_s}{\lambda_m}$ (b) $\frac{\lambda_m}{\bar{V}\lambda_s}$

(c) $\frac{\lambda_s}{\bar{V}\lambda_m}$ (d) $\frac{\bar{V}\lambda_m}{\lambda_s}$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**262. It is desired to concentrate a 20% salt solution (20kg of salt in 100kg of solution) to a 30% salt solution in an evaporation. Consider a feed of 300kg/min at 30°C. The boiling point of solution is 110°C, the latent heat of vaporization is 2100 kJ/kg and the specific heat of the solution is 4kJ/kg K. The rate at which heat has to be supplied (in KJ/min) to the evaporator is:**

(a) $3.06 \times 10^5$ (b) $6.12 \times 10^5$

(c) $7.24 \times 10^5$ (d) $9.08 \times 10^5$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $3.06 \times 10^5$

# 03.

# Mass Transfer

## A. Diffusion

**1. Diffusivity coefficient of a binary gas mixture varies with pressure P as:**

(a) P (b) $P^2$

(c) $P^{-0.5}$ (d) $P^{-1}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**2. Schmidt number is given by:**

(a) 2 V/D (b) D/V

(c) V/D (d) 2 D/V

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**3. Sherwood number is given by:**

(a) 3/2 $h_m$ L/D (b) 1/2 $h_m$ L/D

(c) 3$h_m$ L/D (d) $h_m$ L/D

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**4. Fick's law is given by the formula:**

(a) $N_b = -D_{bc}d\ C_b/dx$

(b) $N_b = -2D_{bc}d\ C_b/dx$

(c) $N_b = -3D_{bc}d\ C_b/dx$

(d) $N_b = -4D_{bc}d\ C_b/dx$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**5. Mass transfer rate between two fluid phases does not necessarily depend on the_____of the two phases.**

(a) Chemical properties

(b) Physical properties

(c) Degree of turbulence

(d) Interfacial area

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**6. Estimate the diffusion coefficient for ammonia in air at 25 degree celsius temperature and one atmospheric pressure. For ammonia, Molecular weight = 17, Molecular volume = 25.81 $cm^3$/gm mole. For air, Molecular weight = 29:**

(a) 0.4880 $cm^2/s$ (b) 0.3880 $cm^2/s$

(c) 0.2880 $cm^2/s$ (d) 0.1880 $cm^2/s$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**7. In the expression for molar flux $N_A = (N_A + N_B)C_A/C - D_{AB}dC_A/dz$, the terms representing bulk flow and molecular diffusion are, respectively:**

(a) $(N_A + N_B)\ C_A/C,\ D_{AB}dC_A/dz$

(b) $(N_A - N_B)\ C_A/C,\ D_{AB}dC_A/dz$

(c) $(N_A + N_B)\ C_A/C,\ (N_A + N_B)\ C_A/C$

(d) $(D_{AB}dC_A/dz,\ D_{AB}dC_A/dz)$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**8. The dimension of diffusion coefficient is given by:**

(a) $MLT^{-2}$ (b) $L^2T^{-1}$

(c) $LT^{-1}$ (d) $ML^{-2}T$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**9. Unit of molecular diffusivity '$D_{AB}$' is**

(a) m/s (b) $m^2/s$

(c) $m/s^2$ (d) $\frac{mol}{m^2 s}$

**TNPSC AE-2018**

**Ans. (b)**

**10. Mass transfer coefficient 'k' is related with molecular diffusivity $D_{AB}$ as $K \alpha D^n_{AB}$ what is the value of 'n'?**

(a) – 1 (b) 0

(c) 0.5 (d) 1

**TNPSC AE-2018**

**Ans. (d)**

**11. _____occurs when the mean free path of the molecule is greater than the diameter of the catalyst pore**

(a) Molecular Diffusion

(b) Surface Tension

(c) Knudsen diffusion

(d) Reaction

**TNPSC AE-2019**

**Ans. (c)**

**12. If mass diffusivity in a mixture is equal to the thermal diffusivity, then the Lewis number is**

(a) 1 (b) 0

(c) > 1 (d) < 1

**TNPSC AE-2019**

**Ans. (a)**

**13. The ratio of the diffusion coefficient in a gas to that in liquid is of orders of**

(a) $10^2$ (b) $10^5$

(c) $10^{-2}$ (d) $10^{-5}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**14. Diffusion coefficient in gaseous mixture is proportional to (Where T, is temperature, K)**

(a) $T^{3/2}$ (b) $T^{-1/2}$

(c) $T^{-3/2}$ (d) $T^{1/2}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**15. For steady state molecular diffusion of gas A through non-diffusing gas B, the ratio $\frac{NA}{(N_A + N_B)}$ is**

(a) $\frac{1}{2}$ (b) 0

(c) $\infty$ (d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**16. Fick's second law of diffusion in one dimension is**

(a) $J_A = -D_{AB}\frac{\partial C_A}{\partial z^2}$ (b) $J_A = -D_{AB}\frac{\partial C_A}{\partial z}$

(c) $\frac{\partial C_A}{\partial \theta} = -D_{AB}\frac{\partial^2 C_A}{\partial z^2}$ (d) $\frac{\partial C_A}{\partial \theta} = D_{AB}\frac{\partial^2 C_A}{\partial z^2}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**17. For ash layer control, time required for complete conversion ($\tau$), consider $A_g + bB_{(s)} \rightarrow C_g + D_s$**

**Where $\rho_B$ = molar density**

**R = Radius of particle**

**b = Stiochiometric coefficient**

**$D_e$ = Diffusivity**

**CAg = Concentration**

(a) $\frac{\rho_B R^2}{6bD_e CAg}$ (b) $\frac{\rho_B^2 R^2}{12bD_e CAg}$

(c) $\frac{R^2}{6bD_e CAg}$ (d) $\frac{\rho_B^2 R^2}{12bD_e}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**18. The equation used for diffusion process for a smaller pore size which is less than the normal mean free path is**

(a) Chapman -Enskog equation

(b) Stokes- Einstein equation

(c) Knudsen diffusion equation

(d) Lennard-Jones potential

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**19. Relationship between effective diffusivity ($D_E$), molecular diffusivity ($D_M$) and Knudsen diffusivity ($D_K$) is given as"**

(a) $D_E = D_M / D_K$

(b) $D_E = D_M \times D_K$

(c) $(1/D_E) = (1/D_M) + (1/D_K)$

(d) $D_E = D_M + D_K$

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**20. If the pressure of the gas is increased the value of diffusivity**

(a) increases

(b) decreases

(c) remains constant

(d) Any of the other option

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**21. Mass transfer coefficient of liquid is**

(a) affected more by temperature than that for gases

(b) affected much less by temperature than that for gases

(c) not affected by temperature

(d) None of the above

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**22. The diffusivity has same dimension as**

(a) absolute viscosity

(b) kinematic viscosity

(c) density

(d) concentration

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**23. Which one of the following is not a diffusional process?**

(a) Mass transfer

(b) Heat transfer

(c) Momentum transfer

(d) Molecular diffusion

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**24. According to the surfACF renewal theory, the unit of fractional rate of surfACF renewal is**

(a) $m^2s^{-2}$ (b) $m^2s^{-1}$

(c) $ms^{-1}$ (d) $s^{-1}$

**GATE-2018**

**Ans. (d)**

**25. The Sherwood number ($Sh_L$) correlation for laminar flow over a flat plate of length L is given by**

**$Sh_L = 0.664\ Re_L^{0.5} Sc^{1/3}$**

**Where $Re_L$ and Sc represent Reynolds number and Schmidt number, respectively,**

**This correlation, expressed in the form of Chilton-Colburn $j_D$ factor, is**

(a) $j_D = 0.664$

(b) $j_D = 0.664\ Re_L^{-0.5}$

(c) $j_D = 0.664\ Re_L$

(d) $j_D = 0.664\ Re_L^{0.5} Sc^{2/3}$

**GATE-2017**

**Ans. (b)**

**26. The Knudsen diffusivity is proportional to**

(a) $T^{-1/2}$ (b) $T^{1/2}$

(c) T (d) 1/T

**RPSC ACF-2011**

**Ans. (b)**

**27. A solid coated on the surfACF of a flat plate dissolves in a liquid flowing past the plate at an angle of incidence, Diffusion of the solute occurs through the boundary layer formed on the plate. In such a case, the thickness of the momentum boundary layer is expected to be greater than the concentration boundary layer, if Schmidt Number, $S_c$ is**

(A) >1 (B) <1

(C) 1 (D) 0.005

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**28. Unit of mass velocity is**

(A) Kg / hour

(B) Kg /$m^2$ hour

(C) Kg/$m^2$

(D) Kg/$m^3$ hour

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**29. Component A is diffusing in a medium B. The flux $N_A$ relative to a stationary point is equal to the flux due to molecular diffusion if.**

(A) Mass transfer is accompanied by reaction

(B) Diffusion of A is in stagnant medium B

(C) Molecular mean free path is high

(D) There is equimolar counter diffusion

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**30. Lewis number is the ratio of**

(a) Thermal diffusivity to mass diffusivity

(b) Mass diffusivity to momentum diffusivity

(c) Momentum diffusivity to thermal diffusivity

(d) Momentum diffusivity to mass diffusivity

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**31. Eddy diffusivities $v_t$ and $a_t$ are———**

(a) not fluid properties and their values depend on flow conditions

(b) fluid properties but their values do not depend on flow conditions

(c) fluid properties and their values depend on flow conditions

(d) not fluid properties and their values do not depend on flow conditions

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**32. The surfACF renewal frequency in Danckwerts' model of mass transfer is given by ($k_l$; mass transfer co- efficient, m/s); $D_A$ – Diffusivity, $m^2/s$)**

(a) $\sqrt{k_L^2 D_A}$ (b) $k_L^2 D_A$

(c) $\frac{k_L^2}{D_A}$ (d) $\frac{k_L}{D_A^2}$

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**33. The binary diffusivity in gases and liquids vary respectively as**

**A. $T^{3/2}$ and T**

**B. T and $T^{3/2}$**

**C. $\sqrt{T}$ and $T^{3/2}$**

**D. $T^{3/2}$ and $\sqrt{T}$**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**34. The Lewis number of a mixture is defined as**

**A. $N_{Pr}/N_{Sc}$**

**B. $N_{Pr}\ N_{Sc}$**

**C. $N_{Sc}/N_{Pr}$**

**D. $C_S\ (N_{Sc}/N_{Pr})$**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**35. Which one of the following is the unit of mass diffusivity?**

(a) $m^3/s$ (b) $m^2/s$

(c) m/s (d) None

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**36. Consider the following two cases for a binary mixture of ideal gases A and B under steady state conditions. In Case 1, the diffusion of A occurs through non-diffusing B. In Case 2, equimolal counter diffusion of A and B occurs. In both the cases, the total pressure is 100 kPa and the partial pressures of A at two points separated by a distance of 10 mm are 10kpa and 5 kPa. Assume that the Fick's first law of diffusion is applicable. What is the ratio of molar flux of A in Case 1 to that in Case 2?**

(a) 0.58 (b) 1. 08

(c) 1.58 (d) 2.18

**GATE-2016**

**Ans. (b)**

**37. Mass transfer coefficient, k according to penetration theory various with mass diffusivity as**

(a) D ^ (0.5)

(b) D

(c) 1/D

(d) D ^ (1.5)

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**38. Match the following:**

| | | |
|---|---|---|
| **A. Reynolds Analogy** | **P.** | **$k / \rho C_p$** |
| **B. Colburn Analogy** | **Q.** | **St = f/2** |
| **C. Momentum diffusivity** | **R.** | **St. $Pr^{2/3}$ = f/2** |
| **D. Thermal diffusivity** | **S.** | **Kinematic viscosity** |

**Where f – friction factor, k-Thermal conductivity, ρ-density, $C_p$–heat capacity at constant pressure.**

(a) A-Q, B-R, C-P, D-S (b) A-Q, B-R, C-S, D-P

(c) A-R, B-S, C-Q, D-P (d) A-S, B-P, C-R, D-Q

**ISRO-12.02.2016**

**Ans. (b)**

**39.Match the following:**

| | |
|---|---|
| **A. Diffusion** | **P. Momentum diffusivity /Mass diffusivity** |
| **B. Schmidt Number** | **Q. Movement of particles** |
| **C. Film theory** | **R. Sc/Pr** |
| **D. Lewis Number** | **S. Mass transfer coefficient (k) α Diffusivity ($D_{AB}$)** |

(a) A-Q, B-P, C-S, D-R (b) A-Q, B-P, C-R, D-S

(c) A-P, B-Q, C-R, D-S (d) A-S, B-R, C-P, D-Q

**ISRO-12.02.2016**

**Ans. (c)**

**40. The units of diffusivity are :**

(a) m/s (b) $m^2/s$

(c) $kmol/m^2.s$ (d) s/m

**AP PSC IOF-2009**

**Ans. (b)**

**41. For steady state molecular diffusion of gas 'A' through non-diffusing B, the rate $N_A/(N_A + N_B)$ is :**

(a) 0 (b) 1

(c) ∞ (d) 1/2

**AP PSC IOF-2009**

**Ans. (b)**

**42. The Knudsen diffusivity is dependent on :**

(a) the molecular velocity only

(b) the pore radius of the catalyst only

(c) the molecular velocity and pore radius

(d) the molecular mean free path

**AP PSC IOF-2009**

**Ans. (c)**

**43. Mass transfer coefficient is defined as :**

(a) Flux = coefficient/concentration difference

(b) Coefficient = flux/concentration difference

(c) Coefficient = flux × concentration

(d) flux = concentration difference/coefficient

**AP PSC IOF-2009**

**Ans. (b)**

**44. A binary distillation column is operating with a mixed feed containing 20 mol% vapour. If the feed quality is changed to 80 mol% vapour, the change in the slope of the q-line is ——**

**GATE-2014**

**Ans. (3.6 to 3.9)**

**45. A spherical ball of benzoic acid (diameter = 1.5 cm) is submerged in a pool of still water. The solubility and diffusivity of benzoic acid in water are 0.03 $kmol/m^3$ and $1.25 \times 10^{-9}$ $m^2/s$ respectively. Sherwood number is given as $Sh = 2.0 + 0.6\ Re^{0.5}\ Sc^{0.33}$. The initial rate of dissolution (in kmol/s) of benzoic acid approximately is**

(a) $3.54 \times 10^{-11}$ (b) $3.54 \times 10^{-12}$

(c) $3.54 \times 10^{-13}$ (d) $3.54 \times 10^{-14}$

**GATE-2014**

**Ans. (b)**

**46. A spherical storage vessel is quarter-filled with toluene. The diameter of the vent at the top of the vessel is 1/20th of the diameter of the vessel. Under the steady state condition, the diffusive flux of toluene is maximum at**

(a) the surface of the liquid
(b) the mid-plane of the vessel
(c) the vent
(d) a distance 20 times the diameter of the vent away from the vent

**GATE-2014**

**Ans. (c)**

**47. A spherical naphthalene ball of 2 mm diameter is subliming very slowly in stagnant air at 25°C. The change in the size of the ball during the sublimation can be neglected. The diffusivity of naphthalene in air at 25°C is $1.1\times10^{-6}$ m²/s.**

**The value of mass transfer coefficient is $B\times10^{-3}$ m/s, where B (up to one decimal place) is _____**

**GATE-2015**

**Ans. (1.0 to 1.2)**

**48. For a binary mixture of components A and B, $N_A$ and $N_B$ denote the total molar fluxes of components A and B, respectively. $J_A$ and $J_B$ are the corresponding molar diffusive fluxes. Which of the following is true for equimolar counter-diffusion in the binary mixture?**

**(1) $N_A + N_B = 0$ and $J_A + J_B \neq 0$**
**(2) $N_A + N_B \neq 0$ and $J_A + J_B = 0$**
**(3) $N_A + N_B \neq 0$ and $J_A + J_B \neq 0$**
**(4) $N_A + N_B = 0$ and $J_A + J_B = 0$**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (d)**

**49. Wilke-Chang correlation for dilute solutions of concentration of A into B of non-dissociating solutes upto about ±10% is:**

(a) $D_{AB} = 7.4\times10^{-8}\dfrac{(\Psi_B M_B)^{\frac{1}{2}} T}{\mu \hat{V}_A^{\,0.6}}$

(b) $D_{AB} = 4.4\times10^{-8}\dfrac{(\Psi_B)^{\frac{1}{4}} T}{\mu \hat{V}_A^{\,0.4}}$

(c) $D_{AB} = 7.4\times10^{8}\dfrac{(\Psi_B)^{\frac{1}{2}} T}{\mu \hat{V}_A^{\,0.2}}$

(d) $D_{AB} = 4.4\times10^{8}\dfrac{(\Psi_B)^{\frac{1}{2}} T}{\mu \hat{V}_A^{\,0.2}}$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**50. The transport of one constituent from a region of higher concentration to that of a lower concentration is known as:**

(a) Mass Diffusion (b) Mass Transition
(c) Mass Transfer (d) Heat Transfer

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**51. In diffusion, when vapour pressure of water in air equals that of the liquid, the air is saturated and vaporization:**

(a) Persists (b) Initiates
(c) Ceases (d) Preserves

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**52. Ficks second law of diffusion in one dimension is:**

(a) $J_A = -D_{AB}\dfrac{\partial C_A}{\partial z}$

(b) $\dfrac{\partial C_A}{\partial \theta} = -D_{AB}\dfrac{\partial^2 C_A}{\partial z^2}$

(c) $J_A = -D_{AB}\dfrac{\partial^2 C_A}{\partial z^2}$

(d) $\dfrac{\partial C_A}{\partial \theta} = D_{AB}\dfrac{\partial^2 C_A}{\partial z^2}$

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** $\dfrac{\partial C_A}{\partial \theta} = D_{AB}\dfrac{\partial^2 C_A}{\partial z^2}$

**53. Molecular diffusivity of a liquid:**

(a) Increases with temperature
(b) Decreases with temperature
(c) May increase or decrease with temperature
(d) Is independent of temperature

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Increases with temperature

**54. Mass transfer coefficient according to penetration theory varies with mass diffusivity as:**

(a) $D^{0.5}$ (b) D
(c) 1/D (d) $D^{1.5}$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $D^{0.5}$

## B. Absorption

**55. The equation helps us to find the number of trays in an absorber theoretically is:**

(a) Frenske's Equation
(b) Rayleigh's Equation
(c) Kremser's Equation
(d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**56. Packing height in the packed tower = NTU × _____**

(a) HTU (b) HETP

(c) NTU (d) HETP + HTU

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**57. According to Raoult's law, for a pure component solution the partial pressure is equal to:**

(a) Total pressure

(b) Vapor pressure

(c) Atmos pressure

(d) Mole fraction of respective phase

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**58. The unit of packing factor is**

(a) $m^{-1}$ (b) $m^2 / m^3$

(c) m (d) dimension less

**TNPSC AE-2018**

**Ans. (d)**

**59. The relation between the gas, liquid contactor terminologies NTU, HTU, HETP and number of stages (N) is**

(a) HETP = NTU×HTU×N

(b) $HETP = \frac{NTU \times HTU}{N}$

(c) $HETP = \frac{NTU \times N}{HTU}$

(d) $HETP = \frac{HTU \times N}{NTU}$

**TNPSC AE-2018**

**Ans. (b)**

**60. Operating velocity of gas in a packed column absorber is**

(a) 40 – 50% of flooding velocity

(b) 65 – 90% of flooding velocity

(c) 100% of flooding velocity

(d) 120 – 150% of flooding velocity

**TNPSC AE-2018**

**Ans. (b)**

**61. Raoult's law is a good approximation for mixtures of**

(a) Solids

(b) Similar compounds

(c) Water and alochol

(d) Non- ideal gases

**TNPSC AE-2019**

**Ans. (b)**

**62. Which of the following characteristics of solvent used in absorption reduces the liquid film resistance to mass transfer ?**

(a) Low viscosity of liquid solvent

(b) High volumetric solubility of solute gas in liquid solvent

(c) Low vapor pressure of liquid solvent

(d) Low solubility of non-solute gases in liquid solvent

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**63. In a binary solution of component A and B if component A exhibits negative deviation from Raoult's law, then component B**

(a) Exhibitgs negative deviation from Raoults law

(b) Exhibits positive deviation from Raoult's law

(c) Obeys Roults's law

(d) Either (b) or (c) above

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**64. The slope of operating line in an absorption column is_____.**

**(L and V are flow rates of the liquid and gas, respectively)**

(a) 1 (b) $\frac{L}{V+L}$

(c) $\frac{V}{L}$ (d) $\frac{L}{V}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**65. Which of the following is used as solvent of the absorption of $H_2S$ from natural gas ?**

(a) Monoethanol amine

(b) Sulfuric acid

(c) Liquid sulfur dioxide

(d) Water

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**66. The methods used to desorb the solute in the gas tripper are**

(a) Pressure increase and temperature increase

(b) Pressure reduction and temperature increase

(c) Pressure reduction and temperature

(d) Pressure rise and temperature reduction

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**67. The optimum value of the absorption factor lies n the range of**

(a) 5 – 10 (b) 1.25 – 2.0

(c) 2 – 4 (d) 0.25 – 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**68. Raoult's Law is given by**

(a) $P = p_1 + p_2 + p_3$

(b) $p_A = p_A^{\circ}\ x_A (p_A^{\circ} = \text{vap.}p_r, p_A = \text{par.}p_r)$

(c) $V = V_1 + V_2 + V_3$

(d) $p_A = y_A\ .\ P\ (P = \text{Tot.}\ p_r)$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**69. Which of the following operation is not dependent on the vapour-liquid equilibrium data?**

(a) Distillation (b) Absorption

(c) Extraction (d) Sublimation

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**70. The letter '$\alpha$' in individual co-efficient $k_y\alpha$ and $k_x\alpha$ and the overall coefficients $K_y\alpha$ and $K_x\alpha$ used in the design of adsorption column refers to**

(a) interfacial area per unit volume

(b) unit volume per interfacial area

(c) cross-sectional area

(d) specific surface area

(e) answer not known

**TNPSC AE-2022**

**Ans. (a)**

**71. Liquid redistributors are necessary in packed towers to avoid**

(a) flooding (b) weeping

(c) channelling (d) coning

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**72. For gas phase controlled mass transfer, which one is not true?**

(a) The gas is readily soluble in liquid

(b) The gas is sparingly soluble in liquid

(c) The equilibrium curve is flat

(d) Overall resistance to mass transfer is equal to the resistance on gas side

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**73. If the operating line coincides with the equilibrium curve, which one of the following is not true for absorbers?**

(a) The solvent rate is minimum

(b) The number of plates is infinity

(c) The L/G ration is maximum

(d) The driving force becomes zero

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**74. The vapour pressures of toluene are 6.811 kPa at 310 K and 24.15 kPa at 340 K. Assuming that the variation of the vapour pressure P with temperature T may be described by the expression**

$$\log(P) = A - \frac{B}{T}$$

**Where P is in kPa. What are the values of A and B?**

(a) A = 7.059, B = 1930 K

(b) A = 5.307, B = 1930 K

(c) A = 6.971, B = 7780 K

(d) A = 5.307, B = 7780 K

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**75. Absorption coefficient is maximum for**

(a) aluminium plate

(b) steel

(c) furnace refractories

(d) carbon lamp black

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**76. For absorption of $H_2S$ from a mixture with hydrocarbon vapour into an aqueous alkanolamine solution, the liquid phase mass transfer resistance is**

(a) significantly higher than that of the gas phase

(b) negligible compared to that of the gas phase

(c) equal to that of the gas phase

(d) dependent on the gas phase mass transfer resistance

**GATE-2018**

**Ans. (b)**

**77. Two packed towers are designed for the same mass velocity of the gas. The first has liquid and gas flow rates of 30 kg/s and 1.2 kg/s, respectively, while the corresponding flow rates in the second tower are 67.5 kg/s and 1.8 kg/s. The ratio of the design diameter of the wider tower to that of the narrower tower is**

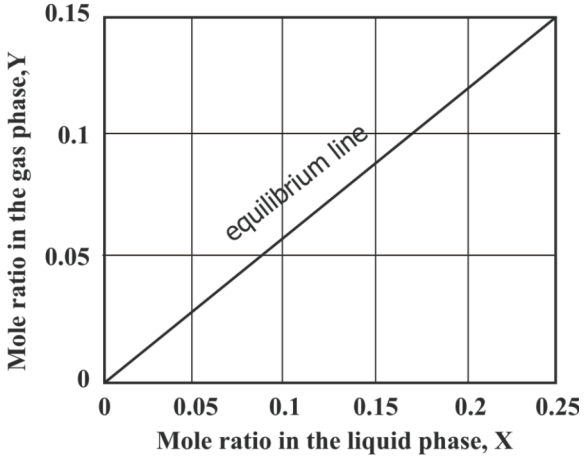
(a) 2 (b) 1.8

(c) 1.5 (d) 1.225

**GATE-2018**

**Ans. (d)**

**78. In a countercurrent stripping operation using pure steam, the mole ratio of a solute in the liquid stream is reduced from 0.25 to 0.05. The liquid feed flowrate, on a solute-free basis, is 3 mole/s. The equilibrium line for the system is given in the figure below.**




**The MINIMUM flowrate of pure steam for this process, rounded to 1 decimal plACF, is _____mol/s.**

**GATE-2017**

**Ans. (3.9 to 4.1)**

**79. The composition of vapour entering a tray in a distillation column is 0.47 The average composition of the vapour leaving the tray is 0.53. The equilibrium composition of the vapour corresponding to the liquid leaving this tray is 0.52. All the compositions are expressed in mole fraction of the more volatile component.**

**The Murphree efficiency based on the vapour phase, rounded to the nearest integer, is ____ %**

**GATE-2017**

**Ans. (119 to 121)**

**80. Consider steady state mass transfer of a solute A from a gas phase to a liquid phase. The gas phase bulk and interfACF mole fractions are $y_{A,G}$ and $y_{A,i}$, respectively. The liquid phase bulk and interfACF mole fractions are $x_{A,L}$ and $x_{A,i}$ respectively. The ratio $\frac{(x_{A,i} - x_{A,L})}{(y_{A,G} - y_{A,i})}$ is very close to zero This implies that mass transfer resistance is**

(a) negligible in the gas phase only
(b) negligible in the liquid phase only
(c) negligible in both the phases
(d) considerable in both the phases

**GATE-2017**

**Ans. (b)**

**81. The gas absorption from dilute solution under isothermal conditions give:**

(a) Straight equilibrium line
(b) Straight operating line
(c) Straight equilibrium and operating line both
(d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**82. Flooding results in**

(a) Low efficiency (b) High efficiency
(c) High gas velocity (d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**83. The total vapor pressure of two immiscible liquids is equal to**

(a) $P = x_1 \gamma_1 P_1^{vap} + x_2 \gamma_2 P_2^{vap}$
(b) Same as (a) but $\gamma_1 = 1 = \gamma_2$
(c) Same as (b) but $x_1 = 1 = x_2$
(d) None of the above

**RPSC ACF-2011**

**Ans. (b)**

**84. Absorption factor is defined as**

(a) mGL (b) mG/L
(c) G/mL (d) L/mG

**RPSC ACF-2011**

**Ans. (d)**

**85. Which of the following solutions will follow Raoult's Law most closely?**

(A) A solution of Benzene, Toluene and O-xylene
(B) 35% solution Camphor in Water
(C) 35% solution $NH_3$ in Water
(D) A Solution of polar organic compounds (Not of homologs of a series)

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**86. _____ of a component gas which is present in a mixture of gases is the pressure that would be exerted by that component gas if it alone were present in the same volume and at the same temperature as the mixture.**

(a) Vapour pressure
(b) Total pressure
(c) Partial pressure
(d) Equilibrium pressure

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**87. A packed tower is designed to recover 98% $CO_2$ from a gas mixture containing 10% $CO_2$ and 90% air using water calculate the height of the tower if HTU is 1 meter NTU is 12.2**

(a) 12.2 m (b) 11.2 m
(c) 10.98 m (d) 11.96 m

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**88. Calculate the height of the column of bromine that would exert the same pressure as atmospheric pressure ($10^5$ pascal)? Assume the density of bromine 3120 kg/$m^3$.**

(a) 2.3 m (b) .23 m
(c) 3.2 m (d) 1 m

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**89. Which one of the following is used in gas absorption?**

(a) Perforated tray tower (b) Wetted wall tower
(c) Bubble cap tower (d) Packed tower

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**90. Benzene is removed from air by absorbing it in a non-volatile wash-oil at 100 kPa in a countercurrent gas absorber .Gas flow rate is 100 mol/min, which includes 2 mol/min of benzene. The flow rate of wash-oil is 50 mol/min. Vapour pressure of benzene at the column conditions is 50 kPa. Benzene forms an ideal solution with the wash-oil and the column is operating at steady state. Gas phase can be assumed to follow Raoult's law. Neglect the change in molar flow rates of liquid and gas phases inside the column. For this process, the value of the absorption factor (upto two decimal plACFs) is**

(a) 0.02 (b) 2.02
(c) 1.02 (d) 0.22

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**91. $H_2S$ is being absorbed in a gas absorber unit. The height of the transfer unit based on the overall mass transfer co-efficient on the gas side is 0.4 m. The equilibrium data is given by, y = 1.5 x. The bulk concentration of $H_2S$ has to be reduced from 0.05 to 0.001 mole fraction in the gas side. The height of the tower (in meters) corresponding to an operating line given by, y = 5x + 0.001 is**

(a) 2.0 (b) 1.56
(c) 1.0 (d) 0.56

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**92. In an absorption column, the flooding velocity for random packing is___that for stacked/regular packing.**

(a) Greater than
(b) Smaller than
(c) Equal to
(d) Either Greater than, or Smaller than; depends on the type of packing

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**93. The Lewis relation for air-water humidification is given by (ky : mass transfer co-efficient of moisture in air; $h_G$ : heat transfer co-efficient; $C_S$: heat capacity of vapour gas mixture )**

(a) $\frac{h_G^2}{k_y C_S} = 1$ (b) $\frac{k_y C_S^2}{h_G} = 1$

(c) $\frac{h_G}{k_y C_S} = 1$ (d) $\frac{k_y^2 h_G}{C_S} = 1$

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**94. "The equilibrium value of the mole fraction of the gas dissolved in a liquid is directly proportional to the partial pressure of that gas above the liquid surfACF".**

**This statement is known as**

**A. Raoult's law**
**B. Henry's' law**
**C. Amgat's law**
**D. Dalton's law**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**95. According to Chilton–Colburn analogy for mass transfer, $N_{st} N_{sc}^{2/3}$ is equal to**

**A. f** **B. f/2**
**C. 2f** **D. 1/f**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**96. For absorption of solute gas A in solvent B with chemical reaction of first order in A and B, the Hatta number is defined as**

**A.** $\frac{K_1}{(K_{AB} C_B D_A)^{1/2}}$ **B.** $\left[\frac{K_{AB} C_B D_A}{K_1^2}\right]^2$

**C.** $\frac{(K_{AB} C_B D_A)^{1/2}}{K_1^2}$ **D.** $\frac{(K_{AB} C_B D_A)^{1/2}}{K_1}$

**Where $K_{AB}$ = Reaction rate constant, $K_1$ = Liquid phase mass transfer coefficient of Physical adsorption, $D_A$ = diffusion coefficient of A in liquid, $C_B$ = Concentration of B**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**97. For an absorber, both equilibrium and operating lines will be straight for,**

(a) Concentrated solution and non-isothermal operation
(b) Dilute solution and non-isothermal operation
(c) Dilute solution and isothermal operation
(d) Concentrated solution and isothermal operation

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**98. In a process, a soluble material is dissolved from its mixture with an insoluble solid by means of a liquid solvent is called:**

(a) Absorption (b) Adsorption
(c) Leaching (d) Distillation

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**99. Kremser – Brown – sonders equation is used to calculate:**

(a) Height of a transfer unit
(b) Plate spacing in the plate column
(c) Number of ideal plates in the plate column
(d) Number of transfer units

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Number of ideal plates in the plate column

**100. For gas spargers with large gas velocity, bubble diameter in water for orifice diameter ($d_o$) of 0.4 to 1.6 mm is,**

(a) $d_o^{1/3}$ (b) $d_o^{1/5}$
(c) $d_o^{-0.05}$ (d) $d_o$

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**101. Absorption of $SO_3$ in 97% $H_2SO_4$ is**

(a) exothermic
(b) endothermic
(c) not possible
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**102. Addition of a non-volatile solute to a solvent produces a _____ in its solvent.**

(a) freezing point elevation
(b) boiling point depression
(c) vapour pressure lowering
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**103. Absorption tower packing materials, Chempak are made of**

(a) Plastic
(b) Metal
(c) Ceramic
(d) Wood

**Karnataka PSC ADF - 2015**

**Ans. (a)**

**104. Which one of the following is the value of absorption factor A, to obtain higher absorption of solute from its mixture?**

(a) 1
(b) Zero
(c) Less than unity
(d) Greater than unity

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**105. Which of the following equations represents Raoult's law?**

**Where,**

$P_A$ – Vapour Pressure of more volatile Component (A)

$p_A^{-*}$ – Partial Pressure of Component A

$y_A$ – Mole fraction of Component A in vapour phase

$x_A$ – Mole fraction of Component A in liquid phase

(a) $P_A = p_A^{-*} y_A$ (b) $p_A^{-*} = P_A y_A$
(c) $P_A = p_A^{-*} x_A$ (d) $p_A^{-*} = P_A x_A$

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**106. The absorption factor is defined as**

(a) mG/L (b) mL/G
(c) L/mG (d) LG/m

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**107. In case of an absorber, the operating**

(a) line always lies below the equilibrium curve
(b) line can be either above or below equilibrium curve
(c) line always lies above the equilibrium curve
(d) Velocity is more than the loading celocity

**ISRO-12.02.2016**

**Ans. (a)**

**108. L /mG is the expression for**

(a) Absorption factor
(b) Slope of operating line in absorber
(c) slope of operating line in stripper
(d) Brinkmann Number

**ISRO-12.02.2016**

**Ans. (a)**

**109. Isopiestic solutions have same**

(a) Vapour pressure
(b) Osmotic pressure
(c) Freezing point
(d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**110. Which of the following is NOT correctly matched?**

(a) Raoult's law – Distillation
(b) Duhring rule – Evaporation
(c) Henry's law – Absorption
(d) Planck's law – Crystallization

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**111. When the liquid phase and vapour phase of a binary system obeys Raoult's law and Dalton's law respectively, the relative volatility is the ratio of**

(a) vapour pressure of component A to that of component B
(b) vapour pressure of component A to the total pressure
(c) vapour pressure of component A to the partial pressure of A
(d) partial pressure of component A to the total pressure

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**112. The absorption factor is defined as :**

(a) L/(mG) (b) G/(mL)
(c) mL/G (d) LG/m

**AP PSC IOF-2009**

**Ans. (a)**

**113. Assuming the mass transfer coefficients in the gas and the liquid phases are comparable, the absorption of $CO_2$ from reformer gas ($CO_2$ + $H_2$) into an aqueous solution of diethanolamine is controlled by**

(a) gas phase resistance
(b) liquid phase resistance
(c) both gas and liquid phase resistances
(d) composition of the reformer gas

**GATE-2014**

**Ans. (a)**

**114. Which ONE of the following statements is CORRECT for the surface renewal theory?**

(a) Mass transfer takes place at steady state
(b) Mass transfer takes place at unsteady state
(c) Contact time is same for all the liquid elements
(d) Mass transfer depends only on the film resistance

**GATE-2014**

**Ans. (b)**

**115. Packed towers are preferred for gas-liquid mass transfer operations with foaming liquids because**

(a) in packed towers, high liquid to gas ratios are best handled
(b) in packed towers, continuous contact of gas and liquid take place
(c) packed towers are packed with random packings
(d) in packed towers, the gas is not bubbled through the liquid pool

**GATE-2014**

**Ans. (d)**

**116. Consider a steady state mass transfer process between well-mixed liquid and vapour phases of a binary mixture comprising of components A and B. The mole fractions of component A in the bulk liquid ($x_A$) and bulk vapour ($y_A$) phases are 0.36 and 0.16, respectively. The mass transfer coefficients for component A in liquid and vapour phases are 0.1 mol/($m^2$.s) and 0.05 mol/ ($m^2$.s), respectively. The vapour-liquid equilibrium can be approximated as $y_A = 2x_A$, for $x_A$ less than 0.4. The mole fraction of A in the liquid at the interface (up to two decimal places) is_____**

**GATE-2015**

**Ans. (0.20 to 0.24)**

**117. Benzene is removed from air by absorbing it in a non-volatile wash-oil at 100 kPa in a counter-current gas absorber. Gas flow rate is 100 mol/min, which includes 2 mol/min of benzene. The flow rate of wash-oil is 50 mol/mm. Vapor pressure of benzene at the column conditions is 50 kPa. Benzene forms an ideal solution with the wash-oil and the column is operating at steady state. Gas phase can be assumed to follow ideal gas law. Neglect the change in molar flow rates of liquid and gas phases inside the column.**

**For this process, the value of the absorption factor (up to two decimal places) is_____**

**GATE-2015**

**Ans. (0.95 to 1.05)**

**118. The ratio of slope of operating line $\left(\frac{L}{V}\right)$ to the equilibrium line, m is called:**

(a) Adsorption factor (b) Absorption factor
(c) Dew point (d) Boiling point

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**119. The upper limit of velocity in a sieve-tray column is determined by:**

(a) Weeping point (b) Flooding point
(c) Clogging point (d) Leaking point

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

## C. Distillation

**120. If the McCabe-Thiele assumptions are valid, the molal flow rates of liquid stream in the enriching and stripping sections are equal for:**

(a) Subcooled liquid feed
(b) Saturated liquid feed
(c) Saturated vapor feed
(d) Super heated vapor feed

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Saturated vapor feed

**121. Which method will produce distillate richer in more volatile component from the same binary feed:**

(a) Single stage flash vaporization
(b) Differential distillation
(c) Both will produce distillate of the same average composition
(d) Single stage flash vaporization for $\alpha>1$ and differential distillation for $1 < \alpha < 2$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Differential distillation

**122. In order to find the total number of theoretical trays we usually add one stage additional for:**

(a) Condenser
(b) Reboiler
(c) Condenser or Reboiler
(d) No additional trays

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**123. At Total reflux:**

(a) Residue is nil (b) Distillate is nil
(c) Reflux is nil (d) All of the mentioned

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**124. At minimum reflux ratio for a given separation:**

(a) Number of plates is zero
(b) Number of plates is infinity
(c) Minimum number of the theoretical plates is required
(d) Separation is most efficient

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**125. Fenske's equation for determining the minimum number of theoretical stages in distillation column holds good, when the :**

(a) Relative volatility is reasonably constant
(b) Mixture shows negative deviation from ideality
(c) Mixture shows positive deviation from ideality
(d) Multicomponent distillation is involved

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**126. The section above the feed tray in a fractionator is known as_____**

(a) Enriching section (b) Rectifying section
(c) Feed section (d) All of the mentioned

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**127. The section below the feed tray in a fractionator is known as_____**

(a) Enriching section
(b) Rectifying section
(c) Feed section
(d) All of the mentioned

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**128. The Enriching section operating line intercept at origin in McCabe-Thiele method when:**

(a) $R = 0$
(b) $R = \infty$
(c) $R = 1$
(d) $R = 1.5$ to 2 times $R_{min}$

**where, R = reflux ratio and $R_{min}$ = Minimum reflux ratio.**

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**129. If there is no reflux to a fractionating column then:**

(a) Large condenser size is needed
(b) Less reboiler size needed
(c) Minimum number of trays
(d) None of the mentioned

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**130. A binary mixture A – B forms an azeotrope with a boiling point of 71.8 ºC at 1 bar pressure, the azeotropic composition being 55 mole percent 'A'. The pure component vapor pressures at 71.8 ºC are A = 0.50 Bar; B = 0.70 bar. What is the activity coefficient of ' A' in the liquid assuming the vapor to be an ideal.**

(a) 1 (b) 1.5
(c) 2 (d) 2.5

**TNPSC AE-2018**

**Ans. (c)**

**131. Entrainer used in azeotropic distillation to**

(a) Form a new azeotrope of low relative volatility with one of the constitutes of the mixture
(b) Form a low boiling azeotrope with one of the constituents of the mixture
(c) Have high latent heat of vaporization
(d) Have high viscosity to provide high tray efficiency

**TNPSC AE-2019**

**Ans. (b)**

**132. Flash distillation is not effective in separating components of comparable**

(a) Purity (b) Density
(c) Volatility (d) Viscosity

**TNPSC AE-2019**

**Ans. (c)**

**133. In distillation column, the top product quality is maintained by manipulating.**

(a) flow rate of feed
(b) flow rate of reflex stream
(c) column pressure
(d) liquid holdup

**TNPSC AE-2019**

**Ans. (b)**

**134. Fenske-Underwood equation is used for determining**

(a) Diameter of the column
(b) Efficiency of the tray column
(c) Minimum number of theoretical plates
(d) Efficiency of the packed column

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**135. For McCabe - Thiele method, the rectifying and stripping operating lines coincide with the diagonal line at**

(a) No reflux and no reboiling
(b) Total reflux
(c) No reboiling
(d) No reflux

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**136. in a perfectly mixed flow model, the ratio of the murphree plate efficiency to point efficiency is**

(a) 0 (b) << 1
(c) >1 (d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**137. An equimolar mixture of A and B (A being more volatile) is flash distilled continuously at a feed rate of 100 kmol/h such that the liquid product contains 40 mole% of A. If the relative volatility is 6, then vapor product in kmol/h is**

(a) 45 (b) 10
(c) 25 (d) 20

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**138. If y* = x, the relative volatility for a binary system is always**

(a) One except at x = 0 or 1
(b) Zero
(c) Infinity
(d) One

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**139. If the feed to a distillation column is a saturated liquid, then the slope of the feed line**

(a) < 1 (b) 0
(c) > 1 (d) ∞

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**140. A batch rectifier is usually operated under _____ conditions for an initial period of time prior to withdrawal of distillate product.**

(a) Sub-cooled (b) Partial-reflux
(c) Total-reflux (d) Super saturated
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**141. Flash vapourization is carried out by**

(a) cooling the feed at constant pressure
(b) increasing the pressure on the feed stream
(c) reducing the pressure on the feed stream
(d) increasing the temperature on the feed stream
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**142. The condition at which the liquid hold up starts to increase, as judged by a change in the slope of the pressure drop line, is called**

(a) Channeling (b) Loading point
(c) Flooding (d) Short-circuting
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**143. The minimum tray spacing in distillation column of diameter less than 3 ft is normally**

(a) 6 inch (b) 12 inch

(c) 18 inch (d) 24 inch

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**144. Which of the following efficiency can be greater than 100% ?**

(a) Point efficiency

(b) Overall efficiency

(c) Murphree plate efficiency

(d) None of the other option

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**145. Stefan boltzman constant in $W/m^2k^4$ is**

(a) $3.669 \times 10^{-9}$ (b) $4.669 \times 10^{-9}$

(c) $5.669 \times 10^{-8}$ (d) $6.669 \times 10^{-9}$

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**146. A 100 kmol mixture of benzene and toluene containing 60 mol% benzene is separated into top product (D) containing 80 mol% benzene and a bottom product (B) containing 20 mol% benzene. What is the amount of top product (D) ?**

(a) 50 kmol (b) 25 kmol

(c) 33.3 kmol (d) 66.7 kmol

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**147. For a saturated liquid-vapour mixture, if the liquid flow rates inside the enriching and the stripping section of a fractionator are 30 mol/hr and 20 mol/hr respectively and also the feed rate is 10 mol/hr, find the feed condition.**

(a) Saturated vapour

(b) Saturated liquid

(c) Superheated vapour

(d) 50% vapour

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**148. For the nth tray (counted from the top of distillation column), the Murphree tray efficiency is given by**

(a) $\frac{Y_{n+1} - Y_n}{Y^*_n - Y_{n-1}}$ (b) $\frac{Y_n - Y_{n-1}}{Y^*_n - Y_{n-1}}$

(c) $\frac{Y_{n-1} - Y_n}{Y_{n+1} - Y_{n-1}}$ (d) $\frac{Y^*_n - Y_{n-1}}{Y^*_n - Y_{n+1}}$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**149. If the distillation is operating at total reflux, the distillation column requires minimum**

(a) reboiler load

(b) condenser load

(c) number of plates

(d) diameter of column

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**150. The flooding velocity in a plate column operating at 1 atm pressure is 3 m/s. If the column is operated at 2 atm pressure under otherwise identical conditions, the flooding velocity will be**

(a) $3/\sqrt{2}$ (b) 3/2

(c) 1 (d) 3/4

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**151. Alcohol is dehydrated using _____ distillation.**

(a) extractive (b) steam

(c) azeotropic (d) molecular

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**152. A binary distillation column is designed by McCabe-Thiele method to get a distillate mole fraction of 0.9. The enriching section operating line has an intercept with y-axis at 0.3 mole fraction. The ratio of liquid to vapour molar flow rate in the enriching section is ___ (rounded off to third decimal plACF).**

**GATE-2018**

**Ans. (0.65 to 0.68)**

**153. The vapor phase composition and relative volatilities (with respect to n-propane) on an ideal try of a distillation column are**

| Component | Methane | Ethane | n-Propane |
|---|---|---|---|
| Mole fraction in vapour | 0.12 | 0.28 | 0.60 |
| Relative volatility | 10 | 4 | 1 |

**The mole fraction of n-propane in the liquid phase, rounded to 2 decimal p**

**GATE-2017**

**Ans. (0.86 to 0.90)**

**154. Distillation of a conjugate solution of two partially miscible liquids always results in**

(a) The lower boiling compound as the final distillate.

(b) The higher boiling compound as the final distillate.

(c) Depends on the relative proportions of the two existing phases
(d) None of the above

**RPSC ACF-2011**

**Ans. (b)**

**155. In a continuous distillation column as the reflux ratio is increased, the overhead product purity**
(a) Increases
(b) Decreases
(c) May increase, or decrease, depending on the system
(d) Remains constant

**RPSC ACF-2011**

**Ans. (a)**

**156. Steam distillation is usually employed for**
(a) Insoluble liquids
(b) A component which is heat sensitive
(c) Miscible liquids
(d) Both (a) and (b)

**RPSC ACF-2011**

**Ans. (b)**

**157. Fenske's equation is used to calculate number of plates in a distillation column**
(a) at total reflux
(b) for system having constant relative volatility
(c) at minimum reflux
(d) for both (a) and (b)

**RPSC ACF-2011**

**Ans. (c)**

**158. The distillate flow rate from a distillation column is 100 kmol / hr and the reflux ratio is 2. The flow rate of vapor from the top plate in kmol / h is**
(a) 100 (b) 200
(c) 300 (d) 50

**RPSC ACF-2011**

**Ans. (c)**

**159. If $R_m$ is the minimum reflux ratio the optimum reflux ratio is**
(a) 1.25 to 1.5 $R_m$ (b) 2 $R_m$
(c) 2.5 to 3.5 $R_m$ (d) 5 $R_m$

**RPSC ACF-2011**

**Ans. (a)**

**160. One of the examples of minimum boiling azeotropes is a mixture of**
(a) Benzene and toluene
(b) Hydrochloric acid and water
(c) Ethanol and water
(d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**161. A paper reports VLE data for a binary hydrocarbon mixture. Experimental data shows formation of an azeotrope and the author claims that the liquid behavior can be fitted well by a Regular solution model. Which of the following conclusions may be made about the azeotrope?**
(a) It must be maximum boiling azeotrope
(b) It must be a minimum boiling azeotrope
(c) There is no way to tell from the above statement whether it will be a maximum boiling or minimum boiling azeotrope
(d) The author's claim about the Regular solution model is wrong as Regular solution model forbids formation of any azeotrope.

**RPSC ACF-2011**

**Ans. (d)**

**162. The driving force for separation by distillation is the highest**
(a) at total reflux
(b) at minimum reflux
(c) at an intermediate reflux between the total and the minimum.
(d) at the point of intersection of the enriching section operating line with the equilibrium curve

**RPSC ACF-2011**

**Ans. (a)**

**163. Ponchon-Savarit method of calculation of number of theoretical plates of a distillation column is based on**
(a) Material balance only
(b) Enthalpy balance only
(c) Both material and enthalpy balances
(d) Assumption of consist molal overflow

**RPSC ACF-2011**

**Ans. (c)**

**164. In flash vaporization, the product vapour**
(a) is always in equilibrium with the liquid leaving the flash chamber
(b) is never in equilibrium with the liquid leaving the flash chamber
(c) may be in equilibrium with the liquid leaving the flash chamber and the extent of equilibration depends on the vapor – liquid contact time in the chamber
(d) is 100% pure.

**RPSC ACF-2011**

**Ans. (a)**

**165. At what percentage (by volume) of Alcohol composition it forms an azeotrope with water.**

(A) 90 (B) 91.5
(C) 95 (D) 99
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**166. For the same feed, feed quality and separation (in a distillation column) with increase of total pressure, number of ideal plates will.**

(A) Increase
(B) Decrease
(C) Remains same
(D) Data insufficient, can't be predicted
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**167. In a distillation operation, the heat removed in a condenser**

(a) Remains unaffected with change in reflux ratio
(b) Increase with increase in reflux ratio
(c) Decrease with increase in reflux ratio
(d) And the heat required in re-boiler decrease with increase in reflux ratio

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**168. An ideal plate is defined as the one where,**

(a) The vapour entering stream is in equilibrium with the liquid leaving stream
(b) The vapour and liquid entering streams are in equilibrium
(c) The vapour leaving stream is in equilibrium with the liquid entering stream
(d) The vapour and liquid leaving streams are in equilibrium

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**169. In distillation operation, q factor is zero for**

(a) Cold liquid
(b) Saturated liquid
(c) Saturated vapor feed
(d) Super heated vapor feed

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**170. Which of the following is an example of maximum boiling azeotrope:**

(a) Ethanol-water
(b) Hydrochloric acid-water
(c) Carbon disulfide- ACFtone
(d) All of the above

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**171. Steam traps are used to**

(a) Remove the condensate
(b) Vent the steam
(c) Regulate pressure
(d) None of these

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**172. Considering the operations of tray column the total area of the perforations on the tray is known as ——**

(a) Net area (b) Bubbling area
(c) Hole area (d) Active area

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**173. In Mc-Cabe Thiele method, if q is defined as the moles of liquid flow per mole feed in stripping section, then the slope of feed line is equal to:**

(a) $-q/(1-q)$ (b) $-1/q$
(c) $-(1-q)/q$ (d) $q/(1-q)$

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**174. In a continuous fractionating column, the vapours rising through the rectifying section are completely considered in condenser and collected in accumulator in which a definite liquid level is maintained. Some part of liquid from the accumulator is delivered to the top late of water. This liquid stream is called:**

(a) Bottom product (b) Purge
(c) Overhead product (d) Reflux

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**175. If the distillation is operating at total reflux, the distillation column requires minimum:**

(a) Number of plates (b) Reboiler load
(c) Condenser load (d) Diameter of column

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**176. An ideal flash vaporization is carried out with a binary mixture at constant temperature and pressure. A process upset leads to an increase in the mole fraction of the heavy component in the feed. The flash vessel continues to operate at the previous temperature and pressure and still produces liquid and vapour. After steady state is re-established,**

(a) The amount of vapour produced will increase
(b) The amount of liquid produced will decrease
(c) The new equilibrium compositions of the vapour and liquid products will be different
(d) The new equilibrium compositions of the vapour and liquid products will remain as they were before the upset occurred

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**177. In azeotropic distillation the third component is added**

(a) To form another azeotrope with one of the components
(b) To increase the value of relative volatility of the original two components
(c) To decrease the value of relative volatility of the original two components
(d) To change the value of the vaporization equilibrium constant of the two components

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**178. Find the distillate and the residue rates in mole/hr, if the feed rate is 100 mol/hr; The compositions of feed. Distillate and residue in terms of mole fraction are 0.65, 0.9 and 0.2**

(a) 36,64 (b) 64, 36
(c) 20, 80 (d) 80, 20

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**179. Steam distillation is not recommended to be used, if the**

(a) material cannot be distilled by indirect heating even under low pressure, because of the high boiling temperature,
(b) liquids decompose, if distilled directly at atmospheric pressure
(c) azeotropic mixture is to be separated and the final product is miscible with water.
(d) material to be distilled is thermally unstable or has the tendency to react with other components associated with it, at the boiling temperature.

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**180. According to the Fenske equation, what will be the minimum number of plates required in a distillation column to separate an equimolar binary mixture of components A & B into an overhead fraction containing 99 mole % A and a bottoms fraction containing 98 mole % B ? [Assume that the relative volatility ($\alpha_{AB}$ = 2) does not change appreciably in the column]**

(a) 5 (b) 9
(c) 12 (d) 28

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**181. In a binary mixture containing components A and B, the relative volatility of A with respect to B is 2.5 when mole fractions are used. The molecular weights of A and B are 78 and 92 respectively. If the compositions are however expressed in mass fractions the relative volatility will then be**

(a) 1.18 (b) 2.12
(c) 2.5 (d) 2.95

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**182. As the reflux ratio decreases, the**

**A. Separation becomes more efficient**
**B. Number of plates increases**
**C. Column diameter increases**
**D. None of these**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**183. If f = moles of vapour present per mole of feed, then the slope of feed line is (McCabe–Thiele method)**

**A.** $\frac{1-f}{f}$ **B.** $\frac{f-1}{f}$ **C.** $\frac{-1}{f}$ **D.** $\frac{-f}{1-f}$

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**184. Pick out the correct relationship (where Ri = internal reflux ratio, $R_0$ = external reflux ratio)**

**A. $(1+R_o) = R_o/Ri$**
**B. $(1-R_o) = R_o/Ri$**
**C. $(1+R_i) = R_o/Ri$**
**D. $(1-R_i) = R_o/Ri$**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**185. On moving the feed line (q-line) from saturated liquid feed (vertical position) to saturated vapour feed (horizontal position), if the slope of both operating lines is to be increased, then it will result in**

**A. Greater degree of separation with fixed number of trays**
**B. Increased reboiler load**
**C. Increased reflux ratio**
**D. None of these**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**186. In a distillation process, point efficiencies and Murphree efficiencies are equal when,**

(a) Liquid in plug flow
(b) Liquid back mixed and uniform
(c) Gas in plug flow
(d) Gas back mixed and uniform

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**187. In a sieve tray tower, for high tray efficiency,**

(a) Shallow liquid pool with low gas velocity
(b) Shallow liquid pool with high gas velocity
(c) Deep liquid pool with low gas velocity
(d) Deep liquid pool with high gas velocity

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**188. Which mass transfer operation is employed for industrial production of Argon gas?**

(a) Leaching from rocks
(b) Fractional distillation of air
(c) Cryogenic distillation of natural gas
(d) Adsorption from natural gas

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**189. Fill in the blanks of the following sentence by choosing any one of the following options. In order to separate the components in binary mixture using distillation, _____ number of trays are required at minimum reflux ratio and _____ number of plates are required for total reflux.**

(a) Minimum, infinite (b) Infinite, single
(c) Infinite, minimum (d) Zero, minimum

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**190. 100 moles of equimolar mixtures of benzene and toluene are fed to a distillation column. The top product contains 95 mole % benzene. The bottom product is 60 moles. What is the % recovery of benzene?**

(a) 95% (b) 5%
(c) 76% (d) 60%

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**191. A binary distillation column is to be designed using McCabe Thiele method. The distillate contains 90 mol% of the more volatile component. The point of intersection of the q-line with the equilibrium curve is (0.5, 0.7). The minimum reflux ratio (rounded off to the first decimal place) for this operation is—**

**GATE-2016**

**Ans. (0.98 to 1.02)**

**192. When a multistage try tower uses a minimum reflux ratio it implies?**

(a) Infinite trays and maximum reboiler heat load
(b) minimum trays and minimum reboiler heat load
(c) infinite trays and minimum reboiler heat load
(d) minimum trays and maximum reboiler heat load

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**193. The feed to a binary distillation column has 40 mole% vapour and 60 mole% liquid. Then the slope of the q-line in the McCabe-Thiele plot is**

(a) –1.5 (b) –0.6
(c) 0.6 (d) 1.5

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**194. Murphree efficiency is defined as**

(a) Ratio of no. of theoretical stages required to produce a given separation to the no. of stages actually necessary to produce the same separation
(b) Ratio of actual change in the average vapour composition accomplished by a given plate to the change in average vapour composition, if the vapours leaving the plate were in equilibrium with the liquid leaving plate
(c) Ratio of no. of stages actually necessary to produce the given separation to the no. of theoretical stages required to produce the same separation
(d) Ratio of change in average vapour composition, if the vapours leaving the plate were in equilibrium with the liquid leaving plate to the actual change in the average vapour composition accomplished by the plate.

**ISRO Scientist Engineer-06.08.2017**

**Ans. (b)**

**195. Vapour liquid equilibrium relation for an ideal binary system is given by,**

$y^*_A = \frac{\alpha_{AB} x_A}{1+(\alpha_{AB}-1)x_A}$. **Here $x_A$ and $y^*_A$ are the mole fractions of species A in the liquid and vapour, respectively. The relative volatility ($\alpha_{AB}$) is greater than unity.**

**The liquid mole fraction $x_A$ at which the maximum difference between the equilibrium vapour mole fraction and liquid mole fraction occurs is**

(a) $\frac{1}{\left(1+\sqrt{\alpha_{AB}}\right)}$ (b) $\frac{0.75}{\left(1+\sqrt{\alpha_{AB}}\right)}$

(c) $\frac{0.5}{\left(\sqrt{\alpha_{AB}+1}\right)}$ (d) $\frac{0.75}{\left(\sqrt{\alpha_{AB}+1}\right)}$

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**196. Entrainer used in azeotropic distillation of binary mixture,**

(a) Forms no azeotrope with the original substance
(b) Forms a minimum-boiling azeotrope with one of the feed components
(c) Should be of very high viscosity to provide high tray efficiencies
(d) Should have very high latent heat of vapourisation

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**197. Azeotropic distillation is employed to separate**

(a) high boiling mixture
(b) constant boiling mixture
(c) mixture with every high relative volatility
(d) heat sensitive materials

**ISRO-12.02.2016**

**Ans. (b)**

**198. If reflux to a distillation column ($L_0$) is 100 moles / hr, when the overhead product rate (D) is 50 moles / hr, the reflux ratio is**

(a) 2 (b) 0.5
(c) 5000 (d) 20

**ISRO-12.02.2016**

**Ans. (a)**

**199. Arrange the q-values of the following feeds to a distillation column in increasing order.**

**1. Subcooled liquid solution**
**2. Saturated liquid solution**
**3. Saturated vapour mixture**
**4. Saturated liquid-saturated vapour mixture**

**Select the correct answer from the code given below,**

(a) 3, 2, 4 and 1
(b) 3, 4, 2 and 1
(c) 1, 2, 4 and 3
(d) 1, 4, 2 and 3

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**200. Match List –I with List –II and Select the correct answer using the code given below lists.**

| List-I | List-II |
|---|---|
| A. Packed column | 1. Lower pressure drop |
| B. Valve trays | 2. Large turndown ratio |
| C. Batch distillation | 3. Dew and bubble point calculation |
| D. Multi-component distillation | 4. Rayleigh equation |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 2 | 1 | 3 |
| (b) | 1 | 3 | 4 | 2 |
| (c) | 4 | 3 | 1 | 2 |
| (d) | 1 | 2 | 4 | 3 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**201. In the McCabe-Thiele method, if the x-coordinate of the point of intersection of the q-line and vapour-liquid equilibrium curve is greater than the x- coordinate of the feed point, then thermal condition of the feed is**

(a) Superheated vapour
(b) Liquid below bubble point
(c) Saturated vapour
(d) Saturated liquid

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**202. Which of the following is NOT correctly matched?**

(a) Underwood equation – Minimum reflux ratio
(b) Fenske equation – Minimum number of plates
(c) Gilliland equation – Actual number of real plates
(d) Vapour velocity at flooding – Column diameter

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**203. Fenske equation for determining the minimum number of plates in distillation column holds good**

(a) When the relative volatility of the mixture is reasonably constant.
(b) When the mixture to be separated shows negative deviation from ideality
(c) When the mixture to be separated shows positive deviation from ideality
(d) For multi component distillation

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**204. The Boiling Point (BP) of an azeotropic mixture of two liquids is**

(a) Always higher than the BP of two liquids
(b) Same as that of the higher BP liquid
(c) Same as the lower BP liquid
(d) May be lower or higher than the BP of component liquids

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**205. Rayleigh's equation is applicable to**

(a) flash distillation
(b) differential distillation
(c) molecular distillation
(d) none of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**206. The operating range and efficiency of bubble cap trays are.**

(a) law
(b) wide and low
(c) wide and high
(d) dependent upon nature of mixture

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**207. What is the reflux ratio at total reflux?**

(a) Zero (b) Infinity
(c) Unity
(d) Cannot be predicted

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**208. Distillation separates components of a liquid mixture based on their difference in :**

(a) Density (b) Diffusivity
(c) Boiling points (d) Specific heats

**AP PSC IOF-2009**

**Ans. (c)**

**209. In binary distillation, the separation of the components is easier if the relative volatility ($\alpha$) is :**

(a) $\alpha << 1$ (b) $\alpha >> 1$
(c) $\alpha = 1$ (d) $\alpha = 0$

**AP PSC IOF-2009**

**Ans. (b)**

**210. In distillation under minimum reflux condition, number of theoretical stages would be :**

(a) one (b) minimum
(c) infinite (d) 15

**AP PSC IOF-2009**

**Ans. (c)**

**211. Diameter of the distillation column is set by :**

(a) Number of theoretical plates
(b) Allowable vapour velocity
(c) Feed rate to the column
(d) Static submergence

**AP PSC IOF-2009**

**Ans. (b)**

**212. Identify the WRONG statement amongst the following:**

**(1) Steam distillation is used for mixtures that are immiscible with water.**
**(2) Vacuum distillation is used for mixtures that are miscible with water.**
**(3) Steam distillation is used for mixtures that are miscible with water.**
**(4) Vacuum distillation columns have larger diameters as compared to atmospheric columns for the same throughput.**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**213. q is defined as mole of liquid flow in stripping section resulting from introduction of each mole of feed. Its value for feed partially vapour is:**

(a) $q = 0$ (b) $q = 1$
(c) $0 < q < 1$ (d) $q > 1$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**214. When a dephlegmator is used, the liquid reflux is:**

(a) having the same composition as the over head product
(b) not having the same composition as the over head product

(c) having a zero composition as the over head product
(d) having an infinity composition as the over head product

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**215. The reflux at the top of the distillation column is at its:**

(a) Boiling point (b) Melting point
(c) Critical point (d) Triple point

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**216. The relative volatility of a binary mixture at the azeotropic composition is:**

(a) Zero
(b) Infinity
(c) Unity
(d) Either less than or more than unity

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Unity

## D. Extractions (liquid-liquid)

**217. Penicillin is recovered from the fermentation broth by**

(a) Distillation
(b) Selective absorption
(c) Adsorption
(d) Liquid extraction

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

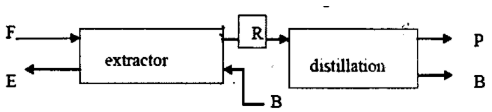
**218. For the extraction of oil from lemon grass, the technique to be adopted is**

(a) Extractive distillation (b) Steam distillation
(c) Vacuum distillation (d) Flash distillation

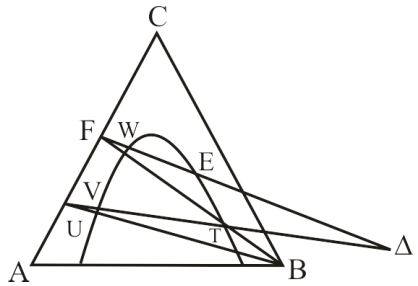
**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**219. A multi-stage counter-current liquid-liquid extractor is used to separate solute C from a binary mixture (F) of A and C using solvent B. Pure solvent B is recovered from the raffinate R by distillation, is shown in the schematic diagram below.**




**Locations of different mixtures for this process are indicated on the triangular diagram below. P is the solvent-free raffinate, E is the extract, F is the feed and Δ is the difference point from which the mass balance lines originate. The line PB intersects the binodal curve at U and T. The lines PΔ and FB intersect the binodal at V and W respectively.**




**The raffinate coming out of the extractor is represented in the diagram by the point:**

**(1) T** **(2) U**
**(3) V** **(4) W**
(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (b)**

**220. In solvent extraction operation, the selectivity of solvent:**

(a) Decreases with decrease in temperature
(b) Increases with decrease in temperature
(c) Is independent of temperature
(d) Increases only linearly with decrease in temperature

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Decreases with decrease in temperature

**221. Solvent lean phase are known as:**

(a) Extract
(b) Raffinate
(c) Residue
(d) None of the mentioned

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**222. For all useful liquid-liquid extraction operation, the selectivity of the solvent must be**

(a) Less than or equal to one
(b) More than zero
(c) Less than one
(d) More than one

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**223. Solvent B is used to extract solute, C from a given geed, containing, A and C. If C is more soluble is A than in B, the distribution coefficient for C will be**

(a) $> 1$ (b) $>> 1$
(c) $< 1$ (d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**224. A mixture of para and ortho-nitro benzoic acids can be separated by solvent extraction operation using**

(a) Water as solvent
(b) Chloroform as solvent
(c) Both chloroform and water
(d) None of the above

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**225. In liquid-liquid extraction, the selectivity of the plait point is**

(a) More than - but less than 12
(b) 0
(c) ∞
(d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**226. Identify the separation method based on the relative solubilities in different solvents that are used in the manufacture of lube oil**

(a) Distillation (b) Absorption
(c) Extraction (d) Adsorption
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

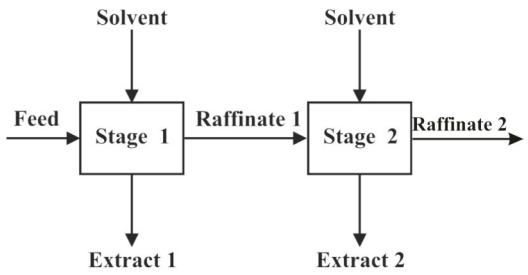
**227. Which of the following extractors are especially useful for liquids with very small density difference and very short residence time?**

(a) Centrifugal Extractors
(b) Mixer-Settler Extractors
(c) Pulsed Extractors
(d) Packed Extraction tower
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**228. It is decided to extract A from a feed containing 20 mol% A and 80 mol% B in two ideal cross-current stages as shown below, using equal amount of pure solvent C in each stage**




**Components B and C are immiscible. 60% of A in the feed is extracted in stage 1. The equilibrium relation is given by Y* = 1.5 X where,**

**X = moles of A per mole of B in raffinate**

**Y* = moles of A per mole of C in extract in equilibrium with raffinate**

**The mol % of A in raffinate from stage 2 is ____ (rounded off to second decimal plACF).**

**GATE-2018**

**Ans. (3.75 to 3.95)**

**229. Which of the following conditions are valid at the plait point?**

**P) Density difference between the extract and raffinate phases is zero**
**Q) Interfacial tension between the extract and raffinate phases in zero**
**R) Composition difference between the extract and raffinate phases is zero**

(a) P and Q only (b) Q and R only
(c) P and R only (d) P, Q and R

**GATE-2017**

**Ans. (d)**

**230. Heats sensitive materials with very latent heat of vaporization may be economically separated using**

(a) Liquid extraction (b) Distillation
(c) Evaporation (d) Absorption

**RPSC ACF-2011**

**Ans. (a)**

**231. Choose the best combination of properties for a good solvent for extraction out of the following: (i) high selectivity (ii) low selectivity (iii) high viscosity (iv) low viscosity (v) large distri bution coefficient (vi) small distri bution coefficient (vii) high interfacial tension (vii) low interfacial tension**

(a) (i), (ii), (v), (vii) (b) (i), (iii), (v), (vi)
(c) (i), (iii), (v), (vii) (d) (i), (ii), (iv), (vii)

**RPSC ACF-2011**

**Ans. (a)**

**232. The solvent used in liquid extraction should not have high latent heat of vaporization because**

(a) The pressure drops and hence the pumping cost will be very high
(b) It cannot be recovered by distillation
(c) Its recovery cost by distillation maybe prohibitively high
(d) It will decompose while recovering by distillation

**RPSC ACF-2011**

**Ans. (c)**

**233. In equilateral triangular coordinate a point on any side of the triangle represents**

(a) A pure component (b) A binary mixture
(c) A ternary mixture (d) None of the above

**RPSC ACF-2011**

**Ans. (b)**

**234. The operation which is employed for the removal of caffeine from coffee is**

(a) adsorption
(b) crystallization
(c) solvent extraction
(d) fractional distillation

**RPSC ACF-2011**

**Ans. (c)**

**235. Extraction of Coffee from its seed is done by.**

(A) Liquid-liquid Extraction
(B) Leaching
(C) Extractive distillation
(D) Steam distillation

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**236. Which of the following is true for a Bollman extractor?**

(a) Is a static bed leaching equipment
(b) Is a centrifugal extractor
(c) Uses both co current and counter current extraction steps
(d) Use only counter current extraction

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**237. In liquid extractor, the treated mixture is called**

(a) Solute (b) Raffinate
(c) Diluent (d) Extract

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**238. Ballman extractor**

**A. Is a static bed leaching equipment**
**B. Is used for extraction of oil from oil seed**
**C. Is a centrifugal extractor**
**D. Employs only counter current extraction**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**239. Solvent B is used to extract solute C from a given feed containing A and C. If solute C is more soluble in A than in B, the distribution coefficient for C will be**

**A. $\gg 1$ B. 1**
**C. <1 D. >1**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**240. Penicillin is separated from fermented broth by**

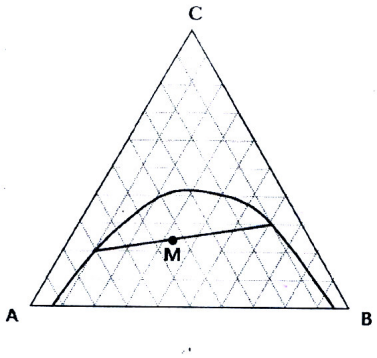
**A. Ternary azeotropic distillation**
**B. Evaporation in calandria**
**C. Simple distillation**
**D. Extraction with amyl or butyl ACFtate**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**241. Solute C is extracted in a batch process from its homogenous solution of A and C, using solvent B. The combined composition of the feed and the extracting solvent is shown in the figure below as point M, along with the tie line passing through it. The ends of the tie line are on the equilibrium curve.**




**What is the selectivity for C?**

(a) 3.5 (b) 7
(c) 10.5 (d) 21

**GATE-2016**

**Ans. (c)**

**242. An excellent solvent which has very high selectivity for aromatics and low volatility that is used for extraction aromatics from catalytic reformate is,**

(a) Cyclo hexane (b) Dimethyl formamide
(c) Diethyl Ether (d) Sulfolane

**ISRO Scientist Engineer-06.08.2017**

**Ans. (d)**

**243. Choose the best combination of properties for a good solvent for extraction out of the following:**

**A. High selectivity**
**B. Low selectivity**

**C. High viscosity**
**D. Low viscosity**
**E. Large distribution coefficient**
**G. High interfacial tension**
**H. Low interfacial tension**

(a) A, D, E, G (b) B, C, F, H
(c) A, D, F, H (d) B, C, E, G

**ISRO-12.02.2016**

**Ans. (a)**

**244. Sides of the equilateral triangular co-ordinates (on which ternary liquid system is plotted) represent**

(a) A pure component
(b) A binary mixture
(c) A ternary mixture
(d) Partially miscible ternary mixture

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**245. Interfacial area in an extractor is defined as**

(a) Surface area of drop/volume of drop
(b) 6 × hold-up/diameter of drop
(c) Hold-up/volume of drop
(d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

## E. Humidification

**246. A binary feed consisting of 25 mol% liquid and 75 mol% vapour is separated in a staged distillation column. The mole fraction of the more volatile component in the distillate product is 0.95. The molar flow rate of distillate is 50% of the feed flow rate and the MeCabe-Thiele method can be used to analyze the column. The q-line intersects the operating line of the enriching section at (0.35, 0.5) on the x-y diagram. The slope of the stripping section operating line (up to one decimal place) is_____**

**GATE-2015**

**Ans. (1.3 to 1.5)**

**247. In a humidity chart, the slanting lines running downward and to the right of saturation line are called:**

(a) Humidity lines
(b) Humid volume lines
(c) Adiabatic-cooling lines
(d) Temperature lines

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**248. For air at 100% saturation, the dry bulb temperation and wet bulb temperature are related as:**

(a) Dry bulb > wet bulb
(b) Dry bulb = wet bulb
(c) Wet bulb > Dry bulb
(d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (b)**

**249. If the percent humidity of air (30°C, total pressure 100KPa) is 24% and the saturation pressure of water vapor at that temperature is 4KPa, the percent relative humidity and the absolute humidity of air are:**

(a) 25.2, 0.0062
(b) 25, 0.0035
(c) 20.7, 0.0055
(d) 18.2, 0.0035

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 25.2, 0.0062

**250. Humid volume $V_H$ is the total volume of a unit mass of**

(a) Vapor laden gas at 1 atm and gas temperature
(b) Vapor laden gas at 1 atm and room temperature
(c) Gas plus its accompanying vapor at 1 atm and room temperature
(d) Vapor free gas plus whatever vapor it may contain at 1 atm and the gas temperature

**TNPSC AE-2019**

**Ans. (d)**

**251. Humidification involves mass transfer between a pure liquid phase and a fixed gas, which is**

(a) Soluble in the liquid
(b) Insoluble in the liquid
(c) At a fixed temperature
(d) Non-ideal in nature

**TNPSC AE-2019**

**Ans. (b)**

**252. For a vapor-gas mixture, the ratio of percentage saturation to relative saturation**

(a) May be equal or more than or less than unity depending upon temperature and pressure
(b) Is always equal to unity
(c) Is always less than unity
(d) s always more than unity

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**253. If air is heated at constant pressure which of the following does NOT change?**

(a) Relative humidity
(b) Humid heat
(c) Dew point
(d) Wet-bulb temperature

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**254. _____ is the volume of a mixture of air and accompanying water vapor per kg of dry air.**

(a) Relative humidity
(b) Absolute humidity
(c) Psychrometric volume
(d) Humid heat
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**255. The logarithm of the vapor pressure of the substance is plotted as the ordinate against the logarithm of the vapor pressure of the reference at the same temperature as the abscissa in**

(a) Hausbrandt Chart
(b) Duhring Plot
(c) Cox Chart
(d) Psychrometric Chart
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**256. If atmospheric temperature and the dew point are same, than the relative humidity is**

(a) zero (b) 50%
(c) 100% (d) No relation

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**257. Which of the following statement is true**

(a) Dew point is dependent of dry bulb temperature
(b) Dew point is independent of dry bulb temperature
(c) Dew point is dependent on both dry bulb temperature and wet bulb temperature
(d) Dew point is dependent of dry bulb temperature and independent on wet bulb temperature

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**258. If the relative humidity is 90% and temperature is 25 $^{o}$C, the water evaporates from the surface of a water body at a rate of 1.0 kg/m$^2$-h. The RH that will lead to an evaporation rate of 3.0 kg/m$^2$-h, with other conditions remaining same, is**

(a) 30% (b) 50%
(c) 60% (d) 70%

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**259. The percentage humidity is less than the relative humidity only at _____% humidity.**

(a) 0 (b) 100
(c) 0 &100 (d) <0

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**260. Condensation of a vapour under saturation condition means removal of**

(a) latent heat (b) sensible heat
(c) super heat (d) None of the above

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**261. The humidity of air at a dry bulb temperature of 65 $^{o}$C is 0.025 kg water/kg dry air. The latent heat of vaporization of water at 0 $^{o}$C is 2500 kJ/kg. The psychrometric ratio of air is 0.95 kJ (Kg dry air)$^{-1}$ K$^{-1}$ Considering 0 $^{o}$C as reference temperature, the enthalpy of air (in kJ/kg) at its adiabatic saturation temperature of 35 $^{o}$C is ___ (rounded off to two decimal plACF).**

**GATE-2018**

**Ans. (124 to 125)**

**262. When the vapor pressure of a liquid equals atmospheric pressure, the temperature of the liquid equals.**

(a) 100$^{o}$ C
(b) Boiling point
(c) The vaporization point
(d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**263. Dry bulb temperature of gas is**

(a) Less than the wet bulb temperature
(b) More than the wet bulb temperature
(c) Equal to the wet bulb temperature
(d) None of the above

**RPSC ACF-2011**

**Ans. (b)**

**264. Relative humidity is the ratio of**

(a) Actual humidity to the saturation humidity at the gas temperature

(b) Partial pressure of the vapor to the vapor pressure of the liquid at the gas temperature

(c) Mass of vapor to the mass of vapor free gas

(d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**265. Distillation is particularly important in**

(a) Petroleum refining

(b) Manufacture of sulfuric acid

(c) Vegetable oil hydrogenation

(d) All (a), (b) and (c)

**RPSC ACF-2011**

**Ans. (a)**

**266. Generally tower packing are made of**

(a) Quartz (b) Coke

(c) Clay (d) Stone ware

**RPSC ACF-2011**

**Ans. (c)**

**267. If $P_A$ is the partial pressure of Vapour A in a vapour-gas mixture and $P_s$ is the saturation vapour pressure of liquid A, condensation of vapour-gas just begins when.**

(A) $P_A > P_S$

(B) $P_A < P_S$

(C) $P_A = P_S$

(D) None of above

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**268. Dew point of a gas-water mixture decreases with,**

(a) Increasing pressure

(b) Decreasing pressure

(c) May increase or decrease

(d) No change with pressure

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**269. The cooling effect in a cooling tower can be increased by,**

(a) Increasing the barometric pressure

(b) Increasing the humidity of entering air

(c) Increasing the air velocity over the wet surfACF

(d) All of the above

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**270. The humidity of a gas is generally defined as**

(a) The weight of water per mole of moisture-free gas

(b) The weight of moisture-free gas per unit weight of water.

(c) The weight of water per unit weight of moisture-free gas

(d) All the above

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**271. When partial pressure is equal to its equilibrium vapour pressure it is termed as**

(a) Super saturated vapour

(b) Saturated vapour

(c) Ideal vapour

(d) Equilibrium vapour

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**272. At a climatic station, air pressure is measured as 100 kPa, air temperature as 25° C, and the wet-bulb, or dew-point, temperature as 19° C. Calculate the relative humidity as the saturated vapor pressure at 25° C is given to be 2733 Pa & actual vapor pressure is 2126 Pa.**

(a) 0.97 (b) 0.77

(c) 0.67 (d) 0.87

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**273. If the percent humidity of air (30 °C), total pressure 100 kPa) is 24% and the saturation pressure of water vapor at that temperature is 4 kPa, the percent relative humidity and the absolute humidity of air are**

(a) 20,7,0.0055 (b) 25, 0.0035

(c) 24,75,0.0062 (d) 18.2,0.0035

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**274. Fog is an example of colloidal system of**

**A. Solid dispersed in gas**

**B. Solid dispersed in liquid**

**C. Liquid dispersed in gas**

**D. Gas dispersed in liquid**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**275. The term range in a cooling tower refers to the difference in temperature of the,**

(a) Cold water leaving the tower and wet bulb temperature of the surrounding air

(b) Hot water entering the tower and cooled water leaving the tower

(c) Hot water entering the tower and dry bulb temperature of the surrounding air
(d) Hot water entering the tower and wet bulb temperature of the surrounding air.

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**276. The ratio of actual partial pressure of water in air to partial pressure of water at saturation is**
(a) Saturated humidity (b) Relative humidity
(c) Molal humidity (d) Humidity

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**277. For what value of Lewis number, the wet-bulb temperature and adiabatic saturation temperature are nearly equal?**
(a) 0.33 (b) 0.5
(c) 1 (d) 2

**GATE-2016**

**Ans. (c)**

**278. Which one of the following statements is true with regards to the percentage saturation (P.S) and relative saturation (R.S)?**
(a) P.S = R.S
(b) P.S > R.S
(c) P.S < R.S
(d) P.S and R.S are not inter related

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**279. Steady state temperature reached by a small amount of liquid evaporating into a large amount of unsaturated vapour-gas mixture is**
(a) dry bulb temperature (b) dew point
(c) adiabatic saturation temperature
(d) wet bulb temperature

**ISRO-12.02.2016**

**Ans. (d)**

**280. With reference to the humidification operation, which of the statements is/are correct?**
**1. Wet-bulb temperature of an unsaturated moist air at a given pressure is always lower than its dew point temperature.**
**2. Relative humidity of an unsaturated moist air at a given pressure increases during sensible heating process.**
**Select the correct answer using the code given below:**
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**281. Humid volume may be increased by increasing**
(a) temperature at constant humidity
(b) humidity at constant temperature
(c) both (a) and (b)
(d) neither (a) nor (b)

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**282. If the specific heats of a gas and vapour are 0.2 and 1.5 kJ/kg.K respectively and the humidity is 0.01 the humid heat in kJ/kg.K is :**
(a) 0.31 (b) 0.107
(c) 0.017 (d) 0.215

**AP PSC IOF-2009**

**Ans. (d)**

**283. The Lewis relation for air-water humidification is given by :**
(a) $\frac{h_G^2}{k_y C_S} = 1$ (b) $\frac{k_y C_S^2}{h_G} = 1$
(c) $\frac{k_y^2 h_G}{C_S} = 1$ (d) $\frac{h_G}{k_y C_S} = 1$

**AP PSC IOF-2009**

**Ans. (d)**

**284. Number of theoretical stages for an ideal binary system can be estimated by using :**
(a) Ponchon-Savarit method
(b) McCabe-Thiele method
(c) Sherwood method
(d) Fenske-Underwood method

**AP PSC IOF-2009**

**Ans. (b)**

**285. An ideal plate is defined as one where :**
(a) The vapor and liquid leaving streams are in equilibrium
(b) The vapor and liquid entering streams are in equilibrium
(c) The vapor leaving stream is in equilibrium with the liquid entering stream
(d) The vapor entering stream is in equilibrium with liquid leaving stream

**AP PSC IOF-2009**

**Ans. (d)**

## F. Drying

**286. Rotary driers are used :**
(a) To make milk powder
(b) To make synthetic detergent powder
(c) To make free-flowing granular fertilizers
(d) To dry sticky materials

**AP PSC IOF-2009**

**Ans. (c)**

**287. A wet solid of 100 kg is dried from a moisture content of 40 wt% to 10 wt%. The critical moisture content is 15 wt% and the equilibrium moisture content is negligible. All moisture contents are on dry basis. The falling rate is considered to be linear. It takes 5 hours to dry the material in the constant rate period. The duration (in hours) of the falling rate period is ——**

**GATE-2014**

**Ans. (1.1 to 1.3)**

**288. The portion of the water in the wet solid that cannot be removed by the inlet air due to humidity is called:**
(a) Equilibrium moisture
(b) Free moisture

(c) Relative humidity
(d) Unbound moisture

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**289. If ,**

**$W_s$ =Weight of dry solid**
**$S_s$ = Specific gravity of solids**
**$\gamma$ = unit weight of water**
**$W_f$ = Weight of fixed solids (non volatile)**
**$S_f$ =Specific gravity of fixed solids**
**$W_v$ = Weight of volatile solids**
**$S_v$ = Specific gravity of volatile solids**

**The specific gravity of solid matter in sludge can be computed from the relationship:**

(a) $\frac{W_s}{S_s\gamma} = \frac{W_f}{S_f\gamma} + \frac{W_v}{S_v\gamma}$ (b) $\frac{W_s}{S_v\gamma} = \frac{S_s\gamma}{S_f\gamma} - \frac{W_v}{W_f}$

(c) $\frac{S_s\gamma}{S_f\gamma} = \frac{W_s}{W_v} - \frac{S_v\gamma}{W_f}$ (d) $\frac{W_v}{W_f} = \frac{W_f}{S_f\gamma} + \frac{W_v}{S_v\gamma}$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**290. 100 kg of wet solids are to be dried from 80% to 10% mixture (by weight). The mass of moisture removed in kg is:**

(a) 78 (b) 82
(c) 98 (d) 56

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**291. Constant rate period is that drying period during which:**

(a) The moisture content of the substance remains constant
(b) The rate of vaporization per unit of drying surface area is constant
(c) The rate of vaporization increases with time
(d) The rate of vaporization decreases with time

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** The rate of vaporization per unit of drying surface area is constant

**292. The additional operation requires for drying gas and liquid is_____**

(a) Absorption (b) Adsorption
(c) Humidification (d) De-humidification

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**293. If the equilibrium vapour pressure is lower than pure liquid pressure then the moisture content is_____**

(a) Bound moisture
(b) Unbound moisture
(c) Equilibrium moisture
(d) None of these

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**294. Which of the following is not a continuous drier?**

(a) Drum drier (b) Spray drier
(c) Tunnel drier (d) Tray drier

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**295. For estimating the drier size it is necessary to know_____**

(a) Time of drying
(b) Heat of drying
(c) Speed of drying
(d) All of the mentioned

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**296. Lyophilization is a process of drying**

(a) of heat sensitive products
(b) of suspended particles
(c) of non-porous solids
(d) of solids and pastes

**TNPSC AE-2018**

**Ans. (a)**

**297. The moisture contained by a substance which exerts an equilibrium vapor pressure equal to that of pure liquid at the same temperature is known as**

(a) equilibrium moisture (b) Bound moisture
(c) Unbound moisture (d) Free moisture

**TNPSC AE-2018**

**Ans. (c)**

**298. 1000 kg of wet solids are dried from 60% to 20% moisture (by weight). The mass of moisture removed in kg is.**

(a) 520 (b) 200
(c) 400 (d) 500

**TNPSC AE-2018**

**Ans. (d)**

**299. Match the items in the left hand column (List - I ) with the appropriate items in the right hand column (List - II) and select the correct answer using the code given below:**

| **List-I** | **List-II** |
|---|---|
| **A.** Solutropic system | **1.** Crystallization |
| **B.** Equilibrium moisture content | **2.** Distillation |
| | **3.** Liquid-liquid extraction |
| | **4.** Drying |

**Code:**

| | **A** | **B** |
|---|---|---|
| (a) | 1 | 4 |
| (b) | 3 | 4 |
| (c) | 2 | 3 |
| (d) | 1 | 2 |

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**300. The temperature at which a vapor-gas mixture becomes saturated, when cooled at constant volume is called**
(a) Dry-bulb temperature
(b) Bubble point
(c) Dew point
(d) Wet bulb temperature

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**301. If the moisture content of the wet solid on dry basis is twice the moisture content on wet basis, then the moisture content on dry basis should be _____ kg moisture/kg dry solid.**
(a) 1.5 (b) 1.0
(c) 2.0 (d) 0.5

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**302. It takes 6 hours to dry a wet solid from 50% moisture content to the critical moisture content of 15%. How long will it take to dry the solid to 10% moisture content under the same drying conditions? (the equilibrium moisture content of solid is 5%)**
(a) 94 min (b) 15 min
(c) 71 min (d) 51 min

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**303. The vacuum dryers are suitable for**
(a) Highly viscous materials
(b) Heat sensitive materials
(c) Low molecular weight materials
(d) High molecular weight materials

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**304. Find the time taken to dry 50% to 40% of moisture (dry basis) if critical moisture content is 30%. (If the weight of the wet solid is 5 kg and the product of area and the constant rate is 0.5 kg/min)**
(a) 20 sec (b) 40 sec
(c) 60 sec (d) 80 sec

**Haryana PSC Poly Lect.-12.06.2022**
**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**305. A fiberboard sheet (1.5 m × 2.0 m × 15 mm) is being dried by suspending it horizontally in a current of hot, dry air. The edges are insulated so that drying takes plACF only from the top and bottom surfACFs. The wet sheet weighing 16 kg with initial moisture content of 60% loses moisture at a constant rate of $1.25 \times 10^{-5}$ kg $m^{-2}$ $s^{-1}$ unitl the moisture content falls to 30%. All moisture contents are on dry basis. The time required for drying during constant rate period (in hour) is ____ (rounded off to third decimal plACF).**

**GATE-2018**

**Ans. (11.0 to 11.3)**

**306. Moisture content on dry basis is 0.5 . Calculate moisture content on wet basis.**
(a) 3.0 (b) 0.75
(c) 0.33 (d) 1

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**307. Moisture content of a material at which the constant rate period ends and the falling rate period starts is called**
(a) Bound moisture content
(b) Unbound moisture content
(c) Free moisture content
(d) Critical moisture content

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**308. Double drum dryers are effective for:**
(a) Moderately heavy slurries
(b) Solution of salt
(c) Slurries of abrasive solids
(d) Solutions with limited solubility

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**309. Lyophilization is also known as:**
(a) Liquid extraction (b) Crystallization
(c) Flash-drying (d) Freeze-drying

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**310. During drying at constant rate period, the rate of drying _____.**
(a) Decreases
(b) Remains constant
(c) Becomes zero
(d) Increases

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**311. Choose any one of the following term that represents the process of concentrating orange juice.**
(a) Crystallizer
(b) Vacuum dryer
(c) Continuous dryer
(d) Evaporator

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**312. 2000 kg of wet solids containing 60% solids by weight are fed to a tray dryer where it is dried by hot air. The product finally obtained is found to contain 4% moisture by weight. What is the amount of product obtained?**
(a) 1350 kg (b) 1200 kg
(c) 1250 kg (d) 1850 kg

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**313. 1000 kg of wet solids are to be dried from 60% to 20% moisture (by weight). The mass of moisture removed in kg is**

(a) 100 kg (b) 200 kg
(c) 500 kg (d) 400 kg

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**314. Drying operation under vacuum is carried out to**

(a) reduce drying temperature
(b) increase drying temperature
(c) dry materials having very high unbound moisture constent
(d) dry materials having high bound moisture content

**ISRO-12.02.2016**

**Ans. (a)**

**315. The rate of drying during constant-rate period**

(a) is unaffected by the air humidity
(b) increases with increased air humidity
(c) decreases with increased air humidity
(d) none of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**316. Length to diameter ratio of most rotary drier is in the range of :**

(a) 1 to 2 (b) 4 to 10
(c) 10 to 20 (d) 20 to 30

**AP PSC IOF-2009**

**Ans. (b)**

## G. Adsorption

**317. In case of physical adsorption, heat of adsorption is of the order of _____ kcal/kg. mole.**

(a) 100000 (b) 1000
(c) 100 (d) 10000

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**318. In decaffeination of coffee, the caffeine is separated from $CO_2$ by adsorption on**

(a) Silica gel (b) Activated carbon
(c) Alumina (d) Zeolite

**TNPSC AE-2018**

**Ans. (b)**

**319. At a given equilibrium pressure, the concentration of adsorbed gas on adsorbent solid**

(a) remains constant with change in temperature
(b) increases with increased temperature
(c) decreases with increased temperature
(d) decreases with decreasing temperature

**TNPSC AE-2018**

**Ans. (c)**

**320. The unit of Freundlich capacity factor in the Freundlich adsorption isotherm equation is**

(a) $(mg/g).\left(\frac{L}{mg}\right)^{1/n}$ (b) $(mg/g)^{1/n}.\left(\frac{L}{mg}\right)$
(c) $(mg/g)^{1/n}.\left(\frac{L}{mg}\right)^{1/n}$ (d) $(mg/g).\left(\frac{L}{mg}\right)^{n}$

**TNPSC AE-2018**

**Ans. (a)**

**321. Reactivation of carbon involves**

(a) The carbon is heated to drive off the adsorbed organic material
(b) The carbon is heated to recover the adsorbed capacity
(c) The carbon is heated to increase its porosity
(d) The carbon is heated to increases the surface active area

**TNPSC AE-2018**

**Ans. (a)**

**322. Chemisorption is**

(a) A reversible phenomenon
(b) An irreversible phenomenon
(c) Same as 'Van Der Waals' Phenomenon
(d) Characterised by adsorption of heat

**TNPSC AE-2019**

**Ans. (b)**

**323. Langmuir equation is associated with**

(a) Diffusion (b) Absorption
(c) Drying (d) Adsorption

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**324. BET apparatus is used to determine the**

(a) specific surface area of a catalyst
(b) activity of the catalyst
(c) pore diameter
(d) porosity of the catalyst
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**325. When hysterisis is observed, the desorption equilibrium pressure**

(a) is always equal to that obtained by adsorbtion
(b) is always higher than that obtained by adsorbtion
(c) is always lower than that obtained by adsorbtion
(d) may be equal to or higher than that obtained by adsorbtion
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**326. In a batch adsorption process, 5g of fresh adsorbent is used to treat 1 liter of an aqueous phenol solution. The initial phenol concentration is 100mg/liter. The equilibrium relation is given by**

$$q^* = 1.3C$$

**Where $q^*$ is the amount of phenol adsorbed in mg of phenol per gram of adsorbent; and C is the concentration of phenol in mg/liter in the aqueous solution.**

**When equilibrium is attained between the absorbent and the solution, the concentration of phenol in the solution, rounded to 1 decimal plACF, is _____ mg/liter.**

**GATE-2017**

**Ans. (13.0 to 13.5)**

**327. Freundlich equation applies to adsorption of solute from**
(a) Concentrated solutions
(b) Dilute solutions, over a small concentrated range
(c) Gaseous solutions at high pressure
(d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**328. Chemisorption (chemical adsorption) is**
(a) Same as Van der Waal's adsorption
(b) Characterized by adsorption of heat
(c) Also called activated adsorption
(d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**329. For bulk separations adsorption becomes competitive with distillation when the relative volatility is**
(a) Less than about 1.25
(b) Between 1.5 and 2.0
(c) More than 2.0
(d) Between 2 and 2.5

**RPSC ACF-2011**

**Ans. (a)**

**330. Adsorption of a gas onto a solid**
(a) is an exothermic process
(b) is an endothermic process
(c) does not product any heat effect
(d) may be exothermic or endothermic, depending on the system

**RPSC ACF-2011**

**Ans. (a)**

**331. At 30°C, the amounts of acetone adsorbed at partial pressures of 10 and 100 mmHg are 0.1 and 0.4 kg acetone/kg activated carbon, respectively. Assume Langmuir isotherm describes the adsorption of acetone on activated carbon. What is the amount of acetone adsorbed (in kg per kg of activated carbon) at a partial pressure of 50 mmHg and 30°C?**
(a) 0.23 (b) 0.25
(c) 0.30 (d) 0.35

**GATE-2016**

**Ans. (c)**

**332. Adsorption on activated carbon is to be used for reducing phenol concentration in wastewater from 0.04 mol/l to 0.008 mol/l. The adsorption isotherm at the operating temperature can be expressed as $q = 0.025C^{1/3}$; where q is the phenol concentration in solid (mol/g solid) and C is the phenol concentration in water (mol/l). The minimum amount of solid (in grams) required per liter of wastewater (up to one decimal place) is_____**

**GATE-2015**

**Ans. (6.3 to 6.5) :**

**333. When hysteresis is observed, the desorption equilibrium pressure:**
(a) Is always equal to that obtained by adsorption
(b) Is always higher than that obtained by adsorption
(c) Is always lower than that obtained by adsorption
(d) May be equal to or higher than that obtained by adsorption

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Is always lower than that obtained by adsorption

## H. Leaching

**334. Extraction of coffee from its seed is done by**
(a) Leaching
(b) Absorption
(c) Extractive distillation
(d) Steam Distillation

**TNPSC AE-2019**

**Ans. (a)**

**335. Leaching at high temperature is favorable due to several reasons. One of the following is false**
(a) Increased viscosity liquid
(b) Increased solubility of solute
(c) Increased diffusivity in solvent
(d) High concentration in leach -liquor

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**336. Which of the following operations does not involve leaching?**
(a) Dissolving gold from ores
(b) Dissolving pharmACFutical products from bark or roots
(c) Dissolving sugar from the cells of the best
(d) Removing nicotine from its water solution by kerosene

**RPSC ACF-2011**

**Ans. (d)**

**337. Which of the following device is used for leaching of minerals from finely divided materials?**
(a) Percolation tank
(b) Filter press
(c) Pachuca tank
(d) Dorr classifier

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**338. Leaching of coarse solid lumps is also termed as**
(a) Liquid-Liquid extraction
(b) Dissolution
(c) Percolation
(d) Evaporation

**ISRO-12.02.2016**

**Ans. (c)**

**339. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).**
**Assertion (A):** The rate of leaching is higher at higher temperature.
**Reason (R):** The viscosity of liquid is lower and diffusivity larger at higher temperatures.
**Choose the correct answer form the code given below.**

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(c) (A) is true, but (R) is false
(d) (A) is false, but (R) is true

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**340. In solid-liquid extraction, if the solvent dissolves very little of solute, the**
(a) Solvent of low latent heat of vaporization should be used
(b) Solvent of low freezing point should be used
(c) Large quantity of solvent is required to extract the solute
(d) None of the above

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**341. In a counter-current leaching system, the ratio of solvent discharged in the overflow to that in the underflow**
(a) Varies with number of units
(b) Is constant
(c) Varies with the inlet flow rate
(d) Varies with outlet flow rate

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

## I. Crystallization

**342. The number of crystal forms based on the angles of the individual faces are**
(a) 3 (b) 4
(c) 7 (d) 9

**TNPSC AE-2018**

**Ans. (c)**

**343. The retained mother liquor is separated from the crystals by**
(a) Evaporation (b) Drying
(c) Adsorption (d) Centrifuging

**TNPSC AE-2019**

**Ans. (d)**

**344. The sequence of stages in the evolution of a crystal is**
(a) Embryo → Cluster → Nucleus → Crystal
(b) Cluster → Embryo → Nucleus → Crystal
(c) Nucleus → Cluster →Embryo → Crystal
(d) Nucleus → Embryo → Cluster → Crystal
(e) Answer not known

**TNPSC AE-2022, 2019**

**Ans. (b)**

**345. Heuristic rules is used in**
(a) Distillation
(b) Evaporation
(c) Crystallisation
(d) Liquid-liquid extraction

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**346. The volume of atoms in a unit cell divided by total volume of the unit cell is known as**
(a) atomic number
(b) atomic packing factor
(c) coordination number
(d) crystal index

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**347. How many atoms are there per unit cell in a body-centred cubic lattice system?**
(a) 2 (b) 3
(c) 4 (d) 6

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**348. Using $\Delta L$ law calculate the size of the sugar crystal if the growth rate of crystal is 300 μm/s in a time interval of 40s.**
(a) 1200 μm (b) 7.5μm
(c) 8 μm (d) 30 μm

**Karnataka PSC ADF - 2015**

**Ans. (a)**

**349. In Oslo cooling crystallizer, the main control variable(s) is /are**
(a) The feed rate
(b) The feed temperature
(c) The circulation rate
(d) All of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**350. Rate of crystal growth is defined as the rate**
(a) At which the volume of crystal increase with time
(b) At which the face of a crystal moves in a direction perpendicular to the face
(c) At which mass of crystal increases with time
(d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**351. A trigonal crystal has**
(a) Three unequal length but equally inclined axes
(b) Three equal length and equally inclined axes
(c) Three equal length but unequally inclined axes
(d) Three unequal length and unequally inclined axes

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**352. Which one of the following chemical industries is utilizing the process of Cyclization?**
(a) Organic chemicals (b) Synthetic rubber
(c) Dyes (d) Petroleum

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**353. The McCabe $\Delta L$ law states that the:**
(a) olar heats of vaporization of components are nearly equal
(b) Linear crystal growth rate depends on the degree of super saturation
(c) Linear crystal growth rate does not depend on the crystal size
(d) Linear crystal growth rate depends on the crystal size

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Linear crystal growth rate does not depend on the crystal size

# 04.

# Advance Seperation Process

## A. Fundamental of separation and equilibrium separation

**1. The separation technique used for separation of multi component mixtures of gas or liquids is**

(a) Ion exchange (b) Absorption
(c) Chromatography (d) Adsorption

**TNPSC AE-2018**

**Ans. (c)**

**2. The term used for separations carried out at high pressure with very fine particles and high flow rates is**

(a) GC (b) GPC
(c) LC (d) HPLC

**TNPSC AE-2018**

**Ans. (d)**

**3. Which one of the following is called salting?**

(a) Increasing the solubility of solute by increasing temperature
(b) Cooling/temperature reduction of saturated solution
(c) Evaporating portion of a solvent
(d) Adding a third component to generate supersaturation

**TNPSC AE-2019**

**Ans. (d)**

**4. _____theory describes the membrane as a porous film into which both water and solute (ion) dissolve.**

(a) Solution-diffusion (b) Surface-capillary
(c) Polarization (d) Donnan exclusion

**TNPSC AE-2019**

**Ans. (a)**

**5. Identify the process from the following, one which could not be done by pervaporation technique?**

(a) Removal of water from organic solvents
(b) Adjustment of $H_2$/CO ratio in synthesis gas
(c) Removal of organics from water
(d) Dehydration of ethanol-water azeotrope
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**6. Permeance if a solute through a membrane is defined as :**

(a) The ratio of permeability to the membrane thickness
(b) The ratio of molecular weight of the polymer to the thickness of the membrane
(c) The ratio of membrane thickness to the solute diffusivity
(d) The ratio of molar transmembrane flux to the thickness of the membrane

**AP PSC IOF-2009**

**Ans. (d)**

## B. Membrane sepration

**7. Flat sheet membranes for reverse osmosis are usually used in**

(a) Spiral-wand modulus (b) frame modulus
(c) Tubular modulus (d) Hollow modulus

**TNPSC AE-2018**

**Ans. (a)**

**8. The advantages of reverse osmosis is**

(a) Separation takes places at room temperature
(b) Separation takes place at elevated pressures
(c) Phase change is involved
(d) None of these

**TNPSC AE-2018**

**Ans. (a)**

**9. The separation process in which one or more components of a liquid mixture diffuse through a selective membrane evaporate under low pressure on the downstream side is known as**

(a) Pervaporation (b) Permeation
(c) Crystallization (d) Dialysis

**TNPSC AE-2018**

**Ans. (a)**

**10. The technique used to obtain pure water from dilute aqueous solutions is**

(a) Osmosis (b) Reverse osmosis
(c) Dialysis (d) Filtration

**TNPSC AE-2018**

**Ans. (b)**

**11. The separation accomplished mainly by including large molecules or collodial partial form pores of membrane is**

(a) Reverse osmosis (b) Ultra filtration
(c) Micro filtration (d) Dialysis

**TNPSC AE-2018**

**Ans. (b)**

**12. The_____ process is based on the difference in rates of diffusional transport through membranes**

(a) Thermal diffusion (b) Permeation
(c) Dialysis (d) Osmosis

**TNPSC AE-2018**

**Ans. (c)**

**13. The process of using a ion-selective membrane and a potential gradient to speed migration of ions through membranes is**

(a) Reverse Osmosis (b) Ultra Filtration
(c) Dialysis (d) Electro dialysis

**TNPSC AE-2018**

**Ans. (d)**

**14. The flux through a dense payment film is inversely proportional to its.**

(a) Thickness
(b) Temperature
(c) Pressure
(d) Concentration gradient

**TNPSC AE-2018**

**Ans. (a)**

**15. The separation of Uranium isotopes using hexa fluorides can be done by**

(a) Adsorption (b) Leaching
(c) Separation (d) Extraction

**TNPSC AE-2018**

**Ans. (c)**

**16. The technique used in separation of products from bioreactors is**

(a) Super critical fluid extraction
(b) Ion exchange
(c) Permeation
(d) Osmosis

**TNPSC AE-2018**

**Ans. (b)**

**17. Separation processes that work by virtue of difference in time of movement through some medium under the impetus of any driving force are collectively called as**

(a) Pressure governed processes
(b) Temperature governed processes
(c) Rate governed processes
(d) Composition governed processes

**TNPSC AE-2018**

**Ans. (c)**

**18. The action which transforms a mixture of substance into 2 or more products which differ from each other in composition is**

(a) Reaction (b) Separation
(c) Mixing (d) Sublimation

**TNPSC AE-2018**

**Ans. (b)**

**19. Reverse osmosis system in house hold appliances usually operates in the range.**

(a) 10 to 20 psi (b) 20 to 30 psi
(c) 50 to 70 psi (d) 80 to 100 psi

**TNPSC AE-2019**

**Ans. (c)**

**20. Solutions having the same osmotic pressure are known as_____solutions.**

(a) isobaric (b) isotonic
(c) isometric (d) isosonic

**TNPSC AE-2019**

**Ans. (b)**

**21. The packing density of a hollow fibre membrane is in the range of_____**

(a) 50 to 100 $m^2/m^3$
(b) 100 to 300 $m^2/m^3$
(c) 300 to 500 $m^2/m^3$
(d) 500 to 9000 $m^2/m^3$

**TNPSC AE-2019**

**Ans. (d)**

**22. Microfiltration process is carried out with the help of membranes having pore size range of_____**

(a) 0.1 to 10 microns
(b) 10 to 100 microns
(c) 0.001 to 0.01 microns
(d) 0.01 to 0.1 microns

**TNPSC AE-2019**

**Ans. (a)**

**23. Reverse osmosis is used as an alternate to**

(a) Leaching (b) Crystallization
(c) Evaporation (d) Adsorption

**TNPSC AE-2019**

**Ans. (c)**

**24. Driving force across a membrane in Dialysis is**

(a) pressure difference
(b) size difference
(c) concentration difference
(d) activity difference

**TNPSC AE-2019**

**Ans. (c)**

**25. Pervaporation, a separation process which is used to separate**

(a) one or more components of a gas mixture
(b) one or more components of a liquid mixture
(c) a mixture of gas and liquid
(d) one or more components of a liquid solution

**TNPSC AE-2019**

**Ans. (b)**

**26. An ideal dialyser has a clearance of_____ml/min for a low molecular weight solute like urea.**

(a) 50-100 ml/min (b) 100-200 ml/min
(c) 200-300 ml/min (d) 400-500 ml/min

**TNPSC AE-2019**

**Ans. (a)**

**27. _____pressure is the principal during force responsible for the transport of materials in nano filtration process.**

(a) osmotic (b) hydraulic
(c) transmembrane (d) pneumatic

**TNPSC AE-2019**

**Ans. (c)**

**28. Which design of membrane module have poor resistance to mechanical damage?**

(a) Plate and frame (b) Spiral wound
(c) Tabular (d) Hollow fibre

**TNPSC AE-2019**

**Ans. (c)**

**29. Metal membranes are stable at a temperature range of _____ $^{o}$C.**

(a) 500-800 $^{o}$C (b) 1000 – 1500 $^{o}$C
(c) 2000-2500 $^{o}$C (d) < 500 $^{o}$C

**TNPSC AE-2019**

**Ans. (a)**

**30. The charge on the Ion-exchanger depends on the _____ of the solvent.**

(a) pH (b) Concentration
(c) Temperature (d) Solubility
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**31. Which of the following membrane materials is used for Reverse Osmosis application?**

(a) Polybenzimidazole
(b) Polypropylene
(c) Polyimide
(d) Polysulphone
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**32. In dialysis, the flux of each solute is proportional to the**

(a) Temperature gradient
(b) Velocity gradient
(c) Concentration gradient
(d) Momentum gradient
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**33. Which one is not a zeolite based membrane?**

(a) Alumina supported silicate
(b) Rare earth doped slicate
(c) ZSM 5
(d) Graphene oxide
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**34. The membranes which are generally made with the skin on the outside, and a bundle with thousands closely packed fibres sealed in a metal cylinder is**

(a) Frame – fiber membrane
(b) Tubular – fiber membrane
(c) Spiral wound fiber membrane
(d) Hollow - fiber membrane
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**35. A process for selectively removing low molecular weight solutes from a solution by allowing them to diffuse into a region of lower concentration is**

(a) Filtration (b) Dialysis
(c) Pervaporation (d) Permeation
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**36. The total _____ depends on the number of charged functional groups per gram of dry ion exchanger (or) per ml of swollen gel.**

(a) ionic capacity
(b) non-ionic capacity
(c) concentration
(d) metallic capacity
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**37. Cation exchange resins include**

(a) weak base resins with Ammonium groups
(b) strong base resins with Ammonium groups
(c) strong acid resins with Sulphuric acid groups
(d) weak acid resins with Sulphuric acid groups
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**38. The method used in recovery of metals from dilute solutions is**

(a) crystallization (b) permeation
(c) osmosis (d) ion exchange
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**39. The difference in permeablities through membranes is used to separate solutions containing large molecules or colloids in the process of**

(a) Gas Permeation (b) Ultrafiltration
(c) Electrolysis (d) Osmosis
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**40. Typically, microfiltration retains matters of diameter between_____ and _____ microns**

(a) 0.02 and 10 microns
(b) 10 and 50 microns
(c) 0.001 and 0.01 microns
(d) 51 and 100 microns
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**41. The complete rejection of solute by the membrane which leads to a higher concentration at membrane surface than in bulk solution is known as**

(a) Osmosis
(b) Dialysis
(c) Concentration polarization
(d) Diffusion
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**42. Recovery of salts or sugars from natural products or colloidal solutions can be achieved by**

(a) Chromatography
(b) Micro filtration
(c) Ultra filtration
(d) Reverse osmosis
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**43. Impurities in the size range of 1-1000 micro-meters are suggested to be treated by**

(a) Ultra-filtration (b) Micro-filtration
(c) Nano-filtration (d) Reverse Osmosis

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**44. In which of the following membrane process-driving force is not pressure gradient**

(a) Microfiltration (b) Ultrafiltration
(c) Reverse osmosis (d) Dialysis

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**45. Ion exchange process is similar to**

(a) Absorption (b) Extraction
(c) Adsorption (d) Leaching

**RPSC ACF-2011**

**Ans. (c)**

**46. The mechanism of separation in reverse osmosis :**

(a) Screening (b) Diffusion
(c) Adsorption (d) Extraction

**AP PSC IOF-2009**

**Ans. (b)**

**47. The range of particles separated in ultrafiltration :**

(a) 0.5 μm – $10^{-3}$ μm
(b) 1 – 1000 nm
(c) 0.5 μm – μm
(d) 1 cm – $10^{-3}$ mm

**AP PSC IOF-2009**

**Ans. (c)**

**48. Separation of uranium isotopes is done by :**
(a) Ultrafiltration
(b) Adsorptive separation
(c) Gas separation by porous membranes
(d) Cyclone separator

**AP PSC IOF-2009**

**Ans. (c)**

**49. Selective removal of low molecular weight solutes from a solution is called :**
(a) Dialysis
(b) Reverse osmosis
(c) Ion-exchange
(d) Molecular distillation

**AP PSC IOF-2009**

**Ans. (b)**

## C. Super critical fluid extraction

**50. Solubility in a super critical fluid are strong functions of**
(a) Compositions (b) Mobility of ions
(c) Temperature (d) Osmosis

**TNPSC AE-2018**

**Ans. (c)**

**51. A chief disadvantage of supercritical fluid extraction is because of it requires**
(a) a high density (b) a high viscosity
(c) a high temperature (d) a high pressure

**TNPSC AE-2019**

**Ans. (d)**

**52. Which among the following does not relate to solubility and crystal size of super saturation process?**
(a) Kelvin Equation
(b) Gibbs-Thompson Equation
(c) Ostwald Equation
(d) Pensky- Randall Equation
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**53. Solutes can be recovered from a super critical fluid by changing**
(a) ion exchange properties
(b) temperature
(c) permeability
(d) diffusivities
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**54. The technique used in recovery of sensitive biological products is**
(a) permeation technique
(b) per vaporation
(c) super critical fluid extraction
(d) dialysis
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**55. In super critical extraction, the commonly used solvent is**
(a) $H_2O$ (b) $C_2H_5OH$
(c) $CCl_4$ (d) $CO_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**56. Useful solvent for supercritical fluid extraction used in food processing industry is**
(a) Carbon dioxide
(b) Sulphur dioxide
(c) Hydrogen
(d) Oxygen

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**57. The solubility of a solute in a supercritical fluid :**
(a) decreases
(b) increases
(c) remains the same
(d) all of the above

**AP PSC IOF-2009**

**Ans. (b)**

**58. An attractive supercritical solvent used for extraction of phyto-chemicals :**
(a) Water (b) Ethylene
(c) Carbon dioxide (d) Methane

**AP PSC IOF-2009**

**Ans. (c)**

**59. Compounds separated from solids or liquids with solvent held at pressure and temperature above critical point of solvent, the process is termed as:**
(a) Extraction
(b) Supercritical fluid leaching
(c) Supercritical fluid extraction
(d) Adsorption

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

## D. Bioseperation

**60. In a biopharmaceutical industry, chromatography is a technology used in _____ processing.**
(a) inoculum (b) downstream
(c) pretreatment (d) fermentation

**TNPSC AE-2019**

**Ans. (b)**

# 05.

# Process Equipment Design

## A. Plant Design (Basic design procedure, cost consideration, factors affecting vessel design)

**1. Mohr stress diagram is a graph of**
(a) Shear stress versus angle of repose
(b) Shear stress versus pressure
(c) Angle of repose versus angle of internal friction
(d) Shear stress versus shear rate.
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**2. For a given fluid, as the pipe diameter increases, the pumping cost**
(a) decreases
(b) increases
(c) remains the same
(d) may increase or decrease, depending upon whether the fluid is Newtonian or non-Newtonian

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**3. In which design stage, "cost-and-profit potential" of an established chemical process is determined?**
(a) Preliminary design
(b) Quick estimate design
(c) Detailed estimate design
(d) Firm process design

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**4. Pilot plant is necessary for the design and setting up of**
(a) Mixers (b) Cyclone separator
(c) Centrifugal pump (d) Reactor

**ISRO Scientist Engineer-06.08.2017**

**Ans. (b)**

**5. Process synthesis technology involves the following application**
(a) Heat Exchanger Network (HEN)
(b) Combined Heat and Power (CHP)
(c) Reactor system optimisation
(d) All of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

## B. Equipment Design (Spherical and Cylindrical vessel, Supports and Heads)

**6. The Longitudinal stress $\sigma_i$ and the circumferential stress $\sigma_h$, in a cylindrical vessel are related by**
(a) No relation exists (b) $\sigma_h = \sigma_{\frac{1}{2}}$
(c) $\sigma_h = \sigma_i$ (d) $\sigma_h = 2\sigma_1$

**TNPSC AE-2019**

**Ans. (d)**

**7. Annealing is carried out in order to**
(a) relieve stress
(b) decrease softness
(c) decrease ductility
(d) increase stress
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**8. The corrosion occurring at the contact area between materials under load subjected to vibration and slip is known as**
(a) Galvanic corrosion
(b) Crevice corrosion
(c) Fretting corrosion
(d) Intergranular corrosion
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**9. The circumferential (hoop) stress in a thin walled cylindrical vessel under internal pressure is**
(a) $\frac{pd}{2t}$ (b) $\frac{pd}{t}$
(c) $\frac{pd}{3t}$ (d) $\frac{pd}{4t}$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**10. The term 'knuckle radius' is associated with**
(a) flat heads
(b) torispherical heads
(c) hemispherical heads
(d) conical heads

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**11. In a chemical process plant, utilities include steam, water, compressed air, electricity, oxygen, fuel gases etc. The cost of utilities in a common chemical process plant is roughly what percentage of the total product cost?**

(a) 1 to 5 (b) 10 to 20
(c) 25 to 35 (d) 35 to 40

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**12. The joint efficiency of seamless pipe is**

(a) 1 (b) 1.5
(c) 0.5 (d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**13. Skirt support is most suitable for**

(a) Small horizontal vessels
(b) Large horizontal vessels
(c) Tall vertical vessels
(d) Thick walled small vertical

**RPSC ACF-2011**

**Ans. (c)**

**14. Bracket supports are the most suitable for supporting _____vessels.**

(A) Thick walled vertical
(B) Horizontal
(C) Thin spherical
(D) None of these

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**15. The testing pressure of storage tanks and pressure vessels designed as per Indian standard codes should be about___times the design pressure.**

(A) 1.5 to 2 (B) 3 to 4
(C) 4 to 5 (D) >5

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**16. Of the pressure vessels with same thickness but different diameters, which one with stands higher pressure?**

(a) larger dia vessel
(b) strength of the vessel is same irrespective of the diameter
(c) larger dia long vessel
(d) smaller dia vessel

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**17. During design of pressure vessel, in circumferential application for any thickness, the joint efficiency for "double welded butt joint" with single "V" groove will be**

(a) 0.65 - 0.85 (b) 0.45 - 0.70
(c) 0.35 - 0.55 (d) 0.75 - 0.95

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**18. In the designing of cylindrical pressure vessel, if P is design pressure, D is mean diameter and t is the thickness of the shell, then longitudinal stress $f_a$ will be**

(a) $f_a = \frac{PD}{2t}$ (b) $f_a = \frac{4PD}{t}$
(c) $f_a = \frac{2PD}{t}$ (d) $f_a = \frac{PD}{4t}$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**19. The "Knuckle radius" is associated with**

(a) Conical heads
(b) Flat heads
(c) Torispherical heads
(d) Hemispherical heads

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**20. _____ head is the most economical for cylindrical vessels designed for operating at high pressure (>15 atm).**

(a) Hemispherical (b) Conical
(c) Ellipsoidal (d) Dished

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**21. A cylindrical storage tank can have a self supported conical roof, if its diameter is**

(a) < 50 m
(b) between 20 to 50 m
(c) > 50 m
(d) < 20 m

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**22. In a thin walled cylindrical vessel of thickness (t) with inside radius (r) and internal gauge pressure is p. If the hoop stress is $\sigma_h$ and longitudinal stress is $\sigma_l$, then which one is correct?**

(a) $\sigma_h = \frac{Pr}{2t}, \sigma_l = \frac{Pr}{t}$
(b) $\sigma_h = \frac{Pr}{4t}, \sigma_l = \frac{Pr}{t}$
(c) $\sigma_h = \frac{Pr}{t}, \sigma_l = \frac{Pr}{2t}$
(d) $\sigma_h = \frac{Pr}{t}, \sigma_l = \frac{Pr}{4t}$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**23. In a cylindrical vessel subjected to internal pressure, the longitudinal stress ($\sigma_1$), and the circumferential stress ($\sigma_h$) are related by**

(a) $\sigma_h = \sigma_L / 3$ (b) $\sigma_h = 2\sigma_L$

(c) $\sigma_h = 2\sigma_L$ (d) $\sigma_h = \sigma_h = ½ \sigma_L$

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**24. The storage tank for volatile liquid will have a roof which is of**

**A. Flat type**

**B. Conical type**

**C. Hemispherical**

**D. Floating head**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**25. Normal screwed fittings are used for pipes up to a diameter of……inches.**

**A. 1.5**

**B. 3**

**C. 5**

**D. 9**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**26. The most commonly used joint in underground pipeline is the**

(a) sleeve joint (b) coupling

(c) flange (d) expansion joint

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**27. Spherical shaped pressure vessel is considered to be the most ideal, because**

(a) withstand higher pressure

(b) fabrication very easy

(c) designed without wind load considerations

(d) supported very easily

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**28. Generally, no corrosion allowance in wall thickness of a part is required, if the thickness is more than**

(a) 5 mm (b) 10 mm

(c) 20 mm (d) 30 mm

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**29. Under operating conditions of process vessel, the maximum working pressure is usually expressed in :**

(a) gauge pressure (b) absolute pressure

(c) vacuum pressure (d) deferential pressure

**AP PSC IOF-2009**

**Ans. (a)**

**30. Bracket supports are most suitable for :**

(a) Spherical vessels (b) Vertical vessels

(c) Horizontal vessels (d) Storage vessels

**AP PSC IOF-2009**

**Ans. (b)**

**31. Two 30 cm diameter steel pipes are connected by :**

(a) Threaded joints (b) Union couplings

(c) Flanges (d) Sleeves

**AP PSC IOF-2009**

**Ans. (c)**

**32. The ends of a cylindrical vessel can be closed by a :**

(a) Flat plate

(b) Hemispherical head

(c) Torispherical head

(d) Any one of the above

**AP PSC IOF-2009**

**Ans. (d)**

**33. For a 25 mm hole drilled in plates to be riveted, the shank diameter of the rivet should be**

(a) 25 mm

(b) 23 mm

(c) 26 mm

(d) 27 mm

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**34. A head of a vessel comprises of straight flanged section, corner torus section and central dished section. Crown radius and knuckle radius in a head is related respectively to the**

(a) Central dished section and corner torus section

(b) Corner torus section and central dished section

(c) Central dished section and straight flange section

(d) Straight flange section and corner torus section

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**35. In hydrostatic testing of welded pipe (for leakage, strength etc), the ratio of minimum hydrostatic test pressure to internal design pressure is**

(a) 0.5 (b) 1.5

(c) 7.5 (d) 20

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**36. Work out the thickness of a standard torispherical head for a cylindrical pressure vessel under following conditions:**

**Design pressure = 1.43 N/mm$^2$**
**Design stress = 85 N / mm$^2$**
**Joint efficiency, J = 1**
**Crown radius, $R_C = D_i = 1.5$ m**
**Knuckle radius = 6 percent $R_C$**
**The correct answer is**

(a) 18 mm (b) 20 mm
(c) 22 mm (d) None of these

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**37. In an ellipsoidal head with major axis 2a and minor axis 2b, the circumferential stress will be negative (compressive) if:**

(a) $a/b > 2$
(b) $a^2/b^2 > 2$
(c) $a/b > 1$
(d) $a^2/b^2 > 4$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $a^2/b^2 > 2$

**38. When can a cylindrical storage tank have a self supported conical roof:**

(a) If its diameter is more than 50 m
(b) If its diameter is less than 20 m
(c) Whatever is the diameter
(d) If the thickness of the roof is more than that of the cylindrical shell

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** If its diameter is less than 20 m

**39. For a cylindrical internally pressurized vessel, which of the following closure types would withstand highest pressure if each closure is of the same material and thickness:**

(a) Hemispherical
(b) Ellipsoidal (2:1)
(c) Conical
(d) Flat plate

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Hemispherical

**40. The volume capacity of 2:1 ellipsoidal head is:**

(a) $\pi/24 \times Da^3$
(b) $\pi/12 \times Da^3$
(c) $\pi/18 \times Da^3$
(d) $\pi/6 \times Da^3$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $\pi/24 \times Da^3$

**41. The thickness of a flat plate required to resist a given pressure load is:**

(a) Directly proportional to the effective plate diameter
(b) Directly proportional to the square root of the internal pressure
(c) Inversely proportional to the square root of the maximum allowable stress
(d) All options are correct

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** All options are correct

**42. In a spherical vessel subjected to internal pressure, the longitudinal stress $\sigma_L$ and the circumferential stress $\sigma_n$ are related by:**

(a) $\sigma_n = \sigma_L$
(b) $\sigma_n = 2\sigma_L$
(c) $\sigma_n = \sigma_{L/2}$
(d) $\sigma_n = \sqrt{\sigma_L}$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $\sigma_n = \sigma_L$

## C. Material of Construction (Selection and characteristics)

**43. Sulphuric acid containing 93.19% $H_2SO_4$ is transported in tanks made of**

(a) Steel (b) Iron
(c) Copper (d) Zinc

**TNPSC AE-2018**

**Ans. (a)**

**44. Glass is**

(a) a crystalline solid
(b) a undercooled liquid
(c) a solid having a definite melting point
(d) a solid of volatile oxides

**TNPSC AE-2018**

**Ans. (b)**

**45. Pilling-Bedworth ratio is the ratio of**

(a) the volume of the metal consumed to the volume of the oxide formed
(b) the mass of the metal consumed to the mass of the oxide formed
(c) the volume of the oxide formed to the volume of the metal consumed
(d) the mass of the oxide formed to the mass of the metal consumed

**TNPSC AE-2018**

**Ans. (c)**

**46. The percentage of carbon is the least in**

(a) White cast iron (b) Grey cast iron
(c) Wrought iron (d) Stainless steel

**TNPSC AE-2018**

**Ans. (c)**

**47. Rock formed entirely of plagioclase feldspar is known as**

(a) Bauxite (b) Magnatite
(c) Anorthosite (d) Illite

**TNPSC AE-2019**

**Ans. (c)**

**48. Heat exchangers are never made up of**

(a) Carbon steel (b) Copper
(c) Stainless steel (d) Lead

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**49. Fibre composite used for replacing screws for bone fracture and joint replacement is made of**

(a) Carbon (b) Polymer
(c) Fiberglass (d) Aramid
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**50. The raw materials for steel manufacture are**

(a) cast iron, wrought iron and air/oxygen
(b) cast iron and air/oxygen
(c) pig iron and air/oxygen
(d) wrought iron and air/oxygen
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**51. Which of the following is more suitable for the construction of refractories ?**

(a) High porosity bricks
(b) Low porosity bricks
(c) Bricks with high thermal expansion
(d) Bricks with low thermal conductivity
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**52. Pressure of Cobalt in steel improves is**

(a) Corrosion resistance
(b) Tensile strength
(c) Cutting ability
(d) Refractory resistance
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**53. Duralumin mainly composed of**

(a) Al and Cu (b) Al and Zn
(c) Al and Ni (d) Al and Mg

**WB PSC Scientific Assistant-2019**

**Ans. (a)**

**54. Aluminium is suitable as a material of construction for vessels handling**

(a) 80% $H_2SO_4$
(b) 98% $H_2SO_4$
(c) HCl of all concentrations
(d) $HNO_3$ of concentration greater than 85%

**RPSC ACF-2011**

**Ans. (d)**

**55. Which one of the following materials contains copper?**

(a) SS 321 (b) Inconel 0.05%
(c) Monel (d) Karbate

**RPSC ACF-2011**

**Ans. (c)**

**56. Impellers of 98% sulfuric acid pump are made of**

(a) Mild steel
(b) Cast iron
(c) Lead
(d) Illium (a nickel - chrome alloy)

**RPSC ACF-2011**

**Ans. (d)**

**57. Material of construction for heat exchanger tubes handling sea water is**

(a) Mild steel
(b) Stainless steel
(c) Cupro-nickel alloys (70% Cu, 30% Ni)
(d) Aluminium

**RPSC ACF-2011**

**Ans. (c)**

**58. The major material of construction in a soda ash plant is**

(a) Cast iron (b) PVC
(c) Graphite (d) Stainless steel

**RPSC ACF-2011**

**Ans. (b)**

**59. The corrosion resistance of stainless steel is due to the**

(a) Formation of a thin oxide film on the surface
(b) Formation of graphitic carbon on the surface
(c) Formation of an inert Fe – C – Fe type complex chain
(d) None of the foregoing

**RPSC ACF-2011**

**Ans. (a)**

**60. Monel is an alloy of**

(a) Iron and silicon
(b) Copper and chromium
(c) Nickel and Molybdenum
(d) Copper and nickel

**RPSC ACF-2011**

**Ans. (d)**

**61. Which of the following metals is called metallic glass?**

(a) Lead (b) Molybdenum
(c) Tantalum (d) Titanium

**RPSC ACF-2011**

**Ans. (c)**

**62. In the processing of steel, nitriding is done. Read the following statement and choose the correct option.**

**Statement-I: Nitriding is done for softening of surface of steel.**

**Statement-II: Nitriding is done at 510°C (approx.) for hardening the surface of steel.**

(a) I and II are correct

(b) None is correct
(c) Only II is correct
(d) Only I is correct

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**63. The most resistant material to alkaline corrosion is**

(a) duriron (b) carbate
(c) aluminium (d) nickel

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**64. For handling of fuming sulphuric acid, a suitable material of construction is**

(a) Nickel (b) Stainless steel 304
(c) Carbon steel (d) Monel

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**65. A minimum corrosion allowance for carbon and low alloy steel, where severe corrosion is not expected, should be**

(a) 1.0 mm (b) 4.0 mm
(c) 3.0 mm (d) 2.0 mm

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**66. Monel is an alloy containing**

(a) Ni and Cu in the ratio of 1 : 2
(b) Cu and Fe in the ratio of 2 : 1
(c) Ni and Fe in the ratio of 1 : 2
(d) Ni and Cu in the ratio of 2 : 1

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**67. "Teflon" polymer has significant contribution in material of construction. It is represented by**

(a) $CH_2 = CH - F$ (b) $C_2F_2$
(c) $C_2F_4$ (d) $CF_4$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**68. Nitriding of steel is done in an electronic furnace, kept at about**

(a) $1700^{\circ}C$ (b) $125^{\circ}C$
(c) $510^{\circ}C$ (d) $2200^{\circ}C$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**69. For handling, concentrated hydrochloric acid, a suitable material a construction is**

(a) Cast iron (b) Hastelloy
(c) Stainless steel (d) Aluminium

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**70. In process design, "Guest theory" represent the theory of failure. If $f_{RS}$ is resultant shear stress, and $f_y$ is yield stress in tension test, then condition of failure will be**

(a) $f_y = \frac{1}{2} f_{RS}$ (b) $f_y = (f_{RS})^{1/2}$
(c) $f_y = (f_{RS})^2$ (d) $f_y = 2\, f_{RS}$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**71. Tensile strain $\sigma_t$ plays an important role in equipment design consideration. If $f_t$ is tensile stress, then in $\sigma_t = \frac{f_t}{E}$, E will represent**

(a) modulus of rigidity
(b) modulus of efficiency
(c) modulus of direct elasticity
(d) modulus of economy

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**72. Corrosion resisting materials are the alloys containing chromium as a base alloyed with**

(a) iron, molybdenum, tungsten
(b) iron, vanadium, tungsten
(c) iron, vanadium, molybdenum
(d) iron, nickel, cobalt

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**73. Brass is composed of**

(a) Cu – Zn (b) Al – Fe
(c) Mg – C (d) Cu – Sn

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**74. A suitable material of construction to use with fuming sulphuric acid**

(a) Nickel
(b) Carbon steel
(c) Stainless steel type 304
(d) Monel

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**75. The most abundant metal in the earth's crust is**

**A. Copper**
**B. Aluminum**
**C. Silver**
**D. Silicon**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**76. Approximate percentage of Nickel in stainless steel**

(a) 8 (b) 18
(c) 28 (d) 38

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**77. Which acid is used to the greatest extent for metal pickling?**

(a) Sulphuric acid (b) Hydrochloric acid
(c) Nitric acid (d) Phosphoric acid

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**78. A material is called ductile if it can be**

(a) drawn into wire
(b) hammered into a thin sheet
(c) fractured without deformation
(d) made lustrous by heating it

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**79. Which is an amorphous material?**

(a) Glass (b) Mica
(c) Brass (d) Cast iron

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**80. The main purpose of galvanizing iron sheets is to**

(a) harden the surface
(b) increase the glossiness and luster
(c) prevent the action of water
(d) prevent the action of oxygen

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**81. Mercury is transported in metal containers made of**

(a) aluminium (b) lead
(c) iron (d) nickel

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**82. Glass is corroded by**

(a) fluorine (dry or wet)
(b) $H_2SO_4$
(c) $H_3PO_4$
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**83. Atmospheric corrosion of metals results from their**

(a) slow oxidation (b) fast oxidation
(c) fast hydration (d) slow hydration

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**84. Which among the following is not a carbon polymorph?**

(a) Diamond (b) Pearl
(c) Graphite (d) Fullerene

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**85. As per classification of materials, 'fiber glass' may be classified under**

(a) metals (b) ceramics
(c) composites (d) polymers

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**86. The softest material in Mohs scale is**

(a) talc (b) gypsum
(c) rubber (d) iron

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**87. Teflon is corroded by**

(a) hydrochloric acid (10%)
(b) hydrochloric acid (95%)
(c) sulphuric acid
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**88. A material is able to retain the deformation permanently by virtue of its**

(a) elasticity (b) plasticity
(c) ductility (d) malleability

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**89. Which one is a non-magnetic material?**

(a) Cobalt (b) Nickel
(c) Zinc (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**90. Which metal is protected by the layer of its own oxide?**

(a) Iron (b) Silver
(c) Calcium (d) Aluminium

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**91. Match the following:**

| | |
|---|---|
| **A. Brass** | **P. Alloy of nickel and copper (2:1)** |
| **B. Bronze** | **Q. Alloy of copper, tin and zinc** |
| **C. Gun metal** | **R. Alloy of copper and zinc** |
| **D. Monel** | **S. Alloy of copper and tin** |

(a) A-R, B-P, C-Q, D-S
(b) A-S, B-Q, C-P, D-R
(c) A-R, B-S, C-Q, D-P
(d) A-Q, B-R, C-S, D-P

**ISRO-12.02.2016**

**Ans. (c)**

**92. Rasching rings made of porcelain are NOT suitable for dealing with.**

(a) Concentrated HCl
(b) Concentrated $HNO_3$
(c) Concentrated $H_2SO_4$
(d) Concentrated alkali

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**93. Monel metal is an alloy of**
(a) Molybdenum and nickel
(b) Copper and nickel
(c) Molybdenum and aluminium
(d) Molybdenum and zinc

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**94. Composition for austenitic stainless steel is**
(a) 13 - 20% Chromium, < 0.1% Carbon, Nickel - nil
(b) 18 - 20% Chromium, > 7% Nickel
(c) 10 - 12% Chromium, 0.2- 0.4%% Carbon, Nickel - upto 2%
(d) None of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**95. Presently most used metal in the world is**
(a) Aluminium (b) Gold
(c) Steel (d) Silver

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**96. Rust of iron is**
(a) Powdered iron (b) Ferrous oxide
(c) Hydrated ferric oxide (d) None of these

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**97. Which property of nickel alloys makes them ideal for high temperature application with corrosive environment?**
(a) Oxidation resistance
(b) Corrosion resistance
(c) Oxidation and corrosion resistance
(d) None of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**98. The process of protection of iron by coating with zinc is called**
(a) Tempering (b) Galvanizing
(c) Nitriding (d) Smelting

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**99. _____ are used as corrosion inhibitor for iron and steel in aqueous solution.**
(a) Phosphates (b) Chromates
(c) Sulphates (d) Bi-carbonates

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**100. The tower, used for production of acetaldehyde by hydration of acetylene in sulphuric acid solution in mercuric sulphate, is made of**
(a) Cast iron
(b) Aluminium
(c) Rubber lined mild steel
(d) Lead lined mild steel

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**101. Steel towers for storage of oleum are lined with**
(a) Lead (b) Acid proof bricks
(c) Rubber (d) Not lined

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**102. Composite materials are classified based on**
(a) Type of matrix
(b) Size and shape of reinforcement
(c) Both (a) and (b)
(d) None

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**103. The process of Galvanization can refer as the process in which:**
(a) The steel is melted in high temperature to provide ductility
(b) The metal is dipped in an acid for the fixed length of time, rinsed with water and stored for further use
(c) The metal surface is covered with a thin coat of zinc for protection against corrosion
(d) A thin coat of one metal over the surface of other metal or non metal to alter the surface properly

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**104. Protection of iron by coating it with zinc is called:**
(a) Nitriding (b) Smelting
(c) Galvanizing (d) Tempering

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Galvanizing

**105. What is the effect of increasing carbon content in the steel:**
(a) Decreases its strength
(b) Increases its ductility
(c) Increases its strength
(d) None of these

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Increases its strength

**106. Stainless steel contains iron, carbon and which of the following elements**
(a) Aluminium (b) Chromium
(c) Zinc (d) Lead

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Chromium

**107. What is the purpose of galvanizing iron sheets:**
(a) To increase its luster
(b) Prevent action of oxygen
(c) Prevent action of water
(d) To harden the surface

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Prevent action of water

**108. Tensile strength is maximum in which of the following iron:**
(a) Pig iron (b) Nodular cast iron
(c) Grey cast iron (d) White cast iron

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Nodular cast iron

**109. To get bronze, copper is alloyed with:**
(a) Zinc (b) Aluminium
(c) Beryllium (d) Tin

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Tin

**110. Which of the following materials is most suitable for fabrication of pressure vessels:**
(a) Aluminum alloy (b) Plastic
(c) Plain carbon steel (d) Copper

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Plain carbon steel

# 06.

# Material and Energy Balance

## A. Units and Dimensions

**1. The kilogram (kg) is the mass of a _____cylinder kept at the International Bureau of weight and measures.**

(a) Platinum/iridium (b) Plutonium/iridium
(c) Neobium/iridium (d) Cadmium/iridium

**TNPSC AE-2019**

**Ans. (a)**

**2. One (short) ton is approximately equal to**

(a) 1007 kg (b) 950 kg
(c) 907 kg (d) 1000 kg

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**3. 1000 $lb_m$. ft/min is equal to X kg. cm/$s^2$, where**

(a) X = 3.73 (b) X = 3.93
(c) X = 3.63 (d) X = 3.83

**RPSC Lect.-15.01.2016**

**Ans. (*)**

**4. Unit of specific heat capacity**

(a) J/kg ºC (b) Cal mole/K
(c) J mole /ºC (d) Cal g/ K

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**5. Which of the following is not equivalent to standard atmospheric pressure?**

(a) 1.01325 Pa (b) 1.0135 bar
(c) 760 mm Hg (d) $1.01325 \times 10^5$ $N/m^2$

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**6. One mole of a compound contains**

(a) one molecule of the substance
(b) one atom of the substance
(c) $6.023 \times 10^{23}$ molecules
(d) $22.4 \times 10^3$ molecules

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**7. Pressure p = $2.1 \times 10^3$ dyne /$cm^2$ is:**

(a) 210 pa (b) 210 kpa
(c) 210 Mpa (d) 210 mPa

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**8. The temperature of 335 K in a material is:**

(a) 541°R (b) 571°R
(c) 603°R (d) 643°R

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

## B. Basic chemical calculation

**9. How many moles are there in 256 g of oxygen?**

(a) 2 (b) 16
(c) $9.4 \times 10^{24}$ (d) $22 \times 10^3$

**TNPSC AE-2018**

**Ans. (b)**

**10. 1 kmol of glycerin contains**

(a) 2/3 k atom of carbon (b) 1 k atom of carbon
(c) 3k atom of carbon (d) 2 k atom of carbon

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**11. If 30g of NaOH is present in a litre of solution, then the molarity of the solution is**

(a) 30 M (b) 1 M
(c) 0.75 (d) 40 M
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**12. pH of 0.098 M HCl solution is**

(a) log(0.098) (b) – log(0.098)
(c) – 0.012 (d) – log(0.012)
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**13. 0.01 $MK_2Cr_2O_7$ solution is**

(a) 0.01 (N) (b) 0.05 (N)
(c) 0.06 (N) (d) 0.006 (N)

**WB PSC Scientific Assistant-2019**

**Ans. (c)**

**14. Value of specific heat ratio for monoatomic gases is**

(a) 1 (b) 1.44
(c) 2 (d) 1.66

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**15. One litre of an alcohol (density = 810 kg/$m^3$) is mixed with one litre of water (density = 1005 kg/$m^3$) to form a solution (density = 960 kg/$m^3$), find the total volume of the solution ?**

(a) 1.99 litre (b) 1.89 litre
(c) 2.04 litre (d) 2.13 litre

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**16. Of the following, _____ is a correct statement of Boyle's law.**

(a) P V = constant (b) P/V = constant

(c) V/P = constant (d) n/P = constant

**RPSC ACF-2011**

**Ans. (a)**

**17. Instead of natural gas, air (also assume ideal) is stored at same bullet under identical conditions. What would be the molar density of air? Average mol wt of natural gas = 19.50 kg/kmol and that of air = 28.8 kg/kmol.**

(a) 30.46 (b) 1.523

(c) 44.9 (d) 20.6

**RPSC ACF-2011**

**Ans. (a)**

**18. The temperature at which, a real gas obeys the ideal gas law, over a range of pressure is called.**

(A) Critical temperature

(B) Inversion temperature

(C) Reduced temperature

(D) Boyle temperature

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**19. The weight fraction of methanol in an aqueous solution is 0.64. The mole fraction of methanol $X_m$ satisfies**

(A) $X_m < 0.5$ (B) $X_m = 0.5$

(C) $0.5 < X_m < 0.64$ (D) $X_m \geq 0.64$

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**20. Area of an equilateral triangle is $16\sqrt{3}$ cm$^2$. Its perimeter is**

(a) 16 cm (b) 32 cm

(c) 24 cm (d) $4\sqrt{3}$ cm

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**21. Dalton's law**

(a) $V = V_A + V_B + V_C + ....$

(b) $PV = nRT$

(c) $PV = mRT$

(d) $p = P_A + P_B + P_C + ....$

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**22. 1 gram-mole of material at standard conditions (0°C, 760 mm Hg) occupies**

(a) 22.4 ml (b) 22.4 m$^3$

(c) 22.4 m$^3$/gram (d) 22.4 litre

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**23. A solution has 25% w/w Sodium Chloride in 10 grams of water. What is the mole fraction of Sodium Chloride?**

(a) 0.05 (b) 0.06

(c) 0.07 (d) 0.08

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**24. Acidity or alkalinity of a solution is expressed by its pH value, which is defined as**

**A. Log 1/[$H^+$]** **B. -Log 1/[$H^+$]**

**C. 1/Log [$H^+$]** **D. -1/Log [$H^+$]**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**25. Compounds having same molecular formula but different structural formulae are called**

**A. Allotropes** **B. Isomers**

**C. Isotopes** **D. Isobars**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**26. An element 'A' form cation by loss of two electrons and 'B' gain two electrons to form Anion. Then formula of Ionic compound formed is**

**A. $A^2B$**

**B. $AB^2$**

**C. AB**

**D. $AB^3$**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**27. A solution with reasonably permanent pH is called a/an**

(a) non-ideal solution (b) colloidal solution

(c) buffer solution (d) ideal solution

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**28. The number of gram equivalent of solute dissolved in one litre of solution is called its**

(a) normality (b) molarity

(c) molality (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**29. "The total volume occupied by a gaseous mixture is equal to the sum of the pure component volume." This is**

(a) Dalton's law (b) Amagat's law

(c) Gay-Lussac law (d) Avogadro's law

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**30. When chemical reaction occurs in a process, then usually the number of independent material balances and the number of atomic species**

(a) is different

(b) is equal

(c) may or may not matter

(d) is irrelevant

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**31. The weight fraction of methanol in an aqueous solution is 0.64. The mole fraction of methanol 'X_(m)' satisfies**

(a) X_(m) < 0.5 (b) X_(m) = 0.5

(c) 0.5 < X_(m) < 0.64 (d) X_(m) >= 0.64

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**32. The measure of concentration of a small amount of a solute in a liquid is expressed as parts per million (ppm) is numerically equal to**

(a) 1 kg/*l* (b) 1 g/*l*

(c) 1 kg/kg (d) 1 g/1000 kg

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**33. Avogadro's number is equal to :**

(a) $6.023 \times 10^{23}$ molecules/kg.mole

(b) $6.023 \times 10^{23}$ molecules/gm.mole

(c) $6.023 \times 10^{26}$ molecules/gm.mole

(d) $6.023 \times 10^{5}$ molecules/kg.mole

**AP PSC IOF-2009**

**Ans. (b)**

**34. 1 gm mole of an alcohol weighs 74 grams contains 48 grams of carbon, 10 grams of hydrogen and 16 grams of oxygen. Its molecular formula is :**

(a) $C_3H_{21}OH$ (b) $\begin{matrix} C_4H_9OH \\ \uparrow \\ C_4H_9OH \end{matrix}$

(c) $C_3H_5OH$ (d) $C_2H_{33}OH$

**AP PSC IOF-2009**

**Ans. (b)**

**35. Decomposition efficiency ($\eta_D$) of an electrolytic cell used for producing NaOH is defined as**

(a) $\eta_D$ = (grams of NaOH produced/grams of NaCl decomposed) × 100

(b) $\eta_D$ = (grams of NaOH produced/grams of NaCl charged) × 100

(c) $\eta_D$ = (gram equivalents of NaOH produced/gram equivalents of NaCl charged) × 100

(d) $\eta_D$ = (theoretical current to produce one gram equivalent/actual current to produce one gram equivalent) × 100

**GATE-2014**

**Ans. (c)**

**36. The weight of $6.022 \times 10^{23}$ molecules of $CaCO_3$ will be:**

(a) 75 g

(b) 100g

(c) 125 g

(d) 150 g

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**37. One litre of milk weighs 1.035 kg. The butter fat content is 4 % by volume and has a density of 875 kg /$m^3$. The density of the fat-free 'skimmed' milk will be: ($1m^3=10^3$ litres)**

(a) 1042 kg/$m^3$ (b) 1124 kg/$m^3$

(c) 1242 kg/$m^3$ (d) 1424 kg/$m^3$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**38. The number of moles of $K_2CO_3$ for 117 kg of K will be:**

(a) 4.5 kmol (b) 3.5 kmol

(c) 2.5 kmol (d) 1.5 kmol

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**39. When a metal is dipped into a solution containing its own ions:**

(a) Reversible electrode is formed

(b) Irreversible cell is formed

(c) A half cell is formed

(d) pH Electrode is formed

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**40. The average molecular weight of air is:**

(a) 79 (b) 21
(c) 29 (d) 1

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**41. The volume percent of a component in a gas mixture equals to its mole percent. This is strictly valid for:**

(a) Ideal gases
(b) Real gases
(c) Ideal as well as Real gases
(d) Neither Real nor Ideal gases

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Ideal gases

## C. Material balance without chemical reaction

**42. PURGE STREAM is a stream**

(a) That is recycled to improve conversion
(b) Bled off to remove accumulation of inerts in recycle stream
(c) That skip one or two stages
(d) That changes the equilibrium of the reaction

**TNPSC AE-2018**

**Ans. (b)**

**43. The stream that skips one or more stages of the process and goes directly to another downstream stage is called as**

(a) Recycle stream (b) Bypass stream
(c) Purge stream (d) Secondary stream

**TNPSC AE-2018**

**Ans. (b)**

**44. Fresh feed of 100 mol/hr is passed into a reactor followed by a separator. The product is removed from the separator and unreached feed is recycled. If single pass conversion of A $\rightarrow$ B is 20% assuming A and B both pure, the rate of recycle (in mol/hr) is**

(a) 300 (b) 400
(c) 500 (d) 600

**TNPSC AE-2018**

**Ans. (b)**

**45. It is required to make 100 kg of a solution containing 40% salt by mixing solution A containing 25% salt and solution B containing 50% salt. The mass in kilograms of solution A required is**

(a) 40 (b) 60
(c) 75 (d) 25

**TNPSC AE-2018**

**Ans. (a)**

**46. "Recycle ratio" refers to**

(a) Gross feed stream / recycle feed stream
(b) Recycle stream / fresh feed stream
(c) Fresh feed stream / recycle stream
(d) Recycle stream / net product outlet stream
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**47. A continuous crystallizer produces 2000 kg/hr of $Al_2(SO_4)_3.18H_2O$ (density = 100 kg/m$^3$) and residence time of magma in crystallizer is 2 hours. What would be volume of crystals in crystallizer?**

(a) 20 m$^3$ (b) 20 Litres
(c) 2 m$^3$ (d) 40 m$^3$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**48. If the approximate molar composition of air is 79% $N_2$ and 21% $O_2$, then the mass composition of air is**

(a) 76.7% $N_2$ and 23.3% $O_2$
(b) 76.1% $N_2$ and 23.9% $O_2$
(c) 78.1% $N_2$ and 21.9% $O_2$
(d) 77.7% $N_2$ and 22.3% $O_2$

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**49. An aqueous salt-solution enters a crystallizer operating at steady state at 25°C. The feed temperature is 90°C and the salt concentration in the feed is 40 weight %. The salt crystallizes as a pentahydrate. The crystals and the mother liquor leave the crystallizer. The molecular weight of the anhydrous salt is 135. The solubility of the salt at 25 °C is 20 weight %**

**The feed flow rate required for a production rate of 100 kg/s of the hydrated salt, rounded to the nearest integer, is _____ kg/s**

**GATE-2017**

**Ans. (198 to 202)**

**50. Natural gas containing $CH_4$ (77%), $CO_2$ (4%), $C_2H_6$ (5%) and rest $N_2$ is compressed to a pressure of 2 atm. What is the partial pressure (atm) of $CH_4$?**

(a) 1.54 (b) 0.46
(c) 2.39 (d) 1.0

**RPSC ACF-2011**

**Ans. (a)**

**51. The molar composition of a gas is 10% $H_2$, 10%$O_2$, 30%$CO_2$ and balance $H_2O$. If 50% $H_2O$ condenses, the final mole percent of $H_2$ in the dry gas will be.**

(A) 10% (B) 5%
(C) 18.18% (D) 20%
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**52. The ratio of the quantity of a reactant present in reactor feed of a recycling operation to the quantity of that same reactant entering the operation as fresh feed is known as:**

(a) By pass ratio (b) Recycle ratio
(c) Combined feed ratio (d) Purge ratio

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**53. The molecular weights of two components, A and B are 20 and 40, respectively. The relation between the weight fraction of A ($W_A$) and mole fraction or A ($x_A$) in their binary mixture is**

(a) $W_A = \frac{x_A}{2 - x_A}$ (b) $W_A = \frac{2x_A}{2 - x_A}$

(c) $W_A = \frac{1}{2 - x_A}$ (d) $W_A = \frac{x_A}{1 - x_A}$

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**54. A cellulose solution contains 5.2% cellulose by weight in water. How many kilograms of 1.2% solution are required to dilute 100 kg of 5.2% solution to 4.2%?**

(a) 23.3 (b) 33.3
(c) 66.6 (d) 83.3

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**55. A mixture contains 23 g of ethanol and 36 g of water. Some acetic acid is added to the mixture and its mole fraction becomes 0.5. How much acetic acid was added?**

(a) 25g (b) 75 g
(c) 100g (d) 150g

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**56. Molar flow rate of $NH_3$ when it is flowing at 112 lit/min through a non-reacting continuous reactor is …..moles/min.**

**A. 5** **B. 10**
**C. 0.5** **D. 1**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**57. Which among the following is a unit process?**

(a) Distillation
(b) Hydrogenation of oils
(c) Absorption
(d) Humidification

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**58. Recycling in a chemical process facilitates**

(a) increased yield
(b) enrichment of product
(c) heat conservation
(d) All of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**59. A stream bled off to remove an accumulation of inerts or unwanted material in the process is called**

(a) bypass stream (b) purge stream
(c) recycle stream (d) process stream

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**60. Limiting reactant for a chemical equation is the reactant that is**

(a) present in excess of the stoichiometric amount
(b) present exactly as per stoichiometric amount
(c) present in the smallest stoichiometric amount
(d) not having any relation with stoichiometry

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**61. Which of the following is a unit operation?**

(a) Sublimation (b) Sulphonation
(c) Pyrolysis (d) Hydrolysis

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**62. How many molecules of nitrogen will be present in 28g of nitrogen at standard temperature and pressure?**

(a) 1 molecule
(b) 2 molecules
(c) 6.023× 10^(23) molecules
(d) 12.046× 10^(23) molecules

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**63. Which of the following ratios defines the recycle ratio in a chemical process?**

(a) recycle stream/fresh feed stream
(b) gross feed stream/recycle feed stream
(c) recycle stream/gross feed stream
(d) None of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**64. The molar composition of a gas is 10% 'H_(2)', 10% 'O_(2)', 30% 'CO_(2)' and balance 'H_(2)O', If 50% 'H_(2)O' condenses, the final mole % of 'H_(2)' in the gas on a dry basis will be**

(a) 10%
(b) 5%
(c) 18.18%
(d) 20%

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**65. The objective of recycling in chemical processing is :**

(a) To increase yield (b) To enrich a product
(c) To conserve heat (d) All of the above

**AP PSC IOF-2009**

**Ans. (d)**

**66. Purging is done to control :**

(a) Building of impurities in the system
(b) Heat losses
(c) The reaction
(d) Input of the reactants

**AP PSC IOF-2009**

**Ans. (a)**

**67. The molar composition of a gas is 10% $H_2$, 10% $O_2$, 30% $CO_2$ and balance $H_2O$. If 50% $H_2O$ condenses, the final mole percent of $H_2$ in the dry gas will be ;**

(a) 10 (b) 5
(c) 19.2 (d) 20

**AP PSC IOF-2009**

**Ans. (d)**

**68. Molar composition of air is 79% $N_2$ and 21% $O_2$. Its weight composition is :**

(a) 77% $N_2$ and 23% $O_2$
(b) 78% $N_2$ and 22% $O_2$
(c) 80% $N_2$ and 20% $O_2$
(d) 50% $N_2$ and 50% $O_2$

**AP PSC IOF-2009**

**Ans. (a)**

**69. Recycle ratio R is:**

(a) $\frac{\text{Reactor Volume}}{\text{Reservoir Volume}}$

(b) $\frac{\text{Reservoir Volume}}{\text{Reactor Volume}}$

(c) $\frac{\text{Volume of fluid leaving at the reactor outlet}}{\text{Volume of fluid returned at the reactor inlet}}$

(d) $\frac{\text{Volume of fluid returned at the reactor inlet}}{\text{Volume of fluid leaving at the reactor outlet}}$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**70. $Na_2O$ content of lye containing 73 % of caustic soda will be nearly:**

(a) 47.8 % (b) 56.6 %
(c) 65.4 % (d) 74.2 %

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**71. The weight of lime prepared by heating 95% pure limestone of 200 kg will be:**

(a) 102.8 kg (b) 104.6 kg
(c) 106.4 kg (d) 108.2 kg

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**72. Pure $O_2$ is mixed with air to produce an enriched air containing 50 vol% of $O_2$. The ratio of moles of air to $O_2$ used is:**

(a) 1.72 (b) 0.58
(c) 0.5 (d) 0.2

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**73. Air constains 79 mol% $N_2$ and 21 mol% $O_2$. its average molecular weight is equal to:**

(a) 28 (b) 32
(c) 28.84 (d) 27

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** 28.84

**74. The vapor pressure of water is given by $InP^{Sat} = A - (5000/T)$, where A is a constant, $P^{Sat}$ is vapor pressure in atm and T is temperature in K. The vapor pressure of water in atm at 50°C is approximately:**

(a) 0.07 (b) 0.09
(c) 0.11 (d) 0.13

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** 0.13

## D. Material Balance with chemical reation

**75. Ammonia is catalytically oxidised by oxygen to give nitric acid and water. How much ammonia and oxygen by volume are required to manufacture 63 g of nitric acid at STP**

(a) 1 *l* of $NH_3$ and 2 *l* of $O_2$
(b) 17 *l* of $NH_3$ and 32 *l* of $O_2$
(c) 22.4 *l* of $NH_3$ and 44.8 *l* of $O_2$
(d) 22414 *l* of $NH_3$ and 44828 *l* of $O_2$

**TNPSC AE-2018**

**Ans. (c)**

**76. Pure carbon id completely burnt in oxygen. The flue gas analysis is 85% $CO_2$, 10% CO and 5% $O_2$. The percent excess oxygen used is**

(a) 10 (b) 0
(c) 5 (d) 6.125

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**77. Orsat analysis shows 10% $H_2$, 10% $O_2$, 30% CO and balance $H_2O$. Mole percent of $H_2$ in gas on dry basis is**

(a) 17% (b) 10%
(c) 33% (d) 20%

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**78. The quantities of the desired compounds which are formed in the reaction will be determined by the quantity of**

(a) Excess reactant (b) Limiting reactant
(c) Neutral reactant (d) Molar reactant

**ISRO Technical Asstt.-25.02.2018**

**Ans. (*)**

**79. Carbon black is produced by decomposition of methane**

**$CH_4$ (g) $\rightarrow$ 4C(s) + $2H_2$ (g)**

**The single pass conversion of methane is 60%. If fresh feed is pure methane and 25% of the methane exiting the reactor is recycled, then the molar ratio of fresh feed stream to recycled stream will be?**

(a) 9 (b) 10
(c) 0.9 (d) 90

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**80. 1.6 kg of an organic compound on combustion gave 4.4kg $CO_2$. The percentage of carbon in organic compound is about**

(a) 75% (b) 30%
(c) 45% (d) 60%

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**81. The amount of air or oxygen required for bringing a process into complete combustion is called**

(a) theoretical air or oxygen
(b) excess air or oxygen
(c) practical requirement
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**82. After the combustion of a hydrocarbon fuel, 1 mole of $CO_2$, 0.5 mole of $O_2$, 5 mole $N_2$ and 2 mole of $H_2O$ were leaving the furnace. The fuel and air entered the furnace at 298 K. The heat of combustion at 298 K is 200 kJ. If the specific heat values in J/mole K respectively of the flue gases are 26.5, 25.6, 27.3 and 30.38, the flame temperature will be**

(a) 845.45 K (b) 547.45 K
(c) 1143.45 K (d) 870.45 K

**Karnataka PSC ADF - 2015**

**Ans. (*)**

**83. In production of $SO_3$, 150 kgmol of $SO_2$ and 250 kgmol of $O_2$ are fed to a reactor. The product stream is found to contain 90 kgmol of $SO_3$. What is the percentage conversion of $SO_2$?**

(a) 90 (b) 80
(c) 70 (d) 60

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**84. The reactant which is present in less than its stoichiometric proportion is called the :**

(a) Excess reactant (b) Net reactant
(c) Limiting reactant (d) Reacted reactant

**AP PSC IOF-2009**

**Ans. (c)**

**85. Methane is completely burned with air. The maximum possible volume percent of $CO_2$ (dry basis) in the flue gas is :**

(a) 11.7 (b) 21
(c) 44 (d) 28

**AP PSC IOF-2009**

**Ans. (c)**

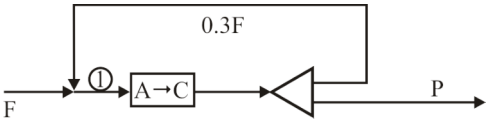
**86. 100 mol/h of butane ($C_4H_{10}$) and 5000 mol/h of air are fed into a combustor. The percentage excess air used was :**

(a) 50.6 (b) 20
(c) 30 (d) 61.6

**AP PSC IOF-2009**

**Ans. (*)**

**87. The schematic diagram of a steady state process is shown below. The fresh feed (F) to the reactor consists of 96 mol% reactant A and 4 mol% inert I. The stoichiometry of the reaction is A → C. A part of the reactor effluent is recycled. The molar flow rate of the recycle stream is 0.3 F. The product stream P contains 50 mol% C. The percentage conversion of A in the reactor based on A entering the reactor at point 1 in the figure (up to one decimal place) is_____**




**GATE-2015**

**Ans. (44 to 47)**

**88. When 1 g mol of $C_7H_{16}$ are mixed with 12 g mol $O_2$, to produce $CO_2$ and $H_2O$. The limiting reactant will be:**

(a) $C_7H_{16}$ (b) $O_2$
(c) $CO_2$ (d) $H_2O$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**89. Which of the following is a formation reaction:**

(a) $H_2O + SO_2 \rightarrow H_2SO_4$
(b) $4HCl + O_2 \rightarrow 2H_2O + 2Cl_2$
(c) $H_2O + CO_2 \rightarrow H_2CO_3$
(d) $C + \frac{1}{2}O_2 + 2H_2 \rightarrow CH_3OH$

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** $C + \frac{1}{2}O_2 + 2H_2 \rightarrow CH_3OH$

**90. The maximum adiabatic flame temperature is attained when the fuel is burned with:**

(a) The theoretically required amount of air
(b) The theoretically required amount of pure oxygen
(c) More than the theoretically required amount of pure oxygen
(d) Less than the theoretically required amount of pure oxygen

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** The theoretically required amount of pure oxygen

**91. A sample of natural gas containing 80% Methane and the rest Nitrogen ($N_2$) is burnt with 20% excess air, with 80% of the combustibles producing $CO_2$ and the remainder going to CO, the orsat analysis in volume percent is:**

(a) $CO_2$ : 6.26, CO : 1.56, $O_2$ : 3 : 91, $H_2O$ : 15.66, $N_2$ : 72.60
(b) $CO_2$ : 7.42, CO : 1.86, $O_2$ : 4.64, $N_2$ : 86.02
(c) $CO_2$ : 6.39, CO : 1.60, $O_2$ : 3.99, $H_2O$ : 15.96, $N_2$ : 72.06
(d) $CO_2$ : 7.60, CO : 1.90, $O_2$ : 4.75, $N_2$ : 85.74

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $CO_2$ : 7.42, CO : 1.86, $O_2$ : 4.64, $N_2$ : 86.02

## E. Energy Balance

**92. Kopp's rule is useful for the determination of**

(a) Molar heat capacities of gases
(b) Heat capacities of solids
(c) Activation energy
(d) Heat capacities of gases

**TNPSC AE-2018**

**Ans. (b)**

**93. Estimates of latent heat of vapourization for pure liquids at their normal boiling points are given by**

(a) Hess's law (b) Kirchoff 's law
(c) Trouton's rule (d) Laplace law

**TNPSC AE-2019**

**Ans. (c)**

**94. It is found that enthalpy change at 298 K during burning of 1 mol of CO (g) under constant pressure of 1 bar is – 293.0 kJ/mol. For chemical reaction, $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$, the standard enthalpy (heat) of reaction at 298 K is**

(a) – 849.0 kJ/mol (b) – 566.0 kJ/mol
(c) –283.0 kJ/mol (d) – 141.5 kJ/mol

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**95. For estimation of heat capacity of solid compound, one can use**

(a) Trouton's equation (b) Claypeyron
(c) Kopp's rule (d) Gibb's equation

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**96. At standard conditions**

$N_2 + 2O_2 \rightarrow 2NO_2 \quad \Delta H^\circ = -90$ KJ/mol

$NO + \frac{1}{2}O_2 \rightarrow NO_2 \quad \Delta H^\circ = -11$ KJ/mol

**The standard heat of formation of NO in KJ/mol is**

(a) – 78 (b) 78
(c) – 34 (d) – 101
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**97. The formation of the oxide ion ($O^{2-}$) in the gas phase from the oxygen atom requires first an exothermic and then an endothermic step as shown below:**

$$O_{(g)} + e^- \rightarrow O^-_{(g)}, \Delta H^0 = -141 KJmol^{-1}$$

$$O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)}, \Delta H^0 = +780 KJmol^{-1}$$

**Thus, process of formation of $O^{2-}$ in the gas phase is unfavourable even though $O^{2-}$ is isoelectronic with neon. It is due to the fact that,**

(a) Oxygen is more electronegative.
(b) addition of electron in oxygen results in larger size of the ion.
(c) electron repulsion outweighs the stability gained by achieving noble gas configuration.
(d) $O^-$ ion has comparatively smaller than oxygen atom.

**WB PSC Scientific Assistant-2019**

**Ans. (c)**

**98. It is desired to concentrate a 20% salt solution (20kg of salt in 100kg of solution) to a 30% salt solution in an evaporator. Consider a feed of 300 kg/min at 30°C. The boiling point of the solution is 110°C, the latent heat of vaporization is 2100 kJ/kg, and the specific heat of the solution is 4 kJ/(kg. K). The rate at which heat has to be supplied (in kJ/min) to the evaporator is**

(a) $3.06\times10^5$ (b) $6.12\times10^5$
(c) $7.24\times10^5$ (d) $9.08\times10^5$

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**99. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).**

**Assertion (A):** Significant combustion of coke takes place only if it is heated at higher temperature in presence of air.

**Reason (R):** $C + O_2 \rightarrow CO_2$ is an endothermic reaction.

**Choose the correct answer from the code given below.**

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(c) (A) is true, but (R) is false
(d) (A) is false, but (R) is true

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**100. $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$, $\Delta H = -13.66$ kcal. In the reaction heat of formation of one mole of $H_2O(l)$ is :**

(a) –13.66 kcal (b) 27.32 kcal
(c) 6.83 kcal (d) –6.83 kcal

**AP PSC IOF-2009**

**Ans. (d)**

**101. Two elemental gases (A and B) are reacting to form a liquid (C) in a steady state process as per the reaction A+B → C. The single-pass conversion of the reaction is only 20% and hence recycle is used. The product is separated completely in pure form. The fresh feed has 49 mol% of A and B each along with 2 mol% impurities. The maximum allowable impurities in the recycle stream is 20 mol%. The amount of purge stream (in moles) per 100 moles of the fresh feed is ————**

**GATE-2014**

**Ans. (9.99 to 10.01)**

**102. Carbon monoxide (CO) is burnt in presence of 200% excess pure oxygen and the flame temperature achieved is 2298K. The inlet streams are at 25 °C. The standard heat of formation (at 25 °C) of CO and $CO_2$ are –110 kJ $mol^{-1}$ and –390 kJ $mol^{-1}$, respectively. The heat capacities (in J $mol^{-1}K^{-1}$) of the components are**

$$C_{po_2} = 25 + 14 \times 10^{-3}T$$

$$C_{pco_2} = 25 + 42 \times 10^{-3}T$$

**where, T is the temperature in K. The heat loss (in kJ) per mole of CO burnt is ————**

**GATE-2014**

**Ans. (32.0 to 38.0)**

**103. At standard conditions,**

$$N_2 + 2O_2 \rightleftharpoons 2NO_2 \ \Delta G^0 = 100 \text{ kJ/mol}$$

$$NO + \tfrac{1}{2} O_2 \rightleftharpoons NO_2 \ \Delta G^0 = -35 \text{ kJ/mol}$$

**The standard free energy of formation of NO in kJ/mol is:**

(a) 15 (b) 85
(c) 30 (d) 170

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 85

## F. Fuels and combustion

**104. Major constituent of Biogas is**

(a) Methane (b) Ethane
(c) Propane (d) Butane

**TNPSC AE-2019**

**Ans. (a)**

**105. A mixture of $H_2$ and Co (Hydrogen and Carbon monoxide) is called_____ gas.**

(a) biogas (b) synthesis gas
(c) water gas (d) natural gas

**TNPSC AE-2019**

**Ans. (b)**

**106. For the case of a fuel gas undergoing combustion with air, if the air/fuel ratio is increased, the adiabatic flame temperature will**

(a) Not change
(b) Increase
(c) Increase or decrease, depending on fuel
(d) Decrease

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**107. By appropriate conversion processes, biomass can be converted into**

(a) LPG (b) Producer gas
(c) Ammonia (d) Urea
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**108. Which of the following is/are combustible?**

(a) $H_2$, $CCl_4$ (b) CO, $CCl_4$
(c) $CCl_4$, $CH_4$ (d) CO, $CH_4$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**109. The correct sequence of increasing calorific value is**

(a) producer gas < natural gas < water gas
(b) natural gas < producer gas < water gas
(c) producer gas < water gas < natural gas
(d) water gas < natural gas < producer gas

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**110. Efficiency of combustion is judged by the _____ the flue gas.**

(a) % of $CO_2$ in (b) Color
(c) Temperature (d) % of $O_2$ in

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**111. Junker's calorimeter is used to determine the calorific value of**

(a) gaseous fuel (b) liquid fuel
(c) pulverised coal (d) solid fuel

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**112. Pick the correct statement from the following:**

(a) Synthesis gas is a mixture of CO, $CO_2$ and $H_2$
(b) Synthesis gas is a mixture of $CO_2$ and $H_2$
(c) Synthesis gas is a mixture of $O_2$ and $H_2$
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**113. For complete combustion of pure methane with theoretical amount of dry air arrange the mole fractions of the following combustion products in increasing order**

**1.Carbon dioxide 2.Water vapour**
**3.Oxygen 4. Nitrogen**

**Select the correct answer from the code given below:**

(a) 3, 2, 1 and 4 (b) 3, 1, 2 and 4
(c) 2, 3, 4 and 1 (d) 2, 3, 1 and 4

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**114. The gas which contributes the maximum to the heating value of natural gas is :**

(a) CO (b) $H_2$
(c) $C_2H_4$ (d) $CH_4$

**AP PSC IOF-2009**

**Ans. (d)**

**115. Select the WRONG statement regarding water gas shift converters from the list below.**

**(1) Inter-stage cooling is provided between the two stages of shift converters.**
**(2) Usually high temperature shift (HTS) reactor has a iron-based catalyst and low temperature shift (LTS) reactor has a copper based catalyst.**
**(3) HTS reactor is followed by LTS reactor.**
**(4) LTS reactor is followed by HTS reactor**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (d)**

**116. Combustion reaction of fuels is a/an_____ reaction.**

(a) Autocatalytic (b) Exothermic
(c) Endothermic (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**117. Ignition temperature of fuel in air is _____that in pure oxygen.**
(a) More than
(b) Less than
(c) Equal to
(d) Either none or less; depends on the type of fuel

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**118. Air/gas ratio for complete combustion will be the highest for:**
(a) LPG
(b) Gobar gas
(c) Coke oven gas
(d) Carbureted water gas

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**119. Presence of nitrogen in combustion air does not reduce the :**
(a) Amount of heat liberated
(b) Flame temperature
(c) Flue gas temperature
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**120. Largest constituents of blast furnace gas is :**
(a) $N_2$ (b) CO
(c) $CO_2$ (d) $H_2$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**121. The heat of combustion of a fuel:**
(a) Is equal to the heat of formation
(b) Is always negative
(c) Can't be known without calculating it
(d) Is always positive

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**122. Blast furnace gas is a very poisonous gas because of its predominantly high_____ content.**
(a) $H_2O$ (b) $CO_2$
(c) CO (d) $CH_4$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**123. The combustion reaction, $C + O_2 \rightarrow CO_2$, is :**
(a) Exothermic (b) Endothermic
(c) Autocatalytic (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**124. High excess air in combustion of fuels results in :**
(a) Increased fuel consumption
(b) Incomplete combustion
(c) Smoky flame
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**125. Use of excess of combustion air in the combustion of fuels results in :**
(a) Heat losses
(b) Long flame
(c) Condensation of water vapour from the fuel gas
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**126. Which of the following is called "Blue gas" ?**
(a) Coke oven gas (b) Water gas
(c) Natural gas (d) Producer gas

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**127. Incomplete combustion of a fuel is characterized by the high _____ in the flue gas.**
(a) Smoke
(b) Temperature
(c) Oxygen
(d) Carbon monoxide

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**128. With increase in calorific value of fuels, their adiabatic flame temperature:**
(a) Increases
(b) Decreases
(c) Remain unchanged
(d) May increase or decrease depends on quality of products of combustion

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**129. Producer gas mainly contains:**
(a) CO, $CO_2$, $N_2$, $H_2$ (b) CO, $H_2$
(c) $H_2$, $CH_4$ (d) $C_2H_2$, $CO_2$, $H_2$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**130. Which one of the following temperature range can be maintained for fusion point of ash in the furnace for the manufacturing of producer gas?**
(a) 800°C –1000°C (b) 900°C –1250°C
(c) 1000°C –1500°C (d) 1600°C –1950°C

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**131. A coal has an ultimate analysis of carbon: 50%, 6% $N_2$, 18% $H_2$, 2%S, $O_2$: 24%. On what basis this analysis has been given?**
(a) Dry Basis
(b) Proximate Basis
(c) Mineral matter free basis
(d) Dry mineral matter free basis

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**132. During the combustion of fuel gas, if the air/fuel ratio is increased, the adiabate flame temperature will:**
(a) Not change
(b) Increase
(c) Increase or decrease
(d) Decrease

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

# 07.

# Chemical Engineering Thermodynamics

## A. Thermodynamic system and properties

**1. Which is a state function?**

(a) Work (b) Pressure
(c) Specific volume (d) Temperature

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**2. How many degree of freedom does the following system posses ? "Liquid water in equilibrium with its vapour"**

(a) one (b) two
(c) three (d) four

**TNPSC AE-2019**

**Ans. (a)**

**3. Which one is not a variable of phase-rule ?**

(a) temperature (b) pressure
(c) phase composition (d) molecular structure

**TNPSC AE-2019**

**Ans. (d)**

**4. The unit of temperature in Kelvin (K) equal to 1/273.16 of the thermodynamic temperature of**

(a) Triple point of mercury
(b) Dual point of water
(c) Triple point of water
(d) Freezing point of mercury

**TNPSC AE-2019**

**Ans. (c)**

**5. The density of gases (i) with increase in pressure and it (ii) with increase in temperature.**

(a) (i) decreases (ii) increases
(b) (i) increases (ii) increases
(c) (i) increases (ii) decreases
(d) (i) decreases (ii) decreases

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**6. A control volume refers to**

(a) an isolated system
(b) a fixed region in a space
(c) a closed system
(d) a definite mass

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**7. Snow is formed by**

(a) Freezing of liquid water
(b) Cooling of ice
(c) Crystallization of solid particles in a vapor
(d) Either (a) of above

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**8. AT a fixed temperature and pressure, a liquid mixture of benzene and toluene is in equilibrium with its vapor, the degree of freedom is**

(a) 3 (b) zero
(c) 2 (d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**9. In _____, both mass and energy can flow across the boundary.**

(a) open system (b) closed system
(c) insulated system (d) isolated system
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**10. Gibbs phase rule relates degree of freedom (F), number of components (C) and number of phases (P) by following equation:**

(a) $F - P - C - 2 = 0$ (b) $F - P + C - 2 = 0$
(c) $F + P - C + 2 = 0$ (d) $F + P - C - 2 = 0$

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**11. An azeotropic mixture of ethanol and water is to be separated in a distillation column using benzene as an entrained. At the column operating conditions, two liquid phases are formed on a tray. The degree (s) of freedom of the system for the choice of intensive properties at equilibrium is (are) _____**

**Ans. (1 to 1 2 (or) to 2)**

**12. Consider the following properties :**

**(a) temperature**
**(b) specific gravity**
**(c) chemical potential**
**(d) volume**

**The option which lists ALL the intensive properties is**

(a) P (b) P and Q
(c) P, Q and R (d) P, Q, R and S

**GATE-2018**

**Ans. (c)**

**13. A sparingly soluble gas (solute) is in equilibrium with a solvent at 10 bar. The mole fraction of the solvent in the gas phase is 0.01. At the operating temperature and pressure, the fugacity coefficient of the solute in the gas phase and the Henry's law constant are 0.92 and 1000 bar, respectively. Assume that the liquid phase obeys Henry's law.**
**The MOLE PERCENTAGE of the solute in the liquid phase, rounded to 2**

**GATE-2017**

**Ans. (0.90 to 0.92)**

**14. What is the degree of freedom of a system consisting of a gaseous mixture of carbon dioxide and nitrogen?**
(a) 0 (b) 1
(c) 2 (d) 3

**RPSC ACF-2011**

**Ans. (d)**

**15. First empirical temperature scale is**
(a) Fahrenheit scale
(b) Reaumer scale
(c) Celsius scale
(d) Thermodynamic celsius scale

**RPSC ACF-2011**

**Ans. (a)**

**16. At absolute zero temperature, all substances have same**
(a) Heat capacity (b) Crystalline structure
(c) Entropy (d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**17. The degree of freedom (F) of non-reacting system is equal to**
(a) 2 –n+N (b) 2 +n+N
(c) 2 –N–n (d) n+N–2
Where n = number of phases
N = number of chemical species

**RPSC ACF-2011**

**Ans. (a)**

**18. The triple point of water occurs at 0.00602 atm &**
(a) 0.01 ºC (b) 444.6 ºC
(c) 100 ºC (d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**19. The value of $C_P$ and $C_v$ for a monatomic gas are**
(a) 5 and 3 (b) 3.987 and 1.987
(c) 0.66 and 1.987 (d) None of the above

**RPSC ACF-2011**

**Ans. (a)**

**20. The triple point of most fluids is close to**
(a) Boiling point (b) Melting point
(c) Critical point (d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**21. A _____ may be defined as one which behaves ideally at all conditions and hence is incapable of condensation to either a liquid or a solid phase.**
(a) Perfect gas (b) Equilibrium gas
(c) Saturated gas (d) Molar gas

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**22. Form of energy which is transferred from one body to another as a result of a difference in temperature is termed as**
(a) Heat
(b) Power
(c) Temperature gradient
(d) LMTD

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**23. A _____ refers to a substance or a group of substances under consideration**
(a) Process (b) Environment
(c) System (d) Species

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**24. Properties which are dependent on mass under consideration are classified as**
(a) Extensive (b) Intensive
(c) Colligative (d) Non-colligative

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**25. In a process the contents of a pressurized gas container are allowed to secape to the atmosphere, which kind of process is it?**
(a) Continuous process (b) Semi batch process
(c) Steady state (d) Batch process

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**26. Pick out the correct statements,**

**(A) A closed system does not permit exchange of mass with its surroundings but may permit exchange of energy,**

**(B) An open system permits exchange of both mass and energy with its surroundings**

**(C) The term microstate is used to characterize an individual whereas macro-state is used to designate a group of micro-states with common characteristics**

**(D) Melting of ice involves increase in enthalpy and a decrease in randomness**

(a) (A), (B) and (C) (b) (B) and (C)

(c) (A) and (C) (d) (D) only

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**27. For a given system, number of variables is 6 and number of equation is 4. What is the number of degrees of freedom?**

(a) 2 (b) 4

(c) 6 (d) 8

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**28. Closed system has…..type of boundary wall**

**A. Impermeable**

**B. Permeable**

**C. Rigid**

**D. None of the above**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**29. Normal temperature and pressure (N,T,P) corresponds to**

**A. 0°C and 760 mm Hg**

**B. 15°C and 760 mm Hg**

**C. 20°C and 760 mm Hg**

**D. 0°C and 1 kgf/cm$^2$**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**30. Gibbs' phase rule is given by**

(a) F + C = P + 2 (b) F – C = P – 2

(c) F = C + P + 2 (d) F = C – P + 2

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**31. The degree of freedom for the reaction**

**$MgCO_{3(S)} \rightarrow MgO_{(S)} + CO_{2(g)}$**

(a) 2 (b) 1

(c) 3 (d) 0

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**32. The number of degrees of freedom for a mixture of ice and water are**

(a) 2 (b) 1

(c) 3 (d) 0

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**33. Specific volume is defined as**

(a) Volume/specific heat capacity

(b) Mass/Volume

(c) Volume/Mass

(d) None of these

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**34. What is the thermodynamic temperature scale in the English system**

(a) Kelvin scale (b) Absolute scale

(c) Fahrenheit scale (d) Rankine scale

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

**35. A system in which mass and energy are not allowed to cross the boundary is called_____**

(a) Closed system (b) Exclusive system

(c) Isolated system (d) Special system

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**36. At absolute zero temperature a gas has**

(a) A temperature of – 273.15 ºC

(b) No kinetic energy

(c) No pressure

(d) All of the above

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**37. The minimum number of degree of freedom for any non-reacting system is**

(a) 0 (b) 1

(c) 2 (d) 3

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**38. What is the effect of increasing pressure and molecular weight on the density of a gas?**

(a) No effect

(b) Density will decrease

(c) Density will decrease with increasing pressure but will not change with increasing molecular weight

(d) Density will increase

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**39. For $CO_2$ – air –water system, how many phases and components are there?**

(a) 1 phase, 3 components
(b) 2 phases, 3 components
(c) 3 phases, 3 components
(d) 1 phase, 1 component

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**40. For steady state system :**

(a) The rate of input is zero
(b) The rate of consumption is zero
(c) The rate of accumulation is zero
(d) The rate of generation is zero

**AP PSC IOF-2009**

**Ans. (c)**

**41. The degree of freedom for a system comprising liquid water in equilibrium with its vapour :**

(a) 2 (b) 1
(c) 4 (d) 3

**AP PSC IOF-2009**

**Ans. (b)**

**42. Gibbs phase rule is :**

(a) C = P – F + 2 (b) F = C – P – 2
(c) P + F = C + 2 (d) P = F – C – 2

**AP PSC IOF-2009**

**Ans. (c)**

**43. From the following list, identify the properties which are equal in both vapour and liquid phases at equilibrium**

P. Density Q.Temperature
R. Chemical potential S. Enthalpy

(a) P and Q only (b) Q and R only
(c) R and S only (d) P and S only

**GATE-2014**

**Ans. (b)**

**44. A property whose value is not an additive and does not vary with the quantity of material in the subsystem is called:**

(a) Intensive property
(b) Extensive property
(c) Chemical property
(d) Physical property

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**45. At a given T and P, a liquid mixture of benzene and toluene is in equilibrium with its vapour. The available degrees of freedom are:**

(a) Zero (b) 1
(c) 2 (d) 3

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

## B. Energy, Work and Heat

**46. Water flows over a waterfall 100 meters in height. Take 1 kilogram of water as the system and assume it does not exchange energy with its surroundings. What is the potential energy of water at the top of the falls with respect to base of the falls?**

(a) 1980.66 Kg $m^2/s^2$ (b) 980.66 Kg $m^2/s^2$
(c) 2980.66 Kg $m^2/s^2$ (d) 3980.66 Kg $m^2/s^2$

**TNPSC AE-2019**

**Ans. (b)**

**47. 8 kJ heat is added to a closed system while its internal energy is decreased by 12kJ. Work done by the system is**

(a) 96 kJ (b) 20 kJ
(c) 4 kJ (d) –20 kJ

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**48. The heat required to raise the temperature of one kg of a substance by 1k is defined as**

(a) Heat capacity
(b) Heat of mixing
(c) Heat of crystallization
(d) Specific heat
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**49. In a throttling process, the pressure of an ideal gas reduces by 50%. If $C_p$ and $C_v$ are the heat capacities at constant pressure and constant volume respectively ($\gamma = C_p/C_v$), the specific volume will change by a factor of**

(a) 2 (b) $2^{1/\gamma}$
(c) $2^{\gamma-1/\gamma}$ (d) 0.5

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**50. The specific heat capacity of air at constant pressure is $C_p$ = 26.693 J/mol-K and T is in kelvin. The heat given off by 1 mol of air, when cooled at 1 atmospheric pressure from 500 $^oC$ to –100 $^oC$, is**

(a) 10.73 kJ (b) 16.89 kJ
(c) 16.01 kJ (d) 18.33 kJ

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**51. A system consists of mixture of ice and water, amount of ice being 0.5 mol and water being .5 mol. What is the sp. heat of the mixture ?**
**(sp heat of water is 18 cal/mol.K and that of ice is 9 cal/mol.)**

(a) $C_{p,mixture} = 0$ (b) $C_{p,mixture} = 18$
(c) $C_{p,mixture} = 9$ (d) $C_{p,mixture} = \infty$

**RPSC ACF-2011**

**Ans. (*)**

**52. When water is vaporized to steam, the total amount of heat added is equal to...**

(a) $u_{steam} - u_{water}$ (b) $h_{steam} - h_{water}$

(c) $S_{steam} - S_{water}$ (d) $h_{steam}$

**RPSC ACF-2011**

**Ans. (d)**

**53. 2 moles of He is mixed with 2g of $H_2$. The molar heat capacity at constant pressure for the mixture is.**

(A) 17R/6 (B) 11R/6

(C) 4R (D) 3R/2

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**54. The total energy at a point comprises of ____ energy.**

(a) potential and kinetic

(b) pressure

(c) internal

(d) All of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**55. The following data are recorded in a flow calorimeter in which water is the test fluid. Flow rate = 4.15 g/sec. $T_1$ = 0°C, $T_2$ = 360°C, $P_2$ = 3 bar, Rate of heat addition from the heater = 12740 Watts. It is observed that water is completely vaporized in the process. The enthalpy of steam at 300°C and 3 bar is**

(a) 3670 J/g (b) 3067 J/g

(c) 3700 J/g (d) 3070 J/g

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**56. During the re-entry of a space vehicle into the earth's atmosphere a large quantity of heat is generated at the surface due to air friction, which is very rapidly removed by a process called:**

(a) Ablation

(b) Abolition

(c) Abrasion

(d) Aeration

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

## C. 1st Law of thermodynamics

**57. In case of an_____process, the temperature of the system increases.**

(a) Isothermal expansion

(b) Isothermal compression

(c) Adiabatic expansion

(d) Adiabatic compression

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**58. From among the following, choose one which is not an exothermic process ?**

(a) Methanol synthesis

(b) Ammonia synthesis

(c) Oxidation of sulphur

(d) Catalytic cracking

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**59. All spontaneous process are:**

(a) Reversible (b) Irreversible

(c) Reversible adiabatic (d) Adiabatic

**TNPSC AE-2018**

**Ans. (b)**

**60. Ist Law of thermodynamics is nothing but the law of conservation of**

(a) momentum (b) energy

(c) mass (d) force

**TNPSC AE-2019**

**Ans. (b)**

**61. No heat transfer between the system and its surroundings is called_____ process.**

(a) Isothermal process (b) Adiabatic process

(c) Isobaric process (d) Isochoric process

**TNPSC AE-2019**

**Ans. (b)**

**62. In a polytropic process ($_PV^{\delta}$ = constant), $\delta = 0$ ; it means**

(a) an adiabatic process

(b) an isothermal process

(c) a reversible process

(d) an isobaric process

**TNPSC AE-2019**

**Ans. (d)**

**63. When a gas is expanded from high pressure region to low pressure region, temperature change occurs, this phenomena is related with**

(a) Gibbs-Duhem equation

(b) Gibbs-Helmholtz equation

(c) Third law of thermodynamics

(d) Joule-Thompson effect

**TNPSC AE-2019**

**Ans. (d)**

**64. The work obtained when one mole of an ideal gas expands to twice, its volume is**

(a) $RT^2$ (b) RT ln (2)

(c) RT (d) RT ln (1/2)

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**65. Match List – I with the List – II and select the correct answer using the code given below.**

| List -I | | List -II | |
|---|---|---|---|
| **A** | **Enthalpy (H)** | **1** | **H – TS** |
| **B** | **Helmholtz free energy** | **2** | **U + PV** |
| **C** | **Gibbs free energy (G)** | **3** | **U – TS** |

**Code :**

| | **A** | **B** | **C** |
|---|---|---|---|
| (a) | 3 | 1 | 2 |
| (b) | 2 | 3 | 1 |
| (c) | 2 | 1 | 3 |
| (d) | 1 | 2 | 3 |

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**66. In a closed system, isenthalpic expansion of an ideal gas with constant ratio of specific heats (γ) is represented by a straight line plot of *l*n (P) versus *l*n (V) with a slope equal to**

(a) –1 (b) γ

(c) $\gamma/(\gamma-1)$ (d) –γ

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**67. Which of the following fundamental thermodynamics equations is not correct?**

(a) dU = T dS – P dV

(b) dH = T dS + V dP

(c) dA = – P dV + S dT

(d) dG = V dP – S dT

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**68. Calculate the difference between the molar internal energy and the molar enthalpy for a perfect gas at 298.15K.**

(a) 35.859 J $mol^{-1}$ (b) 0 J $mol^{-1}$

(c) 2.4790 J $mol^{-1}$ (d) 8.3145 J $mol^{-1}$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**69. For a spontaneous change in a system at constant temperature and pressure, which of the following statements is true?**

(a) $\Delta G < 0$

(b) $\Delta G = 0$

(c) $\Delta G > 0$

(d) There is no restriction on the value of ΔG

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**70. Under isothermal condition, a vertical tube of length L = 100 m contains a gas of molecular weight equal to 60. The pressure and temperature at the top of the tube are 100 kPa and 25 $^oC$ respectively. Consider the universal gas constant and acceleration due to gravity as 8.314 J $mol^{-1}$ $K^{-1}$ and 9.81 $ms^{-2}$ respectively. If the gas is ideal, the pressure (in kPa) at the bottom of the tube will be ____(rounded off to third decimal place).**

**GATE-2018**

**Ans. (102.37 to 102.44)**

**71. Water is heated at atmospheric pressure from 40°C to 80°C using two different processes. In process I, the heating is done by a source at 80°C. In process II, the water is first heated from 40°C to 60°C by a source at 60°C, and then from 60°C to 80°C by another source at 80°C. Identify the correct statement.**

(a) Enthalpy change of water in process I is greater than enthalpy change in process II

(b) Enthalpy change of water in process II is greater than enthalpy change in process I

(c) Process I is closer to reversibility

(d) Process II is closer to reversibility

**GATE-2017**

**Ans. (d)**

**72. The Joule – Thompson expansion coefficient (μ) is defined as....**

(a) $\mu = \left(\frac{\partial T}{\partial P}\right)_h$ (b) $\mu = \left(\frac{\partial T}{\partial P}\right)_s$

(c) $\mu = \left(\frac{\partial P}{\partial T}\right)_v$ (d) $\mu = \left(\frac{\partial h}{\partial s}\right)_T$

**RPSC ACF-2011**

**Ans. (a)**

**73. For an ideal gas enthalpy**

(a) Increases with pressure

(b) Decreases with pressure

(c) Independent of changes in pressure

(d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**74. The constant pressure heat capacity is defined as**

(a) $C_P = \left(\frac{\partial h}{\partial T}\right)_v$ (b) $C_P = \left(\frac{\partial h}{\partial T}\right)_P$

(c) $C_P = \left(\frac{\partial h}{\partial T}\right)_T$ (d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**75. The boiling point of a non-homogeneous mixture must be _____ that of any one of its components alone.**

(a) Independent of (b) Equal to

(c) Higher than (d) Lower than

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**76. In a throttling process, the pressure of an ideal gas reduces by 50%. If $C_p$ & $C_v$ are the heat capacities at constant pressure and constant volume respectively ($Y = C_p/C_v$), the specific volume will change by a factor of**

(a) $2^{(Y-1)}/Y$ (b) $2^1/Y$

(c) 0.5 (d) 2

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**77. In a throttling process, the pressure of an ideal gas reduces by 50%. If $C_p$ & $C_v$ are the heat capacities at constant pressure and constant volume respectively ($Y = C_p/C_v$), the specific volume will change by a factor of**

(a) $2^1/Y$ (b) $2^{(Y-1)}/Y$

(c) 2 (d) 0.5

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**78. An ideal gas is initially at a pressure of 0.1 MPa and a total volume of $m^3$. It is first compressed to 1 MPa by a reversible adiabatic process and then cooled at constant pressure to a final volume of 0.2 $m^3$. The total work done (in kJ) on the gas for the entire process ($C_P = 2.5$ R ) is**

(a) –757.2 (b) 5770.

(c) –577.2 (d) 757.2

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**79. 1 litre gas at 760 torr is compressed to 0.8 litre at constant temperature. What is the final pressure of the gas?**

**A. 800 torr**

**B. 860 torr**

**C. 950 torr**

**D. 540 torr**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**80. Methane is stored in a tank of capacity 5.7 × $10^{-2}$ $m^3$ at a pressure of 15 bar and 294 K. The gas is allowed to flow from the tank through a partially opened valve to a gas holder where the pressure is constant at 1.15 bar. When the pressure in the tank dropped to 5 bar, what would be the mass of methane removed under the condition that the process took place slowly with constant temperature.**

(a) 23.32 kg (b) 0.186 kg

(c) 0.373 kg (d) 0.516 kg

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**81. The enthalpy change of mixing of ideal solution, ΔH, is**

(a) $RT \Sigma x_i \ln x_i$ (b) 0

(c) $R \Sigma x_i \ln x_i$ (d) $\Sigma x_i \ln x_i$

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**82. For an ideal gas, $C_p - C_v$ = ?**

(a) $R^2$ (b) R/2

(c) R (d) 2R

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**83. With rise in temperature, the heat capacity of a substance**

(a) increases

(b) decreases

(c) remains unchanged

(d) either (a) or (b) depends on the substance

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**84. An ideal gas is adiabatically and irreversibly compressed from 3 bar and 300 K to 6 bar in a closed system. The work required for the irreversible compression is 1.5 times the work that is required for reversible compression from the same initial temperature and pressure to the same final pressure. The molar heat capacity of the gas at constant volume is 30J $mol^{-1}K^{-1}$(assumed to be independent of temperature); universal gas constant, R is 8.314 J $mol^{-1}$ $K^{-1}$; ratio of molar heat capacities is 1.277. The temperature (in K, rounded off to the first decimal place) of the gas at the final state in the irreversible compression case is—**

**GATE-2016**

**Ans. (371.0 to 375.0)**

**85. Which of the following is a characteristic of an adiabatic process?**

(a) $\Delta U = 0$
(b) $W = 0$
(c) $Q = 0$
(d) $\Delta V = 0$

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**86. Standard enthalpy of vapourisation of a pure liquid at its Normal Boiling Point ($127^oC$) is 50 $kJmol^{-1}$. The internal energy of vapourisation of the liquid at $127^oC$ (in $kJmol^{-1}$) is.....**

(a) 53.3 (b) –53.3
(c) –46.7 (d) +46.7

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**87. The specific heat of water is _____**

(a) 4.187 J/g K (b) 1 kcal/g K
(c) 80 J/mol K (d) All the above

**ISRO Scientist Engineer-06.08.2017**

**Ans. (d)**

**88. Specific heat ratio ($\gamma$) for an ideal gas is given by.**

(a) $C_p / C_v$ (b) $1 + (R / C_v)$
(c) $C_p/(C_p - R)$ (d) All the above

**ISRO-12.02.2016**

**Ans. (a)**

**89. The Joule-Thomson coefficient ($\mu$) for an ideal gas is**

(a) Positive (b) Negative
(c) Infinity (d) Zero

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**90. Match List -I (thermodynamic processes, $PV^{\delta}$ = Constant) with List-II (Value of $\delta$) and select the correct answer using the code given below the lists.**

| List-I | List-II |
|---|---|
| A. Isobaric process | 1. 1 |
| B. Isothermal process | 2. 0 |
| C. Adiabatic process | 3. $\pm\infty$ |
| D. Isochoric process | 4. $\gamma (= C_p/C_v)$ |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 1 | 2 | 4 | 3 |
| (b) | 1 | 2 | 3 | 4 |
| (c) | 2 | 1 | 4 | 3 |
| (d) | 2 | 3 | 4 | 1 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**91. An adiabatic system is one in which there is :**

(a) Exchange of heat between the system and surroundings
(b) No exchange of heat between system and surroundings
(c) System and surroundings will be in thermal equilibrium
(d) None of the above

**AP PSC IOF-2009**

**Ans. (b)**

**92. First law of thermodynamics is mathematically stated as :**

(a) $dQ = du + dw$ (b) $dQ = dw – du$
(c) $dw = dQ + du$ (d) $dH = du + dv$

**AP PSC IOF-2009**

**Ans. (a)**

**93. Which of the following is not a state function ?**

(a) Temperature (b) Pressure
(c) Specific volume (d) Heat

**AP PSC IOF-2009**

**Ans. (d)**

**94. Throttling process is a constant……..process.**

(a) Entropy (b) Enthalpy
(c) Pressure (d) None of the above

**AP PSC IOF-2009**

**Ans. (b)**

**95. A system is said to be isopiestic if there is no :**

(a) temperature change (b) pressure change
(c) volume change (d) entropy change

**AP PSC IOF-2009**

**Ans. (b)**

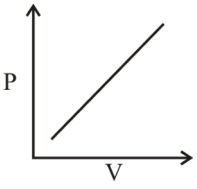
**96. The heat transfer takes place based on :**

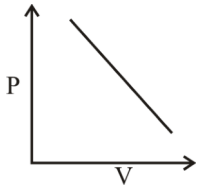
(a) Zeroth law of thermodynamics
(b) Second law of thermodynamics
(c) First law of thermodynamics
(d) Kirchhoff's law of heat transfer

**AP PSC IOF-2009**

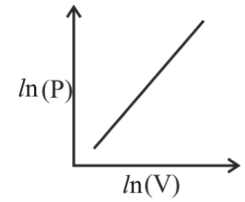
**Ans. (b)**

**97. In a closed system, the isentropic expansion of an ideal gas with constant specific heats is represented by**

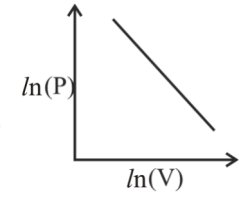
(a)



(b)


(c)




(d)




**GATE-2014**

**Ans. (d)**

**98. An ideal gas is initially at a pressure of 0.1 MPa and a total volume of 2 $m^3$. It is first compressed to 1 MPa by a reversible adiabatic process and then cooled at a constant pressure to a final volume of 0.2 $m^3$. The total work done (in kJ) on the gas for the entire process (up to one decimal place) is _____.**

**Data: R = 8.314 J/mol.K; heat capacity at constant pressure ($C_p$) = 2.5R**

**GATE-2015**

**Ans. (740 to 770)**

**99. For a pure liquid, the rate of change of vapour pressure with temperature is 0.1 bar /K in the temperature range of 300 to 350 K. If the boiling point of the liquid at 2 bar is 320 K, the temperature (in K) at which it will boil at 1 bar (up to one decimal place) is_____.**

**GATE-2015**

**Ans. (309 to 311)**

**100. When air is behaving as an ideal gas with heat capacities ratio of 1.4, the molar heat capacity at constant pressure will be:**

(a) 29.1 J/mol-K
(b) 28.3 J/mol-K
(c) 27.5 J/mol-K
(d) 26.7 J/mol-K

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**101. The first law of thermodynamics is also known as:**

(a) Constant energy principle
(b) Conservation of energy principle
(c) Conservation of entropy process
(d) Constant entropy process

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**102. Consider the following constant-volume heat capacity of a substance:**

$$C_v \equiv \left(\frac{\partial U}{\partial T}\right)_v$$

**What does U represent?**

(a) Molar External Energy
(b) Specific Internal Energy
(c) Specific External Energy
(d) Potential Energy

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**103. For cyclic process:**

(a) $\Delta U = 0$
(b) $Q = 0$
(c) $W = 0$
(d) $\Delta U$, Q and W are all zero

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $\Delta U = 0$

**104. For an ideal gas, $C_p$ and $C_v$ are functions of temperature only and the difference $C_p - C_v$ :**

(a) Varies linearly with temperature
(b) Varies non linearly with temperature
(c) Is always equal to R (Universal gas constant)
(d) Is always equal to zero

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Is always equal to R (Universal gas constant)

## D. 2nd Law of thermodynamic

**105. The thermal efficiency of a heat engine is defined as**

(a) the ratio of the heat source to heat sink
(b) the ratio of the work output to the heat input
(c) the ratio of the energy output to the work input
(d) the ratio of heat output to the work input

**TNPSC AE-2018**

**Ans. (b)**

**106. A Carnot cycle consists of the following steps**

(a) Two isothermal and Two isentropics
(b) Two isobarics and Two isothermals
(c) Two isochorics and Two isobarics
(d) Two isothermals and Two isochorics

**TNPSC AE-2018**

**Ans. (a)**

**107. Second law of thermodynamics deals with**

(a) reversible process only
(b) irreversible process only
(c) direction of energy transfer
(d) entropy

**TNPSC AE-2019**

**Ans. (c)**

**108. The efficiency (%) of a new engine is, if the engine is claimed to be having a power output of 4.5 hp while receiving heat input of 6.25 kW and working between the source and sink temperature limits of 1000 K and 500 K**

(a) 5.37% (b) 50%
(c) 53.7% (d) 5%

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**109. According to second law of thermodynamics, the total entropy change associated with any process must be _____.**

(a) zero (b) infinite
(c) negative (d) positive
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**110. The Coefficient of Performance (COP) of a refrigerator is defined as**

(a) Quantity of heat absorbed per unit of work
(b) Quantity of heat desorbed per unit of work
(c) Quantity of heat radiated per unit of time
(d) Quantity of heat transmitted per unit of time
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**111. In the Claude gas liquefaction process, cooling is done**

(a) by expansion
(b) at constant pressure
(c) at constant temperature
(d) by throttling

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**112. The coefficient of performance (COP) of a Carnot heat pump operating between $T_1$ and $T_2$ (such that $T_1 > T_2$) is given by**

(a) $T_2/(T_1-T_2)$ (b) $T_1/T_2$
(c) $T_1/(T_1-T_2)$ (d) $T_2/T_1$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**113. Ideal refrigeration cycle is**

(a) Same as Carnot
(b) Same as reverse Carnot cycle
(c) Dependent on refrigerant properties
(d) The least efficient of all refrigeration process

**RPSC ACF-2011**

**Ans. (b)**

**114. Air enters an adiabatic compressor at 300 K. The exit temperature for a compression ratio of 3, assuming air to be an ideal gas (= CP/CV =7/5) and the process to be reversible, is**

(a) $300(3^{2/7})$ (b) $300(3^{3/5})$
(c) $300(3^{3/7})$ (d) $300(3^{5/7})$

**RPSC ACF-2011**

**Ans. (a)**

**115. The second law of thermodynamics states that.**

(A) Energy change of a system undergoing any reversible process is zero
(B) It is not possible to transfer heat from lower temperature to higher temperature
(C) Total energy of the system and the surroundings remains constant.
(D) None of these

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**116. A compressor sucks the gas at 1 atm gauge pressure and discharges the gas after compression at 9 atm gauge pressure. What is the compression ratio?**

(a) 10 (b) 9
(c) 5 (d) 4.5

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**117. The thermal efficiency of a reversible heat engine operating between two given thermal reservoirs is 0.4. The device is used either as a refrigerator or as a heat pump between the same reservoirs. The co-efficients of performance as a refrigerator $(COP)_R$ and the Co-efficient of Performance as a heat pump $(COP)_{HP}$ are**

(a) $(COP)_R = (COP)_{HP} = 0.6$
(b) $(COP)_R = 2.5$; $(COP)_{HP} = 1.5$
(c) $(COP)_R = 1,5$; $(COP)_{HP} = 2.5$
(d) $(COP)_R = (COP)_{HP} = 2.5$

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**118. The thermal efficiency of a reversible heat engine operating between two given thermal reservoirs is 0.4. The device is used either as a refrigerator or as a heat pump between the same reservoirs. Then the coefficient of performance as a refrigerator $(COP)_R$ and the coefficient of performance as a heat pump $(COP)_{HP}$ are**

**A. $(COP)_R = (COP)_{HP} = 0.6$**
**B. $(COP)_R = 1.5; (COP)_{HP} = 2.5$**
**C. $(COP)_R = 2.5; (COP)_{HP} = 1.5$**
**D. $(COP)_R = (COP)_{HP} = 2.5$**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**119. A Carnot cycle consists of the following steps:**
**A. Two isobarics and two isothermals**
**B. Two isothermals and two isoentropics**
**C. Two isochorics and two isobarics**
**D. Two isoentropics and two isochorics**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**120. For the same compression ratio and heat input which cycle has maximum thermal efficiency?**
(a) Diesel cycle (b) Otto cycle
(c) Dual cycle (d) None of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**121. Efficiency of a Carnot engine working between temperature $T_1$ and $T_2$ ($T_1 < T_2$) is :**
(a) $\frac{T_2 - T_1}{T_2}$ (b) $\frac{T_2 - T_1}{T_1}$
(c) $\frac{T_1 - T_2}{T_2}$ (d) $\frac{T_1 - T_2}{T_1}$

**AP PSC IOF-2009**

**Ans. (a)**

**122. Freon- 22 is used as a refrigerant, chemically known as :**
(a) Monochloro difluoro methane
(b) Dichloro difluoro methane
(c) Trichloro monofluoro methane
(d) Difluoro methane

**AP PSC IOF-2009**

**Ans. (a)**

**123. Air at 1 bar and 25° C enters a compressor at low velocity, discharges at 3 bar, and enters a nozzle in which it expands to a final velocity of 600 m/s at the initial conditions of pressure and temperature. If the work of compression is 240 kJ /kg of air, the amount of heat removed during the compression will be:**
(a) –90 kJ/kg
(b) –80 kJ/kg
(c) –70 kJ/kg
(d) –60 kJ/kg

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**124. 'The entire power plant, consisting of a compression device, a combustion chamber, and a nozzle' , is a:**
(a) Carnot engine
(b) Diesel engine
(c) Petrol engine
(d) Jet engine

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**125. To maintain the temperature of a solution at 261 K , 1000 kJ of heat per minute is continuously removed from it. The surrounding temperature is 288 K . The least amount of power required will be nearly:**
(a) 6.9 hp (b) 4.7 hp
(c) 3.5 hp (d) 2.3 hp

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**126. With standard notations Carnot engine can be written as:**
(a) $\frac{|Q_H|}{T_H} = \frac{|Q_C|}{T_C}$ (b) $\frac{T_H}{|Q_H|} = \frac{|Q_C|}{T_C}$
(c) $\frac{T_H}{|Q_H|} = \frac{T_C}{|Q_C|}$ (d) $\frac{|Q_H|}{T_H} = \frac{T_C}{|Q_C|}$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**127. If the door of a running refrigerator inside a room is left open then:**
(a) The room will be cooled slightly
(b) The room will be warmed up gradually
(c) The room will be cooled to the temperature inside the refrigerator
(d) The temperature of the room will be unaffected

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** The room will be warmed up gradually

**128. The efficiency of carnot cycle is 1/6. By lowering the temperature of cold reservoir by 65K, it increases to 1/3. The initial and final temperatures of the cold reservoir are:**
(a) 325K, 260K (b) 345K, 280K
(c) 365K, 300K (d) 340K, 275K

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 325K, 260K

## E. 3rd Law of thermodynamics (Entropy)

**129. A system is said to be at equilibrium, if the entropy of the system has reached_____ value.**
(a) Minimum (b) Maximum
(c) Zero (d) None of the above
**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**130. For a spontaneous process, the total entropy of a system and its surroundings always.**
(a) Increases
(b) decreases
(c) does not change
(d) cannot specify as increase of decrease
**TNPSC AE-2018**

**Ans. (a)**

**131. Randomness of a system is a measure of**
(a) specific volume (b) internal energy
(c) entropy (d) enthalpy
**TNPSC AE-2019**

**Ans. (c)**

**132. Entropy change for an irreversible process accounting both system and surroundings together is,**
(a) Positive (b) Negative
(c) Zero (d) Undefinable
(e) Answer not known
**TNPSC AE-2022**

**Ans. (a)**

**133. Which of the following is not a refrigerant?**
(a) Furan (b) Ammonia
(c) Carbon di-oxide (d) Sulfur di-oxide
**RPSC Lect.-15.01.2016**

**Ans. (d)**

**134. A metallic block weighs 5 kg and has an initial temperature of 500 $^{o}$C. The block is dropped in a tank containing 40 kg of water at 25 $^{o}$C and left till the equilibrium is achieved. Specific heat of block material is 0.4 kJ-kg$^{-1}$-K$^{-1}$. The total entropy change in kJ-kg$^{-1}$-K$^{-1}$ is**
(a) –1.87 (b) 0
(c) 1.26 (d) 3.91
**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**135. Melting of ice result in an**
(a) Increases in entropy
(b) Decreases in entropy
(c) Isentropic process
(d) None of these
**RPSC ACF-2011**

**Ans. (a)**

**136. Entropy is a/an.**
(A) State function
(B) Macroscopic property
(C) Extensive property
(D) None of these
(a) A (b) B
(c) C (d) D
**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**137. Calculate the change in enthalpy of carbon dioxide from 30°C to 1500 °C at 100 kPa at constant specific heat of 0.842 kJ/kgK**
(a) 1237.7 kJ/kg (b) 1637.7 kJ/kg
(c) 2237.7kj/kg (d) 2337.7 kJ/kg
**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**138. For a reversible process**
(a) dS = dQ/T
(b) dS > dQ/T
(c) dS< dQ/T
(d) None of the mentioned
**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**139. In which one of the following situations, the entropy change will be negative?**
**A. Air expands isentropically from 6 bar to 3 bar**
**B. Air expands isothermally from 6 bar to 3 bar**
**C. Air is compressed to half the volume at constant pressure**
**D. Heat is supplied to air to constant volume till the pressure becomes double**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**140. As the entropy of the universe is increasing, day by day, the work producing capacity of a heat engine is**
**A. Not changed**
**B. Increasing**
**C. Decreasing**
**D. Data insufficient, can't be predicted**
(a) A (b) B
(c) C (d) D
**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**141. Which among the statements given below is correct definition of third law of the thermodynamics?**

(a) The energy is neither created nor destroyed.
(b) The entropy of a perfect crystal is zero at zero absolute temperature.
(c) Energy can be destroyed at high temperatures.
(d) Entropy of a crystal is maximum at low temperature.

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**142. Entropy is a measure of the _____ of a system.**

(a) temperature changes only
(b) disorder
(c) orderly behaviour
(d) none of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**143. Match the following:**

| | |
|---|---|
| **A. Enthalpy (H)** | **P. dE = dQ – dW** |
| **B. 1st law of thermodynamics** | **Q. Internal Energy (E) + PV** |
| **C. Isobaric process** | **R. Constant temperature process** |
| **D. Isothermal process** | **S. Constant pressure process** |

(a) A-Q, B-P, C-R, D-S (b) A-Q, B-P, C-S, D-R
(c) A-R, B-Q, C-P, D-S (d) A-S, B-R, C-Q, D-P

**ISRO-12.02.2016**

**Ans. (b)**

**144. Entropy is :**

(a) intensive property
(b) derived property
(c) extensive property
(d) fundamental property

**AP PSC IOF-2009**

**Ans. (b)**

**145. The absolute entropy is zero for all perfect crystalline substances at absolute zero temperature. This statement is known as:**

(a) Zeroth law of thermodynamics
(b) First law of thermodynamics
(c) Second law of thermodynamics
(d) Third law of thermodynamics

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Third law of thermodynamics

## F. Thermodynamic relations (Maxwell/Clapeyron equations)

**146. Clayperon Equation deals with the**

(a) effect of an inert gas on vapor pressure
(b) rate of change of vapour pressure with temperature
(c) temperature dependence of heat of phase transition
(d) calculation of $\Delta F$ for spontaneous phase change
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**147. Which of the following equation is not a Maxwell's equation ?**

(a) $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$ (b) $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$

(c) $\left(\frac{\partial V}{\partial P}\right)_T = \left(\frac{\partial S}{\partial T}\right)_P$ (d) $\left(\frac{\partial V}{\partial P}\right)_P = -\left(\frac{\partial S}{\partial T}\right)_T$

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**148. For a state of thermodynamic equilibrium, which are the following criteria will be applicable?**

**(1) Entropy is maximum at constant U & V**
**(2) Gibbs free energy is maximum at constant T & P**
**(3) Gibbs free energy is minimum at constant T & P**
**(4) Entropy is minimum at constant U & V**

(a) 1 & 2 (b) 1 & 3
(c) 2 & 4 (d) 3 & 4

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**149. IN the equation $f_i = \phi_i^{sat} P_i^{sat} \exp\left[V_i^l \frac{(P - P_i)^{sat}}{RT}\right]$, exponential is known as the _____ factor.**

(a) Pressure (b) Poynting
(c) Volume (d) Equlibrium
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**150. $\left[\frac{\partial U}{\partial V}\right]_S$ .....is**

(a) T (b) P
(c) –T (d) –P

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**151. Joule Thomson expansion, μ, is**

(a) $\left[\frac{\partial T}{\partial P}\right]_V$ (b) $\left[\frac{\partial T}{\partial P}\right]_H$

(c) $\left[\frac{\partial T}{\partial V}\right]_P$ (d) $\left[\frac{\partial T}{\partial V}\right]_H$

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**152. Equation used to calculate density is**

(a) Antoine (b) Francis

(c) Hesses (d) Clapeyron

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**153. A gas obeying the Clausius equation of state is isothermally compressed from 5 MPa to 15 MPa in a closed system at 400 K. The Clausius equation of state is $P = \frac{RT}{v - b(T)}$ where P is the pressure, T is the temperature, v is the molar volume and R is the universal gas constant. The parameter b in the above equation varies with temperature as $b(T) = b_0 + b_1T$ with $b_0 = 4 \times 10^{-5}$ $m^3$ $mol^{-1}$ and $b_1 = 1.35 \times 10^{-7}$ $m^3$ $mol^{-1}K^{-1}$. The effect of pressure on the molar enthalpy (h) at a constant temperature is given by $\left(\frac{\partial h}{\partial P}\right)_T = v - T\left(\frac{\partial v}{\partial T}\right)_P$ Let $h_i$ and $h_f$ denote the initial and final molar enthalpies, respectively. The change in the molar enthalpy $h_f - h_i$ ( in J $mol^{-1}$, rounded off to the first decimal place) for this process is—**

**GATE-2016**

**Ans. (399.0 to 400.1)**

**154. The equation dU = TdS – PdV is applicable to infinitesimal changes occurring in**

(a) an open system of constant composition

(b) a closed system of constant composition

(c) an open system with changes in composition

(d) a closed system with changes in composition

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**155. The Clapeyron equation is derived by which of Maxwell's equation ?**

(a) $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$

(b) $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$

(c) $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$

(d) $\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**156. The Maxwell relation corresponding to the thermodynamic relation dH = TdS + VdP is**

(a) $(\partial T/\partial V)_S = -(\partial P/\partial S)_V$

(b) $(\partial V/\partial T)_P = -(\partial S/\partial P)_T$

(c) $(\partial T/\partial P)_S = (\partial V/\partial S)_P$

(d) $(\partial P/\partial T)_V = (\partial S/\partial V)_T$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**157. Which ONE of the following CORRECT for an ideal gas in a closed system?**

(a) $\left(\frac{\partial U}{\partial V}\right)_S V = nR\left(\frac{\partial U}{\partial S}\right)_V$

(b) $-\left(\frac{\partial H}{\partial P}\right)_S P = nR\left(\frac{\partial H}{\partial S}\right)_P$

(c) $\left(\frac{\partial U}{\partial V}\right)_S V = nR\left(\frac{\partial H}{\partial S}\right)_P$

(d) $\left(\frac{\partial H}{\partial P}\right)_S P = nR\left(\frac{\partial U}{\partial S}\right)_V$

**GATE-2014**

**Ans. (d)**

**158. If v, u, s and g represent respectively the molar volume, molar internal energy, molar entropy and molar Gibbs free energy, then match the entries in the left and right columns below and choose the correct option.**

| | |
|---|---|
| P. $-(\partial u/\partial v)_s$ | I. Temperature |
| Q. $(\partial g/\partial P)_T$ | II. Pressure |
| R. $-(\partial g/\partial T)_P$ | III. v |
| S. $(\partial u/\partial s)_v$ | IV. s |

**(1) P-II, Q-III, R-IV, S-I**

**(2) P-II, Q-IV, R-III, S-I**

**(3) P-I, Q-IV, R-II, S-III**

**(4) P-III, Q-II, R-IV, S-I**

(a) 1

(b) 2

(c) 3

(d) 4

**GATE-2015**

**Ans. (a)**

**159. Three identical closed system of a pure gas are taken from an initial temperature and pressure ($T_1$, $P_1$) to a final state ($T_2$ $P_2$), each by a different path. Which of the following is ALWAYS TRUE for the three system? (Δ represents the change between the initial and final states; U, S, G, Q and W are internal energy, entropy, Gibbs free energy, heat added and work done, respectively.)**

**(1) ΔU, ΔS, Q are same**

**(2) W, ΔU, ΔG are same**

**(3) ΔS, W, Q are same**

**(4) ΔG, ΔU, ΔS are same**

(a) 1 (b) 2

(c) 3 (d) 4

**GATE-2015**

**Ans. (d)**

**160. The standard free energy change ($\Delta G^0$), standard entropy change ($\Delta S^0$) and standard heat of reaction ($\Delta H^0$) are related by:**

(a) $\Delta G^0 = \Delta H^0 + T\Delta S^0$

(b) $\Delta G^0 = \Delta H^0 - T\Delta S^0$

(c) $\Delta G^0 = \Delta H^0 + RT\,\Delta S^0$

(d) $\Delta G^0 = \Delta H^0 - RT\Delta S^0$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $\Delta G^0 = \Delta H^0 - T\Delta S^0$

## G. Pure substances

**161. As pressure approaches zero, the ratio of fugacity to pressure (f/P) for a gas approaches to:**

(a) Zero

(b) Infinity

(c) An indeterminate value

(d) Unity

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**162. The value of compressibility factor 'Z' of an ideal gas is**

(a) 0 (b) 1

(c) < 1 (d) > 1

**TNPSC AE-2018**

**Ans. (b)**

**163. At this point, all three phases (solid, liquid and gas) coexist.**

(a) Eutectic (b) Triple

(c) Plait (d) Critical

**TNPSC AE-2018**

**Ans. (b)**

**164. All excess properties becomes_____ as their species approached to pure state.**

(a) super critical (b) sub critical

(c) zero (d) unequal

**TNPSC AE-2019**

**Ans. (c)**

**165. Mollier chart is a**

(a) Pressure Vs Enthalpy plot

(b) Pressure Vs Volume plot

(c) Enthalpy Vs Entropy chart

(d) Temperature Vs Entropy chart

**TNPSC AE-2019**

**Ans. (c)**

**166. For a gas where PVT behaviour is given by $z = 1 + \frac{BP}{RT}$, where B = constant, the fugacity coefficient is**

(a) $\exp\left(\frac{BP}{RT}\right)$ (b) $\frac{BP}{RT}$

(c) $\ln\left(\frac{BP}{RT}\right)$ (d) None of these

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**167. The volume expansivity $\left[\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p\right]$ for an ideal gas is equal to**

(a) $\frac{1}{P}$ (b) T

(c) P (d) $\frac{1}{T}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**168. The measure of the deviation of Gases from the Ideal behaviour is given by**

(a) Daltons law

(b) Compressibility factor

(c) Amagat's law

(d) Avagadro's constant

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**169. $\left(P + \frac{a}{v^2}\right)(v - b) = RT$ is given by**

(a) Virial equation

(b) Vander-Walls equation

(c) Ideal Gas law

(d) Compressibility factor

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**170. The two phases of a pure substance are in equilibrium when each phase has the**

(a) same value of specific gibbs function

(b) same value of driving force

(c) different value of specific gibbs function

(d) different value of mass faction

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**171. For an ideal gas, the compressibility factor**

(a) decreases with pressure raise
(b) is unity at all temperatures
(c) is unity at all concentration
(d) is infinity
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**172. Which of the following is the correct expression for the equation of state of van der Waal's gas?**

(a) $P + \frac{a}{n^2V^2}(V - nb) = nRT$

(b) $\left(P + \frac{na}{n^2V^2}\right)(V - b) = nRT$

(c) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

(d) $\left(P + \frac{n^2a^2}{V^2}\right)(V - nb) = nRT$

**WB PSC Scientific Assistant-2019**

**Ans. (c)**

**173. Which of the following gases most closely behaves as an ideal gas?**

(a) Nitrogen (b) Hydrogen
(c) Helium (d) Oxygen

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**174. For an ideal gas, difference between adiabatic compressibility and isothermal compressibility is**

(a) Positive (b) Negative
(c) Zero (d) Infinite

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**175. Compressibility factor for almost all the gases are approximately same at the**

(a) same pressure
(b) same pressure and temperature
(c) same reduced pressure and reduced temperature
(d) same critical pressure and critical temperature

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**176. $C_p$ of a gas at its critical temperature becomes**

(a) zero
(b) unity
(c) infinity
(d) 7R/2 (where R = Universal gas constant)

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**177. Mollier Diagram is a plot of**

(a) temperature vs enthalpy
(b) temperature vs entropy
(c) enthalpy vs entropy
(d) temperature vs internal energy

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**178. In a closed piston-cylinder system, methane was observed to obey the following equation of state**

**P ( V – nb ) = nRT**

**where b = 0.029 $m^3$/mol. The temperature and volume are 500 $^oC$ and 5 $m^3$ respectively for 100 moles of methane. At this state of the system, the isobaric rate of change of temperature with volume (in $^oC/m^3$) is ____(rounded off to second decimal place).**

**GATE-2018**

**Ans. (367 to 369)**

**179. The pressure of a liquid is increased isothermally.The molar volume of the liquid decreases from $50.45 \times 10^{-6}$ $m^3$/mol to $48 \times 10^{-6}$ $m^3$/mol during this process. The isothermal compressibility of the liquid is $10^{-9}$ $Pa^{-1}$, which can be assumed to be independent of pressure.**

**The change in the molar Gibbs free energy of the liquid, rounded to nearest integer, is ____ J/mol,**

**GATE-2017**

**Ans. (2440 to 2460)**

**180. The volumetric properties of two gases M and N are described by the generalized compressibility chart which expresses the compressibility factor (Z) as a function of reduced pressure and reduced temperature only. The operating pressure (P) and temperature (T) of two gases M and N along with their critical properties (Pc, Tc) are given in the table below.**

| Gas | P(bar) | T (K) | $P_c$ (bar) | $T_c$ (K) |
|---|---|---|---|---|
| M | 25 | 300 | 75 | 150 |
| N | 75 | 1000 | 225 | 500 |

**$Z_M$ and $Z_N$ are the compressibility factor of the gases M and N under the given operating conditions, respectively.**

**The relation between $Z_m$ and $Z_N$ is**

(a) $Z_M = 8Z_N$ (b) $Z_M = 3Z_N$
(c) $Z_M = Z_N$ (d) $Z_M = 0.333Z_N$

**GATE-2017**

**Ans. (c)**

**181. The compressibility factor (Z) of an ideal gas is always**

(a) 0 (b) 1

(c) > 1 (d) < 1

**RPSC ACF-2011**

**Ans. (b)**

**182. Which of the following plots is called Mollier diagram?**

(a) H-S (b) H-P

(c) T-S (d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**183. Van-der Waal's equation of state improved over the ideal gas equation of state with the following assumptions**

(a) Infinitesimal molecular volume

(b) No inter molecular force

(c) Both (a) and (b)

(d) None of the above

**RPSC ACF-2011**

**Ans. (c)**

**184. The value of vanderwaals' constant 'a' for gases $O_2$, $N_2$, $NH_3$ and $CH_4$ are 1.36, 1.39, 4.17 and 2.25 $l^{-1}$. atm.mol$^{-2}$ respectively. The gas which can be most easily liquified is.**

(A) $O_2$ (B) $N_2$

(C) $NH_3$ (D) $CH_4$

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**185. Which of the following is known as Mollier chart?**

(a) Pressure-volume chart

(b) Entropy-enthalpy chart

(c) Temperature-enthalpy chart

(d) Temperature-entropy chart

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**186. In the Vanderwaals equation of state, $(P+ (a/V^2)(V-b) = RT$, the units of 'a' is**

**A. N-m** **B. N-m$^2$**

**C. N-m$^4$** **D. N-m/k**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**187. The catalyst used in Fischer-Tropsch process is**

(a) nickel (b) zinc oxide

(c) alumina (d) thorium oxide

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**188. In general, a gas that exists below its critical temperature is called**

(a) a critical gas (b) a condensate

(c) a vapour (d) a smoke

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**189. With Reference to the thermodynamic system, which of the statements is/are correct?**

**1. A gas obeys P(V−b) – Rt. The work obtained from reversible isothermal expansion of one mole of this gas from an initial molar volume $V_f$ to a final molar volume $V_f$ is RT $l$n $\left(\frac{V_f}{V_i}\right)$**

**2. A car tyre of volume 0.057 m$^3$ is inflated to 300 kPa at 300 K. After the car is driven for ten hours, the pressure in the tyre increases to 330 kPa. Assume air as an ideal gas and $C_v$ for air is 21 J mol$^{-1}$ K$^{-1}$. The change in the internal energy of air in the tyre is 630 J/mol.**

**Select the correct answer using the code given below.**

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**190. Acentric factor for gas is negative for**

(a) $CO_2$ (b) $N_2$

(c) $O_2$ (d) $H_2$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**191. Mollier diagram is a plot of**

(a) Temperature vs Enthalpy

(b) Temperature vs Entropy

(c) Entropy vs Enthalpy

(d) Temperature vs Internal energy

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**192. For ideal gaseous mixtures :**

(a) vol% = mole% = press%

(b) vol% = wt% = mole%

(c) wt% = mole% = press%

(d) Mole fraction of components = mass fraction

**AP PSC IOF-2009**

**Ans. (a)**

**193. The molar density of water vapor at the normal boiling point of water is 33 mol/m$^3$. The compressibility factor under these condition is:**

(a) 0.75
(b) 0.99
(c) 1.25
(d) 1.5

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 0.99

**194. For a pure substance $\left(\frac{\partial s}{\partial p}\right)_H$ is equal to:**

(a) –V/T
(b) T/V
(c) $C_p$/P
(d) $C_v$/P

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** –V/T

**195. For liquid acetone at 20°C and 1 bar, the volume expansivity β is $1.43 \times 10^{-30} C^{-1}$ and the isothermal compressibility k is $62 \times 10^{-6}$ bar 1. If β and k are constant, the pressure generated by heating liquid acetone at constant V from 200C and 1 bar to 300C is:**

(a) 241 bar
(b) 3.4 bar
(c) 2.4 bar
(d) 1.33 bar

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 241 bar

## H. Solution thermodynamics

**196. At 60°C, vapour pressure of methanol and water are 84.562 kPa and 19.953 kPa respectively. An aqueous solution of methanol at 60°C exerts a pressure of 39.223 kPa; the liquid phase and vapour phase mole fractions of methanol are 0.1686 and 0.5714 respectively. Activity co-efficient of methanol is:**

(a) 1.572 (b) 4.238
(c) 3.389 (d) 1.9398

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**197. To test the thermodynamic consistency of data by Redlich-kister method, the area $\int_0^1 \ln \frac{r_1}{r_2} dx_1$ [where $r_1$ and $r_2$ are activity coefficient of component 1 and 2 and $x_1$ is the mole fraction of component 1]**

(a) zero (b) one
(c) two (d) infinity

**TNPSC AE-2018**

**Ans. (a)**

**198. Match the following and select correct answer from the codes given below the lists:**

| List I | List II |
|---|---|
| (a) $y_i P = x_i P_i^{sat}$ | 1. Gibbs-Duhem equation |
| (b) $\ln P^{sat} = A - \frac{B}{T+C}$ | 2. Raoults law |
| (c) $\hat{f}_i^{id} = f_i x_i$ | 3. Antoine equation |
| (d) $\sum_{i=1}^{2} x_i \left[\frac{d \ln r_i}{dx_i}\right]_{T,P} = 0$ | 4. Lewis-Randall rule |

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| (a) | 2 | 1 | 3 | 4 |
| (b) | 2 | 3 | 4 | 1 |
| (c) | 4 | 1 | 2 | 3 |
| (d) | 1 | 2 | 4 | 3 |

**TNPSC AE-2018**

**Ans. (b)**

**199. The molar excess free energy, $G^E$ for a binary liquid mixture at T and P is given by $\frac{G^E}{RT} = AX_1X_2$ Where A is constant. The corresponding equation for $\ln r_1$, where $r_1$ is the activity coefficient of component 1, is**

(a) $Ax_1$ (b) $Ax_2$
(c) $Ax^2_1$ (d) $Ax_2^2$

**TNPSC AE-2018**

**Ans. (d)**

**200. For ideal gases the fugacity is directly proportional to**

(a) Pressure (b) Temperature
(c) Entropy (d) Enthalpy

**TNPSC AE-2018**

**Ans. (a)**

**201. Which of the following is unity for an ideal liquid solution?**

(a) Fugacity (b) Fugacity coefficient
(c) Activity (d) Activity coefficient

**TNPSC AE-2018**

**Ans. (d)**

**202. The fugacity of species 'i' in an ideal gas mixture is equal to its**

(a) atmospheric pressure
(b) partial pressure
(c) latent pressure
(d) liquid pressure

**TNPSC AE-2019**

**Ans. (b)**

**203. Fugacity coefficient of a substance is the ratio of its fugacity to**

(a) mole fraction
(b) activity
(c) pressure
(d) activity coefficient

**TNPSC AE-2019**

**Ans. (c)**

**204. At the equilibrium state, the rate of change of the total Gibb's free energy of the chemical reaction system with reaction co-ordinate at constant temperature and pressure is**

(a) positive (b) infinite
(c) negative (d) zero

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**205. Which of the following is a non-zero quantity for ideal gases?**

(a) Entropy change of mixing
(b) Volume change of mixing
(c) Internal energy change of mixing
(d) Enthalpy change of mixing

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**206. As pressure approaches zero, the ratio of fugacity to pressure (f/p) for a gas approaches to**

(a) an indeterminant value
(b) zero
(c) infinity
(d) unity

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**207. Lewis- Randall rule leads to Raoult's law at**

(a) low temperature
(b) low pressure
(c) very high temperature
(d) very high pressure

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**208. For a pure substance, Gibbs free energy per mole is equal to**

(a) fugacity
(b) latent heat of vaporization
(c) heat capacity
(d) chemical potential

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**209. Which of the following equation is not used for the predication of activity coefficients from experimental data**

(a) Van Laar equation
(b) Van't Hoff equation
(c) Wilson equation
(d) Margules equation

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**210. As per Duhem's theorem, for any closed system formed initially from given masses of prescribed chemical species, the equilibrium state is completely determined when_____ independent variables are fixed period.**

(a) One
(b) Two
(c) Three
(d) Four

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**211. Partial molal quantities are used during study of**

(a) Pure component
(b) Ideal gases
(c) Ideal solutions
(d) Non-ideal mixtures

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**212. Isotonic solutions have the same**

(a) viscosity
(b) normality
(c) molar concentration
(d) specific heat ratio

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**213. Which of the following has the least effect on the solubility of gas in a solvent**

(a) Temperature
(b) Pressure
(c) Nature of gas
(d) Nature of solvent

**RPSC Lect.-15.01.2016**

**Ans. (b)**

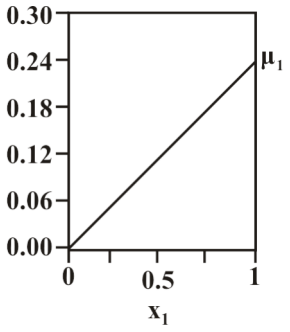
**214. _____ equation predicts the activity coefficient from experimental data.**

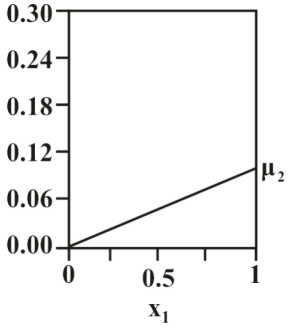
(a) Lewis-Randall (b) Margules
(c) Van Saar (d) Uniquas

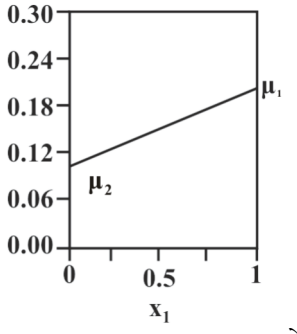
**Haryana PSC Poly Lect.-12.06.2022**

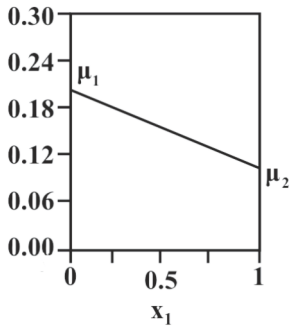
**Ans. (b)**

**215. G denotes the Gibbs free energy of a binary mixture $n_T$ denotes the total number of moles present in the system, $\mu_i$ is the chemical potential of the $i^{th}$ component ($\mu_i \neq 0$ and $\mu_1 > \mu_2$) and $x_i$ is the mole fraction of the $i^{th}$ component. The correct variation of $G/n_T$ (in J/mol) at constant temperature and pressure is given by**

(a)



(b)



(c)



(d)



**GATE-2018**

**Ans. (c)**

**216. The vapour pressure of a pure substance at a temperature T is 30 bar. The actual and ideal gas values of g/RT for the saturated vapour at this temperature T and 30 bar are 7.0 and 7.7, respectively. Here, g is the molar Gibbs free energy and R is the universal gas constant.**

**The fugacity of the saturated liquid at these conditions, rounded to 1 decimal place, is ____ bar.**

**GATE-2017**

**Ans. (14.7 to 15.1)**

**217. For a multi component system, chemical potential is equivalent to**

(a) Molal concentration difference.
(b) Molar free energy of the pure compounds.
(c) Partial molar free energy
(d) Change in molar free energy due to phase change.

**RPSC ACF-2011**

**Ans. (c)**

**218. Temperature affect**

(a) Fugacity (b) Activity co-efficient
(c) Free energy (d) All of these

**RPSC ACF-2011**

**Ans. (d)**

**219. The chemical potential of a species is an ideal solution depends on**

(a) Temperature
(b) Pressure
(c) Composition of solution
(d) All of these

**RPSC ACF-2011**

**Ans. (d)**

**220. A mixture of ideal gases is an ideal solution**

(a) Never
(b) Only at low pressure
(c) Only at low pressure and temperature well above critical point.
(d) Always

**RPSC ACF-2011**

**Ans. (a)**

**221. Excess Gibbs energy of a binary solution is represent by $G^E/RT = Ax_1x_2$. The activity co-efficient of component 1 will be given by**

(a) $\gamma_1 = A x_1^2$ (b) $\gamma_1 = Ax_2^2$
(c) $\gamma_1 = \exp(Ax_1x_2)$ (d) $\gamma_1 = \exp(Ax_2^2)$

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**222. The necessary condition for phase equilibrium in a multiphase system of N components is that the**

**A. Chemical potentials of a given component should be equal in all phases**

**B. Chemical potentials of all component should be same in a particular phase**

**C. Sum of the chemical potentials of any given component in all the phases should be the same**

**D. None of the above**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**223. A 30% by mole methanol-water solution is to be prepared. How many cubic meters of pure methanol (molar volume : $40.727 \times 10^{-6}$ $m^3$/mol) and pure water (molar volume: $18.068 \times 10^{-6}$ $m^3$/mol) are to be mixed to prepare 2 $m^3$ of the desired solution. The partial molar volumes of methanol and water in a 30% solution are $38.632 \times 10^{-6}$ $m^3$/mol and $17.765 \times 10^{-6}$ $m^3$/mol respectively.**

(a) Methanol 1.0 17 $m^3$; Water : 1.053 $m^3$

(b) Methanol 1 $m^3$; Water : 1 $m^3$

(c) Methanol 0. 61 $m^3$; Water : 1.39 $m^3$

(d) Methanol 1.0 17 $m^3$; Water : 0.983 $m^3$

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**224. Free energy change at equilibrium is**

(a) Zero (b) Positive

(c) Negative (d) Indeterminate

**Karnataka PSC ADF - 2015**

**Ans. (a)**

**225. The chemical potential of a pure substance depends upon**

(a) Temperature (b) Pressure

(c) Both (a) and (b) (d) None of the above

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**226. A binary system at a constant pressure with species '1' and '2' is described by the two-suffix Margules equation, $\frac{g^E}{RT} = 3x_1x_2$, where $g^E$ is the molar excess Gibbs free energy, R is the universal gas constant, T is the temperature and $x_1$, $x_2$ are the mole fractions of species 1 and 2, respectively.**

**At a temperature T, $\frac{g_1}{RT} = 1$ and $\frac{g_2}{RT} = 2$ where $g_1$ and $g_2$ are the molar Gibbs free energies of pure species 1 and 2, respectively. At the same temperature, g represents the molar Gibbs free energy of the mixture, For a binary mixture with 40 mole % of species 1, the value (rounded off to the second decimal place) of $\frac{g}{RT}$ is —**

**GATE-2016**

**Ans. (1.63 to 1.67)**

**227. The partial molar enthalpy (in kJ/mol) of species 1 in a binary mixture is given by $\bar{h}_1 = 2 - 60\,x_2^2 + 100\,x_1x_2^2$, where $x_1$ and $x_2$ are the mole fractions of species 1 and 2, respectively. The partial molar enthalpy (in kJ/mol, rounded off to the first decimal place) of species 1 at infinite dilution is —**

**No option**

**GATE-2016**

**Ans. (–58.1 to –57.9)**

**228. The necessary and sufficient condition for equilibrium between two phase is**

(a) concentration of each component should be same in the two phase

(b) the temperature of each phase should be same

(c) the pressure should be same in the two phases

(d) the chemical of each component should be same in the two phases

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**229. Van Laar equation deals with the activity coefficients in**

(a) binary solutions

(b) ternary solutions

(c) azeotropic mixture only

(d) none of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**230. For a mixture of acetic acid and toluene containing 0.486 mol fraction toluene, the partial pressures of acetic acid and toluene are found to be 0.118 bar and 0.174 bar respectively at 343 K. The vapour pressures of pure components at this temperature are 0.269 bar and 0.181 bar for toluene and acetic acid respectively. Calculate the activity coefficient for acetic acid in the mixture based on Lewis - Randall rule**

(a) 1.3 (b) 1.7
(c) 0.65 (d) 0.25

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (c)**

**231. Which of the following is NOT correctly matched?**

(a) Volume change of mixing of ideal gases : Zero

(b) Enthalpy change of mixing of ideal gases : RT + (Internal energy change of mixing of ideal gases)

(c) Gibbs energy change of mixing of ideal gases : $RT\sum_i (y_i ln y_i)$

(d) Entropy change of mixing of ideal gases : $R\sum_i y_i ln\left(\frac{1}{y_i}\right)$

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**232. Mixing of two liquids is a :**

(a) reversible process
(b) irreversible process
(c) isothermal process
(d) a mechanical process

**AP PSC IOF-2009**

**Ans. (b)**

**233. Criterion of a chemical equilibrium is that the total Gibbs free energy change is :**

(a) positive (b) always negative
(c) zero (d) not defined

**AP PSC IOF-2009**

**Ans. (c)**

**234. Consider a binary liquid mixture at equilibrium with its vapour at 25 ºC.**

**Antoine equation for this system is given as $\log_{10} P_1^{sat} = A - \frac{B}{t+C}$ where t is in ºC and p in Torr.**

**The Antoine constants (A, B and C) for the system are given in the following table.**

| Component | A | B | C |
|---|---|---|---|
| 1 | 7.0 | 1210 | 230 |
| 2 | 6.5 | 1206 | 223 |

**The vapour phase is assumed to be ideal and the activity coefficients ($\gamma_i$) for the non-ideal liquid phase are given by**

$$ln(\gamma_1) = x_2^2[2 - 0.6x_1]$$

$$ln(\gamma_2) = x_1^2[1.7 + 0.6x_2]$$

**If the mole fraction of component 1 in liquid phase ($x_1$) is 0.11, then the mole fraction of component 1 in vapour phase ($y_1$) is ——**

**GATE-2014**

**Ans. (0.65 to 0.75)**

**235. Match the following**

| Group - 1 | | Group - 2 | |
|---|---|---|---|
| P. | $\left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_{j\neq i}}$ | I. | Arrhenius equation |
| Q. | $\left(\frac{\partial G}{\partial n_i}\right)_{S,V,n_{j\neq i}}$ | II. | Reaction equilibrium constant |
| R. | $\exp\left(\frac{-\Delta G^0_{reaction}}{RT}\right)$ | III. | Chemical potential |
| S. | $\sum(n_i d\mu_i)_{T,P} = 0$ | IV. | Gibbs-Duhem equation |

(a) Q –III, R – I, S – II
(b) Q –III, R – II, S – IV
(c) P –III, R – II, S – IV
(d) P –III, R – IV, S –I

**GATE-2014**

**Ans. (c)**

**236. Given that molar residual Gibbs free energy, $g^R$, and molar residual volume, $v^R$, are related as $\frac{g^R}{RT} = \int_0^p \left(\frac{v^R}{RT}\right) dP$, find $g^R$ at T = 27°C and P = 0.2 MPa. The gas may be assumed to follow the viral equation of state, z = 1 + BP/RT, where $B = -10^{-4}$ $m^3$/mol at the given conditions (R = 8.314 J/mol.K). The value of $g^R$ m J/mol is:**

**(1) 0.008 (2) –2.4**
**(3) 20 (4) –20**

(a) 1
(b) 2
(c) 3
(d) 4

**GATE-2015**

**Ans. (d)**

**237. A binary mixture of components (1) and (2) forms an azeotrope at 130°C and $x_1 = 0.3$. The liquid phase non-ideality is described by $\ln \gamma_1 = Ax_2^2$ and $\ln \gamma_2 = Ax_1^2$, where $\gamma_1$, $\gamma_2$ are the activity coefficients, and $x_1$, $x_2$ are the liquid phase mole fractions. For both components, the fugacity coefficients are 0.9 at the azeotropic**

**composition. Saturated vapor pressures at 130°C are $P_1^{sat} = 70$ bar and $P_2^{sat} = 30$ bar.**

**The total pressure in bars for the above azeotropic system (up to two decimal places) is**

_____

**GATE-2015**

**Ans. (26.5 to 28.5)**

**238. Which one of following theorems holds true for the statement given below?**

**'For any closed system formed initially from given masses of prescribed chemical species, the equilibrium state is completely determined when any two independent variables are fixed' .**

(a) Gibbs theorem
(b) Duhem's theorem
(c) Liouville's theorem
(d) Goldberg theorem

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**239. In which one of the following approaches a cubic equation of state is used for both vapour and liquid phases?**

(a) Gamma-phi (b) Phi-phi
(c) Alpha-phi (d) Beta-phi

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**240. The pseudocritical property of a component in the mixture is equal to the mole fraction multiplied by critical property of the component. This is known as:**

(a) Kay's rule
(b) Lewis rule
(c) Raoult's law
(d) Henry's law

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Kay's rule

**241. For ideal gases the entropy change of mixing depends on:**

(a) Pressure of the mixture
(b) Temperature of the mixture
(c) Composition of the mixture
(d) Pressure, temperature and composition of the mixture

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Composition of the mixture

**242. For ideal gases the partial molar property is identical with the pure species property at mixture temperature and pressure. This statement is valid for:**

(a) Volume (b) Enthalpy
(c) Entropy (d) Gibbs energy

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Volume

**243. Which of the following conditions is not satisfied by an ideal solution:**

(a) $\Delta H_{mixing} = 0$ (b) $\Delta V_{mixing} = 0$
(c) $\Delta U_{mixing} = 0$ (d) $\Delta S_{mixing} = 0$

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** $\Delta S_{mixing} = 0$

**244. Which of the following thermodynamic functions for a binary solution of species 1 and 2 tends to zero as $x_1 \to 0$:**

(a) $G^{E/RT}x_1x_2$ (b) $G^{E/RT}$
(c) $ln\gamma_1$ (d) $\gamma_1$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $G^{E/RT}$

**245. The fugacity of an ideal gas is equal to its:**

(a) Temperature
(b) Volume
(c) Pressure
(d) Density

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Pressure

**246. A criterion of chemical reaction equilibrium is:**

(a) $\sum_i v_i\mu_i = 0$ (b) $\sum_i v_i\mu_i = 1$
(c) $\sum_i(\mu_i / v_i) = 0$ (d) $\sum_i(\mu_i / v_i) = 1$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $\sum_i v_i\mu_i = 0$

## I. Chemical Reaction Equilibrium

**247. If ΔG (free energy change) for a chemical reaction is very large and negative, then the reaction is**

(a) not feasible
(b) just feasible
(c) very much feasible
(d) ΔG is no measure of feasibility of a reaction

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**248. For a gas phase cracking reaction A → B + C at 300°C, the Gibbs free energy of the reaction at this temperature is ΔG° = – 2750 J/mol. The pressure is 1 bar and the gas phase can be assumed to be ideal. The universal gas constant R = 8.314 J/mol.K. The fractional molar conversion of A at equilibrium is:**

(a) 0.44 (b) 0.50
(c) 0.64 (d) 0.80

**GATE-2015**

**Ans. (d)**

# 08.

# Chemical Reaction Engineering

## A. Basics of chemical Reaction Engg.

**1. _____explains the mechanism of catalysis.**

(a) Thermodynamics
(b) Collision theory
(c) None of the above
(d) Activated complex theory

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**2. Two different first order reactions have rate constants $k_1$ and $k_2$ new at $T_1(k_1 > k_2)$. If temperature is increased from $T_1$ to $T_2$, then new constants become $k_3$ and $k_4$ respectively. Which among the following reaction is correct ?**

(a) $k_1 > k_2 = k_3 = k_4$
(b) $k_1 > k_3$ and $k_2 < k_4$
(c) $k_1 = k_3 = k_4$
(d) $k_1 > k_2 > k_3 > k_4$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**3. What is the value of 'n' if the reaction rate of the chemical reaction A→B, is proportional to $C_A^n$ and it is found that the reaction rate triples, when the concentration of 'A' is increased 9 times?**

(a) 1/2 (b) 1/3
(c) 1/9 (d) 3

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**4. The reaction rate constants at two different temperatures $T_1$ and $T_2$ are related by**

(a) $\ln\frac{K_2}{K_1} = E/R\left(1/T_2 - 1/T_1\right)$

(b) $\ln\frac{K_2}{K_1} = \frac{E}{R}\left(1/T_1 - 1/T_2\right)$

(c) $\exp\left(\frac{K_2}{K_1}\right) = E/R\left(1/T_1 - 1/T_2\right)$

(d) $\exp\left(\frac{K_2}{K_1}\right) = E/R\left(1/T_2 - 1/T_1\right)$

**TNPSC AE-2022, 2018**

**Ans. (b)**

**5. Reactions with very high activation energy are**

(a) very sensitive to temperature
(b) temp insensitive
(c) always irreversible
(d) always reversible

**TNPSC AE-2018**

**Ans. (a)**

**6. Molecularity of an elementary reaction P + Q → R + S is**

(a) 1 (b) 2
(c) 3 (d) 4

**TNPSC AE-2018**

**Ans. (d)**

**7. The unit of frequency factor in Arrhenis equation**

(a) Is same as those of the rate constant
(b) Depend on the order of reaction
(c) Depend on temperature of the reaction
(d) Depend on pressure of the reaction

**TNPSC AE-2018**

**Ans. (a)**

**8. There is no correspondence between stoichiometry and the rate equation in case of _____ reaction.**

(a) Elementary (b) Multiple
(c) Autocatalytic (d) Non-elementary

**TNPSC AE-2018**

**Ans. (d)**

**9. The temperature dependency of the reaction rate constant (k) by the transition state theory is given by**

(a) $k\alpha e^{\frac{-E}{RT}}$ (b) $k\alpha T^{\frac{1}{2}}e^{\frac{-E}{RT}}$
(c) $k\alpha T^2 e^{\frac{-E}{RT}}$ (d) $k\alpha T e^{\frac{-E}{RT}}$

**TNPSC AE-2019**

**Ans. (d)**

**10. Activation energy of a chemical reaction can be determined by**

(a) Changing concentration of reactants
(b) Evaluating rate constant at standard temperature
(c) Evaluating rate constant at two different temperature
(d) Evaluating concentration of catalyst

**TNPSC AE-2019**

**Ans. (c)**

**11. For an elementary reaction $A + 2B \xrightarrow{k} 3C$**

(a) rate of appearance of C = rate of disappearance of A
(b) rate of disappearance of A = rate of disappearance of B
(c) rate of appearance of C = rate of disappearance of B
(d) rate of appearance of C = 3 × rate of disappearance of A
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**12. For a reaction A + 3B → 5C + 7D which one of the following relation is correct?**

(a) $-r_A = -\frac{1}{3}r_B = \frac{1}{5}r_C = \frac{1}{7}r_D$

(b) $-r_A = -3r_B = 5r_C = 7r_D$

(c) $-r_A = -\frac{1}{3}r_B = 5r_C = 7r_D$

(d) $-r_A = \frac{1}{3}r_B = -\frac{1}{5}r_C = -\frac{1}{7}7r_D$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**13. Reaction with high activation energy are**

(a) always non-spontaneous
(b) always spontaneous
(c) fast
(d) slow

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**14. For a chemical reaction, the ratio of rate constant at 500K to that at 400K is 2.5 Given R = 8.314 J-mol$^{-1}$-K$^{-1}$, the value of activation energy (in kJ/mol) is**

(a) 10.5 (b) 12.0
(c) 15.2 (d) 18.4

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**15. A plot of ln k vs 1/T, where k is the rate constant is known as**

(a) Bode diagram (b) van't Hoff plot
(c) Arrhenius plot (d) heat flow diagram

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**16. Rate determining step in a reaction consisting of a number of steps in series is the**

(a) fastest step
(b) slowest step
(c) intermediate step
(d) step with highest rate

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**17. For a chemical reaction, the ratio of rate constant at 500 K to that at 400 K is 2.5. Given R = 8.314 J mol$^{-1}$K$^{-1}$, the value of activation energy (in kJ/mol) is**

(a) 10.5 (b) 12.0
(c) 15.2 (d) 18.4

**GATE-2018**

**Ans. (c)**

**18. What effect does an increase in temperature of 10°C have on the rate of the reaction?**

(a) Halved (b) Multiplied by 1.5
(c) Doubled (d) Tripled

**RPSC ACF-2011**

**Ans. (c)**

**19. Reactions in which the rate equation corresponds to a stoichiometric equation are called**

(a) Elementary reactions
(b) Non-elementary reactions
(c) Heterogeneous reactions
(d) None of the above

**RPSC ACF-2011**

**Ans. (a)**

**20. Fill up the blanks**
**Arrhenius equation shows the variation of ____ with temperature**

(a) Reaction rate (b) Rate constant
(c) Energy of activation (d) Frequency factor

**RPSC ACF-2011**

**Ans. (b)**

**21. By what factor will the rate constant be increased between 298K and 308 K if the activation energy is 35000 cal/mol?**

(a) by a factor 1.35 (b) by a factor 1.92
(c) by a factor 2.79 (d) None of the above

**RPSC ACF-2011**

**Ans. (c)**

**22. For an elementary reaction $2A \xrightarrow{k_1} 2B$**

(a) $-r_A = r_B = k_1C_A^2$ (b) $-r_A = r_B = k_1C_A$
(c) $-r_A = r_B = k_1$ (d) $-r_A = r_B = k_1C_A^{1.5}$

**RPSC ACF-2011**

**Ans. (a)**

**23. For a reaction A + 3B □ 5C + 7D which one of the following is correct?**

(a) $-r_A = -\frac{1}{3}r_B = \frac{1}{5}r_C = \frac{1}{7}r_D$

(b) $-r_A = -3r_B = 5r_C = 7r_D$

(c) $-r_A = -\frac{1}{3}r_B = -5r_C = -7r_D$

(d) $-r_A = \frac{1}{3}r_B = -\frac{1}{5}r_C = -\frac{1}{7}r_D$

**RPSC ACF-2011**

**Ans. (a)**

**24. The rate of reaction between a and b increases by a factor of 50, when the concentration of a is increased 10 folds, the order of reaction with respect to a is:**

(a) 5 (b) 0
(c) 1 (d) 15

**RPSC ACF-2011**

**Ans. (*)**

**25. The rate expression for the reaction between $H_2$ and $Br_2$ to produce HBR ($H_2 + Br_2 \rightarrow HBR$) is given by rate $= \frac{k_1[H_2][Br_2]^{\frac{1}{2}}}{k_2 + \frac{[HBr]}{[Br_2]}}$.**

**The reaction is**

(a) Stochiometric (b) Fundamental
(c) Elementary (d) Non-elementary

**RPSC ACF-2011**

**Ans. (d)**

**26. The dimension of the rate constant k for an n-th order reaction is**

(a) $(time)^{-1}\ (concentration)^{-1}$
(b) $(time)^{1-n}\ (concentration)^{-1}$
(c) $(time)^{-1}\ (concentration)^{1-n}$
(d) $(time)^{-1}\ (concentration)^{-n}$

**RPSC ACF-2011**

**Ans. (c)**

**27. For an isothermal gas phase reaction A→4R, find the value of fractional change in the volume of the system between no conversion and complete conversion of reactant A ($\varepsilon_A$).**

(a) 4 (b) 3
(c) 1 (d) 2

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**28. For a gas reaction at 200 K the rate is reported as $-(dP_A/dt) = 3.66_A^2$ atm/hr**
**What will be the units of rate constant?**

(a) 1/mol (b) 1/sec
(c) mol.sec/L (d) L/(mol.sec)

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**29. A chemical reaction proceeds very slowly and it takes 45 minutes when the temperature is about 70 °C and when the temperature is increased to 85 °C it reaches to same result in just 12 seconds. Find the activation energy for this reaction.**

(a) 449787 J/mol (b) 212.6 J/mol
(c) 758.4 J/mol (d) 948868 J/mol

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**30. Overall order of reaction for which the rate constant has units of $(mol/L)^{-1/2}\ sec^{-1}$ is**

**A. -1/2**
**B. 1/2**
**C. 3/2**
**D. 5/2**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**31. At a given value of E/R (ratio of activation energy and gas constant), the ratio of the rate constants at 500°K and 400°K is 2, if Arrhenious law is used. What will be this ratio, if transition state theory is used with the same value of E/R?**

(a) 1.6 (b) 2.0
(c) 2.24 (d) 2.5

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**32. Rate determining step in a reaction consisting of a number of steps in series is the ——— step.**

(a) Fastest
(b) Slowest
(c) Intermediate
(d) Data insufficient; can't be predicted

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**33. For a homogeneous reaction of $n^{th}$ order, the dimension of the rate constant is given by**

(a) $1/(time)^n$
(b) $(concentration)^{1-n}/(time)$
(c) $(concentration)^{n-1}/(time)$
(d) None of these

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**34. For every 10°C rise in temperature, the rate of chemical reaction doubles. When the temperature is increased from 30 to 70° C, the rate of reaction increases _____ times.**

(a) 8 (b) 12
(c) 16 (d) 32

**ISRO-12.02.2016**
**Karnataka PSC ADF - 2015**

**Ans. (c)**

**35. Hydrogen iodide decomposes through the reaction $2HI \rightleftharpoons H_2 + I_2$. The value of the universal gas constant R is 8.314 J $mol^{-1}$ $K^{-1}$. The activation energy for the forward reaction is 184000J $mol^{-1}$ the ratio (rounded off to the first decimal place) of the forward reaction rate at 600 K to that at 550 K is——**

**GATE-2016**

**Ans. (28.0 to 29.0)**

**36. Overall order of reaction for which rate constant has units of '(mol/L)^ (–3//2) sec ^ (–1)' is**

(a) −3/2 (b) 1/2

(c) 3/2 (d) 5/2

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**37. The rate constant of a reaction depends upon**

(a) time

(b) temperature

(c) weight

(d) mass

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**38. Molecularity of a reaction**

(a) is always equal to the overall order of reaction

(b) may not be equal to the order of reaction

(c) can't have a fractional value

(d) both (b) and (c)

**ISRO-12.02.2016**

**Ans. (d)**

**39. The order of the chemical reaction,** $2A + B \xrightarrow{k} C$**, whose rate equation is given as** $-r_A = k.\ C_A^2.\ C_B$ **is**

(a) 0 (b) 1

(c) 2 (d) 3

**ISRO-12.02.2016**

**Ans. (d)**

**40. From Arrhenius law, a plot of in k Vs 1/T given a straight line with a slope of (– E/ R). The unit of E/R is**

(a) KCal (b) Kcal/ºK

(c) ºK (d) Kcal. $^oC$

**ISRO-12.02.2016**

**Ans. (c)**

**41. With reference to the units/variables, which of the statements is /are correct?**

**1. If $N_C$ is the number of substances present in a stream, then the number of variables needed to specify the condition of the stream completely is $N_C + 2$.**

**2. The rate expression for the reaction of A is given by**

$$-r_A = \frac{k_1 C_A^2}{1 + k_2 C_B^{1/2}}$$

**The units of $k_2$ are $mol^{-1}m^3\ s^{-1}$**

**Select the correct answer using the code given below:**

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**42. Milk is pasteurized if it is heated to 63ºC for 30 min but if it is heated to 74ºC it only needs 15 seconds for the same result. The activation energy of this sterilization process is :**

(a) 422 kJ/mol (b) 500 kJ/mol

(c) 395 kJ/mol (d) 109 kJ/mol

**AP PSC IOF-2009**

**Ans. (a)**

**43. Molecularity of an elementary reaction P+ Q → R + S is :**

(a) 1 (b) 2

(c) 3 (d) 4

**AP PSC IOF-2009**

**Ans. (d)**

**44. The dimensions of the rate constant for $n^{th}$ order homogeneous reaction are :**

(a) $(time)^n$

(b) $(time)^{-1}\ (concentration)^{1-n}$

(c) $(time)^{-n}\ (concentration)^{n-1}$

(d) $(time)\ (concentration)^{n-1}$

**AP PSC IOF-2009**

**Ans. (b)**

**45. The temperature dependency of the rate constant k, of bimolecular reaction by collision theory is given by :**

(a) $k \propto e^{-E/RT}$ (b) $k \propto T^{1/3}e^{-E/RT}$

(c) $k \propto Te^{-E/RT}$ (d) $k \propto T^{1/2}e^{-E/RT}$

**AP PSC IOF-2009**

**Ans. (d)**

**46. Which of the following can change if only the catalyst is changed for a reaction system?**

(1) Enthalpy of reaction

(2) Activation energy

(3) Free energy of the reaction

(4) Equilibrium constant.

(a) 1 (b) 2

(c) 3 (d) 4

**GATE-2015**

**Ans. (b)**

**47. For which reaction order, the half-life of the reactant is half of the full lifetime (time for 100% conversion) of the reactant?**

**(1) Zero order**

**(2) Half order**

**(3) First order**

**(4) Second order**

(a) 1 (b) 2

(c) 3 (d) 4

**GATE-2015**

**Ans. (a)**

**48. An irreversible, homogeneous reaction A → products, has the rate expression:**

$$\text{Rate} = \frac{2C_A^2 + 0.1C_A}{1 + 50C_A},$$

**Where $C_A$ is the concentration of A.**

**$C_A$ varies in the range 0.5 – 50 mol/m$^3$.**

**For very high concentrations of A, the reaction order tends to:**

**(1) 0** **(2) 1**

**(3) 1.5** **(4) 2**

(a) 1 (b) 2

(c) 3 (d) 4

**GATE-2015**

**Ans. (b)**

**49. Which one of the following methods is not suitable for the determination of order of reaction?**

(a) Use of differential rate expression

(b) Use of integral rate expression

(c) Use of reaction rate method

(d) Use of isolation method

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**50. Consider the following reaction between hydrogen and bromine in the chain reaction:**

$$H^{\bullet} + HBr \rightarrow H_2 + Br^{\bullet}$$

**The reaction is called as chain:**

(a) Initiation step

(b) Retardation step

(c) Termination step

(d) Propagation step

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**51. From collision theory, the reaction rate constant is proportional to:**

(a) $\sqrt{T}$

(b) $T^{1/2} \exp(-E/RT)$

(c) $\exp(-E/RT)$

(d) $\exp(-E/2RT)$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $T^{1/2} \exp(-E/RT)$

**52. For a given reaction A + B → C + D the experimentally determined overall order is 2. Then it could be concluded that:**

(a) Reaction is elementary with a molecularity of 2

(b) Molecularity of the reaction is 2 but the reaction may not be elementary

(c) Reaction is elementary but molecularity may not be 2

(d) Reaction is elementary with a molecularity of 4

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Molecularity of the reaction is 2 but the reaction may not be elementary

**53. For the Reaction A + 2B → 3B + C then:**

(a) $r_A = r_B$

(b) $r_A = -r_B$

(c) $r_A = 4r_B$

(d) $4r_A = r_B$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $r_A = -r_B$

## B. Homogenous Systems

**54. Rate of a chemical reaction is independent of the concentration of the reactants for a_____reaction.**

(a) Zero order (b) Third order

(c) Consecutive (d) None of the above

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**55. Which function of [X], plotted against time, will give a straight line for a second order reaction? Where, X → products:**

(a) [X] (b) $[X^2]$

(c) ln[X] (d) 1/[X]

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**56. For n$^{th}$ order reaction $\frac{t_{1/2}}{t_{3/4}}$ depends on (n ≠ 1) :**

(a) Initial concentration only

(b) 'n' only

(c) Initial concentration and 'n' both

(d) Sometimes 'n' and sometimes initial concentration

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**57. For a second order reaction, 2A products, a plot of log $t_{1/2}$ vs log a (where, a is initial concentration) will give an intercept equal to which one of the following ?**

(a) $\frac{1}{k}$ (b) $\log\left(\frac{1}{2k}\right)$

(c) $\log\left(\frac{1}{k}\right)$ (d) $\log k$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**58. For a first order reaction, (A)→ products, the concentration of A changes from 0.1 M to 0.025 M in 40 minutes. The rate of reaction when the concentration of A is 0.01 M is :**

(a) $3.47 \times 10^{-5}$ M/min
(b) $1.73 \times 10^{-4}$ M/min
(c) $1.73 \times 10^{-5}$ M/min
(d) $3.47 \times 10^{-4}$ M/min

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**59. The conversion of a reactant, undergoing a first order reaction, at a time equal to three times the half- life of the reaction is:**

(a) 0.875
(b) 0.5
(c) 0.425
(d) Data insufficient to calculate

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**60. For a certain reaction, A→B, the rate is first order in A at low concentration of A, but zero order at high concentration of A. The possible rate expression for such a case is:**

(a) $-r_A = \dfrac{k_1 C_{A^2}}{1 + k_2 C_A}$

(b) $-r_A = \dfrac{k_1 C_{A^2}}{(1 + k_2 C_A)^2}$

(c) $-r_A = \left(\dfrac{k_1 C_{A^2}}{1 + k_2 C_A}\right)^2$

(d) $-r_A = \dfrac{k_1 C_A}{1 + k_2 C_A}$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**61. The conversion of a reactant, undergoing a first order reaction, at a time equal to three times the half life of the reaction is**

(a) 0.875 (b) 0.5
(c) 0.425 (d) 0.3

**TNPSC AE-2018**

**Ans. (a)**

**62. The half life period ($t_{1/2}$) of a zero order reaction A $\xrightarrow{k}$ products is equal to**

(a) $\dfrac{C_{AO}}{2K}$ (b) $\dfrac{C_{AO}}{K}$
(c) $\dfrac{0.693}{K}$ (d) $1/K$

**TNPSC AE-2018**

**Ans. (a)**

**63. A liquid decomposes by irreversible first order kinetics and the half life period of this reaction is 8 minutes. The times required for 75% conversion of the liquid will be _____ minutes.**

(a) 4 (b) 8
(c) 12 (d) 16

**TNPSC AE-2018**

**Ans. (d)**

**64. For irreversible unimolecular type first order reactions A $\xrightarrow{k}$ product a plot of $-\ln(1 - \lambda A)$ versus time gives a straight line passing through the origin whose slope is _____ where $\lambda A$ is fractional conversion of component 'A' at time 't'**

(a) k (b) –k
(c) 1/k (d) –1/k

**TNPSC AE-2019**

**Ans. (a)**

**65. For a first order reaction the plot of ln $\ln\left(\dfrac{CA_0}{CA}\right)$ Vs time**

(a) is linear and passes through origin
(b) is exponential and passes through origin
(c) is linear but not pass through the origin
(d) is exponential but does no pass through the origin

**TNPSC AE-2019**

**Ans. (a)**

**66. For an isothermal second order aqueous phase reaction A → B in batch reactor, the ratio of time required for 90% conversion to time required for 45% conversion is**

(a) 22 (b) 2
(c) 11 (d) 4

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**67. If the half-life o an irreversible, first order reaction is $t_{1/2}$, the fractional conversion of reactant at time equal to n times $t_{1/2}$ is**

(a) $1 - \dfrac{1}{n \ln 2}$ (b) $1 - \dfrac{1}{e^{2n}}$
(c) $1 - \dfrac{1}{2^{1/n}}$ (d) $1 - \dfrac{1}{2^n}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (*)**

**68. The governing equation to determine the conversion time in a batch reactor for a first order irreversible reaction**

(a) $C_A = \frac{C_{A_0}}{1 + C_{A_0} Kt}$

(b) $C_A = C_{A_0} - Kt$

(c) $C_A = C_{A_0} + Kt$

(d) $C_A = C_{A_0} \cdot e^{-Kt}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**69. A reaction A→3B is conducted in a constant pressure vessel. Starting with pure A, the volume of the reaction mixture increases 3 times in 6 minutes. The fractional conversion is**

(a) 0.33

(b) 0.5

(c) 1

(d) date insufficient, cannot be predicted

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**70. The half life time ($t^{1/2}$) of the reaction, is defined as _____.**

(a) $\frac{(0.5)^{1-n} - 1}{K(n-1)} C_{A_0}^{1-n}$ (b) $\frac{(0.5)^{n-1} - 1}{K(n-1)}$

(c) $\frac{(n)^{1-n} - 1}{K(n-1)}$ (d) $\frac{(n)^{n-1} - 1}{K(n-1)}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**71. The fractional volume change between no conversion and complete conversion for the isothermal gas phase reaction 2A →R, is**

(a) 0.5 (b) – 0.5

(c) 1 (d) 1.5

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**72. For which of the following reaction, reaction rate does not decreases much as the reaction proceeds**

(a) Catalytic (b) Auto-catalytic

(c) Parallel (d) Series

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**73. For the irreversible reactions $A \xrightarrow{k_1} R$ and $A \xrightarrow{k_2} S$, the rate of disappearance of A is given by**

(a) $(k_1 - k_2)C_A$ (b) $(k_1 + k_2)C_A$

(c) $0.5(k_1 + k_2)C_A$ (d) $k_1 C_A$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**74. For which reaction order, the half life of the reactant is half of the full lifetime (time for 100% conversion) of the reactant?**

(a) Zero order (b) Half order

(c) First order (d) Second order

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**75. Half-life period for a first order reaction is**

(a) $t_{1/2} = 1/k$ (b) $t_{1/2} = 0.393/k$

(c) $t_{1/2} = 0.473/k$ (d) $t_{1/2} = 0.693/k$

**RPSC ACF-2011**

**Ans. (d)**

**76. If time required to change to concentration of reactant to half its original value is independent of the initial concentration, the order of reaction is**

(a) 0 (b) 1

(c) 2 (d) 3

**RPSC ACF-2011**

**Ans. (b)**

**77. For isothermal gas phase reaction A → B the fractional change in volume of the system between no conversion and complete conversion is**

(a) 1 (b) 3

(c) 2 (d) 2/3

**RPSC ACF-2011**

**Ans. (a)**

**78. An isothermal gas phase reaction, A→5B is being conducted initially with pure A. The expansion factor for this reaction, $\varepsilon_A$ = →**

(a) 1 (b) 2

(c) 3 (d) 4

**RPSC ACF-2011**

**Ans. (d)**

**79. For a certain gas phase reaction, $-\frac{dC_A}{dt} = K C_A^{\,n}$ The rate of reaction in terms of partial pressures may be expressed as.**

(A) $-\frac{dP_A}{dt} = K P_A^{\,n}$

(B) $-\frac{dP_A}{dt} = KRT P_A^{\,n}$

(C) $-\frac{dP_A}{dt} = K(RT)^{1-n} P_A^{\,n}$

(D) $-\frac{dP_A}{dt} = K(RT)^{n} P_A^{\,1-n}$

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**80. The conversion ($X_A$) for a 2nd order irreversible reaction (Constant volume) $A \xrightarrow{k1} B$ in batch mode is given by**

(A) $\frac{1}{1+K_1C_{Ao}t}$ (B) $\frac{K_1C_{Ao}t}{1+K_1C_{Ao}t}$

(C) $\frac{(K_1C_{Ao}t)^2}{1+K_1C_{Ao}t}$ (D) $\frac{K_1C_{Ao}t}{(1+K_1C_{Ao}t)^2}$

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**81. A 5*l* chemostat is fed fresh medium at 0.2 *l*/min having a substrate concentration of 25g/*l*. At steady state, the outgoing stream has substrate concentration of 2.5 g/*l*. The rate of consumption in (g/ *l* /min) of the substrate in the reactor is.**

(A) 1.2 (B) 0.5
(C) 0.9 (D) 1.5
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**82. Half life period for first order reaction compared to the initial concentration of the reaction is:**

(a) Inversely proportional
(b) Directly proportional
(c) Independent
(d) Exactly half

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**83. In a first order reaction, the time required to reduce the concentration of reactant from 1 mole/liter to 0.5 mole/liter will be___ that required to reduce it from 10 moles/liters to 5 moles/liters in the same valume.**

(a) more than
(b) less than
(c) same as
(d) data insufficient; can't be predicted

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**84. The half-life of a first order liquid phase reaction is 30 s. The rate constant in $min^{-1}$, is**

(a) 0.0231 (b) 0.602
(c) 1.386 (d) 2.0

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**85. For an isothermal second order aqueous phase reaction A→B, the ratio of the time required for 90% conversion to the time required for 45% conversion is**

(a) 2 (b) 4
(c) 11 (d) 22

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**86. The conversion for a second order, irreversible reaction (constant volume)**
**$A \xrightarrow{k_2} B$, in batch mode is given by**

(a) $\frac{1}{1+k_2C_{A0}t}$ (b) $\frac{k_2C_{A0}t}{1+k_2C_{A0}t}$

(c) $\frac{(k_2C_{A0}t)^2}{1+k_2C_{A0}t}$ (d) $\frac{k_2C_{A0}t}{(1+k_2C_{A0}t)^2}$

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**87. In the reaction A+B→R+S, the consumption of A followed the first order and the consumption of B followed zeroth order. What is the overall order?**

**A. 0** **B. 1**
**C. 1.5** **D. 2**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**88. The functional volume change of the system for the isothermal gas phase reaction, 2A → 2B, between no conversion and complete conversion.**

(a) 0.5 (b) 1
(c) 2 (d) 3

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (*)**

**89. Differential method for analyzing the kinetic data is used**

(a) When rate expressions are very simple
(b) For testing complicated mechanisms
(c) When the data are scattered
(d) None of these

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**90. The reaction 2A + B '->' 2C occurs on a catalyst surface. The reactants A and B diffuse to the catalyst surface and get converted completely to the product C, which diffuses back. The steady state molar fluxes of A, B and C are related by**

(a) 'N_(A)' = '2N_(B)'= 'N_(C)'
(b) 'N_(A)' = '–1/2N_(B)'= –'N_(C)'
(c) 'N_(A)' = '2N_(B)'= –'N_(C)'
(d) 'N_(A)' = '–1/2N_(B)'= 'N_(C)'

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**91. The rate of the chemical reaction A→B doubles as the concentration of A doubled. If rate of reaction is proportional to $C_A^n$, where $C_A$ is the concentration of A, then what is the value of n for this reaction?**

(a) 0 (b) 1
(c) 0.5 (d) 2

**ISRO Scientist Engineer-06.08.2017**

**Ans. (b)**

**92. Equilibrium of a chemical reaction as viewed by kinetics is a _____ state.**

(a) dynamic steady (b) static steady
(c) dynamic unsteady (d) all the above

**ISRO-12.02.2016**

**Ans. (a)**

**93. The conversion of a reactant under- going a first order reaction at a time equal to 3 times the half life of the reaction is :**

(a) 0.5 (b) 0.875
(c) 0.45 (d) 0.7

**AP PSC IOF-2009**

**Ans. (b)**

**94. For the gaseous reaction 2A → B where the feed consists of 50 mole% A and 50 mol % inserts, the expansion factor is :**

(a) 1.5 (b) –0.5
(c) –0.25 (d) 0

**AP PSC IOF-2009**

**Ans. (c)**

**95. A homogeneous reaction (R→P) occurs in a batch reactor. The conversion of the reactant R is 67% after 10 minutes and 80% after 20 minutes. The rate equation for this reaction is**

(a) $-r_R = k$
(b) $-r_R = k C_R^2$
(c) $-r_R = k C_R^3$
(d) $-r_R = k C_R^{0.5}$

**GATE-2014**

**Ans. (b)**

**96. In which one of the following reactions, the rate is independent of the concentration of the reactants ?**

(a) Zero-Order Reaction
(b) Fractional- Order Reaction
(c) First- Order Reaction
(d) Second- Order Reaction

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**97. A gets converted to B according to the reaction A → B. If the conversion of A is 50%, what is the mole fraction of A in the exit stream?**

(a) 1/3 (b) 1/2
(c) 1/4 (d) 3/4

**Kerala PSC Lect. 05.02.2015**

**Ans. (b)**

**98. Pure A in gas phase enters a reactor, 50% of this A is converted to B through the reaction A→3B. Mole fraction of A in the exit stream is:**

(a) 1/2 (b) 1/3
(c) 1/4 (d) 1/5

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** 1/4

## C. Design Equation of Reactors

**99. What is the exit conversion of reactant A for a zero order reaction taking place in a CSTR with the following data [K = 1 mol/unit lit, CA = 1 mol/lit, feed flow rate = 0.5 lit/mol and reactor volume = 1 lit]:**

(a) 100% (b) 50%
(c) 200% (d) 75%

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**100. For an isothermal variable volume batch reactor, which of the following relation is applicable for a first order irreversible reaction:**

(a) $X_A = k * t$
(b) $\frac{C_{A_o}}{1+\varepsilon A X A}\left(\frac{dXA}{dt}\right) = k$
(c) $-ln(1 - X_A) = k * t$
(d) $\varepsilon A^* ln(1 - X_A) = k * t$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**101. A zero order reaction is being conducted in a PFR of volume V and fractional conversion of 0.5 is realized. If a second PFR of volume V is now added to the first reactor, the reactant concentration in the fluid exiting the battery of two PFRs will be:**

(a) Zero
(b) 25% of the reactant concentration in the fluid entering the first order
(c) 50% of reactant concentration in the fluid entering the second order
(d) Either (A) or (B)

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (a)**

**102. An autocatalytic reaction A + R → R + R is best conducted in a CSTR provided:**
(a) The level of conversion is high
(b) The level of conversion is low
(c) The feed flow rate is low
(d) The feed flow rate is high

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**103. One liter/min of liquid containing a reactant A at a concentration of 0.2 mol/*l* flows into a continuously operated ideal flow stirred tank reactor. The outflow from the reactor contains A at a concentration of 0.02 mol/*l*. If the volume of the reactor is 2 lit, the rate of reaction of A realized in the reactor is:**
(a) 0.02 mol/*l*. min (b) 0.09 mol/*l*. sec
(c) 0.09 mol/*l*. min (d) 0.18 mol/*l*. min

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**104. The sequence in which three CSTR's of volume 5, 10 and 15. It will be connected in series to obtain the maximum production in a second order irreversiable reaction is**
(a) 15,10,5 (b) 5, 10, 15
(c) 10,5,15 (d) 10, 15, 5

**TNPSC AE-2018**

**Ans. (b)**

**105. For a tubular reactor with space time 'τ' and residence time 'θ' , the following statement holds**
(a) τ and θ are always equal
(b) τ = θ when fluid density changes in the reactor
(c) τ = θ when fluid density remains constant
(d) τ = θ for a non-isothermal reactor

**TNPSC AE-2018**

**Ans. (c)**

**106. The conversion $X_A$ and residence time τ data are collected for zero order liquid phase reaction in a stirred tank reactor. Which of the following will be a straight line?**
(a) $X_A$ vs τ (b) $X_A$ vs In τ
(c) $\frac{X_A}{1-X_A}$ vs τ (d) $X_A(1 - X_A)$ vs. τ

**TNPSC AE-2018**

**Ans. (a)**

**107. In a steady state, CSTR the composition of the end stream**
(a) is same as that in reactor
(b) is different as that in reactor
(c) depends on flow rate
(d) insufficient information

**TNPSC AE-2018**

**Ans. (a)**

**108. The unimolecular type first order reaction in series $A \xrightarrow{k_1} R \xrightarrow{K_2} S$ is treated in CSTR. The optimum residence time $\tau_m$, opt for the reactor is**
(a) $K_1 K_2$ (b) $K_1 + K_2$
(c) $\frac{1}{\sqrt{K_1 K_2}}$ (d) $K_1 / K_2$

**TNPSC AE-2018**

**Ans. (c)**

**109. A ideal plug flow is characterized by**
(a) high capacity
(b) presence of axial mixing
(c) presence of lateral mixing
(d) no mixing in axial and lateral

**TNPSC AE-2019**

**Ans. (c)**

**110. _____is the best example of a reactor with mixed flow of solids**
(a) PFR (b) CSTR
(c) Semi-Batch Reactor (d) Fluidized Bed

**TNPSC AE-2019**

**Ans. (d)**

**111. What is the unit of Liquid hourly space velocity?**
(a) hours (b) $(hour)^{-1}$
(c) $(hours)^2$ (d) $(hours)^{-2}$

**TNPSC AE-2019**

**Ans. (b)**

**112. Space time is equal to residence time in a_____ reactor.**
(a) Constant –volume
(b) Constant-density
(c) Constant-pressure
(d) Constant-temperature

**TNPSC AE-2019**

**Ans. (b)**

**113. The reaction $A \rightarrow B$, $r = KC_A^2$, occurs in CSTR with 90% conversion, If K = 0.5 litre mole$^{-1}$, min$^{-1}$ and $CA_0$ = 2 moles/litre. Estimate the residence time required ?**

(a) 30 minutes
(b) 60 minutes
(c) 90 minitues
(d) 120 minutes

**TNPSC AE-2019**

**Ans. (c)**

**114. MFR followed by PFR of equal volume are connected in series to carry out a first order isothermal liquid phase reaction. The rate constant is 0.2 s$^{-1}$. The space time is t s for both the reactors. The overall fractional conversion of A is**

(a) 0.9 (b) 0.6
(c) 0.8 (d) 0.7

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**115. The reaction A → B is conducted in an isothermal batch reactor. If the conversion of A increases linearly with holding time, then order of the reaction is**

(a) 2 (b) zero
(c) 1.5 (d) 1

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

**116. A first order liquid phase reaction is carried out in an deal CSTR. If a part of outlet stream is recycled back to a reactor, the conversion**

(a) May increase or decrease
(b) Will remain constant
(c) Will decrease
(d) will increase

**UPPSC AE-29.05.2022, Part-I**

**Ans. (b)**

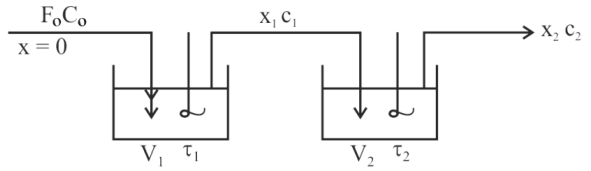
**117. For a mixed flow reactor operating at steady state, the rate of reaction is given by**

(a) $\frac{F_{A_\circ} X_A}{V}$ (b) $\frac{F_{A_\circ}}{dt} - \frac{dC_A}{dt}$
(c) $\frac{-dC_A}{dt}$ (d) $V.F_{A_\circ}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**118. The CSTR in series connection**




**the $\left(\frac{C_2}{C_\circ}\right)$ for a first order reaction is**

**Where $V_1$ and $V_2$ are the volumes of reactor 1 + 2**
**$\tau_1$ and $\tau_2$ are the space times of reactor 1 + 2**

(a) $\frac{C_2}{C_\circ} = \frac{1}{(1+\tau_2 K)}$

(b) $\frac{C_2}{C_\circ} = \frac{1}{(1+\tau_1 K)}$

(c) $\frac{C_2}{C_\circ} = \frac{1}{(1+\tau_1 K)(1+\tau_2 K)}$

(d) $\frac{C_2}{C_\circ} = (1+\tau_1 K)(1+\tau_2 K)$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**119. For a gas phase first order reaction A → 2B carried out in a mixed flow reactor. Which one of the following relation is correct?**

**Where $\tau$ = Space Time**

**$C_{A_0}$ = Initial concentration**

**$X_A$ = Conversion**

**$E_A$ = Fractional volume change**

**K = Rate constant**

(a) $\tau = \frac{C_{A_0} X_A}{(1-X_A)}$

(b) $\tau = \frac{(1+\Sigma_A X_A)X_A}{K(1-X_A)}$

(c) $\tau = C_{A_0}(1-X_A)$

(d) $\tau = \frac{C_{A_0} X_A \Sigma_A}{(1+X_A)}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

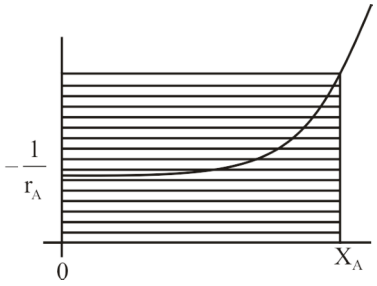
**120. For a given duty and for all positive reaction orders, the size of a mixed reactor is**

(a) smaller than that of PFR
(b) same as that of PFR
(c) larger than that of PFR
(d) insufficient information
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**121. The shaded area is the representation of the following equation.**




(a) $\tau = C_{A_\circ} \int_0^{X_A} \frac{dX_A}{-r_A}$ (b) $\tau = \frac{V}{V_\circ}$

(c) $\frac{\tau}{C_{A_0}} = \frac{X_A}{-V_A}$ (d) $\tau = \frac{C_{A_\circ} \cdot V}{FA_0}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**122. A reactor which has non-uniform and steady concentration is**

(a) Batch reactor
(b) Semi-batch reactor
(c) Plug-flow reactor
(d) Stirred tank reactor

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**123. For reactions carried out in plug flow reactor, dispersion number is**

(a) 0 (zero)
(b) 1
(c) –1
(d) ∞ (infinity)

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**124. For same residence time, maximum conversion will be obtained in which of the following cases**

(a) Single tubular reactor (volume = 2V)
(b) Single stirred tank reactor (volume = 2V)
(c) Two stirred tank reactors (volume = V each)
(d) Stirred tank reactor followed by single tubular reactor (volume = V each)

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**125. For auto-thermal reaction, maximum conversion will be obtained in which of the following cases**

(a) Batch reactor
(b) Semi-batch reactor
(c) Stirred tank reactor
(d) Plug flow reactor

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**126. For autocatalytic reaction, maximum conversion will be obtained in which of the following cases**

(a) Stirred tank reactors in series
(b) Plug flow reactors in series
(c) Plug flow reactor followed by stirred tank reactor
(d) Stirred tank reactor followed by plug flow reactor

**RPSC Lect.-15.01.2016**

**Ans. (d)**

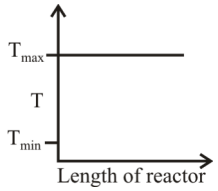
**127. The conversion of a first order liquid-phase A→B in an ideal continuous stirred tank reactor (CSTR) is 50%. If another ideal continuous stirred tank reactor (CSTR) of the same volume is connected in series, then % conversion at the exit of second reactor will be**

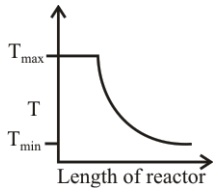
(a) 60 (b) 75
(c) 90 (d) 100

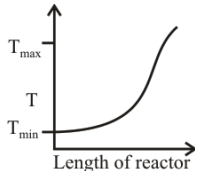
**Haryana PSC Poly Lect.-12.06.2022**

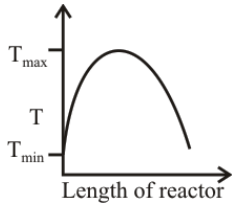
**Ans. (b)**

**128. Consider a reversible reaction in a plug flow reactor. The maximum and minimum permissible temperatures are $T_{max}$ and $T_{min}$ respectively. Which of the following temperature (T) profiles will require the shortest residence time to achieve the desired conversion?**

(a)



(b)



(c)



(d)



**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**129. A set standard stainless steel pipes, each of internal diameter 26.65 mm and 6000 mm length, is used to make a plug flow reactor by joining them in series to carry out degradation of polyethylene. Seven such pipes are required to obtain a conversion of 66% at 450 K. The minimum number of standard 8000 mm long pipes of the same internal diameter to be procured for obtaining at least 66% conversion under the same reaction conditions is _____.**

**GATE-2018**

**Ans. (6 to 6)**

**130. The elementary second order liquid phase reaction $A + B \rightarrow C + D$ is carried out in an isothermal plug flow reactor of 2 $m^3$ volume. The inlet volumetric flow rate is 10 $m^3/h$. The initial concentrations of both A and B are 2 $kmol/m^3$. The rate constant is given as 2.5 $m^3$ $kmol^{-1}$ $h^{-1}$. The percentage conversion of A is ___.**

**GATE-2018**

**Ans. (50 to 50)**

**131. The following gas-phase reaction is carried out in a constant-volume isothermal batch reactor**

**$A + B \rightarrow R + S$**

**The reactants A and B as well as the product S are non-condensable gases. At the operating temperature, the saturation pressure of the product R is 40 kPa.**

**Initially, the batch reactor contains equimolar amount of A and B (and no products) at a total pressure of 100 kPa. The initial concentrations of the reactants are $C_{A,o} = C_{B,o} = 12.5$ $mol/m^3$. The rate of reaction is given by $(-r_A) = 0.08\ C_A\ C_B$ $mol/m^3.s$.**

**The time at which R just starts condensing. rounded to 1 decimal plACF, is _____ s.**

**GATE-2017**

**Ans. (3.9 to 4.1)**

**132. The following liquid phase second-order reaction is carried out in an isothermal CSTR at steady state**

**$A \rightarrow R$ $(-r_A) = 0.005\ C_A^2$ $mol/m^3.hr$**

**Where $C_A$ is the concentration of the reactant in the CSTR. The reactor volume is 2 $m^3$, the inlet flowrate is 0.5$m^3/hr$ and the inlet concentration of the reactant is 1000 $mol/m^3$**

**The fractional conversion, rounded to 2 decimal plACF, is _____.**

**GATE-2017**

**Ans. (0.79 to 0.81)**

**133. For nearly isothermal operation involving large reaction time in a liquid-phaseb increase reaction, the most suitable reactor is**

(a) Stirred tank reactor
(b) Tubular flow reactor
(c) Batch reactor
(d) Fixed bed reactor

**RPSC ACF-2011**

**Ans. (a)**

**134. A homogeneous liquid phase reaction A □ B occurs in a batch reactor and a conversion of 50 % is achieved in one hour. If the same reaction is conducted in a plug flow reactor the spACF time necessary for 50% conversion is**

(a) 1 s
(b) 1 min
(c) 1 hr
(d) Between 1 min and 1 hr

**RPSC ACF-2011**

**Ans. (c)**

**135. An isothermal gas phase reaction, A □ 3B, is being conducted by starting with pure A in a variable volume batch reactor. Initially, the reactor volume is 1L. When 50% conversion of A has been achieved, rector volume is**

(a) 1 L (b) 1.5 L
(c) 2 L (d) 3 L

**RPSC ACF-2011**

**Ans. (c)**

**136. For identical feed composition, flow rate, conversion and for zero order reaction, the ratio of value of mixed reactor to the volume of plug reactor is**

(a) 0 (b) 1
(c) < 1 (d) > 1

**RPSC ACF-2011**

**Ans. (b)**

**137. SpACF time in flow reactor is**

(a) Usually equal to residence time
(b) The reciprocal of the spACF velocity
(c) Both (a) and (b)
(d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**138. The dimension of "spACF velocity" is**

(a) [length] [$time^{-1}$]
(b) [$time^{-1}$]
(c) [length] [time]
(d) none of these; it is dimensionless

**RPSC ACF-2011**

**Ans. (b)**

**139. The elementary liquid phase decomposition reaction $A \xrightarrow{k} 2B$ is to be carried out in a CSTR. The design equation is.**

(A) $k\tau = \dfrac{X_A}{(1-X_A)}$

(B) $k\tau = \dfrac{X_A(1+X_A)}{(1-X_A)}$

(C) $k\tau = \dfrac{X_A}{(1-X_A)^2}$

(D) $k\tau C_{A0} = \dfrac{X_A}{(1+X_A)^2}$

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

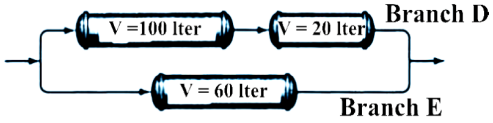
**140. The conversion for a first order liquid phase reaction $A \rightarrow B$, in a CSTR is 50% if another CSTR of the same volume is connected in series, then the percentage conversion at the exit of the second reactor will be.**

(A) 60 (B) 75
(C) 90 (D) 100
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**141. The reactor setup shown in figure consist of three plug flow reactor in two parallel branches. Branch D has a reactor of volume 100 *l*tr followed by a reactor of volume 20 *l*tr. Branch E has a reactor of volume 60 *l*trs. What fraction of the feed should go to branch D?**




(a) 2/3 of feed (b) 1/2 of feed
(c) 3/4 of feed (d) 1/3 of feed

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**142. In case of plug flow reactor when a recycle stream is added to entrance of reactor and as the recycle ratio increases from zero (R = ∞) What kind of effect does it have on reactor?**

(a) The behavior is unaffected by the recycle ratio
(b) The behavior shifts from plug flow to mix flow
(c) The behavior shifts from mix flow plug flow
(d) Percentage conversion starts decreasing

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**143. Following gas phase reaction takes plACF at 500 °C**

**$4PH_3(g) \rightarrow P_4(g)+6H_2(g)$**

**With $rPH_3 = 85\ h^{-1}\ CPH_3$**

**What is the volume of the plug reactor operating at this temperature and 5.5 atm. Giving 80% conversion for a feed of 2.5 kg moles $PH_3$ per hour?**

(a) 1 $m^3$ (b) 0.50 $m^3$
(c) 1.5 $m^3$ (d) 0.75 $m^3$

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**144. Pick out the wrong statement**

(a) In a continuous flow reactor, both the reactants and the products flow out continuously
(b) In a continuous flow reactor, uniform concentration cannot be maintained throughout the vessel even in a well agitated system
(c) In a semi-batch reactor, one reactant is charged batch wise, while the other reactant is fed continuously
(d) In a batch reactor which is exclusively used for liquid phase reactions; temperature pressure and composition may vary with time

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**145. The gas phase reaction A → B + C is carried out in an ideal PFR achieving 40% conversion of A. The feed has 70 mol% A and 30 mol% inerts. The inlet temperature is 300 K and the outlet temperature is 400 K. The ratio of the outlet to inlet molar concentration of A (assuming ideal gas mixture and uniform pressure) is**

(a) 0.60 (b) 0.30
(c) 0.35 (d) 0.47

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**146. The first order gas phase reaction, $A \xrightarrow{k_1} 2B$ is conducted isothermally in batch mode at constant volume. The rate of change of conversion with time is given by**

(a) $dX_A/dt = k_1 (1 - X_A)^2 (1 + 2X_A)$
(b) $dX_A/dt = k_1 (1 - X_A) (1 + 0.5X_A)$
(c) $dX_A/dt = k_1 (1 - X_A)/ (1 + X_A)$
(d) $dX_A/dt = k_1 (1 - X_A)$

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**147. An irreversible first order reaction is being carried out in a CSTR and PFR of same volume. The liquid flow rates are same. The relative conversion will**

(a) be more in CSTR than in PFR
(b) be more in PFR than in CSTR
(c) be same in both cases
(d) depend on the temperature

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**148. The liquid phase reaction, A→ Products is governed by: $r_A = kC_A^{1/2}$ . If the reaction undergoes 75% conversion of A in 10 min in an isothermal batch reactor, then the time (in min) required for the complete conversion of A is**

(a) 40/3 (b) 20
(c) 30 (d) ∞

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**149. In an aqueous solution, reaction P→ Q occurs under isothermal conditions following first order kinetics. The feed rate is 500 $cm^3$/min and concentration of P in the feed is $1.5 \times 10^{-4}$ mol/$cm^3$. The reaction is carried out in a 5 L CSTR. At steady state, 60% conversion is observed. The rate constant (in $min^{-1}$) is**

(a) 0.06 (b) 0.28
(c) 0.21 (d) 0.15

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**150. The conversion for a first order liquid-phase reaction A→ B in a CSTR is 50%. If another CSTR of the same volume is connected in series, then the % conversion at the exit of the second reactor will be**

**A. 60** **B. 75**
**C. 90** **D. 100**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**151. A spACF time of 20 min. means**

**A. 20% conversion is achieved in one min.**
**B. 80% conversion is achieved in one min.**
**C. 100% conversion of the reactant in 20 min.**
**D. One reactor volume of feed at specified conditions is processed in 20 min.**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (d)**

**152. For identical feed composition, flow rate, conversion and for all positive reaction orders the ratio of the volume of mixed reactor to the volume of plug flow reactor**

**A. Is independent of the order of reaction**
**B. Increases linearly with increase in the order of reaction**
**C. Increases with increase in the order of the reaction**
**D. Deceases with increase in the order of reaction**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**153. In an ideal tubular-flow reactor**

(a) There is no mixing in longitudinal direction
(b) Mixing takes plACF in radial direction
(c) There is a uniform velocity across the radius
(d) All (a), (b) and (c)

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**154. For identical flow rate, feed composition and for elementary first order reactions, 'N' equal sized mixed reactors in series with a total volume 'V' gives the same conversion as a single plug flow reactor of volume 'V' for constant density systems. This is true, when the value of 'N' is**

(a) 1 (b) >1
(c) ∞ (d) None of these

**Karnataka PSC ADF - 2015**

**Ans. (c)**

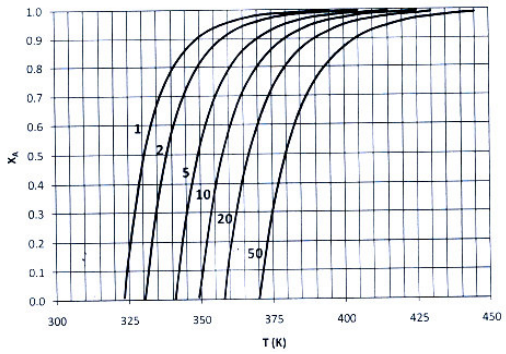
**155. Pick out the wrong statement pertaining to space velocity of flow reactors.**

(a) The unit of space velocity is $(time)^{-1}$.
(b) The space velocity of 5 $hr^{-1}$ means that five reactor volumes of feed at specified conditions are being fed into the reactor every hour.
(c) The space velocity of 5 $hr^{-1}$ means that one fifth reactor volume of feed at specified conditions are being fed into the reactor.
(d) None of these

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**156. A liquid phase irreversible reaction A→B is carried out in an adiabatic CSTR operating under steady state conditions. The reaction is elementary and follows the first order rate law. For this reaction, the figure below shows the conversion ($X_A$) of A as a function of temperature (T) for different values of the rate of reaction ($-\gamma_A$ in mol $m^{-3}s^{-1}$) denoted by the numbers to the left of each curve. This figure can be used to determine the rate of the reaction at a particular temperature, for a given conversion of A**




**The inlet stream does not contain B and the concentration of A in the inlet stream is 5 mol/m$^3$. The molar feed rate of A is 100 mol/s. A steady state energy balance for this CSTR results in the following relation $T = 350 + 25X_A$ where T is the temperature (in K) of the exit stream and $X_A$ is the conversion of A in the CSTR. For an exit conversion of 80% of A, the volume (in m$^3$, rounded off to the first decimal place) of CSTR required is—**

**GATE-2016**

**Ans. (7.9 to 8.1)**

**157. The liquid phase reversible reaction $A \rightleftharpoons B$ is carried out in an isothermal CSTR operating under steady state conditions. The inlet stream does not contain B and the concentration of A in the inlet stream is 10 mol/lit. The concentrations of A at the reactor exit, for residence times of 1 s and 5 s are 8 mol/lit and 5 mol/lit, respectively. Assume the forward and backward reactions are elementary following the first order rate law. Also assume that the system has constant molar density. The rate constant of the forward reaction (in s$^{-1}$, rounded off to the third decimal place) is—**

**GATE-2016**

**Ans. (0.255 to 0.280)**

**158. The gas phase reaction 2A'–>' b is carries out in an isothermal plug flow reactor. The feed consists of 80 mole% A and 20 mole% inert. If the conversion of A at the reactor exit is 50% then 'C_(A)//C_(Ao)' at the outlet of the reactor is**

(a) 2/3
(b) 5/8
(c) 1/3
(d) 3/8

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**159. For the liquid phase zero order irreversible reaction A→B, the conversion of A in a CSTR is found to be 0.3 at a space velocity of 0.1 min$^{-1}$. What will be the conversion for a PFR with a space velocity of 0.2 min$^{-1}$? Assume that all the other operating conditions are the same for CSTR and PFR.**

(a) 0.15 (b) 0.3
(c) 0.6 (d) 0.9

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**160. When the density of the reaction mixture is constant in a chemical reaction, the ratio of the mean residence time to space time is**

(a) >1 (b) <1
(c) 1 (d) 0

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**161. Conversion increases with increase in temperature in case of a _____reaction.**

(a) Autocatalytic
(b) Irreversible
(c) Reversible endothermic
(d) Reversible exothermic

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**162. In a plug flow reactor, there is**

(a) Perfect mixing in the axial direction
(b) A parabolic velocity profile
(c) Perfect mixing in the radial direction
(d) A spread of residence times

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (c)**

**163. In an ideal plug flow (i.e. slug flow, piston flow, tubular unmixed flow) reactor**

(a) there is no mixing in longitudinal direction
(b) mixing takes place in radial or lateral direction
(c) there is a uniform velocity across the radius
(d) all (a), (b) and (c)

**ISRO-12.02.2016**

**Ans. (d)**

**164. The performance equations for constant density systems are identical for :**

(a) PFR and CSTR
(b) PFR and Batch reactor
(c) PFR, Batch and Backmix reactors
(d) CSTR and Batch reactor

**AP PSC IOF-2009**

**Ans. (b)**

**165. Every two minutes one reactor volume of feed is being treated at specified conditions in a CSTR. What is the space time ?**

(a) 1/2 (b) 2

(c) 1 (d) 4

**AP PSC IOF-2009**

**Ans. (a)**

**166. In an ideal mixed reactor at steady state, the :**

(a) composition anywhere in the reactor same

(b) exit stream has same composition as the fluid within the reactor

(c) space time is equal to holding time for a constant density system

(d) all of the above

**AP PSC IOF-2009**

**Ans. (d)**

**167. In order to achieve the same conversion under identical reaction conditions and feed flow rate for a non-autocatalytic reaction of positive order, the volume of an ideal CSTR is**

(a) always greater than that of an ideal PFR

(b) always smaller than that of an ideal PFR

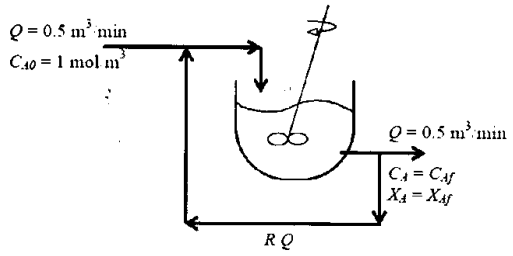
(c) same as that of an ideal PFR

(d) smaller than that of an ideal PFR only for first order reaction

**GATE-2014**

**Ans. (a)**

**168. An isothermal steady state mixed flow reactor (CSTR) of 1 $m^3$ volume is used to carry out the first order liquid-phase reaction A → products. Fresh feed at a volumetric flow rate of Q containing reactant A at a concentration $C_{A0}$ mixes with the recycle stream at a volumetric flow rate RQ as shown in the figure below.**




**It is observed that when the recycle ratio R = 0.5, the exit conversion $X_{Af}$ = 50%. When the recycle ratio is increased to R = 2, the new exit conversion (in percent) will be**

**(1) 50.0 (2) 54.3**

**(3) 58.7 (4) 63.2**

(a) 1 (b) 2

(c) 3 (d) 4

**GATE-2015**

**Ans. (a)**

**169. The performance equations for constant density systems are identical for**

(a) P.F.R. and backmix reactor

(b) P.F.R. and batch reactor

(c) P.F.R., batch reactor and backmix reactor

(d) Batch reactor and backmix reactor

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**170. If for a system density is constant than the performance equations are identical for:**

(a) Batch reactor and plug flow reactor

(b) Batch reactor and CSTR

(c) PFR and CSTR

(d) Batch reactor, PFR and CSTR

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Batch reactor and plug flow reactor

**171. For a given isothermal gas phase reaction A → 3B, the traction change in volume of the system between no conversion and complete conversion is:**

(a) 2 (b) 3

(c) 1 (d) 4

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 2

**172. For an elementary liquid phase decomposition reaction A → 2B which is carried out in a CSTR. The design equation is:**

(a) $K\tau = X_A/(1 - X_A)$

(b) $K\tau = X_A/(1 - X_A)^2$

(c) $K\tau = X_A/(1 + X_A)/(1 - X_A)$

(d) $K\tau C_{A^0} = \dfrac{X_A/(1+X_A)^2}{(1-X_A)^2}$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $K\tau = X_A/(1 - X_A)$

**173. A first order reversible reaction $A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$ occurs in a batch reactor. The exponential decay of the concentration of A has a time constant of:**

(a) $1/k_1$

(b) $1/k_2$

(c) $\dfrac{1}{k_1 - k_2}$

(d) $\dfrac{1}{k_1 + k_2}$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $1/k_1$

## D. Multiple Reaction Systems

**174. For a reaction of the type, $V \xrightarrow{K_1} X \xrightarrow{K_2} Y \xrightarrow{K_3} Z$ the rate of reaction (–rx) is given by:**

(a) $(K_1 + K_1)\, C_x$

(b) $(K_1 + K_2 + K_3)\, C_x$

(c) $K_1C_v - K_2C_x$

(d) $(K_1 - K_2)\, C_x$

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (c)**

**175. The number of moles of an excess reactant per mole of the limiting reactant into the reactor feed is called_____**

(a) Recycle Ratio

(b) Decay Ratio

(c) Reactant Ratio

(d) Yield Ratio

**TNPSC AE-2019**

**Ans. (c)**

**176. If the value of equilibrium constant is very much large (k >> 1.0) then it means _____ is possible and the reaction is _____.**

(a) 100% conversion and irreversible reaction

(b) 50% conversion and reversible reaction

(c) 100 % conversion and reversible reaction

(d) 50% conversion and autocatalytic reaction

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**177. Substance A in a liquid reacts to produce R and S as follows:**

$A \xrightarrow{K_1} R$

$A \xrightarrow{K_2} S$

**$-r_A$ for the above elementary reaction is**

(a) $K_1C_A$

(b) $K_2C_A$

(c) $(K_1 + K_2)C_A$

(d) $\frac{K_1}{K_2}C_A$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**178. To maximize the formation of R in the simultaneous reactions**

**$A + B \rightarrow R$ $\quad r_R = 2C_A^{0.5}C_B^2$**

**$A + B \rightarrow S$ $\quad r_S = 1.4C_A\, C_B$**

**We should have**

(a) Low $C_A$, low $C_B$

(b) Low $C_A$, high $C_B$

(c) High $C_A$, low $C_B$

(d) High $C_A$, high $C_B$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**179. Nitrous oxide reacts with oxygen to form nitrogen dioxide in a reversible reaction given below:**

**$NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$, given that**

**$\Delta_rG^o = -81.2$ kJ mol$^{-1}$.**

**The equilibrium constant at 25 $^oC$ for the reaction is**

(a) $5.91\times10^{-13}$

(b) 28.2

(c) 1.03

(d) $10^{14}$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (*)**

**180. The reversible reaction of t-butyl alcohol (TBA) and ethanol (EtOH) to ethyl t-butyl ether (ETBE) is**

**$TBA + EtOH \rightleftharpoons ETBE + Water$**

**The equilibrium constant for this reaction is $K_c$ = 1. Initially, 74 g of TBA, is mixed with 100 g of aqueous solution containing 46 weight % ethanol. The molecular weights are: 74 g/mol for TBA, 46 g/mol for EtOH, 102 g/mol for ETBE, and 18 g/mol for water.**

**The mass of ETBE at equilibrium, rounded to 1 decimal plACF, is _____ g.**

**GATE-2017**

**Ans. (20.2 to 20.6)**

**181. The rate of forward reaction, at chemical equilibrium is compared to the rate of backward reaction is**

(a) More (b) Less

(c) Equal (d) Equal or less

**RPSC ACF-2011**

**Ans. (c)**

**182. Consider the reaction $N_2 + 3H_2 \xleftrightarrow{K} 2NH_3$. The same reaction can also be written as $\frac{1}{2}N_2 + \frac{3}{2}H_2 \xleftrightarrow{K^*} NH_3$. Where K and K* are the respective equilibrium constants. Then**

(a) $K = 0.5K^*$ (b) $K = K^*$

(c) $K = 2\,K^*$ (d) $K = K^{*2}$

**RPSC ACF-2011**

**Ans. (b)**

**183. For an autocatalytic reaction $A + R \rightarrow R + R$, the rate vs. concentration of A profit is**

(a) linear (b) exponential
(c) parabolic (d) none of these

**RPSC ACF-2011**

**Ans. (c)**

**184. A reaction of type $A \xrightarrow{1st order, K_1} B \xrightarrow{1st order, K_2} C$ conducted in a batch reactor is characterized by.**

(A) Maximum concentration of B, with time
(B) Maximum concentration of C, with time
(C) Both (A) and (B)
(D) Neither (A) nor (B)

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**185. ___explains the equilibrium constant for any chemical reaction**

(A) Henry's law (B) Law of mass action
(C) Hess's law (D) Raoult's law

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**186. For the reversible first order reaction**

**$A \rightleftharpoons R$; $CA = C_A O, C_R = C_{RO} = 0$ at $t = 0$**

**To take plACF isothermally, the equilibrium fraction conversion $x_{Ae}$ of A and equilibrium constant K are related by**

(a) $x_{Ae} = (1 + K)/K$ (b) $x_{Ae} = 1/(1 + K)$
(c) $x_{Ae} = K/(1 + K)$ (d) $x_{Ae} = 1/K$

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**187. A reaction $A + B \rightleftharpoons C + D$ is carried out in a batch reactor. Initially, 1 mole of A and 2 mole of B are fed. At equilibrium, the product C is obtained as 0.6 mole. The equilibrium constant is:**

(a) K = 14/9 (b) K = 9/14
(c) K = 9/4 (d) K = 0.18

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**188. To maximize the formation of R in the simultaneous reactions**

**$A + B \rightarrow R$ $r_R = 2C_A^{0.5}C_B^2$**

**$A + B \rightarrow S$ $r_S = 1.5\, C_A C_B$**

**We should have**

(a) Low $C_A$, Low $C_B$ (b) Low $C_A$ High $C_B$
(c) High $C_A$, Low $C_B$ (d) High $C_A$, High $C_B$

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**189. For a series of reactions $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ having $k_1 << k_2$, the reaction system can be approximated as**

(a) $A \xrightarrow{k_1} B$ (b) $A \xrightarrow{k_2} B$
(c) $A \xrightarrow{k_2} C$ (d) $A \xrightarrow{k1} C$

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**190. Which of the following factors influence a chemical system in accordance with the Le Chatelier's principle?**

**A. Concentration only**
**B. Pressure and concentration**
**C. Concentration, pressure and temperature**
**D. Temperature and pressure**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**191. A reaction sequence is given as follows:**

**$HI \rightarrow H_2 + I_2$**

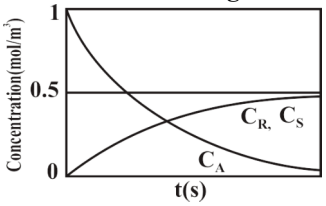
**At equilibrium when more $H_2$ is introduced into the system, which of the following changes occur?**

(a) System moves to new equilibrium state
(b) Equilibrium point remains same
(c) Hydrogen added to system do not react with iodine
(d) None of the above

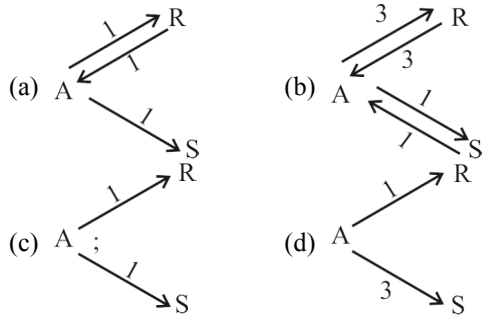
**Karnataka PSC ADF - 2015**

**Ans. (a)**

**192. The variations of the concentrations ($C_A$, $C_R$ and $C_S$) for three species (A, R and S) with time, in an isothermal homogeneous batch reactor are shown in the figure below.**




**Select the reaction scheme that correctly represents the above plot. The numbers in the reaction schemes shown below, represent the first order rate constants in unit of $s^{-1}$**




**GATE-2016**

**Ans. (c)**

**193.** With reference to the irreversible, first-order series reactions,

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$, which of the statements is /are correct?

1. For $k_1 << k_2$, the reaction system can be approximated as $A \xrightarrow{k_1} C$.
2. For $k_2 << k_1$, the reaction system can be approximated as $A \xrightarrow{k_2} C$.

Select the correct answer using the code given below.

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**194.** Consider the following equilibrium equation:

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The density of an equilibrium mixture of $N_2O_4$ and $NO_2$ at 1 atm and 348 K is 1.84 g/L. The value of $k_c$ for equilibrium will be:

(a) 7 (b) 8
(c) 9 (d) 6

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**195.** The sequence in which 3 CSTRS of volumes 10, 15, 20 $m^3$ will be connected in series to obtain the maximum production in a second order irreversible reaction is:

(a) 20, 15, 10 (b) 10, 15, 20
(c) 15, 10, 20 (d) 20, 10, 15

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 10, 15, 20

## E. Design of Multiple Reaction System

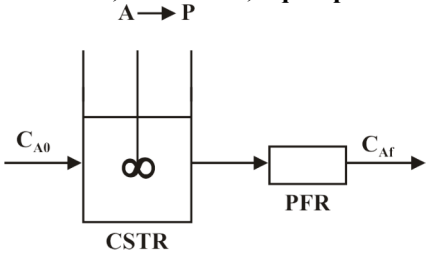
**196.** The conversion for a first order liquid phase reaction A→B in a CSTR is 50%. If another CSTR of the same volume is connected in series, then the % conversion at the exit of the second reactor will be:

(a) 60 (b) 75
(c) 90 (d) 100

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**197.** A CSTR and a PFR of equal volume are connected in series as shown below to carry out a first-order, isothermal, liquid phase reaction




The rate constant is 0.2 $s^{-1}$. The spACF time is 5 s for both the reactors. The overall fractional conversion of A is ____ (rounded off to third decimal plACF).

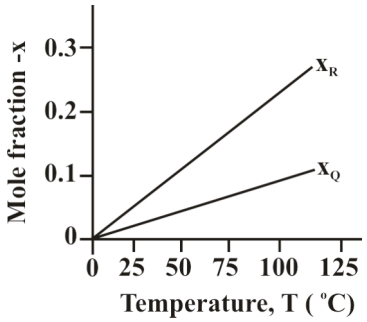
**GATE-2018**

**Ans. (0.80 to 0.84)**

**198.** In a laboratory batch setup, reaction of P over a catalyst was studied at various temperatures. The reactions occurring are

$P \rightarrow 2Q$ ; $P \rightarrow R$

At the end of one hour of operation, the batch contains $x_P$, $x_Q$ and $x_R$ mole fractions of P, Q and R components, respectively. The mole fractions of product components ($x_Q$ and $x_R$) were found to vary linearly with temperature as given in the figure.

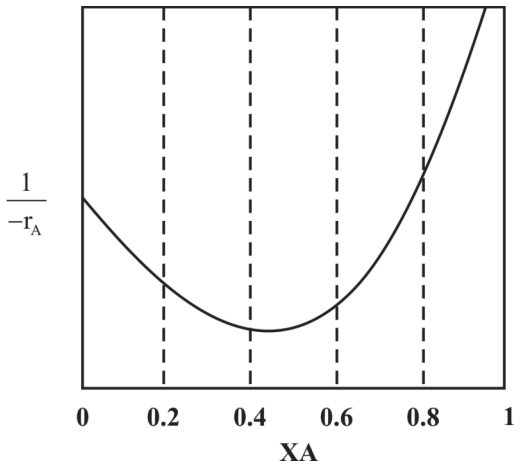



If the yield of Q based on reactant P consumed ($Y_Q$) at 25 °C was found to be 0.40, then the value of $Y_Q$ at 60 °C is ___(rounded off to second decimal plACF)

**GATE-2018**

**Ans. (0.40 to 0.40)**

**199.** The following reaction rate curve is shown for a reaction A → P. Here, $(-r_A)$ and $X_A$ represent reaction rate and conversion, respectively. The feed is pure A and 90% conversion is desired.




Which amongst the following reactor configurations gives the lowest total volume of the reactor(s)?

(a) CSTR followed by PFR
(b) Two CSTR in series
(c) PFR followed by CSTR
(d) A single PFR

**GATE-2017**

**Ans. (a)**

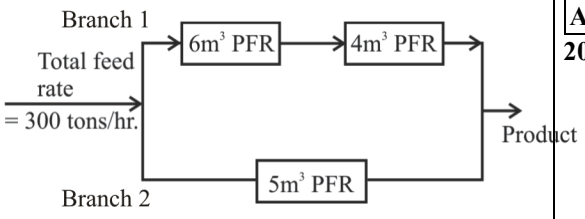
**200. Consider a homogeneous reaction of the type $A \xrightarrow{k_1(1^{st}\ order)} R$ and also $A \xrightarrow{k_2(2^{nd}\ order)} S$. R is the desired product and its concentration is to be maximized by selection of a proper reactor. Which reactor system will you choose in order to get the highest R- concentration (under otherwise uniform conditions)?**

(a) Batch reactor
(b) PFR
(c) Single CSTR
(d) Five CSTRs in series

**RPSC ACF-2011**

**Ans. (c)**

**201. Three plug flow reactors (PFR's) of 4, 5 & 6 $m^3$ volumes are arranged in two branches as shown below in the figure. If the total feed rate is 300 tons/hr, then for the same conversion in each branch, the feed rate through branch II should be _____tons/hr.**



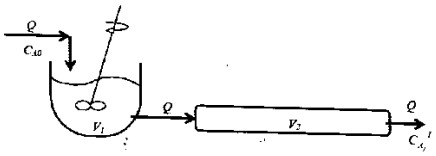

(a) 100 (b) 150
(c) 200 (d) 225

**ISRO Scientist Engineer-06.08.2017**

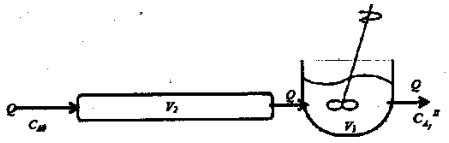
**Ans. (a)**

**202. Consider two steady isothermal flow configurations shown schematically as Case I and Case II below. In Case I, a CSTR of volume $V_1$ is followed by a PFR of volume $V_2$, while in Case II a PFR of volume $V_2$ is followed by a CSTR of volume $V_1$. In each case, a volumetric flow rate Q of liquid reactant is flowing through the two units in series. An irreversible reaction A → products (order n) takes place in both cases, with a reactant concentration $C_{A0}$ being fed into the first unit.**

**Case 1.**



**Case II.**



**Choose the correct option.**

**(1)** $\frac{C_{Af}^{I}}{C_{Af}^{II}} > 1 \text{ for } n = 1$ **(2)** $\frac{C_{Af}^{I}}{C_{Af}^{II}} = 1 \text{ for } n = 1$

**(3)** $\frac{C_{Af}^{I}}{C_{Af}^{II}} < 1 \text{ for } n = 1$ **(4)** $\frac{C_{Af}^{I}}{C_{Af}^{II}} = 1 \text{ for } n > 0$

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (b)**

## F. Hetrogenous System

**203. From the Brunauer-Emmett-Teller plot, the intercept and slope are estimated as $2 \times 10^{-3}$ $cm^{-3}$, $8 \times 10^{-3}$ $cm^{-3}$ respectively. The monomolecular volume of the catalyst is**

(a) 10 $cm^3$ (b) 1 $cm^3$
(c) 100 $cm^3$ (d) 7 $cm^3$

**TNPSC AE-2018**

**Ans. (c)**

**204. For a first- order isothermal chemical reaction in a porous catalyst, the effectiveness factor is 0.3. The effectiveness factor will increase if the**

(a) Catalyst size is reduced or the catalyst diffusivity reduced
(b) Catalyst size is reduced or the catalyst diffusivity is increased
(c) Catalyst size is increased or the catalyst diffusivity is reduced
(d) Catalyst size is increased or the catalyst diffusivity is increased

**TNPSC AE-2018**

**Ans. (b)**

**205. For the first order chemical reaction in a porous catalyst, the theile modulus is 10. The effectiveness factor is approximately equal to**

(a) 1 (b) 0.5
(c) 0.1 (d) 0

**TNPSC AE-2022, 2018**

**Ans. (c)**

**206. Helium -Mercury method is used to determine**

(a) Surface area of the catalyst
(b) Density of the catalyst
(c) Pure volume of the catalyst
(d) Weight of the catalyst

**TNPSC AE-2018**

**Ans. (c)**

**207. In which of the following reaction does have the solid change in size is appreciably?**

(a) $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(S) + 2SO_2(g)$

(b) $Fe_3O_4(s) + 4H_2(g) \rightarrow 3F_2(s) + 4H_2O(g)$

(c) $CaC_2(s) + N_2(g) \rightarrow CaCH_2(s) + C(s)$

(d) $NaNH_2(l) + C(s) \rightarrow NaCN(l) + H_2(g)$

**TNPSC AE-2019**

**Ans. (d)**

**208. The ratio of maximum possible conversion in the film to that of maximum diffusional transport through the film is called_____**

(a) Yield coefficient

(b) Hatta number

(c) Biot number

(d) Thiele modulus

**TNPSC AE-2019**

**Ans. (b)**

**209. The activity of the catalyst lost due to the deposition of carbonaceous material on the surface of catalyst is called_____**

(a) Fouling

(b) Poisoning

(c) Sinking

(d) Carrier

**TNPSC AE-2019**

**Ans. (a)**

**210. Which among the following particle has no influence on the rate of reaction when surface reaction prevails ?**

(a) Porous catalyst particle

(b) Catalyst coated surface

(c) Burning of Droplet of a fuel

(d) Cells and simple living creatures

**TNPSC AE-2019**

**Ans. (c)**

**211. The rate of a catalytic chemical reaction can be increased by**

(a) increasing the size of non-porous pellet

(b) decreasing the temperature

(c) decreasing the concentration

(d) increasing the internal surface area

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**212. Choose Heterogenous non catalytic reaction among following**

(a) Ammonia Synthesis

(b) Nitric acid Production

(c) Burning of coal Particle

(d) Oxidation of $SO_2$ to $SO_3$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**213. For large value of Thiele Modulus $\left(L\sqrt{\frac{K}{D}}\right)$ of a solid catalyzed first order reaction, the effectiveness factor ($\eta$) is equal to**

(a) 1

(b) $L\sqrt{\frac{K}{D}}$

(c) $1+L\sqrt{\frac{K}{D}}$

(d) $1\Big/\left(L\sqrt{\frac{K}{D}}\right)$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**214. If Thiele Modulus is large, _____ limits the reaction.**

(a) External diffusion (b) Internal diffusion

(c) Surface reaction (d) Adsorption

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**215. For gas-solid non catalytic reactions, of change in size, which resistance will not be considered**

(a) film resistance

(b) reaction resistance

(c) ash resistance

(d) overall resistance

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**216. Effectiveness factor of a catalyst pellet is a measure of the**

(a) gas film diffusion resistance

(b) bulk resistance

(c) pore diffusion resistance

(d) chemical reaction resistance

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**217. Which gas is used for determining the surface area of the catalysts**

(a) Nitrogen

(b) Hydrogen

(c) Helium

(d) Carbon di-oxide

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**218. If Thiele modulus is _____, then the pore diffusion resistance in a catalyst may be considered as negligible.**

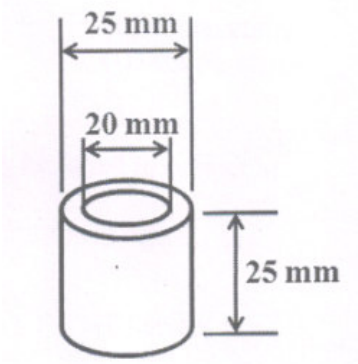
(a) > 0.5 (b) ∞

(c) < 0.5 (d) 0

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

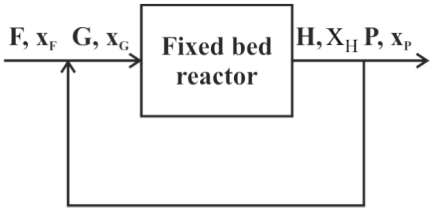
**219. Hydrogenation of benzene is to be carried out using Ni (density = 8910 kg/m$^3$) as catalyst, cast in the form of non-porous hollow cylinders, as shown below. The reaction occurs on all the surfACFs of the hollow cylinder. During an experiment, one such cylinder is suspended in the reactant stream. If the observed rate of reaction is 0.39 mol (m$^2$ of catalyst surfACF)$^{-1}$ min$^{-1}$ then the rate of reaction in mol (kg of catalyst)$^{-1}$ min$^{-1}$ is_____(rounded off to three decimal plACFs).**




**GATE-2018**

**Ans. (0.034 to 0.043)**

**220. The reactant (M) is converted into product (N) in the presence of catalyst in a fixed bed reactor. All the flow rates (F, G, H, P and R) in mol/s, and mole fraction of reactant (M) in these streams ($x_F$, $x_G$, $x_H$, $x_P$, and $x_R$) are shown in the diagram**




**The overall fractional conversion is**

(a) $\frac{x_G G - x_H H}{x_G G}$ (b) $\frac{x_G G - x_P P}{x_G G}$

(c) $\frac{x_F F - x_H H}{x_F F}$ (d) $\frac{x_F F - x_P P}{x_F F}$

**GATE-2018**

**Ans. (d)**

**221. Liquid phase isomerization of O–xylene to p-xylene using a zeolite catalyst was carried out in a CSTR. Three sets of kinetic data at different temperature and stirring speeds were obtained as shown below.**

| | set - A | | | set - B | | | set - C | | |
|---|---|---|---|---|---|---|---|---|---|
| Temperature (K) | 500 | 500 | 500 | 600 | 600 | 600 | 700 | 700 | 700 |
| Stirring speed (rpm) | 1000 | 2000 | 3000 | 1000 | 2000 | 3000 | 1000 | 2000 | 3000 |
| Reaction rate (mol L$^{-1}$s$^{-1}$) | 0.020 | 0.025 | 0.025 | 0.037 | 0.047 | 0.047 | 0.069 | 0.078 | 0.086 |

**The operating condition at which the reaction rate is not controlled by external mass transfer resistance is**

(a) T =500 K; rpm = 3000

(b) T = 600 K; rpm = 1000

(c) T = 700 K; rpm = 1000

(d) T = 700 K; rpm = 2000

**GATE-2018**

**Ans. (a)**

**222. Consider a first order catalytic reaction in a porous catalyst pellet.**

**Given R– characteristic length of the pellet; $D_e$ – effective diffusivity; $k_c$ – mass transfer coefficient; $k_1$ – rate constant based on volume of the catalyst pellet; $C_s$ – concentration of reactant on the pellet surfACF,**

**The expression for Thiele modulus is**

(a) $\frac{k_c R}{D_e}$ (b) $R\sqrt{\frac{k_1}{D_e}}$

(c) $R\sqrt{\frac{k_1 C_s}{D_e}}$ (d) $R\sqrt{\frac{D_e}{k_1}}$

**GATE-2017**

**Ans. (b)**

**223. For a solid-catalyzed gas phase reversible reaction which of the following statements is ALWAYS TRUE?**

(a) Adsorption is rate-limiting

(b) Desorption is rate-limiting

(c) Solid catalyst does not affect equilibrium conversion

(d) Temperature does not affect equilibrium conversion

**GATE-2017**

**Ans. (c)**

**224. Van Laar equation deals with activity coefficients in**

(a) Binary solution

(b) Ternary solution

(c) Azeotropic mixture only

(d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**225. A porous solid catalyst has the following characteristics : particle density = 0.98 g/ $cm^3$; particle diameter = 0.6 mm; total surface area = 50 $m^2$/g. The external surfACF area of the catalyst is**

(a) 50 $m^2$/g (b) 102.04 $m^2$/g
(c) 102.04 $cm^2$/g (d) 102.04 × $10^4$ $cm^2$/g

**RPSC ACF-2011**

**Ans. (*)**

**226. The Thiele modulus for a gas-phase first-order isothermal reaction for a spherical catalyst particle is found to be 2. The catalyst effectiveness factor is**

(a) 0.33 (b) 0.49
(c) 0.80 (d) 0.91

**RPSC ACF-2011**

**Ans. (b)**

**227. An irreversible first order reaction occurs in presence of a spherical porous catalyst particle. The expression for effectiveness factor for a spherical catalyst particle is given by**

(a) $\eta = \frac{3}{\phi}\left(\frac{1}{\tan h\,\phi} - \frac{1}{\phi}\right)$

(b) $\eta = \frac{1}{\phi}\left(\frac{1}{\coth \phi} - \frac{1}{\phi^2}\right)$

(c) $\eta = \frac{3\phi}{\tanh \phi}$

(d) $\eta = 3\tanh\phi - \frac{\cot h\,\phi}{\phi}$

Note: $\phi = R\sqrt{\frac{k}{D_c}}$ R is the radius of the catalyst particle.

**RPSC ACF-2011**

**Ans. (a)**

**228. Which among the following is the actual property of a catalyst?**

(a) It changes the equilibrium constant of the reaction
(b) It increases the equilibrium concentration
(c) It supplies energy to the reaction
(d) It shortens the time to reach equilibrium

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**229. Controlling film coefficient means**

**A. The coefficient which offers the largest resistance**
**B. The coefficient which offers the least resistance**
**C. The coefficient which doesn't offer any resistance**
**D. The average of the film thickness**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**230. For a heterogeneous reaction system, the rate of mass transfer**

(a) Affects the reaction rate
(b) Does not affect the reaction rate
(c) Affects only the reaction rate of fast reactions
(d) Affects only the reaction rate of slow reactions

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**231. Effectiveness factor of a catalyst pellet is a measure of the _____ resistance.**

(a) Pore diffusion (b) Gas film
(c) Chemical reaction (d) None of these

**Karnataka PSC ADF - 2015**

**Ans. (a)**

**232. A catalyst loses its activity due to**

(a) Loss in surface area of the active component.
(b) Agglomeration of metal particles caused by thermal sintering of the solid surface
(c) Covering of the catalytic active sites by a foreign substance.
(d) All (a), (b) and (c)

**Karnataka PSC ADF - 2015**

**Ans. (d)**

**233. What is the Thiele modulus of the solid catalyzed I order reaction, A → products, if the pore diffusion offers negligible resistance to reaction?**

(a) < 4 (b) < 0.4
(c) > 1 (d) > 4

**Karnataka PSC ADF - 2015**

**Ans. (b)**

**234. The equilibrium constant of a catalytic chemical reaction _____ due to the presence of a catalyst.**

(a) Increases
(b) Decreases
(c) Remains unaffected
(d) Unpredictable from the data

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**235. A porous pellet with Pt dispersed in it is used to carry out a catalytic reaction. Following two scenarios are possible.**

**Scenario 1: Pt present throughout the pores of the pellet is used for catalyzing the reaction.**

**Scenario 2: Pt present only in the immediate vicinity of the external surface of the pellet is used for catalyzing the reaction.**

**At a large value of Thiele modulus, which one of the following statements is TRUE?**

(a) Since the reaction rate is much greater than the diffusion rate, Scenario 1 occurs
(b) Since the reaction rate is much greater than the diffusion rate, Scenario 2 occurs
(c) Since the reaction rate is much lower than the diffusion rate, Scenario 1 occurs
(d) Since the reaction rate is much lower than the diffusion rate, Scenario 2 occurs

**GATE-2016**

**Ans. (b)**

**236. For a non-catalytic homogeneous reaction A→B, the rate expression at 300 K is $-\gamma_A$ (mol $m^{-3}s^{-1}$) = $\frac{10C_A}{1+5C_A}$, where $C_A$ is the concentration of A (in mol/$m^3$). Theoretically, the upper limit for the magnitude of the reaction rate ($-\gamma_A$ in mol $m^{-3}s^{-1}$, rounded off to the first decimal place) at 300 K is—**

**GATE-2016**

**Ans. (1.99 to 2.01)**

**237. For a first order isothermal catalytic reaction A'–>' P, occurring in an infinitely long cylindrical pore, the relationship between effectiveness factor, ε and, Thiele modulus φ, is**

(a) ε = 1/φ '^2' (b) ε = φ
(c) ε = 1 (d) ε = 1/φ

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**238. For a first order isothermal chemical reaction in a porous catalyst, the effectiveness factor is 0.3. The effectiveness factor will increase if the**

(a) catalyst size is reduced or the catalyst diffusivity is reduced
(b) catalyst size is reduced or the catalyst diffusivity is increased
(c) catalyst size is increased or the catalyst diffusivity is reduced
(d) catalyst size is increased or the catalyst diffusivity is increased

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**239. A catalyst promoter**

(a) improves the activity of a catalyst
(b) acts as a catalyst support
(c) itself has very high activity
(d) all (a), (b) and (c)

**ISRO-12.02.2016**

**Ans. (a)**

**240. Match List – I and List – II select the correct answer using the code given below the lists.**

| List-I | List-II |
|---|---|
| (a) Porous catalyst | 1. Selectivity |
| (b) Parallel reactions | 2. Shrinking core model |
| (c) Non-ideal tubular reactor | 3. Thiele modulus |
| (d) Gas-solid non-catalytic reaction | 4. Dispersion number |

Code:

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 1 | 4 | 2 |
| (b) | 2 | 1 | 4 | 3 |
| (c) | 1 | 4 | 3 | 2 |
| (d) | 3 | 1 | 2 | 4 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**241. Which of the following is a controlling factor in very fast heterogeneous reaction?**

(a) Heat and mass transfer effects
(b) Pressure
(c) Temperature
(d) Composition of the reactants

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**242. Unreacted core model represents the reaction involving :**

(a) Combustion of coal particles
(b) Roasting of sulphide ores
(c) Manufacturing of $CS_2$ from clements
(d) None of the above

**AP PSC IOF-2009**

**Ans. (b)**

**243. For first order chemical reaction in a porous catalyst the Thiele modulus is 10. The effectiveness factor is nearly equal to :**

(a) 1 (b) 0.5
(c) 0.1 (d) 0

**AP PSC IOF-2009**

**Ans. (c)**

**244. Match the following**

| Group-1 | | Group-2 | |
|---|---|---|---|
| **P.** | Tank in series model | I. | Non-isothermal reaction |
| **Q.** | Liquid-liquid extraction | II. | Mixer-settler |
| **R.** | Optimum temperature progression | III. | PFR with axial mixing |
| **S.** | Thiele modulus | IV. | Solid catalyzed reaction |

(a) P-II, Q-IV, R-I, S-III
(b) P-I, Q-II, R-III, S-IV
(c) P-III, Q-I, R-II, S-IV
(d) P-III, Q-II, R-I, S-IV

**GATE-2014**

**Ans. (d)**

**245. A vapour phase catalytic reaction (Q + R $\rightarrow$S) follows Rideal mechanism (R and S are not adsorbed). Initially, the mixture contains only the reactants in equimolar ratio. The surface reaction step is rate controlling. With constants a and b, the initial rate of reaction ($-r_0$) in terms of total pressure ($P_T$) is given by**

(a) $-r_o = \frac{aP_T}{1+bP_T}$

(b) $-r_o = \frac{aP_T}{1+bP_T^2}$

(c) $-r_o = \frac{aP_T^2}{1+bP_T}$

(d) $-r_o = \frac{aP_T^2}{(1+bP_T)^2}$

**GATE-2014**

**Ans. (c)**

**246. A catalyst slab of half-thickness L (the width and length of the slab >> L) is used to conduct the first order reaction A $\rightarrow$B. At 450K, the Thiele modulus for this system is 0.5. The activation energy for the first order rate constant is 100kJ/mol. The effective diffusivity of the reactant in the slab can be assumed to be independent of temperature, and external mass transfer resistance can be neglected. If the temperature of the reaction is increased to 470K, then the effectiveness factor at 470 K effectiveness factor at 470 K (up to two decimal places) will be _____**

**Value of universal gas constant = 8.314 J/mol. K**

**GATE-2015**

**Ans. (0.7 to 0.9)**

**247. Which one of the following is not the resistance in a gas-liquid reaction?**

(a) Gas film resistance
(b) Liquid film resistance
(c) Magnitude of convection resistance
(d) Bulk liquid resistance

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**248. Which one of the following is not an advantage for three phase catalytic reactors?**

(a) Normal operating conditions
(b) Long catalyst life and high sensitivity
(c) High heat transfer efficiency
(d) Endothermic reactions can be controlled

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**249. Which one of the following catalyst poisons is true for the statement below?**

**'It results due to change in structure of the catalyst. An example of this poison is that, when $SO_2$- air mixture along with moisture is passed over platinum-aluminium catalyst, the water affects the structure of alumina carrier'.**

(a) Deposited poison
(b) Chemisorbed poison
(c) Selectivity poison
(d) Stability poison

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**250. The ratio of the heterogeneous catalytic reaction [A(g)+B(g) $\rightarrow$ c(g)] is:**

$$-r_A = \frac{kk_A P_A P_B}{1+K_A P_A + K_C P_C} \; k_A \text{ and } k_C$$

**are adsorption constants. The rate controlling step is**

(a) Adsorption of A
(b) Surface reaction b/w adsorbed A and adsorbed B
(c) Surface reaction between adsorbed A and B in the gas phase
(d) Surface reaction between A in the gas phase and adsorbed B

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Surface reaction between adsorbed A and B in the gas phase

**251. A catalyst is a substance which:**

(a) Increases the equilibrium concentration of the product
(b) Change the equilibrium constant of the reaction
(c) Supplies energy to the reaction
(d) Decreases the Activation energy

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Decreases the Activation energy

## G. Non-ideal Reactors

**252. What is the dispersion number for a CSTR?**

(a) 0 (b) 1

(c) <1 (d) ∞

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (d)**

**253. Tubular reactor with axial flow falls into which category of model?**

(a) Steady state

(b) Unsteady state

(c) Lumped parameter

(d) Distributed parameter

**TNPSC AE-2018**

**Ans. (d)**

**254. Fluid flow in a real packed bed can be approximated as _____model.**

(a) Plug flow (b) Dispersion

(c) Mixed flow (d) Tanks in series

**TNPSC AE-2018**

**Ans. (b)**

**255. The exit age distribution of fluid leaving a vessel is used to**

(a) study the reaction kinetics

(b) study the extent of non-ideal flow in the venus

(c) study the reaction mechanism

(d) know activation energy of a reaction

**TNPSC AE-2019**

**Ans. (b)**

**256. The vessel dispersion number for plus flow reactor is equal to**

(a) 0

(b) between 2100-4000

(c) between 4000-10,000

(d) ∞

**TNPSC AE-2019**

**Ans. (a)**

**257. When the vessel dispersion number $\left(\frac{D}{uL}\right)$ is approximately zero, then the flow is called_____**

(a) Mixed flow

(b) Plug flow

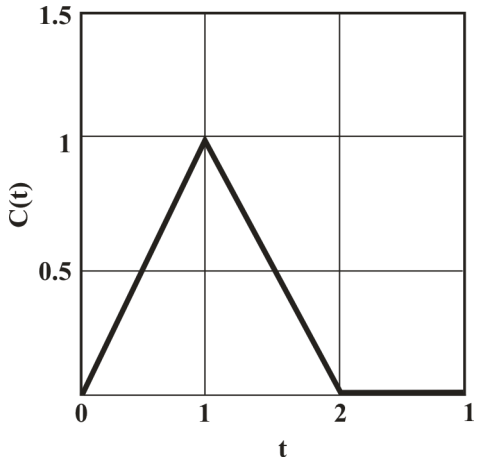
(c) Laminar flow

(d) Transition flow

**TNPSC AE-2019**

**Ans. (b)**

**258. The C-curve measured during a pulse trACFr experiment is shown below. In the figure, C(t) is the concentration of the trACFr measured at the reactor exit in mol/liter at time t seconds.**




**The mean residence time in the reactor, rounded to 1 decimal plACF, is _____ s.**

**GATE-2017**

**Ans. (1.0 to 1.0)**

**259. The dispersion number of perfect mixed flow is:**

(a) 0 (b) > 500

(c) Infinity (d) < 2100

**RPSC ACF-2011**

**Ans. (c)**

**260. The eddy diffusivity for a liquid in a plug flow must be**

(a) 1 (b) 0

(c) Infinity (d) Between to 1

**RPSC ACF-2011**

**Ans. (b)**

**261. The vessel dispersion number of an ideal CSTR is.**

(A) -1 (B) 0

(C) 1 (D) ∞

(a) A (b) B

(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**262. Mean residence time of flowing material in reactor given by the below mentioned relation is known as what?**

$$C_{A0}\int_0^{X_A}\frac{dX_A}{(-r_A)(1+\varepsilon_A X_A)},[hr]$$

(a) SpACF time (b) Reaction time

(c) Holding time (d) Half life

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**263. The exit age distribution of a fluid leaving a vessel is used to study the extent of non-ideal flow in the vessel. What is the value of $\int_0^\infty E.dt$**

(a) 0 (b) ∞
(c) $2\pi$ (d) 1

**Karnataka PSC ADF - 2015**

**Ans. (d)**

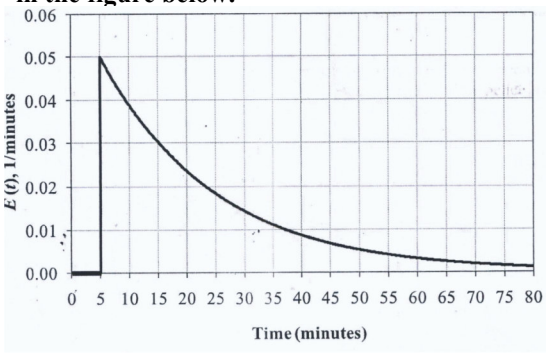
**264. The zero parameter model proposed for describing non-ideal flow is the _____.**

(a) Segregation model
(b) Maximum mixedness model
(c) Both (a) & (b)
(d) Neither (a) nor (b)

**Karnataka PSC ADF - 2015**

**Ans. (c)**

**265. A CSTR has a long inlet pipe. A tracer is injected at the entrance of the pipe. The E-curve obtained at the exit of the CSTR is shown in the figure below.**




**Assuming plug flow in the inlet pipe, the ratio (rounded off to the second decimal place) of the volume of the pipe to that of the CSTR is—**

**GATE-2016**

**Ans. (0.23 to 0.27)**

**266. Which of the following is NOT correctly matched?**

| | |
|---|---|
| (a) Non-ideal tubular reactor | – Tank in series model |
| (b) Zero-order irreversible reaction in batch reactor | – Reaction time for 100% conversion of reactant is equal to two times its half life |
| (c)Arrhenius equation | – The ratio of reaction rate constant to frequency factor is always more than 1 |
| (d) Recycle reactor | – The behavior of recycle reactor changes from plug flow (for zero recycle ratio) to mixed flow (for infinite recycle ratio) |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

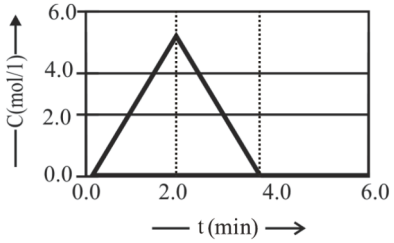
**267. The vessel dispersion number for an ideal CSTR is**

(a) –1 (b) 0
(c) 1 (d) ∞

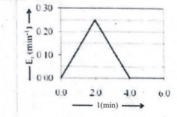
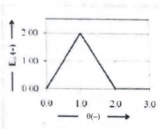
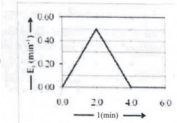
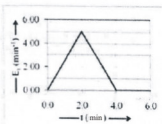
**GATE-2014**

**Ans. (d)**

**268. The impulse response to a tracer pulse experiment for a flow reactor is given below.**




**In the above figure, C is the exit tracer concentration. The corresponding E or $E_\theta$ (normalized E) curve is correctly represented by which of the following choices? Here, θ is dimensionless time.**

(a) 
(b) 
(c) 
(d) 

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**269. Tracer input signal is used to:**

(a) Study the reaction mechanism
(b) Study the extent of non ideal flow in the reactor
(c) Know reaction rate constants
(d) Know Activation energy

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Study the extent of non ideal flow in the reactor

**270. For an ideal CSTR the residence time distribution is given by:**

(a) $\frac{1}{\tau}\exp(-t/\tau)$ (b) $\tau \exp(-t/\tau)$
(c) $\exp(-t/\tau)$ (d) $\frac{1}{\tau}\exp(-\tau/t)$

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** $\frac{1}{\tau}\exp(-t/\tau)$

## H. Non-isothermal Reactions

**271. For an exothermic reaction, the equilibrium constant**

(a) remains unaffected with change in temperature
(b) decrease with increase in temperature
(c) increase with increase in temperature

(d) increase linearly with increase in temperature
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**272. Consider the reaction**

$$A \xrightarrow{k_1(1st\ order)} R \xrightarrow{k_2(1st\ order)} S$$

**R is the desired product. Step (1) has activation energy of 20 kcal/mol and Step (2) has activation energy of 10 kcal/mol. In order to promote the production of R, the reaction should be conducted**

(a) at the lowest possible temperature
(b) at the highest possible temperature
(c) at an intermediate temperature (between the highest and the lowest temperature)
(d) in accordance with a temperature progression; low temperature followed by high temperature

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**273. Reaction A → B is carried out in a reactor operating at steady state and 1 mole/s of pure A at 425°C enters the reactor. The outlet stream leaves the reactor at 325°C The heat input to the reactor is 17 kW. The heat of reaction at the reference temperature of 25°C is 30 kJ/mol. The specific heat capacities (in kJ/mol.K) of A and B are 0.1 and 0.15 respectively.**
**The molar flowrate of B leaving the reactor, rounded to decimal plACFs, is ____ mol/s/.**

**GATE-2017**

**Ans. (0.59 to 0.61)**

**274. Value of heat of formation which indicates that products is least stable, is**

(a) –64.8 Kcal (b) –94 Kcal
(c) 64.8 Kcal (d) 94 Kcal

**RPSC ACF-2011**

**Ans. (d)**

**275. Enthalpy for the reaction $C + O_2 \rightarrow CO_2$ is**

(a) Positive (b) Negative
(c) Zero (d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**276. An exothermic gas phase reaction proceeds according to the equation $3A + 2B \rightleftharpoons 2R$**
**What will be the behavior of equilibrium conversion for this reaction?**

(a) equilibrium conversion is unaffected by the presence of a catalyst
(b) equilibrium conversion decrease on dilution with an inert gas
(c) equilibrium conversion increase with an increase in pressure
(d) equilibrium conversion increases with increase in temperature

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**277. A batch adiabatic reactor at an initial temperature of 373 K is being used for the reaction A → B. Assume the heat of reaction is (–1) kJ/mol at 373 K and the heat capacity of both A and B to be constant and equal to 50 J/mol K. The temperature rise after a conversion of 0.5 will be**

(a) 5ºC (b) 10ºC
(c) 20ºC (d) 100ºC

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (b)**

**278. Peclet number is ratio of the convective mass and diffussive mass transfers in a reactor. It may vary from zero to infinity. For an ideal well mixed reactor, it is**

(a) Zero
(b) Infinity
(c) More than 0 but less than infinity
(d) Nearly equal to zero

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**279. For the reversible exothermic reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, increase of pressure would**

(a) Shift the equilibrium towards right
(b) Give higher yield of $NH_3$
(c) Both (a) and (b)
(d) Neither (a) nor (b)

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**280. When an exothermic reversible reaction is conducted adiabatically the rate of reaction :**

(a) Continuously increases
(b) Passes through a maximum
(c) Continuously a decreases
(d) Passes through a minimum

**AP PSC IOF-2009**

**Ans. (b)**

**281. Exothermic reactions are best carried out in :**

(a) CSTR
(b) CSTR's in series
(c) PFR followed by SCTR
(d) CSTR followed by PFR

**AP PSC IOF-2009**

**Ans. (d)**

## I. Bio-chemical Reactions

**282. Michaelis and Menten guessed the mechanism of which kind of reactions?**

(a) Elementary reactions
(b) Enzyme substrate reaction
(c) Zero order reactions
(d) Reversible reaction

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**283. A bio-catalyst produced by living cells which acts independent of the cell is called :**

(a) Substrate (b) Nutrient
(c) Enzyme (d) Vitamin

**AP PSC IOF-2009**

**Ans. (c)**

**284. The ratio of the substrate concentration required for 75 % of $R_{max}$ to the concentration required for 25 % of $R_{max}$ is:**

(a) 7 (b) 8
(c) 9 (d) 6

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

# 09.

# Chemical Technology

## A. Inorganic Chemicals

**1. The structural formula of Acetic acid is represented by**

(a) $CH_3COOH$ (b) $CH_2COOH$

(c) $C_3H_6O_3$ (d) $C_2H_5OH$

**TNPSC AE-2019**

**Ans. (a)**

**2. Platinum- 10% Rhodium gauze catalyst in used in the manufacture of**

(a) Hydrochloric acid (b) Phosphoric acid

(c) Nitric acid (d) Sulphuric acid

**TNPSC AE-2019**

**Ans. (c)**

**3. The standard solvay process is used for the manufacture of**

(a) Caustic soda

(b) Soda ash

(c) Caustic potash

(d) Soda lime

**TNPSC AE-2019**

**Ans. (b)**

**4. The chemical formula for caustic soda is**

(a) $Na_2CO_3$ (b) NaOH

(c) $NaHCO_3$ (d) NaOCl

**TNPSC AE-2019**

**Ans. (b)**

**5. Solvay process refers to manufacture of**

(a) Sodium Hydroxide

(b) sulphuric acid

(c) Nitric acid

(d) Sodium carbonate

**TNPSC AE-2019**

**Ans. (d)**

**6. Vanadium pentoxide is used as catalyst for manufacture of**

(a) Sulphuric acid (b) Nitric acid

(c) Hydrochloric acid (d) Phosphoric acid

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**7. Washing Soda is**

(a) $Na_2CO_3$ (b) $CaSO_4$

(c) $Na_2CO_3 .10H_2O$ (d) $CaSO_4 \cdot \frac{1}{2}H_2O$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**8. In beer production, the regiment after maturing process, contains yeast, precipitated proteins and tennis is known as**

(a) lager

(b) chill

(c) draff

(d) hops

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**9. A number of metals are available in earth's crust but most abundant elements are:**

(a) Al and Fe (b) Al and Cu

(c) Fe and Cu (d) Cu and Ag

**WB PSC Scientific Assistant-2019**

**Ans. (a)**

**10. Diphenyl carbazide in acetone is used as a reagent for colourimentric estimation of**

(a) Mn (VII) (b) Cr (VI)

(c) Ni (II) (d) Fe (II)

**WB PSC Scientific Assistant-2019**

**Ans. (b)**

**11. For storing 100% sulphuric acid which of the following material will be used**

(a) Ordinary carbon steel (b) Aluminum

(c) SS 304 (d) Lead

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**12. Purging is necessary in the production of ammonia because**

(a) nitrogen is in excess

(b) argon gets accumulated

(c) reactants are not in stoichiometric proportions

(d) conversion is not 100%

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**13. Ammonia separation process is performed by**

(a) absorption process
(b) distillation process
(c) diffusion process
(d) humidification process

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**14. The ammonia ($NH_3$) oxidation process occurs over a catalyst as**

$$4NH_3 + 5O_2 \rightarrow 6H_2O + 4NO$$

**Air is supplied such that oxygen ($O_2$) is 20% in excess of that required for complete conversion of $NH_3$. The mole fraction of $O_2$ in inlet gas mixture ($NH_3$ + air) is _____(rounded off a third decimal plACF)**

**GATE-2018**

**Ans. (0.170 to 0.200)**

**15. Match the equipment in Group-1 with the process in Group-2**

| Group-1 | Group-2 |
|---|---|
| (P) Fluidized bed | (I) Paper-making |
| (Q) Multistage adiabatic reactor with inter-stage cooling | (II) Sodium hydroxide manufacture |
| (R) Fourdrinier machine | (III) $SO_2$ oxidation |
| (S) Diaphragm cell | (IV) Catalytic cracking |

**Choose the correct set of combinations.**

(a) P-IV, Q-III, R-I, S-II
(b) P-IV, Q-III, R-II, S-I
(c) P-III, Q-IV, R-I, S-II
(d) P-III, Q-IV, R-II, S-I

**GATE-2017**

**Ans. (a)**

**16. The purpose of methanation reaction used in ammonia plants is to**

(a) remove CO as it is a catalyst poison
(b) increase the amount of hydrogen
(c) remove sulphur as it is a catalyst poison
(d) utilize methane as a catalyst for ammonia synthesis

**GATE-2017**

**Ans. (a)**

**17. The DCDA (Double Contact Double Absorption) process is used for the manufacture of**

(a) urea (b) sulphuric acid
(c) nitric acid (d) ammonia

**GATE-2017**

**Ans. (b)**

**18. Catalyst used in Fisher-Tropsch process is**

(a) Nickel (b) Zinc oxide
(c) Alumina (d) Thorium oxide

**RPSC ACF-2011**

**Ans. (a)**

**19. In sulfuric acid manufacture, sulfure trioxide is absorbed in98-99% sulfuric acid in a packed absorber. The material of construction of the absorber is usually**

(a) Copper and copper -base alloys
(b) Aluminium
(c) Brick - lined steel or cast iron
(d) Nickel and nickel - base alloys

**RPSC ACF-2011**

**Ans. (d)**

**20. Catalyst employed for oxidation of ammonia to nitric oxide is.**

(A) γ alumina (B) Iron
(C) Platinum-Rhodium (d) Nickel
(D) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**21. Hypo solution is**

(a) Sodium chloride (b) Sodium bisulphite
(c) Sodium thiosulphate (d) Soda ash

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**22. _____ Caustic soda is manufactured by**

(a) Hayber process
(b) Membrane process
(c) Crystallization process
(d) Arhenius process

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**23. Intermediate compound produced during the manufacturing of Sulphuric acid by DCDA process**

(a) Oleum
(b) Alpha-Sulphuric acid
(c) Sulphonal
(d) Sulphomate

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**24. Tri Nitro Phenol is**

(a) PETN (b) RDX acid
(c) Picric acid (d) TNT phenyl

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**25. The catalyst used in the manufacture of sulphuric acid by contact process is**

(a) $V_2O_5$ (b) Cu

(c) $Fe/SiO_2$ (d) $Ni/Al_2O_3$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**26. In contact process the catalytic converter for conversion of $SO_2$ to $SO_3$ should have a feed with $SO_2$ content between**

(a) 7 – 10% (b) 20 – 25%

(c) 13 – 16% (d) 3 – 6%

**UPPSC AE-29.05.2022, Paper-2**
**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**27. What products do we get on electrolysis of saturated brine using steel cathode and graphite anode in electrolytic cell?**

(a) $Cl_2$ & Na (b) $Cl_2$ & $H_2$

(c) $O_2$ & $H_2$ (d) $Cl_2$, $H_2$ & NaOH

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**28. The Oxidation of ammonia to produce nitric acid is an example of**

(a) Homogenous non-catalytic reaction

(b) Homogenous catalytic reaction

(c) Heterogeneous non-catalytic reaction

(d) Heterogeneous catalytic reaction

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**29. DCDA process is the most recent process for the manufacture of**

(a) $HNO_3$ (b) $H_2SO_4$

(c) HCl (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**30. The percent available chlorine in a good commercial sample of bleaching powder is**

(a) 15% to 17% (b) 35% to 37%

(c) 53% to 56% (d) 69% to 71%

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**31. Ammonia synthesis reaction is**

(a) exothermic (b) endothermic

(c) autocatalytic (d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**32. Washing soda is**

(a) $Na_2CO_3$ (b) $Na_2CO_3.H_2O$

(c) $Na_2CO_3.10H_2O$ (d) $NaHCO_3$

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**33. Bromine is used in the preparation of**

(a) fire extinguishing compounds

(b) fireproofing agents

(c) dyes and antiknock compounds

(d) All of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**34. Contact process**

(a) yields acid of higher concentration than chamber process

(b) yields acid of lower concentration than chamber process

(c) is obsolete

(d) eliminates absorber

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**35. India has no elemental sulphur deposits that can be economically exploited. In India, which one of the following industries produces elemental sulphur as a by-product?**

(a) Coal carbonisation plants

(b) Petroleum refineries

(c) Paper and pulp industries

(d) Iron and steel making plants

**GATE-2016**

**Ans. (b)**

**36. Double sulphate of Aluminium or Chromium with monovalent metal is called as**

(a) Illuminite (b) Gypsum

(c) Alum (d) Alumina

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**37. As per US Bureau of Mines, major constituent in the black blasting powder**

(a) Ammonium perchlorate

(b) TNT

(c) RDX

(d) Nitrate of Sodium/Potassium

**ISRO Scientist Engineer-06.08.2017**

**Ans. (d)**

**38. In the Titanium dioxide manufacturing process, chemical used for reducing**

(a) Scrap Titanium tetrachloride

(b) Scrap iron

(c) Scrap Anatase

(d) Scrap Rutile

**ISRO Scientist Engineer-06.08.2017**

**Ans. (a)**

**39. Reaction from Suphur dioxide into Sulphur trioxide can be categorized as**

(a) Exothermic irreversible
(b) Endothermic irreversible
(c) Exothermic reversible
(d) Endothermic reversible

**ISRO Scientist Engineer-06.08.2017**

**Ans. (c)**

**40. Double Contact Double Absorption (DCDA) process is used for the manufacture of which of the following?**

(a) Nitric acid (b) Hydrochloric acid
(c) Sulphuric acid (d) Phosphoric acid

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**41. In caustic soda plant cathode of mercury cell is made of**

(a) Crimped steel wire
(b) Graphite
(c) Activated carbon
(d) Copper

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**42. In industrial nomenclature, alcohol means**

(a) Butanol (b) Propenol
(c) Ethanol (d) Methanol

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**43. What is produced from solvay ammonia soda process?**

(a) Ammonia (b) Sodium carbonate
(c) Calcium carbonate (d) Brine

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**44. Sulphuric acid is mainly used in the_____ industry.**

(a) Fertilizer (b) Steel
(c) Paper (d) Paint

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**45. Salt cake is used in the manufacture of :**

(a) $Na_2SO_4$ (b) $MgSO_4$
(c) $BaSO_4$ (d) $CaSO_4.\ 1/2\ H_2O$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**46. Double Contact Double Absorption (DCDA) process is the most recent process for the manufacture of:**

(a) Nitric acid (b) Sulphuric acid
(c) Ammonium sulphate (d) Hydrochloric acid

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**47. In the Solvay process, the product from the calciner is:**

(a) Light soda ash
(b) Dense soda ash
(c) Sodium bicarbonate
(d) Dehydrated soda ash

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**48. Which of the following is a co-product during the manufacture of caustic soda by electrolysis of brine?**

(a) $Na_2CO_3$
(b) $NaHCO_3$
(c) $Cl_2$
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**49. Which of the following may be viewed as catalyst in the manufacture of soda ash by solvay process ?**

(a) $NH_3$ (b) NaCl
(c) CaO (d) Coke

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**50. Lime and soda ash are added to water to remove:**

(a) Bicarbonate and sulphates of calcium and magnesium
(b) Undesirable taste and odour
(c) Bacteria
(d) Its corrosiveness

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**51. 20% oleum means that in 100 kg oleum, there are 20 kg of:**

(a) $SO_3$ and 80 kg of $H_2SO_4$
(b) $H_2SO_4$ and 80 kg of $SO_3$
(c) $SO_3$ for each 100 kg of $H_2SO_4$
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**52. The raw materials required for the manufacture of DDT are:**

(a) Ethanol and Chlorine
(b) Benzene and Chlorine
(c) Ethanol and Benzene
(d) Benzene and Methane

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**53. Which one of the following catalyst can be used in the manufacturing of Sulphuric Acid by 'Contact Process'?**

(a) Vanadium Pentoxide (b) Palladium
(c) Cobalt (d) Titanium

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**54. Chemical name of caustic soda is:**

(a) Sodium hydroxide
(b) Sodium carbonate
(c) Sodium bicarbonate
(d) Potassium carbonate

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**55. Raw materials for solvay process are:**

(a) Salt, limestone and coke or gas
(b) Ammonia, salt and limestone
(c) Ammonia, limestone and coke
(d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**56. Catalyst used in the manufacture of sulphuric acid by contact process is:**

(a) $Cr_2O_3$
(b) $V_2O_5$
(c) Iron
(d) Oxides of nitrogen

**Kerala PSC Lect. 05.02.2015**

**Ans. (b)**

**57. In the Haber's process for the manufacture of ammonia:**

(a) Finely divided iron is used as catalyst
(b) Finely divided molybdenum is used as catalyst
(c) Finely divided nickel is used as catalyst
(d) No catalyst is required

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Finely divided iron is used as catalyst

**58. Methyl alcohol from wood is manufactured by:**

(a) Fractional distillation
(b) Dehydration
(c) Fermentation
(d) Destructive distillation

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Destructive distillation

## B. Fertilizers

**59. The available $P_2O_5$ is Triple super phosphate is**

(a) 1 to 11% (b) 12 to 33%
(c) 34 to 43% (d) 44 to 51%

**TNPSC AE-2018**

**Ans. (d)**

**60. The maximum percentage content of $N_2$ in Urea is**

(a) 16% (b) 26%
(c) 46% (d) 66%

**TNPSC AE-2018**

**Ans. (c)**

**61. In gas absorption, the point at which the liquid hold up starts to increase as judged by a charge in the slope of the pressure drop line is called**

(a) Loading point (b) Flooding point
(c) Wetting point (d) Hold up point

**TNPSC AE-2019**

**Ans. (a)**

**62. The Hargreaves method is for the manufacture of**

(a) Sodium carbonate
(b) Potassium sulphate
(c) Calcium sulphate
(d) Ammonium sulphate

**TNPSC AE-2019**

**Ans. (b)**

**63. Undesirable side reaction in the product of urea is the formation of**

(a) Ammonium Carbonate
(b) Biuret
(c) Tetrauret
(d) Water
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**64. Triple superphosphate is manufactured by reacting**

(a) phosphate rock with phosphoric acid
(b) phosphate rock with sulphuric acid
(c) phosphate rock with nitric acid
(d) ammonium phosphate with phosphoric acid

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**65. Urea is a nitrogenous fertilizer. Its nitrogen content in wt% is about**

(a) 35 (b) 82
(c) 25 (d) 46

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**66. Maximum nitrogen percentage is in**

(a) ammonium sulphate
(b) liquid ammonia
(c) urea
(d) calcium ammonium nitrate

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**67. Action of phosphoric acid on rock phosphate, produces**

(a) Super phosphate
(b) Diammonium phosphate
(c) Nitrophosphate
(d) Triple super phosphate

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**68. Monoammonium phosphate is represented by the following formula**

(a) $NH_4H_2PO_4$ (b) $(NH_4)_2(PO_4)_2$
(c) $(NH_4)HPO_4$ (d) $(NH_4)_2HPO_4$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**69. Match the product in Group with the nature of the reaction in Group II.**

| Group I | Group II |
|---|---|
| A. Urea | 1. Ammonia and carbon dioxide |
| B. Polyester | 2. Dimethyl terephthalate and ethylene glycol |
| | 3. Ammonia and carbon monoxide |
| | 4. Hexamethylene diamine and adipic acid |

(a) A-1, B – 4 (b) A-3, B – 3
(c) A-3, B – 4 (d) A-1, B – 2

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**70. Pick out the wrong statement**

(a) All the nitrogenous fertilizers are not soluble in water.
(b) A straight fertilizer contains only one nutrient
(c) Calcium cynamide is used as weed killer in onion fields
(d) The phosphorous nutrient makes the plant stem stronger and increases its branches

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**71. Triple superphosphate is made by reacting phosphoric rock with**

(a) phosphoric acid (b) nitric acid
(c) sulphuric acid (d) hydrochloric acid

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**72. Triple superphosphate is manufactured by reacting**

(a) Phosphate rock with phosphoric acid
(b) Phosphate rock with sulphuric acid
(c) Phosphate rock with nitric acid
(d) Ammonium phosphate with phosphoric acid

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**73. Match the technologies in Group 1 with the entries in Group 2 :**

| Group-1 | Group-2 |
|---|---|
| **P. Urea manufacture** | **I. Microencapsulation** |
| **Q. Coal gasification** | **II. Ultra-low sulphur diesel** |
| **R. Controlled release of chemicals** | **III. Shale oil** |
| **S. Deep hydrodesulphurization** | **IV. Prilling tower** |
| | **V. Gas hydrates** |
| | **VI. Gas-solid non-catalytic reaction** |

**1. P-I, Q-V, R-II, S-VI**
**2. P-IV, Q-VI, R-I, S-II**
**3. P-IV, Q-I, R-III, S-II**
**4. P-V, Q-VI, R-IV, S-II**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (b)**

**74. Match the chemicals written on the left with the raw materials required to produce them mentioned on the right.**

| | |
|---|---|
| **(I) Single Superphosphate (SSP)** | **(P) Rock phosphate + Sulfuric Acid + Ammonia** |
| **(II) Triple Superphosphate (TSP)** | **(Q) Brine** |
| **(III) Diammonium Phosphate (DAP)** | **(R) Rock phosphate + Sulfuric Acid** |
| **(IV) Caustic Soda** | **(S) Rock phosphate + Phosphoric Acid** |

**(1) I-Q, II-R, III-S, IV-P**
**(2) I-S, II-P, III-Q, IV-R**
**(3) I-R, II-S, III-P, IV-Q**
**(4) I-S, II-R, III-P, IV-Q**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**75. Which of the following compound represents monoammonium phosphate?**

(a) $NH_3 \cdot HPO_4$ (b) $(NH_3)_2PO_4$

(c) $NH_4 \cdot H_2PO_4$ (d) $NH_2CONH_2$

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**76. Who produced ammonia in Germany during 1904-13?**

(a) Fritz Haber (b) Kent

(c) Riegel (d) None of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**77. A potassic fertiliser contains 50% $K_2O$. What it could be?**

(a) Potassium chloride

(b) A mixture of KCl + NaCl

(c) Potassium sulphate

(d) Potassium permangnate

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**78. Which one of the following is used as a nitrogeneous fertiliser, as a weed killer in the onion fields and for correcting acidic soils?**

(a) Urea

(b) CAN (Calcium Ammonium Nitrate)

(c) Ammonium sulphate

(d) Calcium cynamide

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**79. Phosphoric acid is prepared from :**

(a) Cryolite

(b) Chalcopyrite

(c) Rock phosphate

(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**80. Which one of the following constituent is necessary in fertilizers for the early stages of plant growth to promote development of stems and leaves?**

(a) Phosphorus (b) Potassium

(c) Nitrogen (d) Potash

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**81. Which of the following is a method for production of phosphoric acid?**

(a) Solvay process

(b) Sulphate process

(c) Electric furnace process

(d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

## C. Natural products (Oils/Fats/Sugar/ Paper and pulp)

**82. In cassava starch, the average starch content varies from**

(a) 20 – 30% (b) 10 – 15%

(c) 35 – 45% (d) 50 – 60%

**TNPSC AE-2018**

**Ans. (a)**

**83. Yeast is used in the manufacture of**

(a) Penicillin (b) Streptomycin

(c) Wine (d) Lactose

**TNPSC AE-2018**

**Ans. (c)**

**84. Fourdrinier machine used in the manufacture of**

(a) soap (b) detergent

(c) paper (d) leather

**TNPSC AE-2018**

**Ans. (c)**

**85. The first step in refining of cane-sugar is**

(a) evaporation (b) affination

(c) clarification (d) bleaching

**TNPSC AE-2018**

**Ans. (b)**

**86. The wash liquor from a paper mill contains 2% (by weight) of solids is concentrated in an evaporator to yield product with 20% (by weight) of solids. The quantity of water evaporated per 100 Kg of feed is.**

(a) 20 Kg (b) 80 Kg

(c) 40 Kg (d) 90 Kg

**TNPSC AE-2018**

**Ans. (d)**

**87. Sugar cane contains_____sucrose by weight**

(a) 6 to 8% (b) 11 to 15%

(c) 20 to 25% (d) 2 to 5%

**TNPSC AE-2019**

**Ans. (b)**

**88. Generally, the food pulp is admitted in thickener is**

(a) at the top

(b) slightly below the surface of the liquid

(c) at the middle

(d) near the bottom

**TNPSC AE-2019**

**Ans. (b)**

**89. Soaps form insoluble compounds with the calcium and magnesium ions present in**

(a) Soft water
(b) Heavy water
(c) Hard water
(d) Double distilled water

**TNPSC AE-2019**

**Ans. (c)**

**90. In pulp bleaching process which chemical is used as oxidative bleacher.**

(a) Sodium borohydride
(b) Sodium bisulfite
(c) Sodium dithionante
(d) Hydrogen peroxide

**TNPSC AE-2019**

**Ans. (d)**

**91. The catalyst normally used for hydrogenation of vegetable oil is**

(a) Nickel compound
(b) Iron compound
(c) Aluminium compound
(d) Copper compound

**TNPSC AE-2019**

**Ans. (a)**

**92. Kraft process can also be known as**

(a) Ammonia process
(b) Bisulphite process
(c) Sulphite process
(d) Sulphate process

**TNPSC AE-2019**

**Ans. (d)**

**93. Vegetable oils are recovered from oil seeds by leaching with**

(a) Hexane (b) Hot sulphuric acid
(c) Nitric acid (d) Cold water

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**94. The main raw material for paper and pulp manufacture**

(a) Starch (b) Fatty acids
(c) Sugars (d) Cellulose
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**95. Most commonly used fillers in the paper making process is**

(a) Talc and $TiO_2$
(b) Talc and $SiO_2$
(c) Talc and CaO
(d) Talc and $MnO_2$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**96. Oil is**

(a) a mixture of glycerides of alcohols
(b) a mixture of glycerides of fatty acids
(c) a solid at normal temperature
(d) a ester of alcohols other than glycerine
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**97. Mixtures of glycerides of fatty acids are called as**

(a) Fats and Oils (b) Paints
(c) Pulp (d) Waxes
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**98. Colour of the pulp obtained from K raft process is**

(a) Brown (b) White
(c) Black (d) Yellow
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**99. Sucrose is a**

(a) monosaccharide
(b) disaccharide
(c) trisaccharide
(d) polysaccharide
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**100. Sugar beet contains _____ of sucrose.**

(a) 3 to 6 % (b) 7 to 12 %
(c) 13 to 17 % (d) 20 to 25 %
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**101. In the froth floatation process, pine oil and cresylic acid are used as**

(a) frother (b) collector
(c) depressor (d) conditioner

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**102. For the hydrogenation of oils, _____ is used as a catalyst and _____ is a catalyst poison.**

(a) platinum, sulphur
(b) platinum, oxygen
(c) nickel, sulphur
(d) nickel, oxygen

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**103. The molecular formula of the predominant chemical compound in commercial sugar is**

(a) $C_{12}H_{22}O_{11}$ (b) $C_{12}H_{24}O_{12}$
(c) $C_6H_{10}O_5$ (d) $C_6H_{12}O_6$

**GATE-2018**

**Ans. (a)**

**104. Pine oil used in forth flotation technique acts as a**

(a) Collector (b) Modifier
(c) Frother (d) Activator

**RPSC ACF-2011**

**Ans. (c)**

**105. Black liquor generated during paper manufacture is concentrated in a**

(a) Single effect evaporator.
(b) Single effect evaporator followed by a crystalliser
(c) multiple effect evaporator.
(d) Multiple effect evaporators followed by a crystalliser.

**RPSC ACF-2011**

**Ans. (c)**

**106. Soap solution cannot be used with hard water because.**

(A) Hard water contains sulphate
(B) They form insoluble calcium soap with precipitate
(C) They attract back, the removed dust
(D) None of these

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**107. Kraft process is used for the manufacturing of**

(a) Paper (b) Paint
(c) Explosives (d) Sodium azide

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**108. Product of reaction between fatty acids with alkalis**

(a) Soap (b) Glyceride
(c) Wax (d) Paint

**ISRO Technical Asstt.-25.02.2018**

**Ans. (a)**

**109. Why is salt added in the kettle during soap manufacture?**

(a) To separate soap from lye
(b) To separate the unsaponified fat from soap
(c) To separate glycerin from lye
(d) To separate the metallic soap

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**110. The process that is used in depithing of bagasse is known as?**

(a) Dry method process
(b) Wet method process
(c) Bleaching process
(d) Ritter process

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**111. What temperature and pH are maintained in the diffuser during the production of sugar from sugar cane cossettes?**

(a) 45°C temperature and 4.5 pH
(b) 65°C temperature and 6.5 pH
(c) 25°C temperature and 2.5 pH
(d) 85°C temperature and 8.5 pH

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**112. Which type of perfume is benzyl ACFtate that is generally used in soaps?**

(a) Rose (b) Jasmine
(c) Lilac (d) Musk

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**113. Which of the following is a detergent?**

(a) Polyvinyl alcohol
(b) Benzene hexachloride
(c) Alkyl benzene sulphonate
(d) Cellulose nitrate

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**114. Molasses can be converted into ethyl alcohol by**

(a) Crystallization (b) Distillation
(c) Fermentation (d) Hydrogenation

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**115. Oils and fats are**

(a) higher alcohols
(b) carbohydrates
(c) alkaloids
(d) esters of higher acids

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**116. Which of the following is NOT likely to be a constituent of vegetable oils?**

(a) citric acid (b) glycerol
(c) stearic acid (d) oleic acid

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**117. Essential oils are usually obtained using**

(a) steam distillation
(b) leaching
(c) solvent extraction
(d) extractive distillation

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**118. For the hydrogenation of oils, the commonly used catalyst (x) and catalyst poison (y) are**

| | (x) | (y) |
|---|---|---|
| (a) | nickel | oxygen |
| (b) | palladium | oxygen |
| (c) | platinum | sulphur |
| (d) | nickel | sulphur |

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**119. Conversion of various unsaturated radicals of fatty glycerides into saturated glycerides is defined as**

(a) Saponification (b) Hydrogenation
(c) Esterification (d) Fat filtration

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**120. Two paper pulp plants P and Q use the same quality of bamboo as a raw material. The chemicals used in their digester are as follows:**

| | Plant P | Plant Q |
|---|---|---|
| NaOH | Yes | No |
| $Na_2S$ | Yes | No |
| $Na_2CO_3$ | Yes | Yes |
| $NaHCO_3$ | No | Yes |
| $Na_2SO_3$ | No | Yes |

**Which one of the following statements is CORRECT?**

(a) Plant P and Plant Q both use the Sulfite process
(b) Plant P and Plant Q both use the Kraft process
(c) Plant P uses Sulfite process
(d) Plant P uses Kraft process

**GATE-2016**

**Ans. (d)**

**121. One of the steps during refining of cane sugar consists of addition of hydrated lime to the sugar syrup followed by carbonation of the resulting solution. The purpose of this step is to**

(a) adjust the pH of the syrup
(b) remove the colouring matter from the syrup
(c) reduce the viscosity of the syrup
(d) improve the rate of crystallization of sugar

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**122. Sucrose is a disaccharide consisting of :**

(a) Glucose and maltose
(b) Glucose and fructose
(c) Glucose and galactose
(d) Fructose and galactose

**AP PSC IOF-2009**

**Ans. (b)**

**123. Which one of the following is not likely to be the constituent of vegetable oil ?**

(a) Oleic acid (b) Stearic acid
(c) Citric acid (d) Glycerol

**AP PSC IOF-2009**

**Ans. (c)**

**124. A bio-degradable detergent is one which :**

(a) is manufactured using bio-technology
(b) containing long chain alkyl benzenes
(c) is easily decomposed by microorganisms
(d) is manufactured using organic chemicals

**AP PSC IOF-2009**

**Ans. (c)**

**125. A high grade pulp for paper manufacture :**

(a) Rag pulp (b) Sulphate pulp
(c) Sulfite pulp (d) Mechanical pulp

**AP PSC IOF-2009**

**Ans. (b)**

**126. Bases, used for both, saponification and neutralization reactions, in soap making are**

(a) NaOH and KOH
(b) NaOH and Triethanolamine
(c) KOH and Triethanolamine
(d) KOH, NaOH and Triethanolamine

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**127. The molecular formula of the predominant chemical compound in commercial sugar is**

(a) $C_{12}H_{22}O_{11}$ (b) $C_{12}H_{24}O_{12}$
(c) $C_6H_{10}O_5$ (d) $C_6H_{12}O_6$

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**128. Cellulose content of bamboo and ideal fibrous raw material for the manufacture of paper is _____%.**

(a) 10 (b) 50
(c) 80 (d) 95

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**129. Which is a high grade pulp?**

(a) Rag pulp (b) Mechanical pulp
(c) Sulphate pulp (d) Sulphite pulp

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**130. Which one of the following is not the 'Slurry Reactors'?**

(a) Hydrogenation of fatty acid in presence of supported Nickel catalyst
(b) Hydrogenation of aniline in presence of Nickel supported on clay
(c) Hydrogenation of Glucose in presence of raney Nickel catalyst
(d) Oxidation of ethylene in presence of $PdCI_2$ carbon catalyst

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**131. Which one of the following elements can be used as protective atmosphere to prevent oxidation in metal working and food preservation?**

(a) Oxygen
(b) Nitrogen
(c) Carbon
(d) Hydrogen

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**132. Which of the following is not a source of starch?**

(a) Sorghum (b) Sago
(c) Cassava (d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**133. Sizing material is incorporated in paper to:**

(a) increase its thickness
(b) increase its brightness
(c) increase its strength
(d) none of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**134. The process used for the manufacture of ethyl alcohol from molasses:**

(a) distillation (b) dehydration
(c) fermentation (d) dehydrogenation

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**135. Solvent used for the extraction of oil:**

(a) Acetone
(b) Methyl Ethyl ketone
(c) Hexane
(d) Furfural

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**136. Molasses is the starting material for the production of:**

(a) alcohol (b) oil
(c) fatty acids (d) starch

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**137. Ethyl alcohol from molasses in manufactured by:**

(a) Fractional distillation
(b) Dehydrogenation
(c) Fermentation
(d) Dehydration

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Fermentation

**138. In the sulphate (Kraft) pulp process, the essential chemical reagents used in digester are:**

(a) Caustic soda, sodium sulphide, soda ash
(b) Slaked lime, salt cake, mercaptans
(c) Caustic soda, mercaptans, ethylene oxide
(d) Quick lime, baking soda, dimethyl sulphide

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Caustic soda, sodium sulphide, soda ash

**139. Soap may be prepared by:**

(a) Hydrolysis of tallow
(b) Hydrogenation of vegetable oils
(c) Boiling the vegetable oil or tallow with caustic soda solution
(d) Oxidation of tallow

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Boiling the vegetable oil or tallow with caustic soda solution

**140. Sodium dodecyl benzene sulphate is**

(a) A drug (b) A detergent
(c) An explosive (d) An artificial fiber

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** A detergent

**141. Vanaspati is produced by the process of:**

(a) Hydrolysis (b) Oxidation
(c) Reduction (d) Hydrogenation

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Hydrogenation

## D. Polymers

**142. Flexible foam for mattresses is usually made of**

(a) Polymethane (b) Poly propylene
(c) Polyvinyl chloride (d) Butyl rubber

**TNPSC AE-2018**

**Ans. (a)**

**143. Polyethylene is a polymer obtained by the polymerization of**

(a) ethane (b) ethylene
(c) isoprene (d) butadiene

**TNPSC AE-2018**

**Ans. (b)**

**144. A rubber that is extremely resistant to heat, weathering and ozone attack**

(a) Urethane rubber (b) Hypalon
(c) Natural rubber (d) Chloroprene

**TNPSC AE-2018**

**Ans. (b)**

**145. Teflon is polymer product of**

(a) $C_2F_4$ (b) $CF_4$
(c) $C_2F_2$ (d) $CH_2 = CHF$

**TNPSC AE-2018**

**Ans. (a)**

**146. Epoxy resins come under the category of**

(a) Thermoplastic (b) Thermosetting
(c) Polychloroprene (d) Elastomer

**TNPSC AE-2018**

**Ans. (b)**

**147. Vulcanization make the rubber**

(a) Solube in water (b) Less elastic
(c) Hard and more stiff (d) Soft

**TNPSC AE-2019**

**Ans. (c)**

**148. Carbon black is added to the natural rubber as a**

(a) filler
(b) antioxidant
(c) vulcanizing accelerator
(d) preservative

**TNPSC AE-2019**

**Ans. (a)**

**149. Nitrile rubber is produced by polymerization of**

(a) acrylonitrile and butadiene
(b) acrylonitrile and styrene
(c) arylonitrile and isoprene
(d) isoprene and isobutylene
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**150. Any one of the following is partially depolymerised and contains a high percentage of carbon black, ash and solvent oils**

(a) Reclaimed rubber (b) Hypalon
(c) Urethane rubber (d) Butyl rubber
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**151. Most commonly used rubber vulcanization agent is**

(a) Nickel
(b) Sulfur
(c) Bromine
(d) Chlorine

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**152. Which of the following plastic has lowest cost**

(a) Polythene (b) PVC
(c) Teflon (d) Bakelite

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**153. Zeigler-Natta catalyst is used in the manufacture of**

(a) Vinyl acetate
(b) Styrene
(c) PTFE
(d) Propylene

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**154. Which of the following is used for the manufacture of terephthalic acid**

(a) p-xylene (b) m-xylene
(c) o-xylene (d) Mixed xylenes

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**155. Styrene is produced from ethyl-benzene by**

(a) dehydrogenation
(b) alkylation
(c) oxidation
(d) dehydration

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**156. Antioxidants are added to rubber to protect it from the attack of light, heat and atmospheric ozone. Which of the following is an antioxidant used in rubber?**

(a) Carbon
(b) Alkylated diphenylamine
(c) Thiokol
(d) Magnesium

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**157. Butyl rubber is a copolymer of**

(a) 1-butene with a small amount of isobutene
(b) isobutene with a small amount of 2-methyl butadiene (isoprene)
(c) butadiene with a small amount of propylene
(d) 1-butene with a small amount of butadiene

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**158. Wood and cotton are mostly made up of**

(a) glucose monomers linked up in a polymer
(b) polyisoprene
(c) polysaccharides called cellulose
(d) amino acids in the form of polypeptides

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**159. Choose the correct statement**
**In viscose rayon manufacturing process,**

(a) carbon disulphide used as reactant for xanthate formation is regenerated in a later step
(b) caustic soda used as reactant for steeping of cellulose is regenerated in a later step
(c) sulphuric acid is used in steeping process of cellulose
(d) the spun viscose rayon is hardened in an alkali bath

**GATE-2018**

**Ans. (a)**

**160. Polyvinyl chloride is produced by**

(a) co-polymerization
(b) addition-type kinetics
(c) reacting chlorine with polyethylene
(d) reacting hydrochloric acid with polyethylene

**GATE-2018**

**Ans. (b)**

**161. Match the polymerization processes in Group-1 with the polymers in Group-2**

| Group-1 | Group-2 |
|---|---|
| **(P) Free radical polymerization** | **(I) Nylon 6,6** |
| | **(II) Polypropylene** |
| **(Q) Ziegler Natta polymerization** | |
| **(R) Condensation polymerizations.** | **(III) Poly vinyl chloride** |

**Choose the correct set of combinations.**

(a) P-I, Q-II, R-III (b) P-III, Q-II, R-I
(c) P-I, Q-III, R-II (d) P-II, Q-I, R-III

**GATE-2017**

**Ans. (b)**

**162. Poly Vinyl chloride is a.**

(A) Thermosetting material
(B) Thermoplastic material
(C) Fibrous material
(D) Chemically active material
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**163. Which of the following is not an elastomer?**

(A) Polyisoprene (B) Neoprene
(C) Nitrile-butadiene (D) None of these
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**164. Toluene-di-isocyanate is used in making.**

(A) Dyestuffs
(B) Flexible polyurethane foams
(C) PharmACFutical products
(D) All of the above
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**165. According to Debye-Bueche theory, the viscosity of a polymer solution or melts is proportional to.**

(A) Concentration (B) Molecular weight
(C) Both (A) and (B) (D) None of these
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**166. Vulcanisation of rubber.**

(A) Decreases in tensile strength
(B) Increases its ozone and oxygen reactivity
(C) Increases its oil and solvent resistance
(D) Converts its plasticity to elasticity
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (d)**

**167. Buna-N is a polymer of.**

(A) Butadiene
(B) Butadiene and Styrene
(C) Butadiene and Acrylonitrile
(D) Styrene and Acrylonitrile
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**168. Example for thermosetting resins**

(a) Cellulose derivatives
(b) Polymer resins
(c) Phenolic resins
(d) Modified alkyds

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**169. Fill in the blank:**
**Polyethylene terephthalate is a/an———**
(a) Polyster (b) Inorganic polymer
(c) Unsaturated polyester (d) Polyamide
**HPCL Engineer-01.11.2022**

**Ans. (a)**

**170. Select the correct chemical which is used as asecondary plastisizer for PVC:**
(a) Chlorinated paraffin wax
(b) Dioctyl phthalate
(c) Dibutyl phthalate
(d) Diisobutyl phthalate
**HPCL Engineer-01.11.2022**

**Ans. (a)**

**171. Nitrile rubber is a polymer of butadiene and**
(a) Styrene (b) Methane
(c) Isoprene (d) Acrylonitrile
**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**172. Following statements are related to vinyl chloride.**
**Statement-I: Vinyl chloride is represented by $CH_2 = CH - Cl$.**
**Statement-II: Vinyl chloride is represented by $CH_2 = COOCH = CH - Cl$.**
(a) I and II are correct
(b) I and II are not correct
(c) I is correct
(d) II is correct
**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**173. Why it is difficult to recycle plastics?**
(a) It is very hard
(b) It contain different types of polymer resins
(c) It is adhesive
(d) It comes in different sizes
**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**174. Perspex is**
(a) Acrylic sheet
(b) an elastomer
(c) an alloy of tin and lead
(d) an aluminium foil clad with bakelite
**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**175. Styrene- butadiene rubber is commercially manufactured by**
(a) bulk polymerisation
(b) suspension polymerisation
(c) solution polymerisation
(d) emulsion polymerisation
**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**176. In polyethylene manufacturing, when PE with 90% crystallinity, density of 0.965 g/cc and melting point of the order of 150°C is required, the catalyst employed is,**
(a) Aluminium triethyl combined with titanium dioxide
(b) Aluminum triethyl combined with titanium tetra chloride
(c) Chromic oxide
(d) None of the above
**ISRO Scientist/Engineer-13.03.2016**

**Ans. (b)**

**177. Synthetic polymer membranes are produced by polymerization of a monomer by :**
(a) Condensation polymerization
(b) Addition polymerization
(c) Copolymerization
(d) All of the above
**AP PSC IOF-2009**

**Ans. (d)**

**178. Low density polyethylene (LDPE) is classified as :**
(a) Thermosetting (b) Thermoplastic
(c) Elastomer (d) Polyester fibre
**AP PSC IOF-2009**

**Ans. (b)**

**179. Match the polymer mentioned on the left with the catalyst used for its manufacture given on the right.**

| | |
|---|---|
| **(i) Low density Polyethylene** | **(P) Ziegler-Natta catalyst** |
| **(ii) High density Polyethylene** | **(Q) Traces of Oxygen** |
| **(iii) Polyethylene Terephthalate** | **(R) Butyl Lithium** |
| **(iv) Polyvinyl Chloride** | **(S) Autumony** |

**(1) I-Q, II-R, III-S, IV-P**
**(2) I-S, II-P, III-Q, IV-R**
**(3) I-Q, II-P, III-S, IV-R**
**(4) I-S, II-R, III-P, IV-Q**
(a) 1 (b) 2
(c) 3 (d) 4
**GATE-2015**

**Ans. (c)**

**180. Polyvinyl chloride is produced by**
(a) co-polymerization
(b) addition type kinetics
(c) reacting chlorine with polythene
(d) reacting hydrochloric acid with polyethylene
**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**181. Which of the following is NOT correct?**
(a) Nylon 6-6 is produced by condensation polymerisation
(b) Phenol-formaldehyde resin is thermo-setting polymer
(c) High density polythene (HDPE) is produced by condensation polymerization
(d) Polyethylene terephthalate (PET) is a polyester

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**182. Which of the following is a thermo-setting plastic?**
(a) Milk (b) PVC
(c) Polyethylene (d) Epoxy polymer

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**183. Nitrile rubber is a polymer of**
(a) Acrylonitrile and styrene
(b) Butadiene and styrene
(c) Butadiene and acrylonitrile
(d) Isobutylene and isoprene

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**184. Which of the following material has viscoelastic property?**
(a) Rubber (b) Cork
(c) Glass (d) Graphite

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**185. Thermocole (expanded polystyrene) is NOT used for**
(a) Low temperature thermal insulation, as in refrigerator and air conditioner
(b) Acoustic control and ceiling for building
(c) High temperature thermal insulation in furnaces
(d) Packing of delicate electronic gadgets

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**186. Buna–N is also called**
(a) Butyl rubber (b) Nitrile rubber
(c) Neoprene (d) Thiokol

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**187. Viscose rayon is :**
(a) Cellulose nitrate
(b) Regenerated cellulose nitrate
(c) Regenerated cellulose acetate
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**188. Which one of the following is the disadvantage of an 'Emulsion Polymerization'?**
(a) Low dispersion viscosity
(b) Product of high molecular weight
(c) Film formation at reactor wall
(d) Good heat transfer rate

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**189. Terylene is:**
(a) same as decron (b) a polyester
(c) both (a) and (b) (d) neither (a) nor (b)

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**190. Which of the following is a raw material for the production of nitrile rubber?**
(a) Acrylonitrile (b) Butadiene
(c) both (a) and (b) (d) neither (a) nor (b)

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**191. Caprolactum is produced from:**
(a) Phenol (b) Naphthalene
(c) Pyrieline (d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**192. Raw material for the manufacture of nylon-66:**
(a) Adipic acid (b) Acetic acid
(c) Ethylene (d) Nylon

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**193. Styrene is produced by:**
(a) Catalytic oxidation of cumene
(b) Catalytic oxidation of toluene
(c) Dehydrogenation of ethyl benzene
(d) Catalytic oxidation of xylene

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Dehydrogenation of ethyl benzene

## E. Synthetic Organic chemicals (Pesticides/Insecticides/Paints/Varnishes)

**194. Bismarck brown is mainly used for dyeing**
(a) Wool
(b) Cotton
(c) Leather
(d) Silk

**TNPSC AE-2019**

**Ans. (c)**

**195. Turkey red is a**
(a) sulphur dye (b) mordant dye
(c) solvent dye (d) natural dye
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**196. For many coatings, denaturation begin to occur at**

(a) 25 ºC (b) 35 ºC
(c) 45 ºC (d) 60 ºC

**RPSC ACF-2011**

**Ans. (c)**

**197. White pigment is**

(a) White Zinc (b) White Arsenic
(c) White Lead (d) White Silver

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**198. _____ is widely employed as an adhesive for veneer, plywood, and corrugated and laminated boards where water resistance is not important**

(a) Fevicol (b) ECH
(c) Araldite (d) Native starch

**ISRO Technical Asstt.-25.02.2018**

**Ans. (d)**

**199. Terminology for rapid paint chalking**

(a) Flaking (b) Peeling
(c) Alligatoring (d) Erosion

**ISRO Technical Asstt.-25.02.2018**

**Ans. (c)**

**200. The formation of protective coating by paints and varnishes involves the production of film on surfACFs of the thickness order**

(a) $\frac{1}{1000}$inch (b) 1inch
(c) $\frac{1}{10}$inch (d) $\frac{1}{100}$inch

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**201. Which of the following is NOT an explosive ?**

(a) Nitroglycerine
(b) Cellulose nitrate
(c) Calcium ammonium nitrate
(d) Trinitrotoluene

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**202. Benzene hexachloride (BHC) is represented by**

(a) $C_6Cl_6$ (b) $C_6H_3Cl_3$
(c) $C_6H_6Cl_6$ (d) $C_6H_5$–Cl

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**203. Trinitro Toluene (TNT) is**

**A. Used in glycerin manufacture**
**B. An explosive**
**C. Used in dye manufacture**
**D. In printing industry**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**204. Synthetic glycerine is produced from,**

(a) Naphtalene (b) Phenol
(c) Toluene (d) Propylene

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (d)**

**205. The main difference between a paint and varnish is, varnish has no :**

(a) oil (b) pigment
(c) drier (d) resin

**AP PSC IOF-2009**

**Ans. (b)**

**206. Which of the following material is NOT an insecticide?**

(a) DDT (b) BHC
(c) TNT (d) 2, 4-D

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**207. Mixture of chlorine and sodium bromide acts as a/an**

(a) Insecticides
(b) Analgesic drug
(c) Fire retardant
(d) Hydrogenation catalyst

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**208. Which of the following is an yellow pigment ?**

(a) Titanium dioxide
(b) Ferrous sulphate
(c) Lead chromate
(d) Zinc sulphate

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**209. The temperature in the calcium carbide furnace is_____$^{o}$C.**

(a) 200-300 (b) 700-850
(c) 2000-2200 (d) 4000-4500

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**210. Which oil is preferred for paint manufacture?**

(a) Dry oil (b) Non-drying oil
(c) Semi-drying oil (d) Salinated oil

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**211. Silicon carbide is a/an :**

(a) Adhesive
(b) Abrasive
(c) Type of glass
(d) Brittle material

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**212. Varnish does not contain:**

(a) Thinner
(b) Pigment
(c) Dryer
(d) Anti-skimming agent

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**213. What is the full form of DDT?**

(a) Dichloro diphenyl trichloro ethane
(b) Dichloro diphenyl tetrachloro ethane
(c) Dichloro diphenyl trichloro methane
(d) Dichloro diphenyl tetrachloro methane

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**214. Which of the following is an explosive?**

(a) nitroglycerin (b) trinitrotoluene
(c) cellulose nitrate (d) all of the above

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**215. Tinitro toluene is:**

(a) used in glycerin manufacture
(b) used in dye manufacture
(c) used in pulping process
(d) none of the above

**Kerala PSC Lect. 05.02.2015**

**Ans. (a)**

**216. A glass with high refractive index and used in optical instruments is:**

(a) Pyrex glass (b) Jena glass
(c) Flint glass (d) Crookes glass

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Flint glass

## F. Petrochemicals

**217. Mixed esters of polyhydric alcohols other than glycerin are called**

(a) Waxes (b) Fats
(c) Oils (d) Detergents
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**218. Read the following statements about LPG and choose the correct one.**
**Statement-I: It is gas, Which is heavier than air at NTP.**
**Statement-II: It consist ethyl mercaptan to impart a distinct odor.**

(a) I is not correct
(b) I is correct but II is not correct
(c) I and II are correct
(d) Only II is correct

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**219. Match the petrochemical derivative in Group I with the raw materials in Group II**

| Group I | Group II |
|---|---|
| P. Acrylonitrile | 1. Methane |
| Q. Ammonia | 2. Ethane |
| R. Dodecame | 3. Ethylene |
| | 4. Propylene |

(a) P-1, Q-2, R-1 (b) P-2, Q-1, R-2
(c) P-3, Q-4, R-3 (d) P-4, Q-1, R-4

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**220. Styrene is produced from Ethyl benzene by the process of**

**A. De-hydrogenation**
**B. Oxidation**
**C. Alkylation**
**D. Dehydration**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**221. Chain reaction in $_{92}U^{238}$ will decrease if its mass**

(a) is less than critical mass
(b) is equal to the critical mass
(c) exceeds critical mass
(d) all of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**222. Butadiene is a/an:**

(a) Napthene (b) Aromatic
(c) Olefin (d) Di-olefin

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**223. Petrolactum is:**

(a) Same as petroleum ether
(b) Petroleum cake
(c) A mixture of microcrystalline wax in viscous hydrocarbon liquids
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**224. LPG stands for:**

(a) Liquid petroleum gas
(b) Liquefied petrol gas
(c) Liquid petrol gas
(d) Liquefied petroleum gas

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**225. The chemical formula of vinyl chloride is:**

(a) $CH_3CH_2Cl$
(b) ClCH = CHCl
(c) $CH_2 = CCl_2$
(d) $CH_2 = CHCl$

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** $CH_2 = CHCl$

9.g Petroleum refining

## G. Petroleum refining

**226. Which one of the following it the second most abundant series in most crudes?**

(a) Cycloalkanes (b) Olefin
(c) Iso-alkanes (d) Alkanes

**TNPSC AE-2018**

**Ans. (a)**

**227. The lowest temperature at which a liquid gives enough gives enough vapour to form a ignitable mixture with air is called.**

(a) Flash point (b) Fire point
(c) Ignition temp (d) Combustion

**TNPSC AE-2019**

**Ans. (a)**

**228. Name the catalyst that is used in steam reforming process**

(a) Nickel (b) Cobalt
(c) Iron (d) Molybdenum

**TNPSC AE-2019**

**Ans. (a)**

**229. The detonating properties of petrol are described by its**

(a) Centane number (b) Cloud point
(c) Octane number (d) Smoke point

**TNPSC AE-2019**

**Ans. (c)**

**230. In crude petroleum distillation a dark coloured thick liquid comes out through the bottom of the distillation still is known as**

(a) Diesel (b) Tar
(c) Kerosene (d) Lubricating oil

**TNPSC AE-2019**

**Ans. (b)**

**231. Chromogen is an aromatic body containing a colour giving group commonly called as**

(a) Chromotid (b) Chromophore
(c) Chromosome (d) Auxo chrome

**TNPSC AE-2019**

**Ans. (b)**

**232. Pyrolysis gasoline is obtained from which of the following process**

(a) Hydrocracking
(b) Fluid catalytic cracking
(c) Naphtha cracking
(d) Catalytic reforming

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**233. Desulfurization of liquid fuels in the refinery is carried out using**

(a) Adsorptive desulfurization
(b) Hydro desulfurization
(c) Oxidative desulfurization
(d) Photo desulfurization

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**234. Which of the following is used as feedstock in manufacture of linear alkyl benzene**

(a) Light diesel (b) Light cycle oil
(c) Kerosene (d) Gas oil

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**235. Tertio amyl methyl ether (TAME) is made by etherification of**

(a) Isoprene
(b) Cyclopentadiene
(c) Dicyclopentadiene
(d) 2-methyl butene

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**236. From which of the processess propylene is recovered in the refinery**

(a) Alkylation
(b) Catalytic dewaxing
(c) Catalytic reforming
(d) Fluid catalytic cracking

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**237. In the fluid Catalytic Cracker (FCC), the cracking reaction is ____ and the regeneration reaction is _____ respectively,**

(a) exothermic, endothermic
(b) endothermic, endothermic
(c) exothermic, exothermic
(d) endothermic, exothermic

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**238. The final boiling points of gasoline, diesel, atmospheric gas oil (AGO) and lubricating oils vary as**

(a) gasoline > diesel > AGO > lubricating oils
(b) lubricating oils > AGO > diesel > gasoline
(c) AGO > lubricating oils > diesel > gasoline
(d) lubricating oils > diesel > AGO > gasoline

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**239. Which of the following methods is commonly used to commercially produce ethylene from naphtha?**
(a) Catalytic cracking
(b) Catalytic dehydrogenation
(c) Pyrolysis
(d) Hydrocracking

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**240. Which of the following is undesirable in gasoline but desirable in kerosene?**
(a) Aromatics
(b) Mercaptans
(c) Naphthenic acid
(d) Paraffins

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**241. In connection with petroleum refining, identify the incorrect statement among the following options**
(a) Desalting of crude oil is done before processing it in atmospheric distillation unit
(b) A stream of hydrogen is produced in catalytic reforming of naphtha
(c) Asphalt used for paving is a petroleum product
(d) Cetane number indicates the quality of petrol/motor spirit

**GATE-2018**

**Ans. (d)**

**242. In petroleum industry, which of the flow meters is normally used for heavy liquids characterized by high viscosity?**
(a) Nutating disc meter
(b) Orifice meter
(c) Nozzle
(d) Rotating vane/lobe meter

**RPSC ACF-2011**

**Ans. (d)**

**243. Mercaptans present in light distillates are removed by extraction with**
(a) Concentrated sulfuric acid solution
(b) Concentrated phosphoric acid solution
(c) Aqueous caustic soda solution
(d) None of these

**RPSC ACF-2011**

**Ans. (c)**

**244. The general formulae of olefins is**
(a) $C_nH_{2n-2}$ (b) $C_nH_{2n}$
(c) $C_nH_{2n+2}$ (d) $C_nH_{2n-4}$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**245. Hydrotreating is used for**
**i. removal of water from crude oil**
**ii. removal of sulphur from petroleum fraction**
**iii. removal of nitrogen from petroleum fraction**
**iv. improving octane number of gasoline**
(a) i and ii (b) ii and iv
(c) iii and iv (d) ii and iii

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**246. Petrol is mixed with _____ to form Gasohol.**
(a) Hydrochloric acid (b) Benzene
(c) Lubricating oil (d) Ethyl alcohol

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**247. The proper arrangement of the petroleum fractions, in terms of their boiling points is**
(a) Lubricating oils > diesel > petrol > LPG
(b) Lubricating oils > petrol > diesel > LPG
(c) Petrol > lubricating oil> diesel> LPG
(d) Petrol > diesel > LPG > lubricating oil

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**248. The most important process currently used for industrial production of carbon black from petroleum oil is**
(a) furnACF black process
(b) channel black process
(c) lamp black process
(d) ACFtylene black process

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**249. Crude petroleum consists of**
(a) 84-87 pre-cent carbon and 11-14 per-cent hydrogen
(b) 11-14 per-cent carbon and 84-87 per-cent hydrogen
(c) 54 per-cent carbon and 25 per-cent hydrogen
(d) 70-72 per-cent carbon and 5-7 per-cent hydrogen

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**250. Cetane number of high speed diesel must be greater than and equal to(≥)**
(a) 30 (b) 45
(c) 75 (d) 95

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**251. Petroleum liquid fuels having flash point greater than 66°C is considered as safe during storage and handling. Which of the following has flash point >66°C?**

(a) Naphtha (b) Petrol
(c) Kerosene (d) Heavy fuel oil

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**252. Sweetening of petroleum product means removal of**

(a) Water
(b) Wax
(c) Organic Impurites
(d) Sulphur & its compounds

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**253. Which of the following process is not used for Hydrogen manufacturing?**

(a) Steam-boron process
(b) Steam-hydrocarbon process
(c) Steam-iron process
(d) Electrolysis of water

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**254. The most common method of producing hydrogen from hydrocarbon is known as**

(a) steam reforming (b) partial oxidation
(c) steam blowing (d) complete oxidation

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**255. The main constituent of natural gas is**

(a) $CH_4$ (b) $C_2H_2$
(c) $C_2H_4$ (d) $C_2H_6$

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**256. Which of the following fraction of a crude will have maximum API (i.e., °API)?**

(a) Diesel (b) Gasoline
(c) Atmospheric gas oil (d) Vacuum gas oil

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**257. Kerosine should have**

(a) low smoke point
(b) high smoke point
(c) high aromatic content
(d) low paraffin content

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**258. The catalyst used in catalytic reforming is**

(a) platinum on alumina
(b) nickel
(c) iron
(d) aluminium chloride

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**259. Which of the following is the easiest to crack?**

(a) Paraffin (b) Olefin
(c) Naphthene (d) Aromatic

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**260. 'Sour' crude means the crude oil contains**

(a) wax
(b) paraffins and asphalts
(c) nitrogen compounds
(d) sulphur compounds

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**261. Find the odd one out.**

(a) Digboi Refinery
(b) Numaligarh Refinery
(c) Guwahati Refinery
(d) Bongaigaon Refinery

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**262. A catalytic reforming plant produces hydrogen and benzene from cyclohexane by de-hydro aromatisation. In order to increase the production of hydrogen, the owner plans to change the process to steam reforming of the same feedstock that produces hydrogen and carbon dioxide. Stoichiometrically, what is the maximum ratio of pure hydrogen produced in the proposed process to that in the existing process?**

(a) 1 (b) 2
(c) 5 (d) 6

**GATE-2016**

**Ans. (d)**

**263. Match the industrial processes in Group-1, with the catalyst materials in Group-2**

| Group-1 | Group-2 |
|---|---|
| P. Ethylene polymerisation | I. Nickel |
| R. Petroleum feedstock cracking | II. Vanadium pentoxide |
| Q. Oxidation of $SO_2$ to $SO_3$ | III. Zeolite |
| S. Hydrogenation of oil | IV. Aluminium triethyl with titanium chloride promoter |

(a) P-IV, Q-III, R-II, S-I
(b) P-I, Q-IV, R-III S-II
(c) P-I, Q-II, R-III, S-IV
(d) P-II, Q-III, R-IV, S-I

**GATE-2016**

**Ans. (a)**

**264. In petroleum refining, the process used for conversion of hydrocarbons to aromatics is**
(a) catalytic cracking (b) catalytic reforming
(c) hydro treating (d) alkylation

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**265. Globally, 95% of hydrogen gas is produced through,**
(a) Electrolysis of water
(b) Photolysis of water
(c) Steam methane reforming
(d) None of the above

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (c)**

**266. Arrange the kinematic viscosity of petroleum products produced by crude oil distillation in increasing order:**
**(1) Naphtha (2) Kerosene**
**(3) Diesel (4) Fuel oil**
**Select the correct answer from the code given below.**
(a) 4, 3, 2 and 1 (b) 2, 3, 1 and 4
(c) 1, 2, 3 and 4 (d) 2, 3, 4 and 1

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**267. Minimum plate spacing in most of the petroleum refinery distillation column (dia > 3 ft) is normally :**
(a) 6 inches (b) 12 inches
(c) 34 inches (d) 18 inches

**AP PSC IOF-2009**

**Ans. (d)**

**268. Liquefied Petroleum Gas (LPG) is mainly a mixture of :**
(a) Propane and butane
(b) Methane and ethane
(c) High boiling olefins
(d) High boiling naphthenes

**AP PSC IOF-2009**

**Ans. (a)**

**269. Catalyst used in the catalytic cracking is :**
(a) Silica gel (b) Silica-Alumina
(c) Vanadium pentoxide (d) Nickel-Iron

**AP PSC IOF-2009**

**Ans. (b)**

**270. Catalytic cracking is**
(a) a hydrogen addition process
(b) a carbon rejection process
(c) an exothermic process
(d) a coking process

**GATE-2014**

**Ans. (b)**

**271. In petroleum refining operation, the process used for converting paraffins to naphthenes to aromatics is**
(a) alkylation (b) catalytic reforming
(c) hydro-cracking (d) isomerisation

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**272. Asphalts are**
(a) low molecular weight and low boiling point compounds present in petroleum
(b) desirable in catalytic cracking feed-stock because they produce coke
(c) readily oxidisable and form carbonaceous sludge
(d) all of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**273. Which of the following reactions is undesirable in the production of catalytically reformed gasoline?**
(a) Dehydrogenation of naphthene
(b) Dehydrogenation of lower paraffins
(c) Dehydrocyclisation of higher paraffins
(d) Isomerisation of paraffins

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**274. Before refining of crude oil which operation is performed?**
(a) Filtration
(b) Desalting
(c) Gravity separation and desalting
(d) None of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**275. Gum formation in stored gasoline is mainly due to**
(a) Alkylation of unsaturates
(b) Presence of sulphur
(c) Oxidation and polymerisation of unsaturates
(d) Higher aromatic content

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**276. Pick out the additive property of a lube oil out of the following:**

(a) 'API' gravity
(b) Specific gravity
(c) Viscosity
(d) Flash point

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**277. Aniline point of the diesel is a measure of its_____ content.**

(a) Aromatic (b) Paraffin
(c) Olefins (d) Naphthene

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**278. Presence of sulphur in gasoline:**

(a) Leads to corrosion
(b) Increases lead susceptibility
(c) Decreases gum formation
(d) Helps during stabilization

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**279. The main aim of cracking is to produce :**

(a) Gasoline (b) Lube oil
(c) Petroleum (d) Coke

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**280. Visbreaking :**

(a) Uses natural gas as feed
(b) Is carried out at atmospheric pressure
(c) Produces fuel oil of lower viscosity
(d) Produces gasoline only

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**281. Catalyst used in reforming is the :**

(a) Silica-alumina (b) Silica gel
(c) Vanadium Pentoxide (d) Nickel

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**282. The main reaction in reforming is the :**

(a) Dehydrogenation of naphthenes
(b) Hydrogenation of napthenes
(c) Hydrocracking of paraffins
(d) Saturation of olifins

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**283. Solvent used in the deasphalting process is:**

(a) Furfurol (b) Phenol
(c) Propane (d) Hexane

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**284. Sweetening of petroleum product means the removal of:**

(a) Sulfur and its compounds
(b) Water
(c) Organic impurities
(d) Wax

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**285. Sour crude means the_____ bearing crude.**

(a) Asphalt
(b) Sulphur compounds
(c) Wax
(d) Nitrogen compounds

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**286. Octane number of gasoline is a measure of its:**

(a) Knocking tendency
(b) Ignition delay
(c) Ignition temperature
(d) Smoke point

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**287. Tetra-ethyl lead is added in gasoline to:**

(a) Increase its smoke point
(b) Reduce gum formation
(c) Reduces the pour point
(d) Increase its octane number

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**288. Good quality kerosene should have:**

(a) Low smoke point
(b) High smoke point
(c) High aromatic content
(d) Low paraffin content

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**289. Diesel index is defined as :**

(a) ("API")*(Aniline Point, $^oF$)/100
(b) ("API")*(Aniline Point, $^oC$)/100
(c) ("API")*(100)(Aniline Point, $^oF$)
(d) ("API")*(100)(Aniline Point, $^oC$)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**290. Which of the following has the lowest cetane number ?**

(a) Aromatics (b) i-paraffins
(c) Napthene (d) Olefins

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**291. Flash point of an oil is determined by the:**
(a) Pensky Martens Apparatus
(b) Rams Bottom Apparatus
(c) Saybolt Viscometer
(d) Conradson Apparatus

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**292. Which of the following is not a product of tar distillation?**
(a) Phenol and naphthalene
(b) Anthacene and creosote
(c) Benzol and pitch
(d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**293. Which of the following is there in aviation fuel?**
(a) Diesel (b) Naphtha
(c) Petrol (d) Kerosene

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**294. Electrical desalting of crude petroleum is done for:**
(a) Removal of oleophilic impurities
(b) Removal of oleophobic impurities
(c) Removal of both oleophilic and oleophobic impurities
(d) The purpose other than the removal of oleophobic and oteophilic impurities

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Removal of oleophobic impurities

**295. The desirable reactions in the catalytic reforming of naphtha are:**
(a) Slightly exothermic
(b) Endothermic
(c) Autocatalytic
(d) Highly exothermic

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Endothermic

**296. Heavy vacuum gas oil obtained from vacuum distillation unit is mainly used for:**
(a) Blending component for gasoline
(b) Blending component for kerosene
(c) Feed stock for fluid catalytic cracking unit
(d) Blending component for ATF

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Feed stock for fluid catalytic cracking unit

**297. H/C ratio (by weight) for the same number of carbon atoms is the lowest in:**
(a) Paraffins (b) Napthenes
(c) Olefins (d) Aromatics

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Aromatics

## H. Cement and lime industry

**298. In cement industry the diameter of the rotary kiln range from**
(a) 20 to 30 ft (b) 16 to 22 ft
(c) 15 to 20 ft (d) 6 to 12 ft

**TNPSC AE-2019**

**Ans. (d)**

**299. Consumption of lime in carbonation is approximately**
(a) 0.1 – 0.15% conc.
(b) 1.5 – 2% conc.
(c) 0.15 – 0.2% conc.
(d) 0.15 – 2% conc.

**UPPSC AE-29.05.2022, Part-I**

**Ans. (*)**

**300. Cement manufactured by fusing lime stone and bauxite is**
(a) Portland cement
(b) Hydraulic lime cement
(c) Pozzolana cement
(d) High alumina cement
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**301. Artificial cement made by the calcination of argillACFous limestone is called**
(a) Portland (b) Calcium silicate
(c) Gypsum (d) Aluminium silicate

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**302. Lime stone is fused with ____ to manufacture high alumina cement.**
(a) Bauxite (b) Graphite
(c) Washing soda (d) Sand

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**303. Production of one ton of cement requires about _____ tons of lime stone.**
(a) 0.6 (b) 3.8
(c) 2.2 (d) 1.2

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**304. Purpose of adding fluxing agents during ceramic manufacture**
(a) To maintain the equilibrium
(b) To raise the temperature
(c) To prevent reversible reaction
(d) To lower the temperature

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**305. After feeding raw materials to the kiln till production of cement clinkers, what is the correct order of reaction zones?**
(a) Calcining-Drying-Clinkering
(b) Drying-Calcining- Clinkering
(c) Clinkering-Drying-Calcining
(d) Clinkering-Calcining-Drying

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**306. Cement setting under water employs a/an ____ process.**
(a) hydration (b) decomposition
(c) oxidation (d) reduction

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**307. Penetrating index and softening points are determined for**
(a) gasoline (b) kerosine
(c) bitumen (d) gas oil

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**308. Cement mainly contains**
(a) $CaO$, $SiO_2$, $Al_2O_3$ (b) $MgO$, $SiO_2$, $K_2O$
(c) $Al_2O_3$, $MgO$, $Fe_2O_3$ (d) $CaO$, $MgO$ $K_2O$

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**309. Soundness test of cement is carries out to determine its**
(a) Alumina content
(b) Iron oxide content
(c) Free lime content
(d) Durability under sea water

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**310. The workability of concretes is assessed through**
**1. Slump test**
**2. Compaction factor test**
**3. Setting time of cement**
**4. Le-Chatelier's apparatus**
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 4 and 1

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**311. Portland cement is primarily a mixture of :**
(a) Calcium, magnesium, oxychlorides
(b) Calcium, aluminates and silicates
(c) Calcium and magnesium bicarbonates
(d) Gypsum and fly ash

**AP PSC IOF-2009**

**Ans. (b)**

**312. Calcareous of argillaceous materials are used in the manufacturing of :**
(a) Lead (b) Cement
(c) Carbon disulphide (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**313. Cement mainly contains:**
(a) $CaO$, $SiO_2$, $Al_2O_3$
(b) $MgO$, $SiO_2$, $K_2O$
(c) $Al_2O_3$, $MgO$, $Fe_2O_3$
(d) $CaO$, $MgO$, $K_2O$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

## I. Coal Industry

**314. Coke oven gas is composed of**
(a) $CO$ and $CO_2$
(b) $CH_4$ and $C_2H_4$
(c) $CH_4$ and $CO_2$
(d) $CO$, $N_2$ and $H_2$

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**315. The high temperature carbonization of coal is carried out at**
(a) 900 - 1150ºC
(b) 1250 - 1500ºC
(c) 600 - 800ºC
(d) 175 - 275ºC

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**316. Producer gas is obtained by**
(a) thermal cracking of naphtha
(b) passing steam and air through red hot coke
(c) passing air through red hot coke
(d) passing steam through red hot coke

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**317. Raw material for the manufacture of calcium carbide are**
(a) limestone and coke
(b) limestone and slaked lime
(c) limestone and sand
(d) limestone and caustic soda

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**318. Proximate analysis of determines**
(a) Moisture, ash, sulfur, volatile matter
(b) Moisture, volatile matter, ash, fixed carbon
(c) Moisture, sulfur, nitrogen, fixed carbon
(d) None of these

**RPSC ACF-2011**

**Ans. (b)**

**319. When steam is passed over coal resulting in the endothermic reaction ($C + H_2O$ ..... $CO + H_2$), it is called**

(a) Carbonization of coal
(b) Oxidation of coal
(c) Coalification
(d) Gasification of coal

**RPSC ACF-2011**

**Ans. (d)**

**320. Which is released during the production of clinker?**

(a) $CaCO_3$ (b) CO
(c) $Ca(OH)_2$ (d) $CO_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**321. An orsat analysis of the combustion product of a hydrocarbon liquid fuel gave 10% $CO_2$, 0.6% CO, 5.7% $O_2$ and 79% $N_2$ by volume. The C/H ratio (by wt) of the liquid fuel is**

(a) 6.36 (b) 1.06
(c) 3.18 (d) None

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**322. Following coal has maximum calorific value**

(a) Bituminous (b) Peat
(c) Lignite (d) Anthracite

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**323. The ratio of the percentage carbon obtained in the ultimate analysis and the percent fixed carbon obtained in the proximate analysis is always**

(a) one
(b) between zero and one
(c) less than one
(d) greater than one

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**324. Which of the following cannot be determined by ultimate analysis of coal?**

(a) Total carbon (b) Fixed carbon
(c) Hydrogen (d) Oxygen

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**325. When you completely burn the coal sample in a muffled furnACF at 700ºC – 750ºC and weigh the residue you get**

(a) volatile matter (b) fixed carbon
(c) ash content (d) mineral matter

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**326. When coal is heated in absence of air, it is called**

(a) deoxidation (b) gasification
(c) coalification (d) carbonisation

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**327. A coal that softens and fuses on heating is**

(a) classified (b) carbonised
(c) caking (d) non-caking

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**328. The calorific value is highest for**

(a) producer gas (b) water gas
(c) coke-oven gas (d) blast-furnACF gas

**Assam PSC-CCE-Pre-2015**

**Ans. (d)**

**329. In flue gas analysis by Orsat's apparatus, carbon monoxide is absorbed by**

(a) cuprous chloride
(b) potassium hydroxide
(c) alkaline pyrogallol solution
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**330. Determine the correctness or otherwise of the following Assertion [a] and Reason [r].**

**Assertion:** Significant combustion of coke takes place only if it is heated at higher temperature in presence of air.

**Reason:** $C + O_2 \rightarrow CO_2$ is an exothermic reaction.

(a) Both [a] and [r] are true and [r] is the correct reason for [a]
(b) Both [a] and [r] are true but [r] is not the correct reason for [a]
(c) [a] is correct but [r] is false
(d) Both [a] and [r] are false

**GATE-2014**

**Ans. (b)**

**331. Match the raw materials of Groups 1 and 2 with the final products of Group 3**

| Group I | Group 2 | Group 3 |
|---|---|---|
| $P_1$: Ethylene | $Q_1$: Ammonia | $R_1$: Synthetic fibre |
| $P_2$: Propylene | $Q_2$: 1-Butene | $R_2$: Nylon 66 |
| $P_3$: Adipic acid | $Q_3$: Ethylene glycol | $R_3$: LLDPE |
| $P_4$: Terephthalic acid | $Q_4$: Hexamethylene diamine | $R_4$: Acrylonitrile |

(a) $P_1+Q_2\rightarrow R_3;P_2+Q_1\rightarrow R_4;P_3+Q_4\rightarrow R_2;P_4+Q_3\rightarrow R_1$
(b) $P_1+Q_1\rightarrow R_3;P_2+Q_3\rightarrow R_1;P_3+Q_4\rightarrow R_4;P_4+Q_2\rightarrow R_2$
(c) $P_1+Q_2\rightarrow R_2;P_2+Q_3\rightarrow R_1;P_3+Q_4\rightarrow R_3;P_4+Q_1\rightarrow R_4$
(d) $P_1+Q_1\rightarrow R_4;P_2+Q_2\rightarrow R_3;P_3+Q_4\rightarrow R_2;P_4+Q_3\rightarrow R_1$

**GATE-2014**

**Ans. (a)**

**332. Bomb calorimeter is used to determine the_____ of the coal.**
(a) Sulphur content
(b) Calorific value
(c) Both (A) and (B)
(d) Neither (A) nor (B)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**333. When coal is heated in absence of air, it is called its :**
(a) Deoxidation (b) Gasification
(c) Coalification (d) Carbonization

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**334. Coke oven gas consists mainly of :**
(a) $H_2$ and $CH_4$
(b) CO and $CO_2$
(c) $H_2$ and CO
(d) $CH_4$ and CO

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**335. Spontaneous combustion of coal on storage results due to :**
(a) Inadequate ventilation
(b) Low temperature oxidation
(c) Storage in large heaps with small surface to volume ratio
(d) All (A), (B) and (C)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**336. Which of the following constituents of coal is the most important in the production of coke?**
(a) Moisture
(b) Ash
(c) Volatiles
(d) Carbon

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**337. Coal that softens and fuses on heating is:**
(a) Classified
(b) Carbonized
(c) Caking
(d) Non-caking

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**338. Most of the coking coals are**
(a) Anthracite (b) Bituminous
(c) Lignite (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**339. Grind ability index of coal is 100. It implies that the:**
(a) Coal can be pulverized with great difficulty
(b) Coal can't be easily pulverized
(c) Coal can be easily pulverized
(d) Power consumption in grinding the coal will be very high

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**340. In an integrated steel plant, $NH_3$ present in coke oven gas in normally recovered as:**
(a) $(NH_4)_2SO_4$ (b) $NH_4Cl$
(c) $(NH_4)_2NO_3$ (d) Liquid $NH_3$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**341. Which of the following is not a product of coal tar distillation?**
(a) Anthracite (b) Cresote oil
(c) Carbolic oil (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**342. Coke used for the production of calcium carbide should have:**
(a) Low ash content
(b) Low ignition temperature
(c) High electrical resistivity
(d) All (A), (B) and (C)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**343. Which one of the following gases can be produced by using Bituminous, anthracite coal or coke as raw material?**
(a) Water gas (b) Coke oven gas
(c) Natural gas (d) Synthesis gas

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**344. A coal containing very high percentage of durain is called ……. coal.**
(a) boghead (b) splint
(c) non-banded (d) bright

**Kerala PSC Lect. 05.02.2015**

**Ans. (b)**

**345. Low temperature carbonization process for coal which is generally carried out at a temperature of 600°C has the primary product semi coke or smokeless char whose yield is around:**
(a) 75 to 80% (b) 70 to 75%
(c) 60 to 70% (d) 50 to 60%

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 75 to 80%

**346. Coal tar produced by high temperature carbonization of coal is the main source of:**
(a) Aliphatic compounds
(b) Aromatic compounds
(c) Paraffins
(d) Cycloalkane

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Aromatic compounds

## J. Nuclear Industry

**347. Boiling water reactor and pressurised water reactor are**
(a) Nuclear reactor
(b) Solar reactor
(c) Ocean thermal electric conversion
(d) Biogas reactor

**TNPSC AE-2018**

**Ans. (a)**

**348. Principle of atom bomb**
(a) Nuclear fusion
(b) Nuclear fission
(c) Nuclear decomposition
(d) All the above

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**

**349. $_{92}U^{235}$ has ____ neutrons in the nucleus.**
(a) 235 (b) 216
(c) 143 (d) 314

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**350. Alpha ion is stable**
(a) up to 910°C
(b) above 2000°C
(c) above 1403°C
(d) between 910°C and 1403ºC

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**351. The function of moderators in nuclear reactor is to**
**A. Slow down secondary neutrons**
**B. Absorb secondary neutrons**
**C. Control the chain reaction**
**D. None of these**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**352. The specific type of uranium used to produce nuclear energy most efficiently is**
(a) U-233 (b) U-234
(c) U-235 (d) U-236

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**353. Which is the most commonly used molten metal for cooling of nuclear reactor?**
(a) Calcium (b) Sodium
(c) Mercury (d) Zinc

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**354. In heat conduction with a nuclear heat source :**
**$S_{no}$ = Thermal energy produced at the centre per unit volume**
**b = Constant**
**r = Radius of cylindrical shell**
**$R_f$ = Radius of nuclear fuel rod**
**Thermal energy production per unit volume $S_n$ will be**

(a) $S_{no}\left[1+b\left(\frac{r}{R_f}\right)^2\right]$ (b) $S_{no}{}^2\left[1-b\left(\frac{r}{R_f}\right)^2\right]$

(c) $S_{no}\left[1+b^2\left(\frac{r}{R_f}\right)^{\frac{1}{2}}\right]$ (d) $S_{no}^2\left[1-b^2\left(\frac{r}{R_f}\right)^{\frac{1}{2}}\right]$

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**355. Which is the fuel used in fast breeder reactor at Kalpakam?**
(a) Thorium (b) Plutonium
(c) Uranium (d) Uranium 238

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

## K. Minerals and Ores

**356. Which of the following is an ore of Copper?**
(a) Galena
(b) Hematite
(c) Bauxite
(d) Chalcopyrite

**TNPSC AE-2018**

**Ans. (d)**

**357. Limestone is predominantly**
(a) magnesium carbonate
(b) calcium carbonate
(c) sodium carbonate
(d) potassium carbonate

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**358. The material used to give blue coloration to glass is**
(a) NiO (b) CoO
(c) $FeSO_4$ (d) CdS

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**359. Black powder is a mixture of**
(a) Ammonium chloride, activated carbon and sulphur
(b) Ammonium Nitrate, charcoal and sulphur
(c) Potassium perchlorate, activated carbon and sulphur
(d) Potassium Nitrate, charcoal and sulphur

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**360. Carborundum consists mainly of:**
(a) Bauxite
(b) Silicon carbide
(c) Boron carbide
(d) Calcium Carbide

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

# 10.

# Environmental pollution monitoring & control

## A. Air pollution

**1. Inhalation of _____ particles cause asbestosis.**
(a) asbestos (b) chromium
(c) cadmium (d) Smog
**TNPSC AE-2018**

**Ans. (a)**

**2. Smelters are the main source of _____ , a toxic metal emitted into the atmosphere as particulate matter.**
(a) Cadmium (b) chromium
(c) Magnesium (d) Potassium
**TNPSC AE-2018**

**Ans. (b)**

**3. Cigarette smoking Constitute a major source of _____ in humans.**
(a) Cadmium (b) cobalt
(c) magnesium (d) Potassium
**TNPSC AE-2018**

**Ans. (a)**

**4. The operation of cyclone separator relies on**
(a) Centrifugal force acting on the particle
(b) Diffusion of dust particle
(c) The creation of intimate contact between a stream of gas and a flow of scrubbing liquor
(d) Producing an electric charge on the particle to be collected and then directing it, by electrostatic forces, to the collecting electrodes.
**TNPSC AE-2018**

**Ans. (a)**

**5. The fine dust that is much more hazardous penetrating deep into the lungs and remains there is known as.**
(a) Inhalable dust (b) Respirable dust
(c) particulate dust (d) Pulverizing dust
**TNPSC AE-2018**

**Ans. (b)**

**6. What are the pollutants released while using Air fresheners?**
(a) Tolumide, lanolin
(b) Turpentine, wax
(c) Propylene glycol, morpholine
(d) Cyclopropane, Toluamide
**TNPSC AE-2019**

**Ans. (c)**

**7. Which is the most important reactive intermediate species formed during photochemical process**
(a) Hydroxyl radical
(b) Hypochloride
(c) Carbon monoxide
(d) Hydroperoxyl radical
**TNPSC AE-2019**

**Ans. (a)**

**8. The main pollution sources for Acid Rain are**
(a) Sulphur dioxide ($SO_2$) and Nitrogen oxide ($NO_x$)
(b) Maleic acid and Tartaric Acid
(c) Particulate matter and heavy metal
(d) Carbon monoxide and Ozone
**TNPSC AE-2019**

**Ans. (a)**

**9. In the extended atmosphere, mesosphere is in the range of**
(a) upto 5 km (b) 5 to 45.km
(c) 45 to 80 km (d) above 80 km
(e) Answer not known
**TNPSC AE-2022**

**Ans. (c)**

**10. USHA regards an atmosphere of more than _____ % as oxygen enriched atmosphere.**
(a) 16.4 (b) 17.5
(c) 23.5 (d) 27.6
(e) Answer not known
**TNPSC AE-2022**

**Ans. (c)**

**11. Aerosols are suspension of particles in air with an effective diameter**
(a) Greater than 100 μ m
(b) Less than 1 mm
(c) Between 10 μ m – 100 μ m
(d) Less than 10 μm
(e) Answer not known
**TNPSC AE-2022**

**Ans. (d)**

**12. Impurities in the size range of 0.001-1 micro-meters are considered as**
(a) Dissolved impurities
(b) Colloidal impurities
(c) Suspended impurities
(d) Non-colloidal impurities
**RPSC Lect.-15.01.2016**

**Ans. (b)**

**13. In India, the pH value of rain lies in the range of**

(a) 5 - 6 (b) 6 - 7
(c) 3 - 4 (d) 4 - 5

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**14. Mesophilic temperature range is from**

(a) 0 - 10 ºC (b) 12 - 20 ºC
(c) 25 - 42 ºC (d) 45 - 60 ºC

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**15. For maximum discharge through a chimney, its height should be**

(a) 200 meters
(b) Infinitely long
(c) More than 105.7 meters
(d) Equal to the height of the hot gas column producing draught.

**RPSC ACF-2011**

**Ans. (d)**

**16. If a man exposed to 150-200 ppm $NO_2$ for 3 to 5 weeks what will be the result of it?**

(a) Headache
(b) Death
(c) Bronchiolitis fibrose obliterans
(d) Silo filter disease

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**17. In which of the following theories/hypotheses was it first proposed that in photochemical smog, the time variation in the level of $O_3$, No are explained in terms of chemical reaction?**

(a) Friedlander and Seinfield theory
(b) Sulphate hypothesis
(c) Chlorine hypothesis
(d) Haagen smith theory

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**18. Those fluids which require a gradually increasing shear stress to maintain a constant strain rate and are called**

(a) Pseudoplastic (b) Thixotropic
(c) Dialatant (d) Rheopectic

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**19. In the electrostatic precipitator**

(a) low conductivity material is more easily collected
(b) electrical conductivity has no role
(c) high conductivity material is more easily collected
(d) none of these

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**20. The plume behavior which occurs in the super adiabatic condition with light to moderate wind speed in the presence of large scale thermal eddies are known as**

(a) plume (b) fanning plume
(c) looping plume (d) neutral plume

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**21. Which of the following statement is NOT true for wet scrubbers?**

(a) simultaneous removal of gases and particulates
(b) invisible wet plume
(c) corrosive gases can be neutralized by proper choice of scrubbing liquid
(d) problem of wet sludge disposal

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**22. Following compound is called the secondary air pollutant**

(a) $N_2O_2$ (b) CO
(c) $O_3$ (d) $O_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**23. Which of the following is known as terminal electron accepter in the electron transport chain of aerobic respiration?**

(a) Cytochrome-C (b) Water
(c) NADH (d) $O_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**24. "Siderosis" is a disease caused by the inhalation of _____ dust.**

(a) Coal (b) Aluminium
(c) Iron (d) Silica

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**25. Following is called the particulate pollutants**

(a) methane (b) fly ash
(c) carbon tetrachloride (d) hydrogen sulphide

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**26. Which of the following air pollution control device has the maximum efficiency?**

(a) Electrostatic precipitator
(b) Dynamic precipitator
(c) Spray tower
(d) Wet cyclonic scrubber

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**27. Industrial chimney located near tall building should be at least—— times the height of building.**

(a) 0.5 (b) 2.5
(c) 4 (d) 5.5

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**28. Which of the following particulate collection devices does not discharge the collected particulates as dry solid?**
(a) Cyclone separator
(b) Electrostatic precipitator
(c) Wet scrubber
(d) Fabric filter

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**29. A passenger in an Airplane can see the rainbow in a shape of**
**A. Arc** **B. Cone**
**C. Circle** **D. Sphere**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (c)**

**30. Emission of dense white smoke out of the chimney of a thermal power plant is an indication of the use of**
(a) less air for combustion
(b) correct amount of air for combustion
(c) too much air for combustion
(d) pulverised coal in boilers

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**31. Which of the following fine dust removal equipment is the most efficient?**
(a) bag filter
(b) scrubber
(c) electrostatic precipitator
(d) cyclone separator

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**32. Persons working in cement plants and limestone quarries are more prone to disease like**
(a) Cancer (b) Asthma
(c) Silicosis (d) Fluorosis

**ISRO-12.02.2016**

**Ans. (c)**

**33. In the vicinity of the power plant air contains 15 ppm of $SO_2$. Its mole fraction in the air is :**
(a) $15 \times 10^{-20}$ (b) $15 \times 10^{-6}$
(c) $15 \times 10^{-3}$ (d) 0.15

**AP PSC IOF-2009**

**Ans. (*)**

**34. The upper layer of atmosphere is called the :**
(a) Troposphere (b) Ionosphere
(c) Stratosphere (d) None of the above

**AP PSC IOF-2009**

**Ans. (d)**

**35. Common air pollutants are :**
(a) Oxides of sulfur (b) Oxides of nitrogen
(c) Oxides of carbon (d) Oxides of hydrogen

**AP PSC IOF-2009**

**Ans. (a)**

**36. Equipment used for very fine dust removal from air is :**
(a) Bagfilter
(b) Scrubber
(c) Electrostatic Precipitator
(d) Cyclone separator

**AP PSC IOF-2009**

**Ans. (c)**

**37. Global warming may result in :**
(a) floods
(b) cyclones
(c) fall in food productivity
(d) all of the above

**AP PSC IOF-2009**

**Ans. (d)**

**38. $SO_2$ from air can be removed by :**
(a) Iron oxide (b) Alkaline solutions
(c) Silica gel (d) Activated carbon

**AP PSC IOF-2009**

**Ans. (b)**

**39. Which of the following is a green house gas other than $O_2$ ?**
(a) Methane (b) Nitrous oxide
(c) Chlorofluorocarbons (d) All of the above

**AP PSC IOF-2009**

**Ans. (d)**

**40. Peroxyacyl nitrate (PAN), a pollutant is found in the :**
(a) Automobile exhaust
(b) Power plant exhaust
(c) Nitric acid plant exhaust
(d) Sulfuric acid plant exhaust

**AP PSC IOF-2009**

**Ans. (a)**

**41. Which of the following is the component in PM sampler that determines the size classification of the PM collected on a filter?**
(a) Pump (b) Impactor
(c) Filter (d) Timer

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**42. Which of the following air pollutant is responsible for acid rain?**
(a) $SO_2$ (b) $CO_2$
(c) CO (d) $H_2S$

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**43. For a given particle size, the collection efficiency of a cyclone separator will**
(a) be increased at higher pressure drop
(b) be decreased at higher pressure drop
(c) not be affected by changing pressure drop
(d) none of these

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**44. The causes of desertification are :**

**1. Overgrazing**

**2. Cultivation of marginal lands**

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**45. Thermal pollution is mainly caused due to:**

(a) Sudden change in temperature caused by periodic plant outages
(b) Sudden decrease in temperature of river or lake
(c) Increase of dissolved oxygen (DO) in water bodies
(d) Gradual increase in temperature of earth

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**46. Which of the following is an air pollutant?**

(a) Nitrogen oxides
(b) Sulplur dioxide
(c) Carbon monoxide
(d) All of the above

**Kerala PSC Lect. 05.02.2015**

**Ans. (d)**

**47. Smoke which is formed by combustion of other chemical process contains fine particles of liquid or solid of the size range:**

(a) 0.001 – 0.20 μm
(b) 0.01 – 1.0 μm
(c) 0.1 – 1.0 μm
(d) 1 – 10 μm

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 0.01 – 1.0 μm

**48. Ecosystem refers to:**

(a) A group of individuals of one kind
(b) All population in a given area
(c) A portion of earth where living objects can be found
(d) The community and non living environment working together

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** The community and non living environment working together

**49. Selection of suitable solvents for the removal of gaseous air pollutants is important and from this consideration, sodium hydroxide is widely used one. Which of the pollutants given below cannot be removed by this solvent:**

(a) Sulphur dioxide
(b) Hydrochloric acid
(c) Phosgene
(d) Nitrogen dioxide

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Hydrochloric acid

**50. Which of the following is primary air pollutant:**

(a) Ozone
(b) PAN
(c) Photochemical smog
(d) Particulate matter

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Particulate matter

**51. What of the following is secondary parameter that influence air pollution dispersion:**

(a) Wind direction and speed
(b) Temperature
(c) Mixing Height
(d) Precipitation

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Precipitation

**52. Which is not a combustible gas:**

(a) Carbon monoxide
(b) Carbon dioxide
(c) Hydrogen
(d) Gobar gas

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Carbon dioxide

## B. Water pollution

**53. The $O_2$ starvation characteristic of hemoglobin in blood stream is called as**

(a) Angel man's syndrome (b) Barth syndrome
(c) Blue baby syndrome (d) down's syndrome

**TNPSC AE-2018**

**Ans. (c)**

**54. The principal by product from the anaerobic decomposition of the organic matter in waste water is**

(a) Hydrogen sulfide gas (b) Carbon dioxide gas
(c) Ammonia gas (d) Methane gas

**TNPSC AE-2018**

**Ans. (d)**

**55. Chlorination is done for the removal of**

(a) Sediments (b) Hardness
(c) Bacteria (d) Suspended solids

**TNPSC AE-2018**

**Ans. (c)**

**56. The capacity of anion resins is expressal as**

(a) Milli equivalents per grams of dry hydrogen–form resins
(b) Gram of dry chlorine – form resins
(c) Milli equivalents per gram of dry sodium form resins
(d) Gram of dry nitrogen – form resins

**TNPSC AE-2018**

**Ans. (b)**

**57. Run off coefficient is the**

(a) Ratio of surface run off to the total rain fall on the area in the fixed interval of time.
(b) Ratio of surface run off to the area of catchment in the fixed interval of time.
(c) Ratio of intensity of rain fall to the total rain fall in the fixed interval of time.
(d) Ratio of duration of rain fall to average rain fall

**TNPSC AE-2018**

**Ans. (a)**

**58. Trickling filter with rock or plastic packing for waste water treatment process is**

(a) Non submerged fixed film biological reactor
(b) Suspended growth processes with fixed film packing
(c) Submerged attached growth aerobic processes
(d) Activated sludge processes with fixed film packing

**TNPSC AE-2018**

**Ans. (a)**

**59. The presence of multivalent cations, notably Ca and Mg ions in water causes**

(a) Total solids (b) Hardness
(c) Softness (d) Turbidity

**TNPSC AE-2018**

**Ans. (b)**

**60. Trickle bed reactor is an example for**

(a) Single phase reactor
(b) Two phase reactor
(c) Three phase reactor
(d) Homogeneous reactor

**TNPSC AE-2018**

**Ans. (c)**

**61. What type of reactor used in Aerated Lagoons?**

(a) Complete Mix reactor
(b) Plug flow reactor
(c) Fluidized lead reactor
(d) Bubble columns

**TNPSC AE-2019**

**Ans. (a)**

**62. Name the equation to estimate visibility**

(a) Ergun Equation
(b) Keschmeider Equation
(c) Holland Equation
(d) Antoine Equation

**TNPSC AE-2019**

**Ans. (b)**

**63. Spray ponds are employed for the control of _____ pollution.**

(a) Air (b) Thermal
(c) Noise (d) Water
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**64. The optimum dissolved oxygen in natural waters is**

(a) 1 – 2 ppm
(b) 9 –10 ppm
(c) 4 – 6 ppm
(d) 7 – 9 ppm
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**65. In wastewater treatment process, when the reaction is independent of concentration of the reactants, it is called**

(a) Second order reaction
(b) First order reaction
(c) Zero order reaction
(d) Pseudo First order reaction
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**66. In wastewater treatment during preliminary treatment,**

(a) oil and grease are removed
(b) 50% of total solids are removed
(c) 50% of pathogens are removed
(d) 80% of BOD is removed
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**67. Removal of phosphorous in water is generally done by**

(a) Dephosphorification
(b) Eutrophication
(c) Coagulation
(d) Oxidation
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**68. In the context of chemical process industries, the term BOD is normally associated with the**

(a) Characterization of solids
(b) Characterization of gaseous effluents
(c) Characterization of liquid effluents
(d) Characterization of boiler feed water
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**69. Minamata disease is caused due to**

(a) Mercury
(b) Lead
(c) Phosphorus
(d) Arsenic
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**70. The Oxidation ponds which have a mixture of both aerobic and anaerobic conditions are called as**

(a) Faculative ponds
(b) Lagoons
(c) Polshing ponds
(d) Aerated ponds
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**71. The area required for sludge thickening in activated sludge setting tank can be estimated using the following equation.**

**A – Area required**
**Q – Flow rate**
**$t_w$ – Time to reach desired underflow concentration**
**$H_0$ – Initial height of interface in column**

(a) $A = \frac{Q^2 t_w}{H_0}$

(b) $A = \frac{Q t_w^2}{H_0}$

(c) $A = \frac{Q^2 t_w^2}{H_0}$

(d) $A = \frac{Q t_w}{H_0}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**72. 6 ml of wastewater is diluted to 300 ml distilled water in standard BOD bottle. Initial DO in the bottle is determined to be 8.5 mg/l. DO after 5 days at 20 C is found to be 5 mg/l. $BOD_5$ of wastewater is**

(a) 150 mg/l
(b) 125 mg/l
(c) 175 mg/l
(d) 200 mg/l

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**73. Mercury pollution is mainly due to the emission from**

(a) Petrochemical industry
(b) Thermal power plant
(c) Electroplating industry
(d) Paint industry

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**74. Hazardous waste could be incinerated in**

(a) Pulp and paper industry
(b) Electroplating industry
(c) Cement industry
(d) Petrochemical industry

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**75. Test for determination of $BOD_3$ is carried out at**

(a) 20 ºC (b) 23 ºC
(c) 27 ºC (d) 30 ºC

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**76. Which type of microorganism is responsible to develop muddy odour in water**

(a) Protozoa dinobryon (b) Oscillatoria
(c) Escherichia coli (d) Anabaena

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**77. Why Lagooning process is mainly used?**

(a) For enhancing Global warming
(b) For deforestation
(c) For flow rate reduction in sewer
(d) For sludge disposal

**HPCL Engineer-01.11.2022**

**Ans. (c)**

**78. Which of the following salts is main cause of permanent hardness of water?**

(a) Magnesium bicarbonate
(b) Magnesium carbonate
(c) Magnesium sulphate
(d) None of these

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**79. Maximum permissible limit of Aldrin pollutant in drinking water is**

(a) 0.120 mg/*l* (b) 0.017 mg/*l*
(c) 0.051 mg/*l* (d) 0.232 mg/*l*

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (*)**

**80. What does zero discharge actually refer to?**

(a) there is zero discharge of ions
(b) 100% efficient plant
(c) the reject is partially rejected out
(d) the reject is recycled from every part of the treatment plant

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**81. Presence of nitrogen and phosphorous in waste water discharged in to lakes and ponds causes**

(a) foaming
(b) turbidity
(c) undesirable plant growth
(d) odour nuisances

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**82. Turbidity of water is an indication of the presence of**

(a) floating solids
(b) dissolved gases
(c) suspended inorganic matter
(d) dissolved solids

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**83. The ratio of $O_2$ available to the $O_2$ required for stabilization of sewage is called**
(a) BOD
(b) Bacterial stability factor
(c) Oxygen ion concentration
(d) Relative stability

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**84. Which is used as germicidal in modern time for the treatment of drinking water?**
(a) Bleaching powder
(b) Potassium permanganate
(c) Chlorine
(d) Ozone

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**85. Chloramines are used in water treatment**
(a) for taste control
(b) for weed control in reservoir
(c) as algicides
(d) for disinfection

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**86. Determine the mole fraction of oxygen in water if the concentration of dissolved oxygen is 10.0mg/*l*.**
(a) $5.62 \times 10^{-6}$ (b) $7.8 \times 10^{-6}$
(c) $9.1 \times 10^{-5}$ (d) $1.12 \times 10^{-5}$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**87. Microorganisms and bacteria, present in the water will cause _____ in human and animals.**
(a) indigestion
(b) cancer
(c) intestinal tract infection
(d) brain tumour

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**88. In the design of grit chambers**
(a) maximum velocity of flow is 1 ft/s
(b) length to depth ratio is 500
(c) noise level is important
(d) temperature is important factor

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**89. Determine ultimate BOD for waste water whose 5-day BOD is 200 mg/*l*. The reaction constant $k = 0.23d^{-1}$.**
(a) 297 mg/*l* (b) 278 mg/*l*
(c) 310 mg/*l* (d) 293 mg/*l*

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**90. Dose of chlorine for disinfection of water is about _____ mg/liter.**
(a) 0.01 (b) 1.00
(c) 0.30 (d) 0.10

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**91. What is BOD of a water sample having initial oxygen concentration of 10 ppm and final oxygen concentration of 4 ppm after keeping it in a close bottle for 6 days at 25°C?**
(a) 2.5 (b) 6
(c) 14 (d) 1.5

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**92. The total hardness of soft water as $CaCO_3$ equivalent is**
(a) 50 ppm (b) 100 ppm
(c) 150 ppm (d) 200 ppm

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**93. The following is not used for removing the permanent hardness**
(a) Lime soda process (b) Zeolite process
(c) Demineralization (d) Addition of lime

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**94. The gas, which is produced predominantly in the aerobic digestion of sludge is**
(a) $NO_2$ (b) $CO_2$
(c) $O_2$ (d) $H_2S$

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**95. Chloramines are used in water treatment**
(a) For disinfection
(b) for taste and colour control
(c) as algicides
(d) for weed control in reservoirs

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**96. The main pollutant in the discharged waste water from a petroleum refinery is oil (both in free and emulsified form). Free oil is removed by**
(a) trickling filters
(b) aerated lagoons
(c) biological oxygen pond
(d) gravity separator having oil skimming devices

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**97. There are thirteen metals which are treated as pollutants. Which of the following metal is not a pollutant?**
(a) Mercury (b) Arsenic
(c) Aluminium (d) Lead

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**98. If BOD of river water is high, then it implies that river water is**
(a) Polluted with pesticides
(b) not at all polluted
(c) Highly polluted with inorganic chemicals
(d) Highly polluted with biodegradable organic chemicals

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**99. Septic tanks are primarily used for**
(a) the aerobic decomposition of deposited sewage solids
(b) separation of deposited solids
(c) anaerobic decomposition of deposited solids
(d) separation of oil and grease scums.

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**100. Deaeration of water in its treatment is necessary as it**
(a) minimises its turbidity
(b) helps in controlling its taste and odour
(c) minimises its corrosiveness
(d) None of the above

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**101. The most widely and commonly used coagulant for removal of suspended impurities in water is**
(a) bleaching powder (b) slaked lime
(c) alum (d) copper sulphate

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**102. The resistance of water to the passage of light through it is measure of the**
(a) turbidity (b) colour
(c) hardness (d) dissolved gases

**Kerala PSC Asst. Prof.-2017**

**Ans. (a)**

**103. Biological oxidation ponds remove organic compounds present in the polluted water by :**
(a) Using activities of microbes
(b) Aerobic oxidation
(c) Both (a) and (b)
(d) Anaerobic reaction

**AP PSC IOF-2009**

**Ans. (c)**

**104. The gas from sludge digestion tanks is mainly composed of**
(a) $CH_4$ (b) $N_2$
(c) $H_2S$ (d) CO

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**105. Purge and trap analysis is used for which type of the following analysis ?**
(a) Analysis of metal in water
(b) Analysis of organics in water
(c) Analysis of volatile organics in water
(d) Analysis of semi-volatile organics in water

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**106. Alum is used as a coagulant in water treatment to remove:**
(a) Colour
(b) Turbidity
(c) Bacteria
(d) All (A), (B) and (C)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**107. 'Nitrogenous oxygen demand' (NBOD) is oxygen needed to:**
(a) Convert ammonia to nitrate
(b) Convert nitrogen dioxide to nitrogen trioxide
(c) Denitrifying bacteria to grow
(d) Oxidize organic matters

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**108. A sample of groundwater has 100 mg/ L of $Ca^{2+}$ and 10 mg/L of $Mg^{2+}$. Its total hardness will be nearly:**
(a) 2.4 meq/L
(b) 3.8 meq/L
(c) 4.4 meq/L
(d) 5.8 meq/L

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**109. Trickling filter is used in:**
(a) Filtering fruit juice
(b) Secondary / biological treatment of wastewater
(c) Primary treatment of wastewater
(d) Tertiary treatment of wastewater

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**110. The polluted water requires Chlorine of 8 kg/ day to treat 20,000 cubic metre per day. The residual after 10 minutes contact is 0.20 mg/l. the dosage will be:**
(a) 0.8 mg/l
(b) 0.6 mg/l
(c) 0.4 mg/l
(d) 0.2 mg/l

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**111. Which of the following is a sludge treatment method?**
(a) Sedimentation
(b) Trickling filtration
(c) Incineration
(d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**112. Reagent used for COD determination is:**
(a) Potassium chromate
(b) Iron nitrate
(c) Potassium dichromate
(d) None of these

**Kerala PSC Lect. 05.02.2015**

**Ans. (c)**

**113. Determine total solids from the following data:**
**Weight of sample of sewage = 1000 gm**
**Weight of solids after evaporation of liquid = 0.952 gm**
**Weight of dry residue after ignition = 0.516 gm**
(a) 852 ppm (b) 952 ppm
(c) 752 ppm (d) 516 ppm

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 952 ppm

**114. In order to protect aquatic life in fresh water stream, sewage effluent should never lower the dissolved oxygen less than:**
(a) 5 ppm (b) 10 ppm
(c) 15 ppm (d) 20 ppm

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 5 ppm

**115. A sample of waste water is diluted with specially prepared dilution water with a dilution factor of 150. The contents of dissolved oxygen in the beginning and end of test were found to be 11 ppm and 7 ppm respectively calculate $BOD_5$ (5 days BOD):**
(a) 600 ppm (b) 150 ppm
(c) 700 ppm (d) 250 ppm

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 600 ppm

**116. Colour in the water though harmless is objectionable on grounds of appearance and hence an estimation in domestic water is mandatory. The permissible colour for domestic water on platinum-cobalt scale is:**
(a) 5 mg/liter (b) 25 mg/liter
(c) 50 mg/liter (d) 150 mg/liter

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** 50 mg/liter

## C. Noise pollution

**117. The sound pressure level, expressed in decibels (dB) where $P_1$ is the pressure amplitude of sound and $p_0$ is the reference pressure, is defined as**

(a) $L_p = 10 \log_{10} \left(\frac{p_1}{p_0}\right)^2$ (b) $L_p = \left(\frac{p_1}{p_0}\right)^2$

(c) $L_p = e^{\left(\frac{p_0}{p_1}\right)^2}$ (d) $L_p = 10 \log_{10} \left(\frac{p_0}{p_1}\right)^2$

**TNPSC AE-2018**

**Ans. (a)**

**118. A jet plane during its take off produces _____sound.**
(a) 70 decibel (b) 200 decibel
(c) 150 decibel (d) 100 decibel

**TNPSC AE-2018**

**Ans. (c)**

**119. As per Noise pollution control laws, the ambient noise level in the Industrial Zone during night time is**
(a) 40 dB (b) 45 dB
(c) 55 dB (d) 70 dB

**TNPSC AE-2019**

**Ans. (d)**

**120. Sounds with a very high frequency of _____ is called 'ultrasound'.**
(a) above 20,000 Hz (b) 10,000 Hz
(c) 15,000 Hz (d) 5,000 Hz
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**121. As per Noise Pollution (Regulation and Control) rule, 2000 of India, the day time noise limit for residential zone expressed in dB(A) Leq is**
(a) 55 (b) 50
(c) 75 (d) 65

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**122. Select the frequency audible to human ear from the following.**
(a) 5 Hz (b) 10 Hz
(c) 15000 Hz (d) 25000 Hz

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**123. Planting a series of trees in between the source and receiver of a noise will**
(a) Change the frequency of the solid waste
(b) Increase loudness of the noise at the receiver point
(c) Decrease the sound level of the noise at the receiver point
(d) Increase the sound level of the noise at the receiver point

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

## D. Solid waste (hazards & disposal)

**124. _____ is a chemical decomposition of waste brought about by heating the waste in absence of oxygen.**
(a) Pyrolysis
(b) Incineration
(c) recovery
(d) Biological reprocessing

**TNPSC AE-2018**

**Ans. (a)**

**125. e-waste is**

(a) Waste from nuclear power generation
(b) Eco friendly waste
(c) Hazardous chemical waste
(d) Obsolete electronic waste

**TNPSC AE-2018**

**Ans. (d)**

**126. The Biomass yield is defined as**

(a) $Y = \frac{\text{mass of biomass produced}}{\text{mass of subsrate utilized}}$

(b) Y = mass of biomass produced × mass of substrate utilized

(c) $Y = \frac{COD}{BOD}$

(d) Y = mass of biomass produced + mass of substrate utilized

**TNPSC AE-2019**

**Ans. (a)**

**127. What is the application of "Ryznar Index"**

(a) to estimate the scaling potential
(b) to determine sludge volume
(c) to estimate corrosion rate
(d) to estimate COD

**TNPSC AE-2019**

**Ans. (a)**

**128. The biological decomposition of organic substances in waste controlled conditions is called**

(a) incineration (b) biological oxidation
(c) composting (d) none of these

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**129. Radioactive solid nuclear wastes are disposed off by :**

(a) High temperature incineration
(b) Pathological incineration
(c) Pyrolysis
(d) Underground burial in concrete containers

**AP PSC IOF-2009**

**Ans. (d)**

**130. Which of the following compounds is NOT an electronic waste?**

(a) Pb (b) Argon
(c) Cd (d) Hg

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**131. The term ISWM refers to**

(a) International Solid Waste Management
(b) Integrated Solid Waste Management
(c) Integrated Solid Waste Machine
(d) International Solid Waste Mechanism

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**132. Unit of radioactivity is**

(a) Curie (b) Fermi
(c) Angstorm (d) Candle

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**133. You are appointed as a Planning Engineer for the solid waste handling project. You were given the task to determine the most expensive component of the waste handling project. With the knowledge gained through this process, what is the most expensive component of solid waste handling?**

(a) Collection (b) Storage
(c) Treatment (d) Separation

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**134. In the hierarchy of integrated solid waste management, which gets the highest priority?**

(a) Waste transformation
(b) Waste reduction at source
(c) Landfilling of waste
(d) None of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**135. Which waste is biologically active?**

(a) Municipality solid waste
(b) Ash from thermal power project
(c) Construction and demolition waste
(d) All of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (a)**

**136. Which of the following is likely to be classified as hazardous waste?**

(a) Crop residue
(b) Construction and demolition waste
(c) Effluent treatment plant sludge
(d) Garden waste

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**137. The organic solid waste material may be decomposed by the**

(a) Reduction (b) Oxidation
(c) Sunlight (d) Microorganism

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**138. The pollutional effects of discharging raw waste from a paper mill are :**

**1. Oxygen depletion in the receiving body of w**

**water**

**2. Presence of undesirable colour, odour and taste in water**

**3. Reduced photosynthesis**

**4. Toxicity added to the aquatic life due to the formation of mercaptans, pentachlorophenol and pentachlorophenate**

(a) 1,2,3 and 4 (b) 1,2, and 3 only
(c) 1,3, and 4 only (d) 2 and 4 only

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**139. Waste minimization techniques focus primarily on the activities of :**

(a) Identification and storage
(b) Recycling and reuse
(c) Source reduction and recycling
(d) Land disposal and record keeping

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**140. 'Trash' in Municipal Solid Waste (MSW) is:**

(a) Combustible portion of rubbish
(b) Non combustible portion of rubbish
(c) Garbage residue of food
(d) Construction of auto bodies and industrial waste

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

## E. Pollution control equipments

**141. The most suitable equipment for removing the fine dust particle (<1 μ diameter) from air below its dew point will be a/an.**

(A) Bag-filter
(B) Electrostatic precipitator
(C) Cyclone separator
(D) Wet scrubber

(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**
**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**142. For noise limit of 120 dB, the permissible time will be**

(a) 30 seconds (b) 30 days
(c) 30 hours (d) 30 minutes

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**143. Positive pressure respirators are :**

**1. Self-contained breathing apparatus**
**2. Air-purifying respirators**
**3. Supplied-air respirators**

(a) 1 only (b) 2 only
(c) 1 and 2 (d) 1 and 3

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**144. 'Volatile organic Compounds' (VOCs) are effectively removed by:**

(a) Vacuum filter
(b) Air-stripping tower followed by granular activated carbon filter
(c) Plate and frame filter
(d) Sonicator

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**145. Which of the following device does not discharge collected particulates as a dry solid:**

(a) Cyclone separator
(b) Wet scrubber
(c) Electrostatic precipitator
(d) Fabric filter

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Wet scrubber

**146. The operations of a cyclone separation relies on:**

(a) Diffusion of dust particles
(b) Creation of intimate contact between a stream of gas and flow of scrubbing liquour
(c) Producing electric charge on the particle
(d) Centrifugal force acting on the particle

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Centrifugal force acting on the particle

**147. Which of the following devices of particular collection is the least effective:**

(a) Cyclone separator
(b) Electrostatic precipitator
(c) Wet scrubber
(d) Fabric filter

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Cyclone separator

## F. Fire & Explosion hazards

**148. An example for Class II flammable liquid is**

(a) n – pentane (b) Kerosene
(c) benzene (d) Phenol

**TNPSC AE-2018**

**Ans. (b)**

**149. Submergence and Emergence is the best example for**

(a) Atmospheric Natural hazards
(b) Cumulative Atmospheric Hazards
(c) Terrestrial Natural Hazards
(d) Man Induced Hazards

**TNPSC AE-2019**

**Ans. (c)**

**150. Class –A fire consists of fire due to**

(a) Wood (b) Oil
(c) Transformer (d) Chemicals
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**151. _____ is best suited to extinguish oil or Flammable liquid fire.**

(a) Soda acid
(b) Vaporizing liquid

(c) Foam
(d) Dry Chemical
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**152. Identify the following chemical that will combust on exposure to air**
(a) Barium cyanide (b) Lead amide
(c) Sodium borohydride (d) Phenyl sodium
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**153. Any liquid, whenever the temperature of the liquid is as little as 1 degree less than 100°F. Would be releasing vapour at a rate sufficient to be ignitable, is a**
(a) Flammable liquid
(b) Non-Flammable liquid
(c) Combustible liquid
(d) Organic liquid
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**154. Along with "heat" and "fuel" ____ is also a component of fire triangle.**
(a) $CH_4$ (b) LPG
(c) $O_2$ (d) $N_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**155. The following extinguisher is suitable for cotton or other textile fire**
(a) Water (b) Dry chemicals
(c) Foam (d) Soda acid

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**156. Which fire extinguisher is most suitable for electrical fire?**
(a) Dry powder (b) Foam
(c) Wet chemical (d) $CO_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**157. The probability that a hazard will actually result in an accident is called**
(a) Incident rate (b) Risk
(c) Fault (d) Event

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**158. The Limiting Oxygen Concentration (LOC) of butane ($C_4H_{10}$); lower flammable limit, $LFL_{butane}$ = 1.9(Vol%) is**
(a) 12.35 Vol% (b) 14.35 Vol%
(c) 16.22 Vol% (d) 18.65 Vol%

**UPPSC AE-29.05.2022, Paper-2**
**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**159. Air contains 5 ppm of diethylamine [threshold limit value (TLV)-time- weighted average (TWA) = 10 ppm], 10 ppm of cyclohexanol (TLV-TWA = 50 ppm), and 20 ppm of propylene oxide (TLV-TWA) = 20 ppm). The mixture TLV-TWA will be**
(a) 20.59 (b) 43.75
(c) 22.86 (d) 35.00

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**160. What should you do in an event involving spill of hazardous material during transportation?**
(a) Send someone for help and warn others of the danger
(b) Move leaking material out of the way even if it is unsafe to do so
(c) move your truck/trailer far away from the scene
(d) Try to extinguish any fires

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**161. Eye protection is required for which of the following hazards?**
(a) Flying particles
(b) Liquid chemicals, acids or caustic liquids
(c) Chemical gases or vapours
(d) All of these

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**162. This symbol warns against which of the following hazards?**



(a) Hazardous to both living and non-living material
(b) Dangerous liquid, don't touch
(c) Acidic
(d) Corrosive

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**163. Water is used to extinguish**
(a) Class-A fires (b) Class-B fires
(c) Class-C fires (d) All of these

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**164. Solvents such as Tetrahydrofuran and ethyl ether are particularly hazardous because**
(a) they are flammable
(b) they form explosive peroxides when exposed to air
(c) they penetrate the skin
(d) All of these

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**165. A "Class-D" fire extinguisher can be used to treat fires involving——as fuel sources**

(a) ordinary combustibles (woods, plastics, etc.)
(b) electrical equipment
(c) combustible metals
(d) flammable or combustible liquids

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**166. The hottest part of a flame lies in its**

(a) non-luminous zone
(b) luminous zone
(c) yellow zone
(d) zone of unburnt gases

**Assam PSC-CCE-Pre-2015**

**Ans. (a)**

**167. The three components of a good fire triangle are :**

(a) Pentane–Hydrogen peroxide–A spark
(b) Wood–Metal particles–A spark
(c) Water–Oxygen–A spark
(d) Plastics–Wood–A spark

**AP PSC IOF-2009**

**Ans. (a)**

**168. Fault tree analysis is a method of identifying ways in which hazards can lead to accidents. It mainly works on the logic of :**

(a) Defining top event and working backward toward the various scenarios that can cause an accident
(b) Preparing a detailed process hazards check list
(c) Making an advanced hazards survey
(d) By making a 'what if' analysis

**AP PSC IOF-2009**

**Ans. (a)**

**169. The well known fire triangle represents 3 conditions necessary for fire. They are**

(a) Fuel, Oxidizer, Liquid
(b) Gas, Oxidizer, Solid
(c) Fuel, Heat, Oxygen
(d) Inert gas, Oxidizer, Solid

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**170. Dry pipe fire extinguisher contains**

(a) $NH_3$ (b) $CO_2$
(c) $H_2O$ (d) CO

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**171. Lowest temperature at which a vapor above liquid will continue to burn once ignited is known as**

(a) Flash point (b) Fire point
(c) MOC (d) LOC

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**172. Class C fires are generally referred to**

(a) Ordinary combustible such as wood and paper
(b) Flammable and combustible liquids and gases
(c) Energized electrical equipment
(d) Combustible metals

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**173. As per NFPA (National Fire Protection Association), the background colours of information on health, flammability and reactivity are respectively:**

(a) Red, Yellow and Blue
(b) Blue, Yellow and Red
(c) Red, Blue and Yellow
(d) Blue, Red and Yellow

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**174. The EPA Hazarodous Waste Number for ignitability is :**

(a) D001 (b) D002
(c) D003 (d) D009

**UPSC Poly. Letc.-17.03.2019**

**Ans. (a)**

**175. $LD_{50}$ refers to**

(a) The dose (amount per unit body weight) of a chemical at which first 80% test animal population dies within a period of time by administering chemicals through ingestion
(b) The concentration of a chemical at which first 50 test animal population dies within a period of time
(c) The dose (amount per unit body weight) of a chemical at which 50% of a test animal population dies within a period of time by administering chemicals through ingestion
(d) The dose (amount per unit body weight) of a chemical at which last 50 test animal population dies within a period of time by administering chemicals through ingestion

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**176. Threshold limit Values (TLVs) for slightly toxic classes in dilution ventilation is :**

(a) < 10 ppm
(b) (100 – 500) ppm
(c) > 500 ppm
(d) (10 – 100) ppm

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**177. International Atomic Energy Agency has recommended the maximum permissible radiation dose limit per week by Roentgen Equivalent Man (REM) as :**

(a) 1 (b) 0.5
(c) 0.1 (d) 0.05

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

## G. HAZOP & HAZAN

**178. The situation that, in particular circumstances could lead to the loss to a human being and the loss of an inherent quality suffered by an entity is termed as**

(a) RISK (b) HAZARD
(c) HARM (d) DAMAGE

**TNPSC AE-2018**

**Ans. (b)**

**179. What is FMECA stands for**

(a) Failure Mode Effects and Criticality Analysis
(b) Failure Mode Events and Critical Assessment
(c) Failure Method of Event and Critical Assessment
(d) Fractional Mode Events and Critical Analysis

**TNPSC AE-2019**

**Ans. (a)**

**180. Which Hazard identification techniques uses "Guide words" for Hazard assessment.**

(a) FTA-Fault Tree Analysis
(b) ETA-Event Tree Analysis
(c) HAZOP- Hazards and Operability Studies
(d) PHA-Primary Hazard Analysis

**TNPSC AE-2019**

**Ans. (c)**

**181. Yellow colour in the chemical Hazard NFPA diamond indicates**

(a) Hazard warming
(b) Health hazard
(c) Fire hazard
(d) Reactivity hazard
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**182. Exposure to a chemical can increase the incidence of adverse health effect. It can be determined by _____ process.**

(a) Exposure assessment
(b) Risk characterization
(c) Toxicity assessment
(d) Hazard identification

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**183. A hazard is anything with the _____ for producing ____ .**

(a) Potential, Accident (b) Instinct, Accident
(c) Potential, Safety (d) Instinct, safety

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**184. Main objective of risk assessment is**

(a) to know source of pollutants
(b) remediation of contaminated sites
(c) to evaluate hazard and minimize the risk
(d) hazard management

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**185. Hazard identification mainly focus on**

(a) Chemical pathway
(b) Chemical exposure
(c) Chemical analysis
(d) Chemical source and concentration

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**186. The——Process determines whether exposure to a chemical can increase the incidence of adverse health effect**

(a) Hazard identification
(b) Exposure assessment
(c) Toxicity assessment
(d) Risk characterization

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**187. In which year, Hazards are Operability (HAZOP) studies was developed by Lawley?**

(a) 1968 (b) 1974
(c) 1986 (d) 1972

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

## H. Disaster management

**188. The Chernobyl nuclear disaster occured in the year.**

(a) 1927
(b) 1966
(c) 1986
(d) 1991

**TNPSC AE-2019**

**Ans. (c)**

**189. Three Mile-Island nuclear power plant Leakage in the USA was in the year**

(a) 1977 (b) 1980
(c) 1978 (d) 1979

**TNPSC AE-2019**

**Ans. (d)**

**190. Which chemical released in Bhopal accident in 1984.**

(a) Benzene
(b) Caprolactam
(c) Methyl Isocynate
(d) Polyethylene

**TNPSC AE-2019**

**Ans. (c)**

**191. The Bhopal Gas Tragedy was mainly due to**

(a) Methyl iso butyl carbamate
(b) Methyl iso cyanite
(c) Methyl iso butyl cyanite
(d) Methyl carbamate
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**192. Bhopal gas tragedy was caused by the release of**

(a) EIC (b) BIC
(c) MIC (d) MIG

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**193. Pollutant responsible for 1984 Bhopal tragedy :**

(a) Benzene
(b) Methyl isocyanate
(c) Methyl-ethyl ketone
(d) Phosgene

**AP PSC IOF-2009**

**Ans. (b)**

**194. The leakage of methyl isocyanate (MIC) gas from Union Carbide India Ltd, Bhopal is due to:**

(a) Excess production of MIC
(b) Breakdown of MIC plant
(c) Entry of a large volume of water into MIC storage tank, causing valve to open
(d) Hole was formed on the MIC tank

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

10.i

## I. Environmental Acts

**195. ICRP stands for**

(a) Indian Commission for Radiological Protection
(b) International Commission for Radiological Protection
(c) Indian commission for Radiological Prevention
(d) International Commission for Radiological Prevention

**TNPSC AE-2018**

**Ans. (b)**

**196. Hazardous/polluting chemical industries should have an exclusion zone with a green belt and general public access prohibited around it covering a radius of_____ meter.**

(a) 100 (b) 4000
(c) 400 (d) 1000

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

## J. Safety

**197. The concentration to which workers can be exposed to eight hours per day without any adverse effect is given by**

(a) $LD_{40}$ – TLV (b) TLV – TWA
(c) TLV – STEL (d) C – TLV

**TNPSC AE-2018**

**Ans. (b)**

**198. A non-corrosive substance which can cause skin or long inflammation after repeated contact is a**

(a) Harmful substance (b) Irritant substance
(c) Mutagenic substance (d) Toxic substance

**TNPSC AE-2018**

**Ans. (b)**

**199. During industrial hygine, inorder to determine the study of toxic chemicals _____ data sheet is used.**

(a) Mass safety
(b) Material safety
(c) Operation safety
(d) Risk
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**200. Accident process follow the sequence**

(a) Initiation, propagation, termination
(b) Termination, initiation, propagation
(c) Initiation, termination, propagation
(d) Propagation, termination, initiation

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**201. A 60 kg worker exposed to a particular carcinogen for 5 days per week, 50 weeks per year, over a 25 year. If he breath 20 $m^3$ air per day and carcinogen has a potency factor of 0.02 $(mg/kg\text{-}day)^{-1}$ and average concentration is 0.05 $mg/m^3$ the increment cancer risk is**

(a) $81 \times 10^{-6}$ (b) $18 \times 10^{6}$
(c) $81 \times 10^{6}$ (d) $18 \times 10^{-6}$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**202. For carcinogens Potency Factor (PF) is related to Incremental Lifetime Cancer Risk (ILCR) And Chronic Daily Intake (CDI) as**

(a) $PF = \frac{ILCR}{CDI}$ (b) $PF = ILCR - CDI$
(c) $PF = ILCR + CDI$ (d) $PF = ILCR \times CDI$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**203. OSHA stands for**

(a) Occasional Safety and Health Association
(b) Occupational Safety and Health Administration
(c) Occupational Satisfaction and Health Administration
(d) Occasional Safety and High risk Association

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**204. Process Safety Information (PSI) is needed:**
1. To document formally the as-built/as-modified condition of the plant
2. To provide the necessary data with which to perform the required hazard analysis
3. To communicate information on hazardous substances to employees and others as required by state or federal regulations

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

**UPSC Poly. Letc.-17.03.2019**

**Ans. (d)**

**205. Solar photovoltaic energy sources produce**
(a) AC electricity (b) DC electricity
(c) Voltage (d) Power

**TNPSC AE-2018**

**Ans. (b)**

**206. Tidal energy utilises**
(a) Potential energy (b) Chemical energy
(c) Electrical energy (d) Bio energy

**TNPSC AE-2018**

**Ans. (a)**

**207. Which of the following is not a renewable energy?**
(a) Solar energy (b) Wind energy
(c) Nuclear energy (d) Geothermal energy

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**208. Which form of energy source has the highest potential in Assam compared to all States of India?**
(a) Wind Power (b) Solar Power
(c) Hydro Power (d) Biomass Power

**Assam PSC-CCE-Pre-2015**

**Ans. (c)**

**209. A hydel project that captures the kinetic energy of water without large reservoir or even dam is called**
(a) conventional project
(b) run-of-the-river project
(c) micro-hydro project
(d) conduit project

**Assam PSC-CCE-Pre-2015**

**Ans. (b)**

**210. The released gas from sludge digestion tanks is mainly composed of**
(a) LPG (b) $CH_4$
(c) HCl (d) $N_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (b)**

**211. The maximum power available in the wind (P) is related to the velocity of wind (V) as**
(a) $P \propto V^4$ (b) $P \propto V$
(c) $P \propto V^2$ (d) $P \propto V^3$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**212. Wind energy, transferred to the large sea surface, is stored in waves as**
(a) Chemical energy (b) Electrical energy
(c) Thermal energy (d) Mechanical energy

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**213. The standard value of "Solar constant" is**
(a) 1353 $W/m^2$ (b) 5135 $W/m^2$
(c) 1533 $W/m^2$ (d) 5669 $W/m^2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (a)**

**214. Conversion efficiencies for silicon solar cells range between**
(a) 90 – 95% (b) 42 – 52%
(c) 95 – 99% (d) 10 – 15%

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**215. In the production of ocean energy which form of energy is used?**
(a) Potential energy (b) Wind energy
(c) Solar energy (d) Kinetic energy

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**216. The "Biogas" is the mixture of the following gases**
(a) $C_2H_4$, CO (b) $H_2$, $O_2$
(c) $CH_4$, $CO_2$ (d) $CH_4$, $H_2$

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (c)**

**217. Which ONE of the following statements is CORRECT?**
(a) The major components of biodiesel are triglycerides
(b) Biodiesel is essentially a mixture of ethyl esters
(c) Biodiesel is highly aromatic
(d) Biodiesel has a very low aniline point

**GATE-2014**

**Ans. (b)**

**218. Demerits of non-conventional energy sources are**
(a) Available for free of cost
(b) Produce no or very little pollution
(c) Low gestation period
(d) None of the above

**UP PSC-13.12.2020, Paper-II**

**Ans. (d)**

**219. The ozone layer absorbs following wavelengths of sun's radiation**
(a) 0.80 nm - 1.50 nm (b) 220 nm - 290 nm
(c) 45 nm - 58 nm (d) 6000 nm - 7500 nm

**UP PSC-13.12.2020, Paper-II**

**Ans. (b)**

**220. The function of a windmill is to extract energy from the wind and to produce**
(a) Mechanical energy (b) Thermal energy
(c) Electrical energy (d) Chemical energy

**UP PSC-13.12.2020, Paper-II**

**Ans. (c)**

**221. Hydrothermal reservoirs refer to:**
(a) Hot water stored in a natural dam
(b) Large pools of steam of hot water trapped in porous rocks
(c) Molten rock found below earth's curst
(d) Heat contained in shallow ground

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

# 11.

# Process Engineering Economics

## A. Cost Estimation

**1. Six-tenth factor rule is used for estimating the**
(a) Utilities cost
(b) Piping cost
(c) Equipment cost by scaling
(d) Equipment installment cost

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**2. The major component of Chemical Engineering Plant Cost Index is**
(a) fabricated equipment and machinery
(b) process instruments and control
(c) pumps and compressor
(d) electrical equipment and material

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**3. The total cost ($C_T$) of an equipment in terms of the operating variables x and y is**

$$C_T = 2x + \frac{12000}{xy} + y + 5$$

**The optimal value of $C_T$, rounded to 1 decimal place, is_____.**

**GATE-2017**

**Ans. (90.0 to 93.0)**

**4. The cost of a distillation column in the year 2000 is x rupees. What is the cost of the column is rupees in the year 2010 given the cost indices for the years 2000 and 2010 are 480 and 520 respectively?**
(a) $(520/480)^2$x (b) (480/520)x
(c) (520/480)x (d) $(520/480)^{0.6}$ x

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**5. The cost of two independent process variables $f_1$ and $f_2$ affects the total cost $C_T$ (in lakhs of rupees) of the process as per the following function:**

$$C_T = 100 f_1 + \frac{1000}{f_1 f_2} + 20 f_2^2 + 50$$

**The lowest total cost $C_T$, in lakhs of rupees (up to one decimal place), is _____**

**GATE-2015**

**Ans. (567 to 577)**

**6. During cost estimation, the effect of inflation on equipment cost is taken care by using:**
(a) Lang factor (b) Turn over ratio
(c) Six-tenths rule (d) Cost indices

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Cost indices

## B. Depreciation

**7. P is the investment made on an equipment, S is its salvage value and 'n' is the life of the equipment in years. The depreciation for the $m^{th}$ year by the sum-of-years Digits method will be**
(a) $\frac{P-S}{n}$ (b) $\frac{m}{n}(P-S)$
(c) $\frac{2(n-m+1)}{n(n+1)}(P-S)$ (d) $1-\left(\frac{P}{S}\right)^{1/m}$

**TNPSC AE-2019**

**Ans. (c)**

**8. A furnace installed at a cost of Rs. 24 Lakh is expected to serve its useful life of 5 years. Salvage value of the furnace is Rs. 8 Lakh. The interest rate compounded annually is 8%. The estimated capitalized cost (in Lakhs of rupees) is**
(a) 30 (b) 34.09
(c) 34.9 (d) 58.09

**GATE-2018**

**Ans. (b)**

**9. A machine has an initial cost Rs 10000 and having a service life of 10 years and after 10 years the final salvage value of Rs. 2000. The annual depreciation cost by straight line method is:**
(a) 800 (b) 200
(c) 1000 (d) 400

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**10. A column costs ₹ 5.0 lakhs and has a useful life of 10 years. Using the double declining balance depreciation method, the book value of the unit at the end of five years (in lakhs of ₹) is**
(a) 1.21 (b) 1.31
(c) 1.64 (d) 2.05

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**11. In the straight line method for determining depreciation, it is assumed that the value of the property :**
(a) Decreases exponentially with time
(b) Decreases logarithmically with time
(c) Decreases linearly with time
(d) Remains constant with time

**AP PSC IOF-2009**

**Ans. (c)**

**12. Book value of an equipment is equal to its original price minus:**

(a) Current value of the equipment
(b) Amount of money obtainable from the sale of the equipment
(c) Total depreciation charged
(d) Amount spent on its maintenance

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Total depreciation charged

**13. A column costs Rs. 5.0 lakhs and has a useful life of 10 years. Using the double decling balance depreciation method, the book value of the unit at the end of five years in lakhs of rupees is:**

(a) 1.21 (b) 1.64
(c) 1.31 (d) 2.05

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 1.64

## C. Process Economics

## (Interest, Annuities, Profitability, capitalized cost, pay out period, discounted cash flow)

**14. Return On Equity (ROE) is defined as**

(a) $\text{ROE} = \frac{\text{Stockholders equity}}{\text{net annual profit}} \times 100$

(b) $\text{ROE} = \frac{\text{Net annual profit}}{\text{Stockholders equity}} \times 100$

(c) ROE = net annual profit – stockholder's equity
(d) ROE = Stockholder's equity – net annual profit

**TNPSC AE-2019**

**Ans. (b)**

**15. The total Capital Investment for a chemical plant is Rs. 1,00,00,000 and the working capital is Rs. 10, 00, 000. If a turnover ratio is 1, the gross annual sales will be**

(a) Rs. 90,00,000
(b) Rs. 80, 00,000
(c) Rs. 10,00,000
(d) Rs. 11,00,000

**TNPSC AE-2019**

**Ans. (a)**

**16. The Total Cost of Production (TCOP) is calculated by**

**Where**

**TCOP = Total Cost of Production**
**CCOP = Cash Cost of Production**
**VCOP = Sum of all the Variable Costs of Production minus by product revenues**
**FCOP = Sum of all the Fixed Cost of Production**
**ACC = Annual Capital Charge**

(a) TCOP = (VCOP – FCOP) + ACC
(b) TCOP = (VCOP + FCOP) + ACC
(c) TCOP = (VCOP + FCOP) – ACC
(d) TCOP = FCOP – ACC – VCOP
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**17. Profit is obtained by subtracting_____from revenue**

(a) Total cost
(b) Fixed cost
(c) Operating cost
(d) Book value

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**18. Break-even point is the intersection of**

(a) Total cost and fixed cost
(b) Fixed cost and operating cost
(c) Fixed cost and sales revenues
(d) Total cost and sales revenues

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**19. Effective and nominal interest rates are equal when the interest is compounded**

(a) fortnightly (b) half-yearly
(c) monthly (d) annually

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**20. A five-year asset which has an initial cost of ₹ 2,00,000 with a salvage value of ₹ 20,000, was depreciated by the straight line method. The book value of asset at the end of four years will be**

(a) ₹ 36,000 (b) ₹ 42,000
(c) ₹ 48,000 (d) ₹ 56,000

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**21. Direct cost component of the fixed capital consists of**

(a) contingencies
(b) onsite and offsite costs
(c) labour costs
(d) raw material costs

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**22. With increase in the discounted cash flow rate of return, the ration of the total present value to the initial investment of a given project**

(a) decreases
(b) increases
(c) increases linearly
(d) remains constant

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**23. A company issues bonds worth INR 5,00,000 having a coupon rate of 10% and maturing in 10 years. The company has set up sinking fund method to pay off the bonds. The coupons are to be paid biannually at an interest rate of 6%. The periodic payment of coupon will be in INR**

(a) 40,096.25 (b) 18,607.85
(c) 15,000.00 (d) 17,980.00

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (*)**

**24. In the year 2005, the cost of a shell and tube heat exchanger with 68 $m^2$ heat transfer area was Rs. 12.6 Lakh. Chemical Engineering Index for cost in 2005 was 509.4 and now the index is 575.4 Based on index of 0.6 for capacity scaling, the present cost (in Lakhs of rupees ) of a similar heat exchanger having 100 $m^2$ heat transfer area is estimated to be**

(a) 17.94 (b) 19.94
(c) 20.9 (d) 22.94

**GATE-2018**

**Ans. (a)**

**25. The Annual Fixed Charges (AFC) and Annual Utilities Costs (AUC) of a distillation column being designed are expressed in terms of the reflux ratio (R) as**

**AFC (Rs. Lakh) = $641 \times R^2 - 1796 \times R + 1287 + 1/(R - 1.16)$**

**AUC (Rs. Lakh) = $80 + 62.5 \times R$**

**The reflux ratio ($R_{opt}$) for optimizing the total cost of the distillation column may be found by solving**

(a) $1282 \times R_{opt} - 1796 - 1/(R_{opt} - 1.16)^2 = 0$
(b) $62.5 + 1282 \times R_{opt} - 1796 - 1/(R_{opt} - 1.16)^2 = 0$
(c) $80 + 62.5 \times R_{opt} + 641 \times R_{opt}^2 - 1796 \times R_{opt} + 1287 + 1/(R_{opt} - 1.16) = 0$
(d) $80 + 62.5 \times R_{opt} - 641 \times R_{opt}^2 + 1796 \times R_{opt} - 1287 - 1/(R_{opt} - 1.16) = 0$

**GATE-2018**

**Ans. (b)**

**26. A bond has a maturity value of 20,000 Rupees at the end of 4 years. The interest is compounded at the rate of 5% per year.**

**The initial investment to be made, rounded to the nearest integer, is ____ Rupees.**

(a) (b) j
(c) j (d) j

**GATE-2017**

**Ans. (16300 to 16600)**

**27. The cost of a new pump (including installation) is 24,000 Rupees. The pump has a useful life of 10 years. Its salvage value is 4000 Rupees. Assuming straight line depreciation, the book value of the pump at the end of $4^{th}$ year, rounded to the nearest integer, is _____ P**

**GATE-2017**

**Ans. (15900 to 16100)**

**28. If and amount 100 is paid at the end of every year for 2 years, then what will be the net present value of the annuity at an interest rate of 10?**

(a) 9.91 (b) 9.11
(c) 991 (d) 119

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**29. The time period within which the entire fixed investment on the plant and machineries are earned to recover is known as:**

(a) Depreciation tenure (b) Breakeven point
(c) Capacity ratio (d) Payout period

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**30. Which of the following is a component of working capital investment?**

(a) process equipment
(b) depreciation
(c) plant utilities
(d) maintenance and repair inventory

**UPPSC AE-29.05.2022, Paper-2**

**Ans. (d)**

**31. Discounted cash flow analysis is used to calculate**

(a) The present worth of future earnings
(b) The future worth of present investment
(c) The payback period
(d) The life of the project

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**32. The total capital investment for a chemical plant is ₹ 10,00,000, and the working capital is ₹ 1,00,000. For an asset turnover ratio of 1, the gross annual sale will be**

(a) ₹ 90,000 (b) ₹ 11,00,000
(c) ₹ 10,00,000 (d) ₹ 1,00,000

**RPSC ACF & FRO-24.02.2021**

**Ans. (*)**

**33. Which of the following relationship is not correct is case of a chemical process plant?**

(a) Manufacturing cost = direct product cost + fixed charges + plant overhead costs
(b) General expenses = administrative expenses + distribution & marketing expenses
(c) Total product cost = manufacturing cost + general expenses
(d) Total product cost = direct production cost + plant overhead cost

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**34. Which of the following is not a component of the fixed capital for a chemical plant facility?**

(a) Raw materials inventory
(b) Process equipments
(c) Utilities plants
(d) Contingency

**RPSC ACF & FRO-24.02.2021**

**Ans. (a)**

**35. What will be the total amount available 10 years from now if ₹ 1 is deposited at the present time with nominal interest rate of 11% compounded continuously?**

(a) exp (110) (b) exp (1.1)
(c) 10 exp (0.11) (d) 10 ln (11)

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**36. Break-even point is the point of intersection of**

(a) Fixed cost and total cost
(b) Fixed cost and sales revenue
(c) Total cost and sales revenue
(d) Fixed charges and plant overhead costs

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**37. Heat integration is planned in a process plant at an investment ₹ $2 \times 10^6$. This would result in a net energy savings of 20 GJ per year. If the nominal rate of interest is 15% and the plant life is 3 years, then the breakeven cost of energy, in ₹ per GJ (adjusted to the nearest hundred), is**

(a) 33500 (b) 48300
(c) 54200 (d) 65400

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**38. The point where the total product cost equals the total income**

(a) Threshold point (b) Break-even point
(c) Optimum point (d) Saturation point

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**39. A vertical cylindrical tank with a flat roof and bottom is to be constructed for storing 150 $m^3$ of ethylene glycol. The cost of material and fabrication for the tank wall is Rs 6000 per $m^2$ and the same for the roof and the tank bottom are Rs 2000 and Rs 4000 per $m^2$, respectively. The cost of accessories, piping and instruments can be taken as 10% of the cost of the wall. 10% of the volume of the tank needs to be kept free as vapour space above the liquid storage. What is the optimum diameter (in m) for the tank?**

(a) 3.5 (b) 3.9
(c) 7.5 (d) 7.8

**GATE-2016**

**Ans. (d)**

**40. Terms used in engineering economics have standard definitions and interpretations. Which one of the following statements is INCORRECT?**

(a) The profitability measure 'return on investment' does not consider the time value of money
(b) A cost index is an index value for a given time showing the cost at that time relative to a certain base time
(c) The 'six-tenths factor rule' is used to estimate the cost of an equipment from the cost of a similar equipment with a different capacity
(d) Payback period is calculated based on the payback time for the sum of the fixed and the working capital investment

**GATE-2016**

**Ans. (d)**

**41. Two design options for a distillation system are being compared based on the total annual cost. Information available is as follow:**

| | Option P | Option Q |
|---|---|---|
| Installed cost of the system (Rs in lakhs) | 150 | 120 |
| Cost of cooling water for condenser (Rs in lakhs/year) | 6 | 8 |
| Cost of steam for reboiler (Rs in lakhs/year) | 16 | 20 |

**The annual fixed charge amounts to 12% of the installed cost. Based on the above information, what is the total annual cost (Rs in lakhs/year) of the better option?**

(a) 40
(b) 42.4
(c) 92
(d) 128

**GATE-2016**

**Ans. (a)**

**42. In a manufacturing industry, break-even point occurs when**

(a) The total annual sales equals the fixed costs
(b) The total annual rate of production equals the assigned value
(c) The total annual profit equals the expected value
(d) The total annual product cost equals the total annual sales.

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a)**

**43. Utilities cost in the operation of chemical process plant comes under the :**

(a) Plant overhead cost
(b) Fixed charges
(c) Direct production cost
(d) General expenses

**AP PSC IOF-2009**

**Ans. (c)**

**44. Which is the correct statement of the following ?**

(a) Profit = revenue – operating cost
(b) Profit = revenue – fixed cost
(c) Profit = revenue – book value
(d) Profit = revenue – total cost

**AP PSC IOF-2009**

**Ans. (d)**

**45. Turn over ratio is defined as the ratio of :**

(a) Gross annual sales to fixed capital investment
(b) Net annual sales to operating costs
(c) Rate of production to total product cost
(d) Fixed capital investment to total capital investment

**AP PSC IOF-2009**

**Ans. (a)**

**46. In manufacturing industry, break-even point occurs when :**

(a) The annual sales equals the fixed costs
(b) The total annual product cost equals the total annual sales
(c) The annual profit equals the expected value
(d) All of the above

**AP PSC IOF-2009**

**Ans. (b)**

**47. An investment of Rs. 100 lakhs is to be made for construction of a plant which will take two years to start production. The annual profit from operation of the plant is Rs. 20 lakhs. Then the pay-back time is :**

(a) 5 years (b) 12 years
(c) 7 years (d) 10 years

**AP PSC IOF-2009**

**Ans. (a)**

**48. A bond matures after 5 years and has a maturity value of Rs. 1,000. If the interest rate is 12%, what is the present worth of the bond ?**

(a) Rs. 600 (b) Rs. 567.43
(c) Rs. 473.54 (d) Rs. 670.92

**AP PSC IOF-2009**

**Ans. (b)**

**49. A cash flow of Rs. 12,000 per year is received at the end of each year (uniform periodic payment) for 7 consecutive years. The rate of interest is 9% per year compounded annually. The present worth (in Rs.) of such cash flow at time zero is ———**

**GATE-2014**

**Ans. (60000 to 61000)**

**50. A proposed chemical plant is estimated to have a fixed capital (FC) of Rs. 24 crores. Assuming other costs to be small, the total investment may be taken to be same as FC. After commissioning (at t = 0 years), the annual profit before tax is Rs. 10 crores/years (at the end of each year) and the expected life of the plant is 10 years. The tax rate is 40% per year and a linear depreciation is allowed at 10% per year. The salvage value is zero. If the annual interest rate is 12% the NPV (net present value or worth) of the project in crores of rupees (up to one the decimal place) is_____**

**GATE-2015**

**Ans. (15.0 to 15.5)**

**51. Inflation:**

(a) Affects the time value of money
(b) Refers to the increase in prices of goods and services over time
(c) Concept based upon the fact that physical facilities decline in usefulness with time
(d) Does not affect the amount of money required to purchase goods and services.

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Refers to the increase in prices of goods and services over time

**52. Discounted cash-flow analysis is used to calculate:**

(a) Life of the project
(b) Payback period
(c) Future worth of present investment
(d) Present worth of future earnings

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Present worth of future earnings

**53. Turnover ratio is defined as:**

(a) Fixed capital Investment/gross annual sales
(b) Working capital/Fixed capital investment
(c) Gross Annual sales/Fixed capital investment
(d) Gross Annual sales/Fixed capital investment + working capital

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Gross Annual sales/Fixed capital investment

**54. Project scheduling is done by:**

(a) PERT and CPM (b) Cost estimation
(c) Break even analysis (d) Payback analysis

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** PERT and CPM

**55. For any typical project the cumulative cash flow is zero at:**

(a) End of the design stage
(b) End of the project life
(c) Break even point
(d) Start up

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Break even point

**56. If there is a single lump sum expenditure of Rs. 10 crores and it generates a constant annual cashflow of Rs. 2 crores in each subsequent year, calculate the pay back period (in years) if the scrap value of the capital outlay is zero:**

(a) 10 (b) 5
(c) 20 (d) 1

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** 5

**57. Direct Costs component in the fixed capital is due to:**
(a) Contingency
(b) Onsite and offsite costs
(c) Labour costs
(d) Raw material costs

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Onsite and offsite costs

**58. A series of equal payments made at equal intervals of time like deposit or cost is known as:**
(a) Capital charge factor (b) Future worth
(c) Annuity (d) Perpetuity

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Annuity

**59. The total measurement in a project (whose life is 10 years) is Rs. 10 lakhs and the annual profit is Rs. 1.5 lakhs, then the simple rate of return on investment is:**
(a) 15% (b) 1.5%
(c) 150% (d) 10%

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** 15%

**60. Which of the following is correct:**
(a) Profit = Revenue – Book valve
(b) Profit = Revenue – Total cost
(c) Profit = Revenue – Fixed cost
(d) Profit = Revenue – operating cost

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Profit = Revenue – Total cost

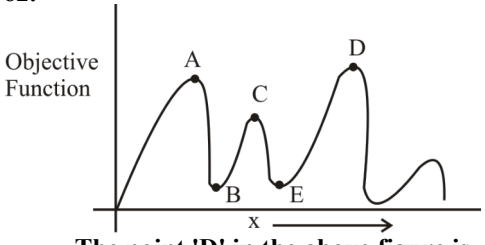
**61. Break even point is reached:**
(a) When total production cost becomes exactly equal to the total income from sale of the product
(b) When operating becomes exactly equal to the total income from sale of all product
(c) When annual operating cost becomes exactly equal to the annual depreciation charge
(d) When total product cost becomes exactly equal to the net profit

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** When total production cost becomes exactly equal to the total income from sale of the product

## D. Optimization is design

**62.**




**The point 'D' in the above figure is**
(a) Local maximum (b) Global maximum
(c) Local minimum (d) Global minimum

**TNPSC AE-2018**

**Ans. (b)**

**63. Major components of economic objective function are**
(a) Inventory costs and capital costs
(b) Capital costs and operating costs
(c) Operating cost and transportation costs
(d) Inventory costs and transportation costs

**TNPSC AE-2018**

**Ans. (b)**

**64. Black box model is known as**
(a) Theoretical model (b) Empirical model
(c) Hybrid model (d) Universal model

**TNPSC AE-2018**

**Ans. (b)**

**65. When a unique solution exists then**
(a) Optimization is needed to obtain a solution
(b) No Optimization is needed to obtain a solution
(c) Equality constraints will beams as inequality constraints
(d) Constraints from a optimal solution

**TNPSC AE-2018**

**Ans. (b)**

**66. The typical design objectives of optimization techniques are**
(a) Minimize cost of production and maximize Return on Investment
(b) Maximize cost of production and minimize Return on Investment
(c) Minimize project net present value and maximize project expense
(d) Maximize total annualized cost and Minimize process yield of mass product

**TNPSC AE-2019**

**Ans. (a)**

**67. Quadratic programming is concerned with the non-linear program of optimizing the quadratic objective function subject to**
(a) Linear equality constraints
(b) Linear inequality constraints
(c) Linear equality or inequality constraints
(d) Non-Linear equality or inequality constraints
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**68. A suitable tool to understand the interface between the process and utility system is**
(a) Composite curve
(b) Grand composite curve
(c) Areas of integrity
(d) Stochastic curve
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**69. Hessian Matrix of an objective function is positive-definite, then stationary point can be called as**
(a) Minimum (b) Maximum
(c) Global minimum (d) Global maximum
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**70. Simplex method begins at when**

(a) all the points are in feasible region
(b) the corner points are only at feasible region
(c) the intermediate points are within the feasible region
(d) an interior points are in the feasible region
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**71. In the relationship between the character of f(x) and the state of H(x). If f(x) is concave, then H(x) is**

(a) Positive - definite
(b) Positive - indefinite
(c) Negative - semidefinite
(d) Negative - definite
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**72. The general form of a constrained optimization problem does not involve**

(a) objective function to be optimized
(b) equality constraints
(c) inequality constraints
(d) auxiliary information of the process
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**73. The value of $X_2$ for the minimization problem $f(X) = 4X_1^2 + 5X_2^2$ subject to $2X_1 + 3X_2 = 6$ by direct substitution is**

(a) 1.286
(b) 1.071
(c) 1.52
(d) 1.77
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**74. An optimization problem of locating oil-wells to identify better production rate can be solved by**

(a) Quadratic programming
(b) Integer Programming
(c) Geometric programming
(d) Linear programming
(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**75. In an optimization problem, the ratio of the partial derivative of objective function to the partial derivative of constraints with respect to any one independent variable is termed as**

(a) Newton's multiplier
(b) Kuhr-Tucker multiplier
(c) Langrange multiplier
(d) Reynolds multiplier
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**76. Formulation of a design problem for specialty product requires**

(a) Physical properties
(b) Chemical properties
(c) Surface properties
(d) Functional properties
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**77. Choose the following condition which is to be satisfied for the local minimum ($x_{min}$) to be called as Global minimum (x*)**

(a) $x^* > x_{min}$ (b) $x^* < x_{min}$
(c) $x^* = x_{min} = 0$ (d) $x^* = x_{min}$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**78. Optimization of one phase of an operation while ignoring some factors that have an effect on the process of plant is known as**

(a) Constrained optimization
(b) Linear optimization
(c) Sub optimization
(d) Quadratic programming optimization
(e) answer not known

**TNPSC AE-2022**

**Ans. (c)**

**79. When the model is non linear in the coefficients, they can be estimated by**

(a) Least square method
(b) Non Linear progression
(c) Linear regression
(d) Non Linear regression
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**80. A polymer plant with a production capacity of 10,000 tons per year has an overall yield of 70%, on mass basis (kg of product per kg of raw material). The raw material costs Rs. 50,000 per ton. A process modification is proposed to increase the overall yield to 75% with an investment of Rs. 12.5 crore. In how many years can the invested amount be recovered with the additional profit?———**

**GATE-2014**

**Ans. (2.55 to 2.70)**

**81. Which of the following method converts organic matter in the solid waste into humus:**

(a) Incineration
(b) Composting
(c) Pulverization
(d) Leaching

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Composting

# 12.

# Process Dynamics Control & Instrumentation

## A. Laplace Transformation

**1. The Laplace transform of the function $e^{at}$ has the form**

(a) $\frac{1}{s-a}$ (b) $\frac{1}{s+1}$

(c) $\frac{1}{s(s+a)}$ (d) $a/s$

**TNPSC AE-2018**

**Ans. (a)**

**2. The Laplace transform of cost is**

(a) $\frac{1}{S^2+1}$ (b) $\frac{1}{S^2-1}$

(c) $\frac{S}{S^2+1}$ (d) $\frac{S}{S^2-1}$

**TNPSC AE-2018**

**Ans. (c)**

**3. Laplace transform of sinh k(t) is**

(a) $\frac{1}{s^2-k^2}$

(b) $\frac{1}{s^2+k^2}$

(c) $\frac{k}{s^2+k^2}$

(d) $\frac{k}{s^2-k^2}$

**TNPSC AE-2019**

**Ans. (d)**

**4. If L(Y) = laplace transform of output variable L(X) = laplace transform of input variable, then transfer function is defined as**

(a) L(X).L(Y) (b) L(X)–L(Y)

(c) L(Y)/L(X) (d) L(X)/L(Y)

**TNPSC AE-2019**

**Ans. (c)**

**5. Laplace transform of unit impulse input is**

(a) 0 (b) $\frac{1}{2}$

(c) 1 (d) 2

**TNPSC AE-2019**

**Ans. (c)**

**6. The Laplace transform of sin t is**

(a) $\frac{1}{S^2+1}$ (b) $\frac{S}{1+S^2}$

(c) $\frac{1}{S^2-1}$ (d) $\frac{S}{S^2-1}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**7. The Laplace transform of $e^{-at}$ cos Kt is**

(a) $\frac{s-a}{(s+a)^2+K^2}$ (b) $\frac{s+a}{(s+a)^2+K^2}$

(c) $\frac{s-a}{(s+a)^2-K^2}$ (d) $\frac{s+a}{(s+a)^2-K^2}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**8. The Lapalce transform of unit step function is**

(a) 1

(b) 0

(c) $\frac{1}{s}$

(d) s

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**9. The Laplace transform of a function is $\frac{s+1}{s(s+2)}$**

**The initial and final values, respectively, of the function are**

(a) 0 and 1 (b) 1 and $\frac{1}{2}$

(c) $\frac{1}{2}$ and 1 (d) $\frac{1}{2}$ and 0

**GATE-2017**

**Ans. (b)**

**10. A liquid flows through an "equal percentage" valve at a rate of $2m^3/h$ when the valve is 10% open. When the valve opens to 20% the flowrate increases to 3 $m^3/h$. Assume that the pressure drop across the valve and the density of the liquid remain constant. When the valve opens to 50%, the flowrate (in $m^3/h$, rounded off to the second decimal place) is—**

**GATE-2016**

**Ans. (10.00 to 10.25)**

**11. The Laplace transform of $e^{at}$ sin (bt) is**

(a) $\frac{b}{(s-a)^2+b^2}$ (b) $\frac{(s-a)}{(s-a)^2+b^2}$

(c) $\frac{(s-a)}{(s-a)^2-b^2}$ (d) $\frac{b}{(s-a)^2-b^2}$

**GATE-2016**

**Ans. (a)**

**12. The Laplace transform of 'e^(at)' sinhat is**

(a) '(a)/(s^2 – 2as + 2a^2))'
(b) '(a)/((s^2 – 2as))'
(c) '(s)/((s^2 + 2as + 2a^2))'
(d) '(s)/((s^2 + 2as))'

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**13. The inverse Laplace transform of the function $F(s) = \frac{1}{s(s+1)}$ is given by:**

(a) f(t) = sin (t) (b) $f(t) = 1 - e^{-t}$
(c) $f(t) = e^{-t}$ (d) $f(t) = e^{-t}$ sin (t)

**ISRO Scientist/Engineer-13.03.2016**

**Ans. (a) :**

**14. For an input forcing function $X(t) = 2t^2$, the Laplace transform of this function is :**

(a) $2/s^2$ (b) $\frac{4}{s^2}$

(c) $\frac{2}{s^3}$ (d) $\frac{4}{s^3}$

**AP PSC IOF-2009**

**Ans. (d)**

**15. For the time domain function $f(t) = t^2$, which ONE of the following is the Laplace transform of $\int_0^t f(t)dt$?**

(a) $\frac{3}{s^4}$ (b) $\frac{1}{4s^2}$

(c) $\frac{2}{s^3}$ (d) $\frac{2}{s^4}$

**GATE-2014**

**Ans. (d)**

## B. Linear open loop system (First order, Second order)

**16. Response of a linear control system for a change in load variables is called**

(a) Frequency response (b) Transient response
(c) Serro problem (d) Regulator problem

**TNPSC AE-2018**

**Ans. (c)**

**17. Mercury in glass thermometer in oil wall is**

(a) First – order system
(b) Second – order system
(c) Zero order system
(d) First – order system with time log

**TNPSC AE-2018**

**Ans. (b)**

**18. The transfer function of the system $\frac{Y}{X} = \frac{1}{S^2+5S+6}$ the roots of the characteristics equation are located,**

(a) To the left of imaginary axis and on real axis
(b) On the imaginary axis
(c) Right of imaginary axis
(d) At the origin

**TNPSC AE-2018**

**Ans. (a)**

**19. The transfer function for second order system is**

(a) $\frac{1}{T^2S^2+2_\varepsilon T_s+1}$ (b) $\frac{1}{T^2S^2+2TS+1}$

(c) $\frac{1}{TS^2+2_\varepsilon T+1}$ (d) $\frac{1}{T^2S^2+2T+1}$

**TNPSC AE-2018**

**Ans. (a)**

**20. Phase lag of first order system is**

(a) $\tan^{-1}(wT)$ (b) $-\tan^{-1}(wT)$
(c) $\pi/2$ (d) 0

**TNPSC AE-2019**

**Ans. (b)**

**21. For two non interaction first order systems connected Si series, the overall transfer function is the_____of individual transfer functions**

(a) Product (b) ratio
(c) sum (d) difference

**TNPSC AE-2019, 2018**

**Ans. (a)**

**22. The amplitude ratio of transportation lag is**

(a) 1 (b) 10

(c) 100 (d) 0.1

**TNPSC AE-2019**

**Ans. (a)**

**23. The time required for the response to first reach its ultimate value is**

(a) Time constant

(b) Response time

(c) Rise time

(d) Period of oscillation

**TNPSC AE-2019**

**Ans. (c)**

**24. Characterization of a dynamic system by a transfer function can be done only for _____ system.**

(a) quadratic (b) cubic

(c) polynomial (d) linear

**TNPSC AE-2019**

**Ans. (d)**

**25. The nature of roots of critically damped system is**

(a) Complex

(b) Real and unequal

(c) Connot be defined

(d) Real and equal

**TNPSC AE-2019**

**Ans. (d)**

**26. A rectangular tank is filled with a valve at the bottom and is used for storing a liquid. The area of cross-section of the tank is $10m^2$ and the flow resistance of the valve (assumed constant) is 0.1 $s/m^2$. The time constant of the tank will be:**

(a) 100 (b) 10.1

(c) 1 (d) 0.1

**TNPSC AE-2019**

**Ans. (c)**

**27. A stream containing dissolved solid is entering continuous at a flow rate of q in a mixing tank of volume V. The time constant of mixing tank is**

(a) $\frac{V}{q}$ (b) $\frac{2V}{q}$

(c) $\left(\frac{V}{q}\right)^{1/2}$ (d) $\left(\frac{V}{q}\right)^{2}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**28. Decay ratio is defined as**

(a) $\exp\left(\frac{-2\pi\xi}{\sqrt{1-\xi^2}}\right)$ (b) $\exp\left(-\frac{-2\pi\xi}{\sqrt{1+\xi^2}}\right)$

(c) $\exp\left(\frac{2\pi\xi}{\sqrt{1-\xi^2}}\right)$ (d) $\exp\left(\frac{2\pi\xi}{\sqrt{1+\xi^2}}\right)$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**29. Transfer function of a transportation lag is**

(a) $G(s)=\frac{Y(s)}{X(s)}=1+e^{-\tau_d s}$

(b) $G(s)=\frac{Y(s)}{X(s)}=\frac{1}{1+e^{-\tau_d s}}$

(c) $G(s)=\frac{Y(s)}{X(s)}=e^{-\tau_d s}$

(d) $G(s)=\frac{Y(s)}{X(s)}=e^{\tau_d s}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**30. The transfer function of the process in $\frac{1}{16S^2+8S+4}$. If a step change is introduced into a system, then the response will be**

(a) Underdamped (b) Overdamped

(c) Critically damped (d) None of these

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**31. the initial value (t = 0) of the unit step response of the transfer function $\frac{(s+1)}{(2s+1)}$ is**

(a) 2 (b) Zero

(c) 1 (d) 1/2

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**32. 1st order pade approximation for Transportation lag is given by**

(a) $1-\tau s/2$ (b) $1+\frac{\tau s}{2}$

(c) $\frac{1+\frac{\tau s}{2}}{1-\frac{\tau s}{2}}$ (d) $\frac{1-\frac{\tau s}{2}}{1+\frac{\tau s}{2}}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**33. The correct expression among the following is**

(a) Decay ratio = (Overshoot)

(b) Decay ratio = $(\text{Overshoot})^2$

(c) Decay ratio = $\frac{1}{\text{Overshoot}}$

(d) Decay ratio = $\frac{1}{(\text{Overshoot})^2}$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**34. The time constant ($\tau$) of a mixing process first order system is**

(a) $\tau = \frac{mc}{hA}$

(b) $\tau = AR$

(c) $\tau = \frac{V}{q}$

(d) $\tau = R.C$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**35. In second order under damped system which of the following is correct?**

(a) Overshoot = $(\text{Decay ratio})^3$

(b) Overshoot = $\frac{1}{\text{Decay ratio}}$

(c) Decay ratio = $\sqrt{\text{overshoot}}$

(d) Decay ratio = $(\text{overshoot})^2$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**36. U-tube manometer is an example for _____ order system.**

(a) $1^{st}$

(b) $2^{nd}$

(c) $3^{rd}$

(d) zero order

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**37. The process that contain a large transport lag ($\exp[-\tau_D S]$) is often used the control strategy by**

(a) First order process

(b) Dead time compensation

(c) Second order process

(d) IMC

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**38. In the second order underdamped system**

(a) Decay ratio = overshoot

(b) Decay ratio = $(\text{overshoot})^2$

(c) Decay ratio = $(\text{overshoot})^{0.33}$

(d) Decay ratio = $(\text{overshoot})^{0.5}$

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**39. An example of an open-loop second order underdamped system is**

(a) liquid level in tank

(b) U-tube manometer

(c) thermocouple in a thermo-well

(d) two non-interacting first order systems in series

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**40. The open-loop transfer function of a unity-gain feedback control system is given by**

$$G(S) = \frac{K}{(s+1)(s+2)}$$

**The gain margin of the system in dB is**

(a) 0

(b) 1

(c) 2

(d) $\infty$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**41. Consider the following transfer function :**

$$G(s) = \frac{3}{(5s+1)^2}$$

**where, the natural period of oscillation is in min. The amplitude ratio at a frequency of 0.5 rad/min is ____(rounded off to second decimal place).**

**GATE-2018**

**Ans. (0.41 to 0.42)**

**42. The decay ratio for a system having complex conjugate poles as**

$\left(-\frac{1}{10}+j\frac{2}{15}\right)$ **and** $\left(-\frac{1}{10}-j\frac{2}{15}\right)$ **is**

(a) $7 \times 10^{-1}$ (b) $8 \times 10^{-2}$

(c) $9 \times 10^{-3}$ (d) $10 \times 10^{-4}$

**GATE-2018**

**Ans. (c)**

**43. Math the items in Column A with the items in Column B**

| Column A | | Column B | |
|---|---|---|---|
| (P) | Pure dead-time | (I) | $\phi = -90^o$ |
| (Q) | Pure capacitive | (II) | $\frac{K_1}{s} - \frac{K_2}{\tau s+1}$ with $K_2 > \tau K_1$ |
| (R) | Inverse response | (III) | $0 < AR < 1$ |
| (S) | First-order process with unit gain | (IV) | $AR = 1$ |

**Here, $\phi$ denotes the phase shift, $K_1$ and $K_2$ the process gains, $\tau$ the time constant, and AR the amplitude ratio.**

(a) P-II, Q-III, R-IV, S-I
(b) P-III, Q-II, R-IV, S-I
(c) P-I, Q-IV, R-II, S-III
(d) P-IV, Q-I, R-II, S-III

**GATE-2018**

**Ans. (d)**

**44. The initial water level in a tank is 4 m. When the valve located at the bottom is opened, the rate of change of water level (L) with respect to time (t) is, $\frac{dL}{dt} = -k\sqrt{t}$, where k = 0.6 $ms^{-3/2}$. The level of water (in m) in the tank at time 0.5 s after opening the valve is _____ (rounded off to second decimal place).**

**GATE-2018**

**Ans. (3.7 to 3.9)**

**45. A first-order process having transfer function, $G_p = \frac{2}{7s+1}$ is controlled by a proportional controller with gain of 3.2. The process time constant is in minutes. Addition of the integral control action with an integral time constant of 5 minutes leads to increase in**

(a) offset
(b) speed of response
(c) order of the closed loops system
(d) proportional band

**GATE-2018**

**Ans. (c)**

**46. The transfer function of a system is**

$$\frac{1}{4s^2+1.2s+1}$$

**For a unit step increase in the input, the fractional overshoot, rounded to 2 decimal places, is ____.**

**GATE-2017**

**Ans. (0.36 to 0.38)**

**47. The 2nd order system with transfer function $4/(S^2+2S+4)$ has a damping ratio of.**

(A) 2 (B) 0.5
(C) 1 (D) 4
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (b)**

**48. What is the name of parameter associated with underdamped response which indicates the time required for the transient's damped oscillations to reach and stay within 2% of the steady-state value?**

(a) Peak time (b) Settling time
(c) Overshoot (d) Rise time

**HPCL Engineer-01.11.2022**

**Ans. (b)**

**49. Suppose that the gain, time constant and dead time of a process with the following transfer function:**

**$G_c(s) = 10 \exp(-0.1s)/(0.5s + 1)$ are known with a possible error of ± 20% of their values. The largest permissible gain Kc of a proportional controller needs to be calculated taking the values of process gain, time constant and dead time as**

(a) 8, 0.6, 0.08 (b) 12, 0.6, 0.12
(c) 8, 0.6, 0.12 (d) 12, 0.4, 0.08

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**50. Step response of a second order system shows 37% overshoot. The value of damping factor is**

(a) 0.37 (b) 0.5
(c) 0.3 (d) 0.2

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**51. Step response of a first order process Gp = 2/(3s + 1) for a unit step change in input is**

(a) $y = 1 - \exp(-t/3)$
(b) $y = 2(1 - \exp(-3t)$
(c) $y = 1 - \exp(-3t)$
(d) $y = 2(1 - \exp(-t/3)$

**RPSC ACF & FRO-24.02.2021**

**Ans. (d)**

**52. Pick out the first order system from the followings:**

(a) Damped vibrator
(b) Interacting system of two tanks in series
(c) Mercury in glass thermometer kept in boiling water
(d) Non-interacting system of two tanks in series

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**53. In feed-back control system G & H denote open-loop and close loop transfer function respectively. The output – input relationship is**

**A. G/(1+H)**

**B. H/(1+G)**

**C. G/H**

**D. H/G**

(a) A (b) B

(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**54. For an open transfer function**

$$G = \frac{k}{(S+1)(S+2)(S+3)}$$

**The number of poles are**

(a) Three (b) Two

(c) One (d) Zero

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**55. The transfer function of a pure dead time system with dead time $\tau d$ is**

(a) $\frac{1}{(\tau ds+1)}$ (b) $\tau ds + 1$

(c) $e^{-\tau ds}$ (d) $e^{\tau ds}$

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (c)**

**56. Signal normally used to study non-ideal flow by stimulus response technique is ____ input.**

(a) Pulse

(b) Step

(c) Both (a) & (b)

(d) Neither (a) nor (b)

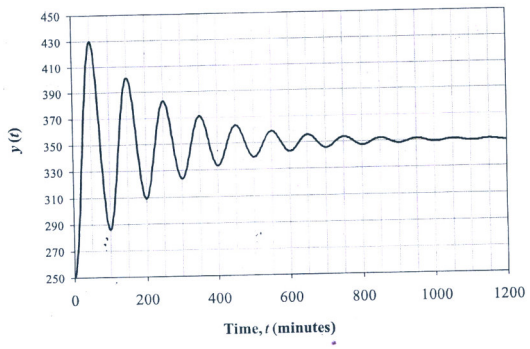
**Karnataka PSC ADF - 2015**

**Ans. (c)**

**57. For a unit step input, the response of a second order system is**

$$y(t) = K_p\left[1 - \frac{1}{\sqrt{1-\varsigma^2}} e^{-\frac{\varsigma\tau}{\tau}} \sin\left(\frac{\sqrt{1-\varsigma^2}}{\tau} t + \varnothing\right)\right]$$

**Where, $K_p$ is the steady state gain, $\zeta$ is the damping coefficient, $\tau$ is the natural period of oscillation and $\varnothing$ is the phase lag. The overshoot of the system is exp $\left(-\frac{\pi\varsigma}{1-\varsigma^2}\right)$.For a unit step input, the response of the system from an initial steady state condition at t = 0 is shown in the figure below.**




**What is the natural period of oscillation ( in seconds) of the system?**

(a) 15.9 (b) 50

(c) 63.2 (d) 100

**GATE-2016**

**Ans. (MTA)**

**58. A system exhibits inverse response for a unit step change in the input. Which one of the following statement must necessarily be satisfied?**

(a) The transfer function of the system has at least one negative pole

(b) The transfer function of the system has at least one positive pole

(c) The transfer function of the system has at least one negative zero

(d) The transfer function of the system has at least one positive zero

**GATE-2016**

**Ans. (d)**

**59. What is the order of response exhibited by a U-tube manometer?**

(a) Zero order (b) First order

(c) Second order (d) Third order

**GATE-2016**

**Ans. (c)**

**60. U-tube type manometer is**

(a) half order system

(b) second order system

(c) zero order system

(d) first order system

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**61. Time constant is the**

(a) Time taken by the controlled variable to reach 63.2% of its full change

(b) Same as transportation log

(c) Same as dead time

(d) Time required by the measured variable to reach 6.32 of its ultimate change

**ISRO-12.02.2016**

**Ans. (a)**

**62. Dead zone in instrumentation is the**

(a) Same as time constant
(b) Same as transportation log
(c) maximum change in the variable that does not change the reading of the instrument
(d) none of these

**ISRO-12.02.2016**

**Ans. (c)**

**63. With reference to the non-interacting/interacting systems, which of the statements is /are correct?**

**1. The response of two interacting first-order systems in series to a unit step change in input is either overdamped or critically damped.**

**2. Process exhibiting underdamped behaviour can be decomposed into two first-order systems in series (interacting or non-interacting).**

**Select the correct answer using the code given below.**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**64. Arrange the damping factor of the following responses of a second-order system to a unit step change in input in increasing order.**

**1. Underdamped response**

**2. Overdamped response**

**3. Critically damped response**

**4. Undamped response**

**Select the correct answer form the code given below.**

(a) 4, 1, 3 and 2 (b) 1, 3, 2 and 4
(c) 4, 3, 1 and 2 (d) 2, 3, 1 and 4

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**65. The time constant, of a first order system with resistance R and capacitance C is :**

(a) R + C (b) R – C
(c) RC (d) $\frac{1}{(RC)}$

**AP PSC IOF-2009**

**Ans. (c)**

**66. A control system has the following transfer function :**

$$F(s) = \frac{(s-1)(s+1)}{s(s-2)(s+4)}$$

**The initial value of the corresponding time function is :**

(a) 1 (b) 1/8
(c) 7/8 (d) –1

**AP PSC IOF-2009**

**Ans. (b)**

**67. A typical example of a physical system with under-damped characteristics is a :**

(a) U-tube manometer
(b) Spring-loaded diaphragm valve
(c) CSTR with first order reaction
(d) Thermocouple kept in a liquid-filled thermowell

**AP PSC IOF-2009**

**Ans. (b)**

**68. When the damping coefficient is unity the system is :**

(a) overdamped (b) critically damped
(c) underdamped (d) highly fluctuating

**AP PSC IOF-2009**

**Ans. (d)**

**69. The general transfer function of a 1st order system is :**

(a) $\frac{1}{\tau s - 1}$ (b) $\frac{A}{\tau s + 1}$
(c) $\frac{A}{\tau s - 1}$ (d) $\tau s + 1$

**AP PSC IOF-2009**

**Ans. (b)**

**70. An input which increases linearly with time is known as :**

(a) Step input (b) Impulse input
(c) Sinusoidal input (d) Ramp input

**AP PSC IOF-2009**

**Ans. (d)**

**71. When a bare thermocouple is covered by a protective sheath, the response becomes :**

(a) Faster and oscillatory
(b) Slower and non-oscillatory
(c) Faster and non-oscillatory
(d) Slower and oscillatory

**AP PSC IOF-2009**

**Ans. (b)**

**72. A process with transfer function , $G_P = \frac{2}{s-1}$ is to be controlled by a feedback proportional controller with a gain $K_c$. If the transfer functions of all other elements in the control loop are unity, then which ONE of the following conditions produces a stable closed loop response?**

(a) $K_c = 0.25$
(b) $0 < K_c < 0.25$
(c) $0.25 < K_c < 0.5$
(d) $K_c > 0.5$

**GATE-2014**

**Ans. (d)**

**73. A step change of magnitude 2 is introduced into a system having the following transfer function**

$$G(s) = \frac{2}{s^2 + 2s + 4}$$

**The percent overshoot is ———**

**GATE-2014**

**Ans. (16.0 to 16.8)**

**74. Assume that an ordinary mercury-in-glass thermometer follows first order dynamics with a time constant of 10 s. It is at a steady state temperature of 0ºC. At time t = 0, the thermometer is suddenly immersed in a constant temperature bath at 100 ºC. The time required (in s) for the thermometer to read 95 ºC, approximately is**

(a) 60 (b) 40
(c) 30 (d) 20

**GATE-2014**

**Ans. (c)**

**75. Integral of the time-weighted absolute error (ITAE) is expressed as**

(a) $\int_0^\infty \frac{|\varepsilon(t)|}{t^2} dt$ (b) $\int_0^\infty \frac{|\varepsilon(t)|}{t} dt$

(c) $\int_0^\infty t|\varepsilon(t)|dt$ (d) $\int_0^\infty t^2|\varepsilon(t)|dt$

**GATE-2014**

**Ans. (c)**

**76. A unit IMPULSE response of a first order system with time constant $\tau$ and steady state gain $K_P$ is given by**

(a) $\frac{1}{K_P\tau} e^{t/\tau}$ (b) $K_P e^{-t/\tau}$

(c) $\tau K_P e^{-t/\tau}$ (d) $\frac{K_P}{\tau} e^{-t/\tau}$

**GATE-2014**

**Ans. (d)**

**77. Which one of the following transfer functions, upon a unit step change in disturbance at t = 0, will show a stable time domain response with a negative intial slope (i.e. slope at t = 0):**

(1) $G(s) = \frac{1}{s+1} - \frac{2}{s+4}$

(2) $G(s) = \frac{1}{s+1} + \frac{2}{s+4}$

(3) $G(s) = \frac{1}{s+1} + \frac{2}{s-4}$

(4) $G(s) = \frac{1}{s-1} + \frac{2}{s-4}$

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (a)**

**78. The transfer function for the disturbance response in an open-loop process is given by $G_d^{open}(s)$. The corresponding transfer function for the disturbance response in a closed-loop feedback control system with proportional controller is given by $G_d^{closed}(s)$. Select the option that is ALWAYS correct {O[G(s)] represents order of transfer function G (s)}:**

**(1)** $O[G_d^{open}(s)] = O[G_d^{closed}(s)]$

**(2)** $O[G_d^{open}(s)] \neq O[G_d^{closed}(s)]$

**(3)** $O[G_d^{open}(s)] \geq O[G_d^{closed}(s)]$

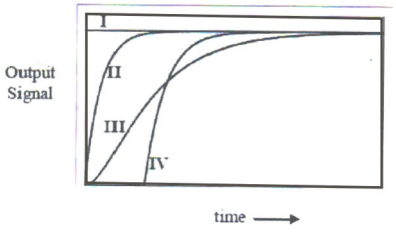
**(4)** $O[G_d^{open}(s)] \leq O[G_d^{closed}(s)]$

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (d)**

**79. Match the output signals as obtained from four measuring devices in response to a unit step change in the input signal.**




**P. Gas chromatograph, with a long capillary tube**
**Q. Venturi tube**
**R. Thermocouple with first order dynamics**
**S. Pressure transducer with second order dynamics**

**(1) P- IV, Q-III, R-II, S-I**
**(2) P-III, Q-I, R-II, S-IV**
**(3) P-IV, Q-I, R-II, S-III**
**(4) P-II, Q-IV, R-III, S-I**

(a) 1 (b) 2
(c) 3 (d) 4

**GATE-2015**

**Ans. (c)**

**80. The unit of time constant of a system is the same as that of:**

(a) Velocity (b) Time
(c) $Time^{-1}$ (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**81. When the damping coefficient is unity, the system is:**

(a) Overdamped
(b) Critically damped
(c) Under damped
(d) Highly fluctuating

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**82. Dead zone is the:**

(a) Same as time constant
(b) Same as transportation lag
(c) Maximum change in the variable that does not change the reading of the instrument
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**83. Response of as system to a sinusoidal input is called_____response.**

(a) Impulse (b) Unit step
(c) Frequency (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**84. Which of the following input increases linearly with time:**

(a) Impulse (b) Ramp
(c) Step (d) Sinusoidal

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Ramp

**85. Damping coefficient (ε) for an under damped second order response is:**

(a) Less than 1
(b) Greater than 1
(c) Equal to 1
(d) Equal to zero

**MPPSC Forest Service Exam-2014**

**Ans. (a) :** Less than 1

## C. Linear closed loop system

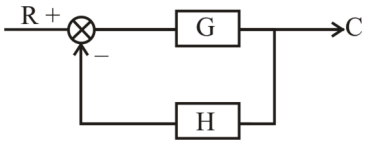
**86. Regular problem means that**

(a) Set point is constant
(b) Load is constant
(c) both Set point and load are constant
(d) Set point and load are variable.

**TNPSC AE-2019**

**Ans. (a)**

**87. The transfer function $\frac{C}{R}$ is**




(a) $\frac{1}{1+GH}$ (b) $\frac{G}{1+GH}$
(c) $\frac{H}{1+GH}$ (d) $\frac{G}{1-GH}$

**TNPSC AE-2019**

**Ans. (b)**

**88. The root locus plot of the roots of the characteristics equation of a closed loop system having the open loop transfer function $\frac{K(s+1)}{2(2s+1)(3s+1)}$ will have a definite number of loci for variation of K from 0 to ∞. The number of loci is**

(a) 1 (b) 3
(c) 4 (d) 2

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

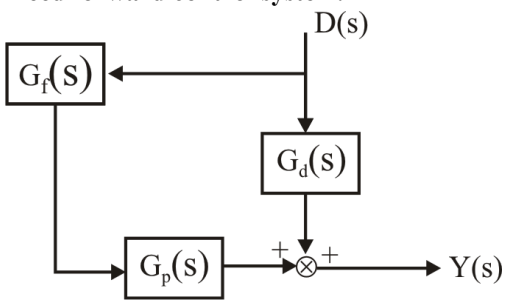
**89. The initial value ($t = 0^+$) of the unit step response of the transfer function $\frac{(s+1)}{(2s+1)}$ is**

(a) 0 (b) 1/2
(c) 1 (d) 2

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**90. Given below is a simplified block diagram of a feed forward control system.**



**The transfer function of the process is $G_P = \frac{5}{s+1}$ and the disturbance transfer function is $G_d = \frac{1}{s^2+2s+1}$. The transfer function of the PERFECT feed forward controller, $G_f(s)$ is**

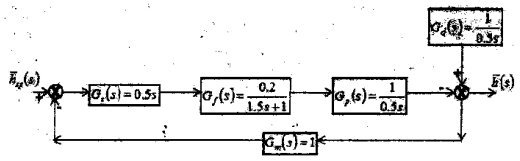
(a) $\frac{-5}{(s+1)^3}$ (b) $\frac{-5}{(s+1)}$

(c) $\frac{-1}{5(s+1)}$ (d) $-5(s+1)$

**GATE-2014**

**Ans. (c)**

**91. The block diagram for a process with feedback control for output deviation variable h is shown in the figure below. All transfer functions are given with pre-factor of s in minutes. A unit step change is made in the set-point at t = 0. The time required for h to reach 50% of its ultimate value, in minutes (up to two decimal places), is: _____**




**GATE-2015**

**Ans. (0.85 to 0.9)**

**92. Response of a linear control system for a change in set point is called :**

(a) Frequency response
(b) Transient response
(c) Serve problem
(d) Regulator problem

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

## D. Controllers

**93. For a tank temperature control system, use of proportional derivative controller would result in the**

(a) decrease of the phase lag for increasing frequencies
(b) increase of the phase lag for increasing frequencies
(c) decrease of the phase lag at all frequencies
(d) increase of the phase lag at all frequencies

**TNPSC AE-2018**

**Ans. (c)**

**94. Offset can be completely eliminated by the use of**

(a) Proportional controller
(b) Proportional derivative controller
(c) Proportional integral controller
(d) Proportional integral derivative controller

**TNPSC AE-2018**

**Ans. (c)**

**95. It excessive oscillations had to be eliminated, _____action will be added.**

(a) Proportional
(b) Integral
(c) Derivative
(d) Proportional Integral

**TNPSC AE-2019**

**Ans. (c)**

**96. Reset rate is defined as**

(a) $\frac{1}{T_I}$ (b) $T_I$

(c) $\frac{1}{T_D}$ (d) $T_D$

**Where $T_I$ = integral time, $T_D$ = Derivative time**

**TNPSC AE-2019**

**Ans. (a)**

**97. Find the ultimate gain and frequency for a proportional controller in case of a process having the transfer function.**

$$G_p(s) = \frac{1}{(4s+1)(2s+1)(s+1)}$$

(a) $w = \left(\frac{7}{8}\right)^{0.5}, K_c = \frac{45}{4}$

(b) $w = \frac{1}{14.5}, K_c = \frac{45}{7}$

(c) $w = 1, K_c = 13$

(d) $w = \left(\frac{7}{6}\right)^{0.5}, K_c \frac{46}{3}$

**UPPSC AE-29.05.2022, Part-I**

**Ans. (d)**

**98. Solenoid value works like**

(a) P controller (b) P- I controller
(c) P- D controller (d) On- off controller

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**99. Which of the following controllers has the least maximum deviation?**

(a) P controller (b) PI controller
(c) PID controller (d) PD controller

**RPSC Lect.-15.01.2016**

**Ans. (d)**

**100. Most expensive controller is**

(a) P controller (b) PI controller

(c) PID controller (d) PD controller

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**101. The open loop transfer function of a process with a proportional controller (gain $K_c$) is**

$$G_{OL} = K_C \frac{e^{-2S}}{S}$$

**Based on the Bode criterion for closed-loop stability, the ultimate gain of the controller rounded to 2 decimal place, is _____.**

**GATE-2017**

**Ans. (0.77 to 0.80)**

**102. The characteristic equation of a closed-loop system is**

$6s^3 + 11s^2 + 6s + (1 + K) = 0$, **Where** $K > 0$

**The value of K beyond which the system just becomes unstable, rounded to the nearest integer, is _____.**

**GATE-2017**

**Ans. (10 to 10)**

**103. Which of the following controllers has maximum offset?**

(a) P-Controller (b) P-D Controller

(c) P-I Controller (d) PID Controller

**RPSC ACF-2011**

**Ans. (a)**

**104. The integral dial on a controller is calibrated in:**

(a) Integral units

(b) Gain units

(c) Percentage

(d) Minutes or repeats /min

**RPSC ACF-2011**

**Ans. (d)**

**105. Which of the following controller has the maximum offset?**

(a) P- controller (b) PI- controller

(c) PD- controller (d) PID- controller

**RPSC ACF-2011**

**Ans. (a)**

**106. Most expensive controller is**

(a) P-controller (b) PI-controller

(c) PID-controller (d) PD-controller

**RPSC ACF-2011**

**Ans. (c)**

**107. Pneumatic controller are used to control upto a distance of about**

(a) 350 m (b) 500 m

(c) 1000 m (d) 5000 m

**RPSC ACF-2011**

**Ans. (*)**

**108. In on-off control, the value of the controler output depends only upon:**

(a) The amount of offset in the system

(b) The sign of the error

(c) The rate at which the error has occurred

(d) How long the error has existed

**RPSC ACF-2011**

**Ans. (b)**

**109. A P & I controller with an error of 9%, a gain of 2.3 and a reset of 0.94 min/repeat in open loop would produce an output signal that would ramp at the rate of:**

(a) 42.7%/min (b) 19.5%/min

(c) 20.7%/min (d) 22.0%/min

**RPSC ACF-2011**

**Ans. (d)**

**110. The transfer function of proportional-derivative (PD) control is:**

(a) $P(s)/\in(s) = K_c + (1 + \tau_D s)$

(b) $P(s)/\in(s) = 1/(K_c + \tau_D s + 1)$

(c) $P(s)/\in(s) = K_c + \tau_D s$

(d) $P(s)/\in(s) = K_c/(\tau_D s + 1)$

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**111. A PI controller with integral time constant of 0.1 min is to be designed to control a process with transfer function**

$$G_p(s) = \frac{10}{s^2 + 2s + 100}$$

**Assume the transfer functions of the measuring element and the final control element are both unity ($G_m = 1$, $G_f = 1$). The gain (rounded off to the first decimal place) of the controller that will constitute the critical condition for stability of the PI feedback control system is—**

**GATE-2016**

**Ans. (2.49 to 2.51)**

**112. Which of the following controllers has maximum offset?**

(a) P controller (b) P-I controller

(c) P-D controller (d) P-I-D controller

**ISRO-12.02.2016**

**Ans. (a)**

**113. Which of the following is NOT correctly matched?**

| | |
|---|---|
| (a) Stability of feedback control system | – Process reaction curve method |
| (b) Proportional control | – Produces an offset for all processes except those with terms 1/5 in their transfer function. |
| (c) Integral control | – Eliminates any offset |
| (d) Derivative control | – Anticipates future errors and introduces appropriate action |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**114. Bode stability method uses**

(a) Close loop transfer function
(b) Open loop transfer function
(c) Either (a) or (b)
(d) Neither (a) nor (b)

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**115. Given below are two statements, one labelled as Assertion (A) and the other as Reason (R).**

**Assertion (A):** For liquid level control, proportional controller cannot be used.

**Reason (R):** Transfer function of liquid level systems includes the term 1/5 for which proportional control does not exhibit offset for unit step changes in the set point.

**Choose the correct answer from the code given below.**

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(c) (A) is true, but (R) is false
(d) (A) is false, but (R) is true

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (d)**

**116. Solenoid valve works like :**

(a) Proportional controller
(b) P-D controller
(c) P-I-D controller
(d) On-off controller

**AP PSC IOF-2009**

**Ans. (d)**

**117. On-off control is a special case of :**

(a) Proportional controller
(b) Proportional integral controller
(c) P-D controller
(d) P-I-D controller

**AP PSC IOF-2009**

**Ans. (a)**

**118. Which of the following controllers has maximum off-set ?**

(a) Proportional controller
(b) P-I controller
(c) P-D controller
(d) P-I-D controller

**AP PSC IOF-2009**

**Ans. (a)**

**119. The transfer function of an ideal proportional controller is :**

(a) $k_C$ (b) $\frac{1}{k_C}$

(c) $1 + k_C$ (d) $\frac{1}{k_C + 1}$

**AP PSC IOF-2009**

**Ans. (b)**

**120. Which of the following controller has maximum offset?**

(a) P controller
(b) P-I controller
(c) P-D controller
(d) P-I-D controller

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**121. Process degree of freedom indicates_____ number of controller be used.**

(a) The maximum (b) The minimum
(c) Both (A) and (B) (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**122. Which one of the following type of a control system is concerned with the controlling a sequence of events rather than regulation or variation of individual variables?**

(a) Connected state control system
(b) Non Discrete-state control system
(c) Discrete-state control system
(d) Digital state control system

**UPSC Poly. Letc.-17.03.2019**

**Ans. (c)**

**123. Which of the following controllers have maximum offset:**

(a) PID (b) PI

(c) PD (d) P

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** P

## E. Closed loop transfer function

**124. For a feedback control system to be stable, the**

(a) Roots of the characteristic equation should be real

(b) Poles of the closed loop transfer function should lie in the left half of the complex plane.

(c) Bode plots of the corresponding open loop transfer function should decrease.

(d) Poles of the closed loop transfer function should lie in the right half of the complex plane.

**TNPSC AE-2019**

**Ans. (b)**

**125. For a closed-loop system, consider the following transfer functions:**

**process $G_P(s)$, controller $G_c(s)$, measuring device $G_m(s)$, and final control element $G_f(s)$**

$$G_p(s)=\frac{2}{7s+1};G_c(s)=2;G_m(s)=1;G_f(s)=1$$

**The offset in the closed loop response due to a unit step change introduced in the set point of the output variable is _____**

**GATE-2018**

**Ans. (0.2 to 0.2)**

**126. In any process application feedback control requires?**

(a) a remote loop (b) a open loop

(c) a closed loop (d) a internal loop

**RPSC ACF-2011**

**Ans. (c)**

**127. Final control element is a**

(a) Valve (b) Switch

(c) Signal (d) None of these

**RPSC ACF-2011**

**Ans. (a)**

**128. For a system with m inputs and n outputs, the number of transfer functions required to relate all outputs to all inputs is**

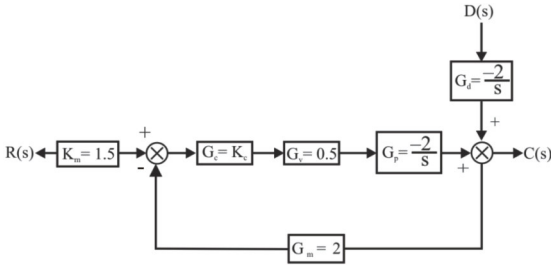
(a) m + n (b) mn

(c) (m + n)/2 (d) (m + n)/mn

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**129. Consider the following block diagram for a closed-loop feedback control system**




**A proportional controller is being used with $K_c = -4$. If a step change in disturbance of magnitude 2 affects the system, then the value of the offset is ——**

**GATE-2014**

**Ans. (0.49 to 0.51)**

## F. Stability analysis

**130. A stable system is one that produces**

(a) a bounded output for unbounded input

(b) a unbounded output for a bounded input

(c) a bounded output for bounded input

(d) a unbounded output for any type of input

**TNPSC AE-2018**

**Ans. (c)**

**131. Number of poles in a system with transfer function $\frac{2S+1}{S^2+3S+1}$**

(a) 0 (b) 1

(c) 2 (d) 3

**TNPSC AE-2018**

**Ans. (c)**

**132. The system is stable. Using the theorems of the Routh test choose the wrong answer**

(a) No change is sign in the first column

(b) No – roots having positive real parts

(c) All the coefficients are positive

(d) Roots having positive real parts

**TNPSC AE-2018**

**Ans. (d)**

**133. The control action is removed from the controller by placing it in manual mode and an open-loop transient is induced by a step change in the signal to the valve and this method of tuning is called as**

(a) Zigler – Nichols Tuning

(b) Cohen and Coon Tuning

(c) Closed – loop Tuning

(d) ITAE Tuning

**TNPSC AE-2019**

**Ans. (b)**

**134. According to Bode stability criterion a system is unstable if the open loop frequency response exhibits an amplitude ratio exceeding unity at frequency for which phase lag**

(a) $0^o$ (b) $45^o$

(c) $90^o$ (d) $180^o$

**TNPSC AE-2019**

**Ans. (d)**

**135. Routh stability method uses for_____loop transfer function.**

(a) Open

(b) Closed

(c) Open and Closed

(d) neither Open nor Closed

**TNPSC AE-2019**

**Ans. (b)**

**136. The important features of Bode Plot for the open loop transfer function of a feedback system is**

(a) $\phi$ is – 180° , where AR = 1

(b) $\phi$ is +180°, where AR = 1

(c) $\phi$ is – 90 °, where AR = 1

(d) $\phi$ is + 90 °, where AR = 1

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**137. The system to provide an algebraic method for determining the stability of a simple feedback control system is called as**

(a) Routh criterion

(b) Rout-Locus

(c) Bude-Stability

(d) Nyquist- Stability

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**138. Overall Amplitude Ratio (AR) is given by log AR =**

(a) $\log AR_1 + \log AR_2 + \ldots\ldots \log(AR)_n$

(b) $\log AR_1 . \log AR_2 \ldots. \log (AR)_n$

(c) $\log AR_1 - \log AR_2 \ldots\ldots - \log (AR)_n$

(d) $\log AR_1 - \log AR_2 + \log AR_3 \ldots - \log (AR)_n$

(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**139. The systems for which the output response is bounded for all bounded input is defined as**

(a) Unstable system

(b) Stable system

(c) Lag system

(d) Feed forward system

(e) Answer not known

**TNPSC AE-2022**

**Ans. (b)**

**140. In Bode stability criterion, amplitude ratio at 180º should be**

(a) 1 (b) <1

(c) >1 (d) 0

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**141. Stability of a control system containing a transportation lag can be best analysed by**

(a) Routh test

(b) Root locus method

(c) Frequency response method

(d) None of these

**RPSC ACF & FRO-24.02.2021**

**Ans. (c)**

**142. According to Bode stability criterion, a system is unstable if the open loop frequency response exhibits an amplitude ratio exceeding unity at frequency for which phase lag is**

(a) 0º (b) 45º

(c) 90º (d) 180º

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**143. Bode diagrams are generated from output response of the system subjected to which of the following input ?**

(a) impulse (b) step

(c) ramp (d) sinusoidal

**AP PSC IOF-2009**

**Ans. (d)**

**144. Bode stability method uses _____loop transfer function.**

(a) Open

(b) Closed

(c) Either (A) or (B)

(d) Neither (A) nor (B)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**145. A stable system is the one:**

(a) For which the output response is bounded for all bounded inputs

(b) Which exhibits an unbound response to a bounded input

(c) Which satisfies the conditions for a serve problem

(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**146. Rooth-Hurwitz procedure is used to determine:**

(a) Roots of the characteristic equation
(b) Check the stability of the closed loop system
(c) Calculate the zeros and poles of the transfer function
(d) None of these

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Check the stability of the closed loop system

## G. Frequency response

**147. The frequency at which maximum amplitude ratios attained is called _____frequency.**

(a) Cornor (b) Resonant
(c) Cross-over (d) Natural

**TNPSC AE-2018**

**Ans. (b)**

**148. The design value of the gain margin should be**

(a) Greater than 1.7 (b) Less than 1.7
(c) Equal to 1.7 (d) Equal to 0.59

**TNPSC AE-2018**

**Ans. (a)**

**149. The ratio of output amplitude to input amplitude in sinusoidal response is**
**Where $\tau$ = Time Constant**

(a) $\sqrt{\tau^2\omega^2+1}$ (b) $\sqrt{\tau^2\omega^2-1}$
(c) $\frac{1}{\sqrt{\tau^2\omega^2+1}}$ (d) $\frac{1}{\sqrt{\tau^2\omega^2-1}}$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**150. The frequency at which maximum amplitude ratio is attained is called**

(a) resonant frequency
(b) corner frequency
(c) cross-over frequency
(d) natural frequency

**RPSC Lect.-15.01.2016**

**Ans. (a)**

**151. Response of a linear control system for a change in set point is called**

(a) frequency response
(b) servo problem
(c) transient response
(d) regulator problem

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**152. Frequency response of a second order system will be sinusoidal when**

(a) Damping coefficient is zero
(b) It is under-damped
(c) It is over-damped
(d) It is critically-damped

**RPSC Lect.-15.01.2016**

**Ans. (b)**

**153. A first order system with unity gain and time constant $\tau$ is subjected to a sinusoidal input of frequency $\omega = 1/\tau$. The amplitude ratio for this system is**

(a) 1 (b) $\frac{1}{2}$
(c) $1/\sqrt{2}$ (d) $\frac{1}{4}$

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**154. A system demonstrates inverse response for a unit step change in the input. Which one of the following is true for such a system?**

(a) The transfer function of the system has at least one negative pole
(b) The transfer function of the system has at least one positive pole
(c) The transfer function of the system has at least one negative zero
(d) The transfer function of the system has at least one positive zero

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (d)**

**155. The frequency at which magnitude of G (jω) H (jω) is unity is known as_____**

(a) Cutoff frequency
(b) Resonant frequency
(c) Phase crossover frequency
(d) Gain crossover frequency

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**156. The frequency at which phase angle of G (j ω) H (J ω) is-180° is knowns as ——**

(a) Resonant frequency
(b) Gain crossover frequency
(c) Gutoff frequency
(d) Phase crossover frequency

**HPCL Engineer-01.11.2022**

**Ans. (d)**

**157. For the first order system, the corner frequency ($\omega_C$) is the frequency corresponding to:**

(a) $\omega.\tau = 1$ (b) $\omega.\tau = 1/\sqrt{2}$

(c) $\omega.\tau = 0$ (d) $\omega.\tau = 1/2$

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**158. Phase angle ($\varphi$) for the sinusoidal response of proportional controller with a transfer function $k_C$ is**

(a) 0° (b) –45°

(c) –90° (d) –30°

**HPCL Engineer-01.11.2022**

**Ans. (a)**

**159. The frequency, at which the amplitude ratio of the underdamped response of a second-order system to a unit step input is largest, is called**

(a) Natural frequency

(b) Resonant frequency

(c) Corner frequency

(d) Crossover frequency

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**160. Consider a control system with the open loop transfer function given by:**

$$G_{OL}(s) = \frac{K_c e^{-0.3s}}{1.5s + 1}$$

**In the above function, pre-factor of s is in minutes and $K_c$ is the gain of proportional controller. The frequency for phase margin of 30°is 4.04 rad/min. The value of $K_c$ for a gain margin of 1.7 (up to one decimal place) is _____**

**GATE-2015**

**Ans. (3.5 to 3.7)**

**161. Gain margin is equal to the :**

(a) Amplitude ratio

(b) Reciprocal of amplitude ratio

(c) Gain in P controller

(d) Gain in P-I controller

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**162. The frequency at which, maximum amplitude ratio is attained is called the_____frequency.**

(a) Corner (b) Resonant

(c) Cross-over (d) Natural

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**163. Phase lag of the sinusoidal response of a first order system is:**

(a) $120^o$ (b) $30^o$

(c) $180^o$ (d) $90^o$

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

## H. Control system with multiple loops

**164. If particular load disturbance occurs frequently in a control process, the quality of control can often be improved by the addition of**

(a) Feed backward control

(b) Feed forward control

(c) Cascade control

(d) Ratio control

**TNPSC AE-2018**

**Ans. (b)**

**165. Master controller in a cascade control system refers to**

(a) Secondary controller (b) Slave controller

(c) Cascade controller (d) Primary controller

**TNPSC AE-2018**

**Ans. (d)**

**166. Processes that contain a large transport lag can be controlled using**

(a) Cascade control system

(b) Feed forward control system

(c) Ratio control system

(d) Smith predictor controller system

**TNPSC AE-2018**

**Ans. (d)**

**167. In a control system configuration involves more than one measurement and one manipulation then the system is**

(a) Multiple loop system (b) Single loop system

(c) Distributed system (d) Simple system

**TNPSC AE-2019**

**Ans. (a)**

**168. The controller used for injection of a fuel-air mixture into furnace is**

(a) Cascade Controller
(b) IMC – Internal Modern Controller
(c) MPC – Model Predictive Controller
(d) Ratio Controller
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**169. During the start-up of the hypochlorate plant you have the opportunity to tune the three mode level controller installed on the inlet separator. Using the Ultimate Sensitivity Method you establish that the system oscillates with a period of 1.26 min. The proportional setting that creates the oscillations is 190%. The optimum settings for good control (One quarter amplitude decay QAD) using the Ziegler and Nichols method will be:**

(a) Proportional band = 317, Reset = 0.630 min/repeat, and 0.157 Rate-min
(b) Proportional band = 422, Reset = 0.504 min/repeat, and 1.180 Rate-min
(c) Proportional band = 380, Reset = 1.59 min/repeat, and 0.210 Rate-min
(d) Proportional band = 238, Reset = 0.840 min/repeat, and 0.140 Rate-min

**RPSC ACF-2011**

**Ans. (*)**

**170. Cascade control employs :**

(a) two feed-forward
(b) two feedbacks
(c) one feedback and one feed-forward
(d) one of the above

**AP PSC IOF-2009**

**Ans. (b)**

**171. Cascade control means:**

(a) Feed forward control
(b) More than feedback loop
(c) On-off control
(d) One feedback loop

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**172. If response of a control system is to be free of offset and oscillation, the most suitable controller is _____ controller.**

(a) Proportional
(b) Proportional-Derivative (PD)
(c) Proportional-Integral (PI)
(d) Proportional-Integral-Derivative (PID)

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**173. Ziegler-Nichols tuning technique is based on:**

(a) Process reaction curve
(b) Real process data from the system response
(c) Open loop frequency response analysis
(d) Closed loop frequency response analysis

**MPPSC Forest Service Exam-2014**

**Ans. (c) :** Open loop frequency response analysis

**174. Optimum controller setting for a PI controller as given by Ziegler Nichols is 0.45 $K_c$ max for which the reset time ($T_R$) is:**

(a) $T_R = P_u$ (b) $T_R = P_u/1.2$
(c) $T_R = P_u/2$ (d) $T_R = P_u/8$

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** $T_R = P_u/1.2$

## I. Instrumentation

**175. Choose the correct one from the following components and the respective conversions**

(a) Converter (pressure to flow rate)
(b) Control valve (current to pressure)
(c) Controller-recorder (current to temperature)
(d) Transducer (temperature to current)

**TNPSC AE-2018**

**Ans. (d)**

**176. Which one of the following is not a basic functional element of a measuring element?**

(a) Transducers
(b) Signal conditioners
(c) Data presentation elements
(d) Calibration element

**TNPSC AE-2018**

**Ans. (d)**

**177. _____Represents the degree of closeness of the measured value to the true value.**

(a) Precision (b) Accuracy
(c) Repeatability (d) Linearity

**TNPSC AE-2019**

**Ans. (b)**

**178. Ability to withstand over loads and operating condition is called as**

(a) Reproducibility (b) Linearity
(c) Accuracy (d) Ruggedness

**TNPSC AE-2019**

**Ans. (d)**

**179. The error is expressed as**

(a) $T_R/T_M$
(b) $T_R + T_M$
(c) $T_R - T_M$
(d) $T_R \times T_M$

**Where $T_R$ = desired temperature $T_M$ = measured temperature**

**TNPSC AE-2019**

**Ans. (c)**

**180. The degree of closeness with which an instrument responds to change in the measured variable is known as**

(a) precision (b) drift
(c) fidelity (d) sensitivity
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**181. The largest range of values of a measured variables to which the instrument does not respond is called as**

(a) Dead zone (b) Drift
(c) Sensitivity (d) Accuracy
(e) Answer not known

**TNPSC AE-2022**

**Ans. (a)**

**182. Which of the following is a static characteristic of instrument?**

(a) Fidelity
(b) Dynamic error
(c) Reproducibility
(d) Speed of Response

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**183. A first-order instrument is**

(a) two tanks connected in interacting mode
(b) vapour pressure thermometer (base type)
(c) two tanks connected in non-interacting mode
(d) U-tube manometer

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (b)**

**184. An LVDT (Linear Variable Differential Transformer) is a transducer used for converting**

(a) displacement to voltage
(b) voltage to displacement
(c) resistance to voltage
(d) voltage to current

**GATE-2017**

**Ans. (a)**

**185. A pressure gauge 0-100 pa has a guaranteed accuracy of 1% of full scale deflection. The limiting error while reading 25 Pa will be**

(a) 1% (b) 2%
(c) 2.4% (d) 4%

**RPSC ACF-2011**

**Ans. (d)**

**186. Active transducer is**

(a) Photo-emissive cell (b) Photo-voltaic cell
(c) Selsyn (d) All of these

**RPSC ACF-2011**

**Ans. (d)**

**187. The resistance of thermistor at 300K is 5000Ω. Its resistance temperature coefficient is 0.04/ºC a measurement with a lead resistance of 10Ω will cause an error of**

(a) 0.025 ºC (b) 0.05 ºC
(c) 0.075 ºC (d) 0.10 ºC

**RPSC ACF-2011**

**Ans. (b)**

**188. Which of the following relates the absorption and evolution of heat at the junction of thermocouple to the current flow in the circuit?**

**A. Seebeck effect**
**B. Peltier effect**
**C. Joule heating effect**
**D. Thompson effect**

(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (b)**

**189. Degree to which an instrument indicates the changes in measured variable without dynamic error is called**

(a) Speed of response (b) Fidelity
(c) Reproducibility (d) Lag

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (b)**

**190. Static characteristics of an instrument is**

(a) response (b) drift

(c) dynamic error (d) time lag

**Kerala PSC Asst. Prof.-2017**

**Ans. (b)**

**191. A 30 m metric chain is found to be 10 cm too short throughout a measurement. If the distance measured is recorded as 300m, what is the actual distance?**

(a) 300.1 m (b) 301.0 m

(c) 299.0 m (d) 310.0 m

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**192. Every measuring instrument has basic elements such as sensor, signal, conditioner, transmitter and indicator. For a mercury-in-glass thermometer, the signal conditioner is**

(a) Glass bulb containing mercury

(b) Capillary tube of the main stem of thermometer

(c) Graduated scale on the outer surface of the stem of thermometer

(d) None of these

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (b)**

**193. The undesirable static characteristics of instruments is**

(a) Accuracy (b) Reproducibility

(c) Drift (d) Sensitivity

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (c)**

**194. Which of the following judges the accuracy of an instrument?**

(a) Dead zone (b) Drift

(c) Static error (d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**195. Degree to which an instrument indicates the changes in measured variable without dynamic error is called its:**

(a) Speed of response

(b) Reproducibility

(c) Fidelity

(d) Static characteristics

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**196. Which of the following is a desirable characteristic of an instrument ?**

(a) High drift

(b) High fidelity

(c) High measuring lag

(d) Poor reproducibility

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**197. Which of the following shows maximum dip effect (indicating reverse direction of temperature change) ?**

(a) Mercury thermometer

(b) Radiation pyrometer

(c) Bimetallic thermometer

(d) Thermocouple

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**198. An aluminium rod of 10 m long at 20°C is expanding between temperature ranges of 0° C to 100° C . If the coefficient of thermal expansion $\gamma = 2.5 \times 10^{-5}$ m/°C, then the expansion of rod will be:**

(a) 20 mm (b) 25 mm

(c) 30 mm (d) 35 mm

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

## J. Measuring instruments (level, pressure, flow, temperature, concentration)

**199. A hot wire anemometer is a device used for measuring:**

(a) Viscosity

(b) Velocity of gases

(c) Pressure of gases

(d) Density of fluids

**OPSC Poly Lect.-2018, Paper-I**

**Ans. (b)**

**200. The accuracy of rotameter ranges from**

(a) ± 0.25% to ± 2% (b) ± 0.5% to ± 1.5%

(c) ± 1% to ± 3% (d) ± 1.5% to ± 3%

**TNPSC AE-2018**

**Ans. (a)**

**201. one inch of water is equal to**

(a) 6.9 k Pa (b) 14.7 Psi
(c) 133 Pa (d) 24 Pa

**TNPSC AE-2018**

**Ans. (d)**

**202. Flat glass gauges are suitable for clean liquids upto pressure of**

(a) 260 $Kg/cm^2$ (b) 260 $N/m^2$
(c) 260 kPa (d) 260 mmHg

**TNPSC AE-2018**

**Ans. (a)**

**203. Diaphragm box method is best suitable for measuring liquid levels in the range**

(a)20' to 250' (b) 20' to 250"
(c) 20" to 250' (d) 20" to 250"

**TNPSC AE-2018**

**Ans. (c)**

**204. The common industrial method of measuring pH is by glass cell and calomel electrodes used with a**

(a) Spectrophotometer (b) Potentiometer
(c) Turbidity meter (d) Refractometer

**TNPSC AE-2018**

**Ans. (d)**

**205. The optical pyrometer is usable in temperature range of_____**

(a) $100^oC$ and $400^oC$
(b) $4000^oC$ and $5000^oC$
(c) $700^oC$ and $3000^oC$
(d) $-700^oC$ and $-3000^oC$

**TNPSC AE-2019**

**Ans. (c)**

**206. Which one of the temperature measuring device can be used in Corrosive Environment, where other sensors cannot be used**

(a) Bimetallic thermometer
(b) Mercury filled thermometer
(c) Resistance temperature detector
(d) Radiation pyrometer

**TNPSC AE-2019**

**Ans. (d)**

**207. The output range of transducer is**

(a) 0-3 ma (b) 7-50 ma
(c) 30-40 ma (d) 4-20 ma

**TNPSC AE-2019**

**Ans. (d)**

**208. Humidity of gases can be measured by**

(a) Pitot tube
(b) Orificemeter
(c) Hygrometer
(d) Barometer

**TNPSC AE-2019**

**Ans. (c)**

**209. Laser Anemometer works on the principle of_____**

(a) Doppler phenomena
(b) Seeback Effect
(c) Raman's Effect
(d) Radiation Effect

**TNPSC AE-2019**

**Ans. (a)**

**210. Which one of the following is suitable for meaning the temperature of red hot moving object (eg. steel ingots on roller table)**

(a) Manometer
(b) Radiation pyrometer
(c) Thermister
(d) Thermometers

**TNPSC AE-2019**

**Ans. (b)**

**211. Match List -I with List - II and select the correct answer using code given below the lists.**

| List -I | | List- II | |
|---|---|---|---|
| A | Temperature | 1 | Hot wire anemometry |
| B | Pressure | 2 | Strain gauge |
| C | Flow | 3 | Chromatographic analyzer |
| | | 4 | Pyrometer |

**Code :**

| | A | B | C |
|---|---|---|---|
| (a) | 4 | 2 | 1 |
| (b) | 1 | 2 | 3 |
| (c) | 1 | 2 | 4 |
| (d) | 4 | 1 | 3 |

**UPPSC AE-29.05.2022, Part-I**

**Ans. (a)**

**212. Which is most suitable to measure a temperature of 2000°C ?**

(a) Platinum resistance thermometer
(b) Glass thermometer

(c) Radiation Pyrometer
(d) None of these

**UPPSC AE-29.05.2022, Part-I**

**Ans. (c)**

**213. Zero pressure on the gauge scale equals**
(a) 14.7 $kg/cm^2$
(b) $10^{-5}$ KPa
(c) 76 cm Hg
(d) $10^{-5}$ $N/m^2$
(e) Answer not known

**TNPSC AE-2022**

**Ans. (c)**

**214. The device used for velocity measurement is**
(a) Barometer
(b) Thermometer
(c) Bolometer
(d) Hotwire Anemometer
(e) Answer not known

**TNPSC AE-2022**

**Ans. (d)**

**215. Hollow Cathode Lamp (HCL) is an important part of**
(a) Atomic Absorption Spectrophotometer
(b) UV-Vis Spectrophotometer
(c) Flame photometer
(d) Colorimeter

**WB PSC Scientific Assistant-2019**

**Ans. (a)**

**216. Pyranometer is used for measuring**
(a) Measuring temperatures from 500 to 1000 ºC
(b) Measuring temperatures from 0 to 500 ºC
(c) Measuring solar radiation flux
(d) Collecting solar energy

**RPSC Lect.-15.01.2016**

**Ans. (c)**

**217. Which one of the following sensors is used for the measurement of temperature in a combustion process (T > 800 $^oC$)?**
(a) Type J thermocouple
(b) Thermistor
(c) Resistance temperature deterctor
(d) Pyrometer

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**218. Gas chromatography is used for the measurement of**
(a) temperature (b) pressure
(c) concentration (d) flow rate

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**219. The working principle of a Pirani gauge is based on**
(a) change in thermal conductivity
(b) change in ionization potential
(c) deformation of elastic body
(d) change in inductance

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (a)**

**220. The flapper nozzle is the primary device for all _____ instruments.**
(a) electronic (b) hydraulic
(c) pneumatic (d) electrical

**Haryana PSC Poly Lect.-12.06.2022**

**Ans. (c)**

**221. Match the variables in Group-1 with the instrument in Group-2**

| Group-1 | Group-2 |
|---|---|
| a) Temperature | I) Capacitance probe |
| b) Liquid level | II) McLeod gauge |
| c) Vacuum | III) Chromatograph |
| d) Concentration | IV) Thermistor |

**Choose the correct set of combination.**
(a) P-IV,Q-III, R-II, S-I
(b) P-I, Q-II, R-IV, S-III
(c) P-IV, Q-I, R-II, S-III
(d) P-III,Q-II, R-I, S-IV

**GATE-2017**

**Ans. (c)**

**222. The property of liquid that measures its resistance to flow is called:**
(a) Capillarity (b) Polarizability
(c) Resistivity (d) Viscosity

**RPSC ACF-2011**

**Ans. (d)**

**223. Which of the following instruments makes use of the standard electrode potentials and the variation of electrode potential with concentration ?**

(a) Polarimeter (b) pH meter
(c) Electrophorous (d) Electrophotometer

**RPSC ACF-2011**

**Ans. (d)**

**224. The series in which the electrode potentials of metals are arranged in an order is known as:**

(a) Electrical conductivity series
(b) Electrode potential series
(c) Electrochemical series
(d) Chemical affinity series

**RPSC ACF-2011**

**Ans. (c)**

**225. A barometer measures**

(a) Absolute pressure
(b) Gauge pressure
(c) Both absolute and gauge pressure
(d) Dynamic pressure

**RPSC ACF-2011**

**Ans. (a)**

**226. Voltmeter is a galvanometer with.**

(a) High resistance (b) Low resistance
(c) Both (a) and (b) (d) Uncertain

**RPSC ACF-2011**

**Ans. (a)**

**227. bimetallic thermometers are used to measure the temperature in the range of**

(a) –20 to 1600ºC (b) 300 to 1100ºC
(c) 800 to 2000ºC (d) –20 to 300ºC

**RPSC ACF-2011**

**Ans. (d)**

**228. Platinum resistance thermometer can be used upto**

(a) 419.58ºC (b) 630.74ºC
(c) 961.93ºC (d) 1064.43ºC

**RPSC ACF-2011**

**Ans. (b)**

**229. Measurement of pressure in ammonia reactor is done by**

(a) Bourdon gauge
(b) U-tube manometer
(c) Pirani gauge
(d) Inclined tube manometer

**RPSC ACF-2011**

**Ans. (a)**

**230. A force digital transducer, measures the pressure range of 0 – 200 N with a resolution of 0.1% of the full scale. The smallest change it can measure is.**

(A) 0.2 N (B) 0.4 N
(C) 0.5 N (D) 1.0 N
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (a)**

**231. Mc-leod gauge is a.**

(A) U tube filled with mercury
(B) U tube filled with low vapour pressure oil
(C) Liquid manometer along with pressure amplifier
(D) None of these
(a) A (b) B
(c) C (d) D

**ISRO Scientist/Engineer-04.06.2022**

**Ans. (c)**

**232. Which of the following thermocouples is capable of measuring a temperature of –50°C?**

(a) Platinum-platinum + rhodium
(b) Chromel –alumel
(c) Iron-constantan
(d) Copper-constantan

**RPSC ACF & FRO-24.02.2021**

**Ans. (b)**

**233. Anemometer is used to measure**

**A. Velocity** **B. Pressure**
**C. Viscosity** **D. Density**
(a) A (b) B
(c) C (d) D

**ISRO Technical Asstt.-04.06.2022**

**Ans. (a)**

**234. Method used for making electrical connections from the resistance thermometer to the measuring instrument is**

(a) Two lead circuit (b) Three lead circuit
(c) Four lead circuit (d) All of these

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (d)**

**235. Anemometer is used to measure**

(a) Velocity (b) Pressure
(c) Viscosity (d) Density

**ISRO Scientist/Engineer-25.02.2018**

**Ans. (a)**

**236. Match the instruments in Group-1 with process variables in Group-2**

| Group-1 | Group-2 |
|---|---|
| P Conductivity meter | I Flow |
| Q Turbine meter | II Pressure |
| R Piezoresistivity element | III Composition |

(a) P-II, Q-I, R-III (b) P-II, Q-III, R-I
(c) P-III, Q-II, R-I (d) P-III, Q-I, R-II

**GATE-2016**

**Ans. (d)**

**237. The resistance of an ideal voltmeter is _____**

(a) Zero (b) Low
(c) High (d) Infinite

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**238. Bimetallic thermometers are used to measure the temperature in the range of**

(a) –20º to 1600ºC (b) 300º to 1100ºC
(c) 800º to 2000ºC (d) –20º to 300ºC

**Kerala PSC Asst. Prof.-2017**

**Ans. (d)**

**239. Gas chromatography is used for measurement of**

(a) temperature (b) pressure
(c) concentration (d) flow rate

**Kerala PSC Asst. Prof.-2017**

**Ans. (c)**

**240. Emf generated in a thermocouple depends on the temperature**

(a) of cold junction only
(b) of hot junction only
(c) difference between hot and cold junctions
(d) difference between cold junction and atmosphere temperature

**ISRO-12.02.2016**

**Ans. (c)**

**241. For measuring the temperature of a red hot furnace, Which is the most suitable instrument?**

(a) Platinum resistance thermometer
(b) Thermocouple
(c) Optical pyrometer
(d) Bimetallic thermometer

**ISRO-12.02.2016**

**Ans. (c)**

**242. Match List-I with List-II and select the correct answer using the code given below the lists.**

| List-I | List-II |
|---|---|
| A. Mercury-in-glass thermometer | 1.Affinity of different constituents to an adsorbent |
| B.Gas Chromatograph | 2.Volumetric expansion of liquids |
| C. Radiation pyrometer | 3. Hydrometer |
| D. Specific gravity of petroleum | 4. Stefan-Boltzmann law |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 1 | 4 | 3 |
| (b) | 1 | 2 | 3 | 4 |
| (c) | 4 | 2 | 1 | 3 |
| (d) | 4 | 3 | 2 | 1 |

**UP PSC AE-13.12.2020, Paper-I**

**Ans. (a)**

**243. Composition on natural gas can be determined by the:**

(a) Chromatograph
(b) Orsat apparatus
(c) Spectrophotometer
(d) None of these

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**244. Pirani gauge is used for the measurement of**

(a) Very high pressure
(b) High vacuum
(c) Liquid level under pressure
(d) Liquid level at atmospheric pressure

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**245. The level of a liquid under pressure can be determined using:**

(a) Bubbler system
(b) Differential pressure manometer
(c) Diaphragm box system
(d) Air trap system

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**246. A barometer measures the_____pressure.**

(a) Absolute
(b) Gauge
(c) Absolute as well as gauge
(d) Dynamic

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (a)**

**247. Which of the following flow metering instruments is an areameter?**

(a) Venturimeter
(b) Rotameter
(c) Pilot tube
(d) Hot wire anemometer

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (b)**

**248. Thermocouple is suitable for measuring:**

(a) Liquid temperature only
(b) Very high temperature only
(c) Very low temperature only
(d) Both high and low temperature

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (d)**

**249. For measuring the temperature of a red hot furnace, which is the most suitable instrument?**

(a) Platinum resistance thermometer
(b) Thermocouple
(c) Optical pyrometer
(d) Bimetallic thermometer

**OPSC Poly Lect.-2018, Paper-II**

**Ans. (c)**

**250. Which one of the following pyrometers is designed to collect the radiation extending from the visible through the infrared wavelengths?**

(a) Broad band Pyrometer
(b) Total Radiation pyrometer
(c) Narrow band pyrometer
(d) Optical Pyrometer

**UPSC Poly. Letc.-17.03.2019**

**Ans. (b)**

**251. In radiation pyrometer the working principle is based on:**

(a) Seebeck effect
(b) Stefan-boltzamann law
(c) Ohm's law
(d) Curie's law

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Stefan-boltzamann law

**252. The principle behind ordinary mercury in glass thermometer is:**

(a) Decrease of vapor pressure with increase in temperature
(b) Increase of vapor pressure with increase in temperature
(c) Increase of electrical resistance with increase in temperature
(d) Increase in volume of mercury with increase in temperature

**MPPSC Forest Service Exam-2014**

**Ans. (d) :** Increase in volume of mercury with increase in temperature

**253. Hot wire anemometer is used for measuring:**

(a) Temperature of gases
(b) Velocity of gases
(c) Pressure of gases
(d) Temperature of liquids

**MPPSC Forest Service Exam-2014**

**Ans. (b) :** Velocity of gases

## K. Valve Characteristics

**254. In a certain process, one needs fluid flow in a given direction and the valve is to open or close by the fluid pressure. Which of the following valves permits fluid flow in one direction only?**

(a) Gate valve
(b) Globe valve
(c) Check valve
(d) Any of the above

**RPSC ACF-2011**

**Ans. (c)**

**255. A globe valve is most suitable for applications in which**

(a) The valve is required to be either fully open or fully closed
(b) Flow control is required
(c) The fluid contains dispersed particles
(d) One way flow is required

**ISRO Technical Asstt.-25.02.2018**

**Ans. (b)**